

Monitoring Racial Bias in Traffic Enforcement with Noisy Proxies of Driver Behavior

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Abstract

Racial bias in policing is well documented. Traffic stops represent the most common type of encounter in which civilians interact with police, which makes this an important setting for investigation of police behavior. Debates in this area frequently rely on evidence from benchmarks chosen for each case to evaluate the racial distribution of police stops. Often, benchmarks such as per capita demographics or not-at-fault drivers in vehicle collisions are used. However, these measures only approximate the intended comparison because they fail to (i) precisely define the causal contrast, (ii) accurately describe the nature of drivers visible to officers, (iii) explain disparities due to potentially biased selection decisions by officers. In this work we formally define racial bias as a violation of an individual fairness criterion: officer decision making should be ignorant of race conditional on the behavior of the civilian, i.e. whether or not they are driving dangerously. This latter, however, is unmeasurable. To address this issue, we build on the growing proximal causal inference literature to make use of cameras, crashes and checkpoints as negative outcome controls for identification, under certain assumptions which may be probed via a sensitivity analysis. We will extend the method’s viability to a host of imperfect data settings which plague policing, e.g. when race is mismeasured or unavailable. We finish with applications of the approach to a variety of U.S. police jurisdictions.

Keywords: benchmarking, policing, partial identification

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1 Introduction

Policing is among the most ubiquitous and discretionary functions of local government. In the United States, millions of civilians encounter police officers each year, most commonly through traffic enforcement. These encounters, though routine, play an outsized role in shaping public trust in law enforcement, and long standing concerns about racial discrimination have made traffic stops a central focus of empirical and legal scrutiny.

Increasingly, claims of racially discriminatory enforcement have been accompanied with statistical evidence. For example, a recent U.S. Department of Justice investigation determined that the Phoenix Police Department engaged in unconstitutional policing practices, reporting that Black drivers were 144% more likely than White drivers to be cited or arrested for low level moving offenses under similar circumstances ([U.S. Department of Justice, 2024](#)). Using data on redlight camera tickets in Phoenix over the study period, this estimate was derived by the following procedure: comparing only drivers on the road near the same camera during the same time of day, day of the week, month and year, the analyst evaluated the odds ratio

$$\frac{\#\{\text{Black in stops}\}}{\#\{\text{White in stops}\}} \bigg/ \frac{\#\{\text{Black on camera}\}}{\#\{\text{White on camera}\}}, \quad (1)$$

and found it equal to 2.44.¹ The principle underlying this comparison is such that racial composition of stopped motorists should resemble the racial composition of drivers who were “at risk” of being stopped under race neutral traffic enforcement, i.e. those committing low level moving offenses ([Ridgeway and MacDonald, 2010](#)). If the camera data can reliably

¹Interpreting an odds ratio of 2.44 as a 144% increase in the relative risk of being stopped is not technically accurate, and the approximation or equality of an odds ratio to the relative risk is valid only under certain conditions. However, it is worth highlighting that results from these studies are often marketed in that manner, a communication problem long identified ([Davies et al., 1998](#); [Holcomb et al., 2001](#); [Zhang and Yu, 1998](#)).

generate this distribution, when (1) is found to differ substantially from unity, it is suggestive of racially biased enforcement.

This empirical procedure is known in the literature as benchmarking, and traffic cameras in Phoenix are an example of a *benchmark*. Benchmarking has become the dominant analytical strategy for evaluating racial bias in traffic enforcement both in academic research and agency practice. Researchers have benchmarked stops against residential populations, licensed drivers, drivers involved in a vehicle collision, among others. Contemporary monitoring reports by police departments continue to rely on these designs, which we summarize in Table 1 along with their academic origins. In a large majority of applications, the odds ratio (1) is computed as a measure of racial bias (where “camera” is replaced by the relevant device).

A critical evaluation of this literature raises two primary statistical questions: which benchmarks most credibly reveal bias, and is the odds ratio the right way to measure that bias? Historically, the literature has primarily been concerned with the former. [Ridgeway and MacDonald \(2010\)](#) define a benchmark to be a measuring device which

“estimates the racial distribution of the individuals who would be stopped
if the police were racially unbiased.” (★)

Definition (★) motivates the aforementioned use of automated traffic cameras as a benchmark, where the “individuals who would be stopped if the police were racially unbiased” states more precisely the notion of being “at-risk” of a police encounter. Now, the commonly used benchmarks in Table 1 do not meet this criterion. Residential population measures include nondrivers. Licensed driver lists do not reveal information on daily driving patterns that could lead to an encounter with the police. While records of civilians in motor vehicle crashes improve on these substantially, as they better describe the driving population, if data determining the fault of the driver is absent, crashes do not isolate the violator (at-risk) population. Of the legacy benchmarks, only field observation plausibly falls under the above

definition because the researchers actually observe dangerous driving. However, this is an impractical benchmark, as large datasets are costly to generate and it remains subject to measurement error.

To our understanding, these problems are known by academics and practitioners, however, Table 1 reveals that methodological development has remained largely unevolved. Early critiques emphasized the importance of isolating at-risk drivers, with recent reviews echoing this point (Ridgeway and MacDonald, 2010; Withrow and Williams, 2015; Smith et al., 2021). Legal authorities rejected crude population denominators as failing to represent the driving population actually encountered by officers (U.S. Court of Appeals for the Seventh Circuit, 2001), and Department of Justice guidance similarly cautioned against simplistic comparisons (McMahon et al., 2002). It is worth noting that the popular veil of darkness test (Grogger and Ridgeway, 2006) was introduced precisely to avoid reliance on external benchmarks for the above reasons by restricting the analysis to stopped motorists.

Concerns regarding the choice of benchmark have been exacerbated by the empirical observation that changing the benchmark can produce wildly inconsistent measures of racial bias on the same stop data, leading to what the literature calls the *denominator problem* (Ayres, 2002). For example, a recent paper by Ratcliffe and Hyland (2025) uses stops by the Philadelphia Police Department to show that the odds ratio in (1) is disturbingly sensitive to the choice of benchmark, obtaining values anywhere between 0.5, indicating anti-White discrimination, and 4, indicating strong racial bias, see Figure 1. Each of the benchmarks considered has been employed in the literature, each with their own defended motivation. To our knowledge, no published work has specified a formal methodology that permits an analyst to rigorously choose which, if any, of those benchmark comparisons to report as credible evidence of discrimination.

This article meets that demand in the literature by tying the notion of a benchmark to a

well understood statistical object. Under the null of no racial discrimination, the rates of officer stops by race should reflect those in the true underlying rate of dangerous driving, since officers are expected to respond to behavior that threatens public safety. However, if race independently influences the decision to stop, the relative prevalence of minorities in stop data is distorted by racial bias. The central difficulty for any empirical study is that the behavior of the motorist observed by the officer is unobserved in the data. If, however, she had access to a variable that predicted dangerous driving but was uncontaminated by race in this sense, she could use it to benchmark officer decision making and recover the baseline rate of dangerous driving against which observed stops should be compared. In the statistics literature, this kind of variable is well studied, and is known as a negative control outcome, or outcome proxy (e.g. [Lipsitch et al., 2010](#)).

A central tenet of this work is the use of negative controls to test the presence, and identify the magnitude of racial discrimination in traffic enforcement across a variety of data environments. We start by introducing the first nonparametric causal model underlying the benchmark test, specifying the unit of analysis and data generating process governing a traffic stop, where unobserved dangerous driving behavior is clearly expressed as a core element of the model. Based on the above intuition, we show how benchmarks arise organically as negative controls within this model, which we then encapsulate in a definition, allowing analysts to be precise about what makes a valid benchmark. We will show the generality and wide applicability of the work by discussing three benchmarks in detail, automated traffic cameras, vehicle collisions and sobriety checkpoints. We define a causal estimand hitherto unspecified in the literature: the causal risk ratio for the effect of race on the decision to stop, conditional on the latent dangerous driving behavior. This estimand targets disparate treatment by race, where we ask if officers treat similarly situated drivers, *who are violating the same law*, differently by race. By specifying which identifying assumptions

tie the causal risk ratio to the odds ratio (1), we formally reveal the fragility of previous attempts of measuring discrimination. Throughout, we provide new identification results for the application of benchmarking to the study of racial bias in policing, accounting for several practical challenges that routinely appear in policing data, including low granularity and volume of policing records, and substitution effects between officer presence on the road and the observation by the benchmark.

A common data quality issue in policing studies is the absence or mismeasurement of race, limiting the computation of racial bias. Standard approaches to address this problem infer race via proxies of it, such as civilian surnames, e.g. using Bayesian Improved Surname Geocoding (Fiscella and Fremont, 2006; McCartan et al., 2025). These methods infer racial proportions by surname in data through the association between surname and race in data from the Census. A largely overlooked complication of this approach is that policing records are nonrandom samples of the census, meaning that unknown correlation between race and behavior which makes a civilian at-risk of being stopped by police distorts the relationship between name and race. We extend the core idea of this work to correct for this selection bias problem with each estimator we propose. We then document applications of these results in a variety of U.S. jurisdictions.

The paper is organized as follows. Section 2, outlines our proposed causal framework for benchmark analysis for three valid choices of benchmark, automated cameras (Section 2.2), checkpoints (Section 2.3), and crashes (Section 2.4). Section 4 describes estimation approaches tailored to each choice of benchmark. Section 5 implements analyses described Section 2, using data on cameras in Chicago (Section 5.1) and crashes in Texas (Section 5.2). Our empirical analysis is followed in Section 6 by a discussion of the modern literature in benchmarking and a prospectus for future research, including via the use of new telematic data sources.

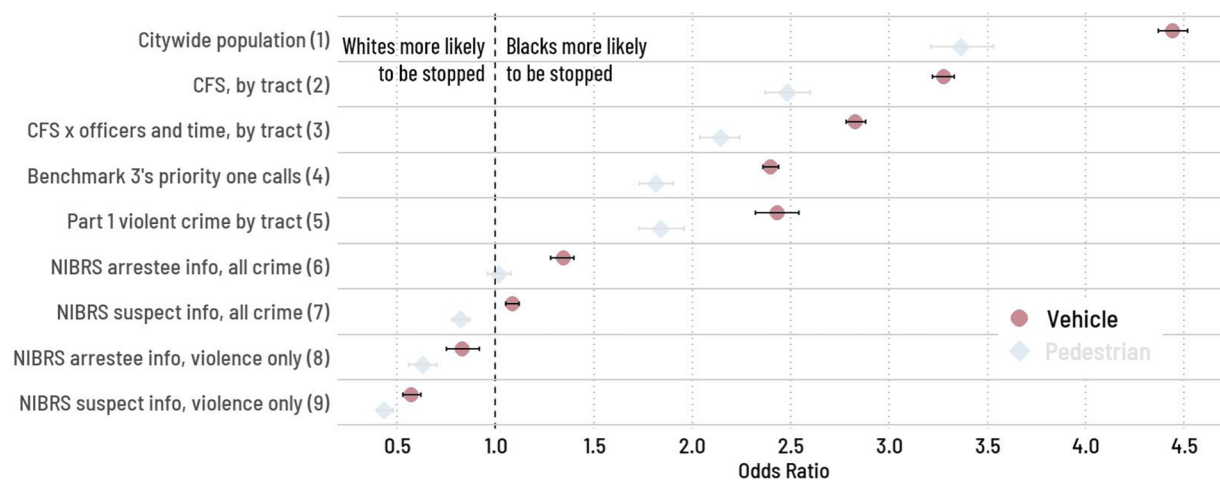


Figure 1: Reproduction of plot from [Ratcliffe and Hyland \(2025\)](#) which shows the sensitivity of the odds ratio to the choice of benchmark. For precise details on the benchmarks compared, consult their article.

Benchmark	Estimand	Academic Origins	Monitoring
Residential population	Share gap; Odds ratio	Smith and Petrocelli (2001) ; Rojek et al. (2004)	Texas Department of Public Safety (2025) ; Nebraska Statistical Analysis Center (2024) ; Virginia Division of Criminal Justice Services (2024) ; Missouri Attorney General (2024)
Licensed drivers	Share gap; Odds ratio	Engel et al. (2002)	Tennessee Comptroller of the Treasury, Office of Research and Education Accountability (2002) ; Metro Nashville Police Department (2018)
Drivers involved in a crash (often not-at-fault)	Share gap; Odds ratio	Alpert et al. (2004) ; Withrow and Williams (2015)	Wolfe et al. (2021) ; Metro Nashville Police Department (2018) ; Illinois Department of Transportation (2024) ; Washington State Patrol (2021) ; Portland Police Bureau (2025)
Field observation (road-side survey)	Share gap; Odds ratio	Zingraff et al. (2000) ; Lange et al. (2005)	Lamberth (1994) ; Lamberth (2003)
Veil of Darkness (VoD)	Odds ratio via logistic regression	Grogger and Ridgeway (2006)	Wolfe et al. (2021) ; Pierson et al. (2020)

Table 1: Benchmarks most commonly used in monitoring traffic enforcement. Here, ‘share gap’ refers to the percentage point difference between the prevalence of minorities in the benchmark and the stop data. Many earlier examples exist but we include contemporary examples to show the persistence of early approaches.

2 A causal framework for benchmark analysis

2.1 Notation, core definitions and estimands

We begin by describing the data generating process (DGP). Ideally, it would be recorded per mile a vehicle moves, whether it was stopped by a police officer, and we would compare officer decision making across civilian racial groups in the same driving context. This is of course not possible. In reality, our data for each vehicle contains only a snapshot of this process, that where a stop occurs. Conceptually, the unit of analysis is the encountered vehicle-mile driven: each unit in the data is a vehicle-mile driven by the motorist that resulted in an encounter with a police officer. Henceforth, we abbreviate ‘encountered vehicle-mile’ to ‘encounter’ or ‘stop’. Let ω denote this unit among all such in some sample space Ω . A vehicle stopped by police has $S(\omega) = s_1$ and $S(\omega) = s_0$ otherwise. The driver of the vehicle has race $R(\omega) \in \{\text{Black, Hispanic, White, other}\}$, where we will take baseline race $r_0 = \text{White}$. Furthermore, we might expect that this vehicle attracted police attention because it was moving in such a fashion to threaten safety on the road. In other words, the driving pattern of the vehicle could be labeled with some level of ‘dangerousness’, which we call $D(\omega) \in \{d_0, d_1\}$, where $D(\omega) = d_1$ indicates dangerous driving and $D(\omega) = d_0$ is safe driving. At this juncture a reader may object to this coarse conceptualization of dangerous driving, however, for benchmarks such as traffic cameras and sobriety checkpoints, we limit the analysis to infractions such as, speeding, bypassing a redlight or driving under the influence, where this binary description of dangerous driving is appropriate. However, when we consider the behaviors that may lead to a crash, indeed a reformulation of $D(\omega)$ is required, on which we will elaborate in Section 2.4.

The *context* in which the stop occurs is contained in a vector $X(\omega)$, such as the place and time of the stop. Factors unique to the individual inducing a correlation between driving



Figure 2: Directed acyclic graphs (DAG) describing the DGP for traffic enforcement. (a) Officer’s decision to stop is a function of driver race, dangerous driving, and context, all of which are correlated through hidden factors U . (b) Identical DGP with added benchmark C capturing race-neutral measurements of dangerous driving (e.g., automated cameras, crash reports, or checkpoints).

behavior, race and the context are contained in $U(\omega)$. The support of a random variable V will be denoted by $\mathcal{S}(V)$. See Figure 2a to visualize this DGP in a directed acyclic graph (Pearl, 2009). Used only to emphasise the unit of analysis, the argument ω will henceforth be suppressed unless that particular emphasis again serves the exposition.

In this article, we work within the potential outcomes framework of causal inference (Rubin, 1974), increasingly adopted in studies of racial bias, which is a causal question (Knox and Mumolo, 2020). The counterfactual $S(r)(\omega)$ represents whether an officer would have stopped a driver of race r in encounter ω . We next define racial bias.

Definition 2.1 (Racial bias). We say an encounter ω is racially biased against minority r relative to a White driver if² $S(r)(\omega) = s_1$ but $S(r_0)(\omega) = s_0$, with the driving behavior $D(\omega)$ and the context $X(\omega)$ held fixed.

²We will frequently omit the phrasing ‘relative to a White driver’ throughout the article, and it should be treated as implicit.

According to the US constitution ([U.S. Congress, 1868](#)), all *similarly situated* civilians are entitled to equal protection under the law irrespective of their features which do not describe their situation, e.g. race, gender, religion, etc. This motivates Definition 2.1. Within the same context $X(\omega)$, for the same behavior $D(\omega)$, an encounter ω initiated by a police officer should have that $S(r)(\omega) = S(r_0)(\omega)$ for all r . Owing to the fundamental problem of causal inference ([Holland, 1986](#)), we can only test the global null of no discrimination via an estimand on the average scale, which also gives a measure of its magnitude. To do so, for each $r \neq r_0$, within context x for dangerous driving behavior d_1 , define $\psi_r^\dagger(x)$ as the conditional causal risk ratio (CRR)

$$\psi_r^\dagger(x) := \frac{\Pr(S(r) = s_1 \mid D = d_1, X = x)}{\Pr(S(r_0) = s_1 \mid D = d_1, X = x)}. \quad (2)$$

We may generalize the effect to an entire jurisdiction by appropriately weighting the two composite terms of the CRR by how often dangerous drivers appear in that context, or

$$\psi_r^\dagger := \frac{\Pr(S(r) = s_1 \mid D = d_1)}{\Pr(S(r_0) = s_1 \mid D = d_1)} = \frac{\sum_x \Pr(S(r) = s_1 \mid D = d_1, X = x)w_x}{\sum_x \Pr(S(r_0) = s_1 \mid D = d_1, X = x)w_x} \quad (3)$$

where $w_x = \Pr(X = x \mid D = d_1)$, supposing X is discrete (otherwise, replace the sum with an integral over the density).³ Throughout the article as we discuss theory for identifying $\psi_r^\dagger$, the target race r will not matter and so we may drop the subscript.

We make three remarks on this choice. First, since the outcome is rare, an estimand on the risk difference scale is hard to interpret, motivating a ratio estimand. Second, in practice we will focus on the subgroup of violators $D = d_1$ to avoid dilution in the calculation of disparities ([Knox and Mummolo, 2025](#)). The vehicle-miles driven which incur a police stop are rare, and, as violation of the law is rare, combining outcomes among the entire

³The risk ratio is a collapsible measure ([Greenland et al., 1999](#); [Huitfeldt et al., 2019](#)), which means there exists a set of weights w'_x such that $\psi^\dagger = \sum_x \psi^\dagger(x)w'_x$, however these weights add unnecessary complexity to the identification strategy. As the two weighting schemes induce the same answer, we do not work with w'_x in this work.

population would misleadingly deflate estimates of discrimination. Additionally, there is selection in the data, as we only see units who were stopped, or captured by the benchmark. Hence, the recorded data is generally only informative for dangerous as opposed to safe driving.⁴

Before moving forward, we should highlight that estimand (2) is not the only definition of racial bias an analyst might consider. For instance, when in litigation, an organization pursuing civil claims against a police department must sometimes show that the department has *harmed* a particular subgroup of civilians (U.S. Department of Justice, Civil Rights Division, 2015), which can be argued via editing $\psi_r^\dagger$ and conditioning on the minority subgroup r

$$\psi_r^\dagger(x) := \frac{\Pr(S(r) = s_1 \mid D = d_1, R = r, X = x)}{\Pr(S(r_0) = s_1 \mid D = d_1, R = r, X = x)}. \quad (4)$$

For our purposes, as (D, X) is sufficient to control for confounding between R and S , $\psi_r^\dagger(x) = \psi_r^\ddagger(x)$. We focus on identifying $\psi_r^\dagger(x)$, as it is mathematically simpler. Another related estimand is the probability of necessity (Pearl, 1999; Tian and Pearl, 2000)

$$\psi_r^\S(x) = \Pr(S(r_0) = s_0 \mid S(r) = s_1, D = d_1, R = r, X = x), \quad (5)$$

which quantifies the extent to which the stop was attributable to race. Such estimands have become of increased interest due to their being a causal translation of the legal ‘but-for’ criterion used to evaluate claims of racial bias (Dawid et al., 2014, 2017; Cuellar, 2022). As the estimand conditions on the confounding variables D and X , it follows that whenever stops biased against White drivers—motorists for whom $S(r_0) = s_1$ and $S(r) = s_0$ (Knox

⁴Conditioning on $D = d_0$ would amount to studying disparities in *false charges*, i.e. how often minority drivers are erroneously pulled over by police relative to White drivers. Generally, due to selection on the outcomes this quantity is difficult to identify: common policing datasets do not have access to the prevalence of safe driving by race within X . However, the framework we develop here will be transportable to more favorable data environments, see Section 6 for a discussion.

et al., 2020)—do not exist we have that it is identified by a simple transformation of the causal risk ratio,

$$\psi_r^{\S}(x) = 1 - \frac{1}{\psi_r^{\dagger}(x)}, \quad (6)$$

which may additionally be reported as evidence of racial bias.

In Sections 2.2 to 2.3, we outline the identification strategy for $\psi^{\dagger}$. As the target estimands condition on the vector (D, X) which is sufficient to control for confounding, the primary empirical challenge for identification is that D is never observed in the data. If we could find a device which identifies units that are likely to exhibit $D = d_1$, then, under certain conditions, we can use this group as a comparison set for officers, and thus for identification. To develop this idea further, recall the definition $(\star)$ from the literature: a variable which captures the group of units in the population for whom we believe justify police attention, i.e. those with $D = d_1$. Intuitively, this device cannot be racially biased among those it identifies as driving dangerously, since this is precisely the behavior we aim to detect of the officers.

This intuition underlies how we should choose benchmarks for analyzing police behavior, so much so that we encapsulate it in a definition.

Definition 2.2 (Benchmark validity). A benchmark C is said to be a valid benchmark for outcome S if

$$\text{Cov}(C, D \mid X) \neq 0 \quad (\text{Relevance}) \quad (\text{B1})$$

$$C \perp\!\!\!\perp R \mid \{D, X\} \quad (\text{Exclusion}) \quad (\text{B2})$$

These conditions simply state mathematically the desirable properties of a benchmark we have discussed above. In other words, the benchmark C is a negative control outcome (NCO) for the decision to stop S (Tchetgen Tchetgen et al., 2024). We change the terminology in this article to better align with the subject matter. Negative control outcomes, or (outcome)

proxies, are used in many domains to detect or correct confounding influences in scientific studies, including both experimental and observational designs (Lipsitch et al., 2010; Arnold and Ercumen, 2016).

NOCs manifest naturally in the benchmarking framework but are nonstandard in their application in several ways. One is mechanical, we are working in a setting where the both the outcome and proxy are subject to selection: all data is conditional on $S = s_1$ or $C = c_1$, which requires special attention. Another is conceptual. The confounder, D , is not a nuisance but an integral element of the substantive question we pose (which is why it appears in the estimand). Rather than detect confounding bias, our goal is to leverage specific properties of the confounder–proxy relationship to detect a specific human bias, discrimination. In so doing, we contribute to the nascent literature on the use proxies in causal inference to detect discrimination, which has been limited to mediation analysis (Liu et al., 2025). We also contribute to the literature on measuring disparities in human decisions by showing that causal identification in observational data is possible even when the latent class in which units of equal merit ought to receive equal treatment is unobserved. Analogous challenges arise with unobserved ability in college admissions, unobserved creditworthiness in lending, and unobserved worker productivity in hiring, among others (Bhutta et al., 2025; Ding and Lehrer, 2014; Kallus et al., 2022).⁵ We briefly note that Han et al. (2024) also

⁵We remark that an assumption for a valid proxy that has become standard in the so-called proximal causal inference literature is the notion of ‘completeness’. This is a technical condition, but in effect, this condition states that the proxy C has enough variability to capture variability in D (Tchetgen Tchetgen et al., 2024). In the case where C and D are both discrete, this condition implies that $|\mathcal{S}(C)| \geq |\mathcal{S}(D)|$. We avoid this technicality in this work because each cases considered falls into at least one of the following categories: (i) a binary confounder with a binary proxy; (ii) $\Pr(C = c \mid D = d)$ is controlled or modeled directly, for both binary and continuous D . Furthermore, we also derive a partial identification result for $\psi^\dagger$, and work has motivated the bounding of causal effects with a single proxy by the express desire to avoid the completeness condition (Park et al., 2024; Ghassami et al., 2026).

emphasize that detecting discrimination rigorously requires an estimand conditional on latent behavior. Their analysis, however, differs from ours in two respects. First, they model the law enforcement process as inherently multistage, incorporating public reporting (such as 911 calls), dispatch, and enforcement, which precludes analysis of traffic enforcement by officers already on patrol. Second, their framework does not leverage external proxies to identify the causal estimand of interest; instead, it imposes structural assumptions on the reporting process and on the role of criminality to derive falsification tests for discrimination.

It remains to ask if benchmarks satisfying Definition 2.2 exist. This is not a trivial question, as the choice should be made in communication with the specific context of the study, akin to the selection of an instrumental variable. Here, we claim that there are three commonly available benchmarks which satisfy Definition 2.2. Variety in the choice of benchmark is useful because data in policing settings can often be hard to acquire—where one of these benchmarks is available, data providing, an estimation of racial bias is possible.

In the sequel, we will show that $\psi^\dagger$ is identified for three benchmarks for which data is commonly available: roadside traffic cameras, vehicle collisions, and checkpoints (for DUIs and equipment violations of HGVs). Each benchmark introduces new challenges which add nuance to the identification strategy, and so we treat each one separately.

2.2 Cameras

Traffic cameras are autonomous devices whose sole purpose is detect violations of the rules of the road wherever they are placed. Broadly, they fall into two types: speeding cameras, and redlight cameras. The former detects excessive speed while the latter a moving violation at a redlight. Cameras function by taking a photograph of the violating vehicle and its license plate, and automatically issuing a ticket for the corresponding violation. Any inference a camera makes is based on the motion of the vehicle, thus, clearly, a camera ticket is

predictive of dangerous driving. Moreover, this mechanism implies that camera tickets may only be associated with race via dangerous driving and context. Thus, cameras satisfy conditions B1 and B2.

2.2.1 Odds ratio: causal?

Leveraging cameras as a valid benchmark, we now discuss under what assumptions we may use them to identify $\psi^\dagger$. We will first connect our framework to current empirical practice by understanding under what assumptions the odds ratio (1) identifies a causal estimand. Then, we will show that many of these assumptions are unnecessary for informative conclusions about the risk ratio, which we illustrate with an interval estimand for the CRR. We keep notation from earlier sections, and additionally let $C \in \{c_0, c_1\}$ denote the binary issuance of ticket from a traffic camera.

In many jurisdictions, the express mission of road traffic enforcement is public safety (e.g., [Texas Department of Public Safety, 2026](#)). As a result, we can think of officers and cameras as two devices designed to extract dangerous drivers from the driving population, and by penalizing them, reduce the risk of injury on the road. Therefore, anyone in officer or camera data should be driving dangerously. If both officers and cameras are unbiasedly sampling from the population of dangerous drivers, within some context X , the distribution of race among the stopped and among the camera ticketed should be equal. This motivates the use of the odds ratio, widely used to capture the association between binary variables, as a measure of disparity. But, for this intuition to go through, we require that indeed officers and cameras are stopping units who were *actually* driving dangerously. Else, our data is contaminated by instances of safe drivers, while the CRR explicitly conditions on dangerous drivers. This observation brings forward our first two ingredients for connecting the odds ratio to a causal quantity.

Assumption A1 (Officers don't lie). Officer precision, $\Pr(D = d_1 \mid S = s_1)$, equals unity.

Assumption A2 (Cameras don't lie). Camera precision, $\Pr(D = d_1 \mid C = c_1)$, equals unity.

That is, everyone whom an officer or a camera cites was definitely driving dangerously at the point of measurement. Throughout the article, whenever either of the above precision rates is less than unity, we will refer to a record in the dataset where $D = d_0$ as an officer or camera *miscapture*, and we will speak of an officer or camera *capturing* a driver when it cites one. We have the following proposition.

Proposition 2.3. *Under Assumptions A1 and A2,*

$$\psi_r^\dagger(x) = \frac{\text{odds}_r[R \mid S = s_1, X = x]}{\text{odds}_r[R \mid C = c_1, X = x]} \quad (7)$$

where we define for discrete V and W of any type, $\text{odds}_v[V \mid W = w] := \Pr(V = v \mid W = w) / \Pr(V = v_0 \mid W = w)$, where v_0 is some chosen reference value in the support of V .

Proof. The proof follows the intuition of the earlier exposition. Within some X , Assumption A1 is equivalent to the implication $S = s_1 \implies D = d_1$, and a similar implication follows from Assumption A2. Therefore

$$\begin{aligned} \frac{\text{odds}_r[R \mid S = s_1, X = x]}{\text{odds}_r[R \mid C = c_1, X = x]} &= \frac{\Pr(R = r \mid S = s_1, X = x)}{\Pr(R = r_0 \mid S = s_1, X = x)} \bigg/ \frac{\Pr(R = r \mid C = c_1, X = x)}{\Pr(R = r_0 \mid C = c_1, X = x)} \\ &\stackrel{\text{A1 \& A2}}{=} \frac{\Pr(R = r \mid S = s_1, D = d_1, X = x)}{\Pr(R = r_0 \mid S = s_1, D = d_1, X = x)} \bigg/ \frac{\Pr(R = r \mid C = c_1, D = d_1, X = x)}{\Pr(R = r_0 \mid C = c_1, D = d_1, X = x)} \end{aligned}$$

By the exclusion condition B2 of the benchmark, we can now drop C from the conditioning set of the terms in the denominator. Thus

$$\begin{aligned} \frac{\text{odds}_r[R \mid S = s_1, X = x]}{\text{odds}_r[R \mid C = c_1, X = x]} &= \frac{\Pr(R = r \mid S = s_1, D = d_1, X = x)}{\Pr(R = r_0 \mid S = s_1, D = d_1, X = x)} \bigg/ \frac{\Pr(R = r \mid D = d_1, X = x)}{\Pr(R = r_0 \mid D = d_1, X = x)} \\ &= \psi_r^\dagger(x) \end{aligned}$$

by Bayes' Rule.⁶

□

Proposition 2.3 is an important result. For the first time in the benchmarking literature, the odds ratio for measuring disparities has been tied to a causal estimand. We can even say something stronger.

Proposition 2.4 (Honesty is a necessity). *Suppose the benchmark C is valid. The conditional odds ratio identifies $\psi^\dagger(x)$ if and only if Assumptions A1 and A2 hold.*

Proof. We prove sufficiency above, we show necessity in Appendix B.

□

Unless the analyst is willing to assume officers and cameras don't lie, the odds ratio cannot be tied to a causal risk ratio as is often attempted. This highlights a potential avenue through which benchmarking strategies in past literature had produced inconsistent estimates of discrimination. We reiterate that *both* the benchmark and the officer data must reflect the population at-risk. Proposition 2.4 shows that even among valid benchmarks, when these measures fail to capture these individuals, the odds ratio is misleading for the causal risk ratio, see the proof in Appendix B.

Above, we showed that the odds ratio is causal within some context X . Often, studies in this area seek to generalize their findings to an entire jurisdiction within some time frame. Aggregating odds ratios is a topic of much discussion in the causal literature, since, unlike risk ratios or risk differences, they are not collapsible (Greenland et al., 1999; Greenland, 2021; Pang et al., 2016; Huitfeldt et al., 2019). That is, absent very restrictive assumptions, there does not generally exist a weighting scheme using which the average of the odds ratios within each stratum produces the marginal odds ratio. Furthermore, the marginal odds ratio is not guaranteed to equal the marginal risk ratio. Marginal odds ratios are commonly

⁶For additional depth, we provide an alternative proof in the Appendix which directly ties this result to the effect restoration methodology of Kuroki and Pearl (2014).

used in the criminology literature and applied practice to describe discrimination. However, their application has largely been informal, despite their being used to fortify significant claims of racial bias. This methodological issue has been known for a long time; prominently described in older reviews such as [Ridgeway and MacDonald \(2010\)](#) with the discussion renewed in recent articles ([Smith et al., 2021, 2022](#)). Neglecting to make tight comparisons between officer stops and benchmarks no longer renders the units ‘similarly situated’, so a measure of discrimination based upon such a comparison is biased.

That said, it is possible to attach a causal interpretation to the marginal odds ratio albeit under very strict assumptions, which are the following:

Assumption A3 (Uniform policing intensity by race). $\Pr(S = s_1 \mid D = d_1, R = r, X = x)$ is constant in x for each r .

Assumption A4 (Uniform camera effectiveness). $\Pr(C = c_1 \mid D = d_1, X = x)$ is constant in x .

Assumptions [A3](#) and [A4](#) are strict homogeneity assumptions on the behavior of officers and cameras across the jurisdiction. [A3](#) says that policing intensity is constant across the *entire* jurisdiction, in a similar vein, [A4](#) implies that cameras make mistakes at the same rate everywhere. These assumptions are sufficient to identify $\psi^\dagger$ with the marginal odds ratio $\text{odds}_r[R \mid S = s_1] / \text{odds}_r[R \mid C = c_1]$ but at the cost of assuming constant treatment effects by context, since Assumption [A3](#) implies that $\psi^\dagger = \psi^\dagger(x)$ for all x . The proof of this result follows readily from the proof of Proposition [2.3](#).

Here, we have shown that only under Assumptions [A1-A4](#) does the marginal odds ratio have a causal interpretation. Now, the pertinent question is whether these assumptions are tenable in practice. First consider Assumption [A1](#), that officers don’t lie. In reality, concerns about officer credibility are well documented: a 2023 report from the Chicago

Office of Inspector General found that the Chicago Police Department continued to employ more than 100 officers with histories of filing false reports, and that structural failures in accountability systems had allowed these false statements to go unpunished or even resulted in some officers being promoted despite lying on the job ([CBS Chicago, 2023](#); [WTTW News, 2025](#)). Second, consider Assumption [A2](#), which treats automated traffic cameras as perfectly accurate measures of dangerous driving. For example, multiple investigations in Chicago have exposed serious issues with camera enforcement systems. A Chicago Tribune analysis of millions of redlight camera citations found that thousands of drivers were tagged for undeserved tickets due to anomalous spikes in camera enforcement, likely resulting from equipment faults or unexplained deviations in ticketing patterns ([Kravets, 2014](#)). Likewise, a 2015 investigation into the city’s speed camera program documented more than \$2.4 million in fines issued improperly—tickets generated when cameras were supposed to be inactive, warning signs were missing or confusing, or the legally required conditions for enforcement were not met ([Kidwell and Epton, 2015](#)). These empirical findings show that cameras are not infallible, and this must be accounted for in the analysis. For both officer and camera miscaptures we do not expect the rates to be high.

Third, considering Assumption [A3](#), it is important to note that police behavior is very heterogeneous in space and time (e.g. [Ba et al., 2021](#)). Traffic enforcement is often highly localized in order to align with departmental incentives, leading to diverse patterns in how officers police different neighborhoods across a city. Additionally, it is natural to expect that as we vary the context, the driving patterns, road conditions and community relationships will also evolve. We cannot reasonably hold Assumption [A3](#) in practice. Lastly, we argue that Assumption [A4](#) is reasonable in general. A violation would require that manufacturing defects or malfunctions in camera systems are systematically related to the locations at which cameras are deployed, so that some locations yield systematically higher or lower

detection probabilities than others for reasons unrelated to conduct. In the empirical setting we present in Chicago, we are not aware of evidence suggesting such correlation. The most threatening violation of this assumption is variation in daylight, as factors such as sunglare during the day and lower visibility at night may have different impacts on the camera’s accuracy. To remedy this issue, one may report results aggregating over either side of sunset separately.

While Assumption A4 can be straightforwardly mitigated, the other assumptions make it difficult to argue for the odds ratio in empirical work as it stands. To make progress, we will consider partial identification of $\psi^\dagger$. Under Assumptions A1-A4 we point identify $\psi^\dagger$. As discussed above, we do not wish to assume that officers’ policing intensity is uniform over X . However, observe that while Assumptions A1 and A2 may not hold in general, we do not expect violations to be large. Consequently, it is instructive to consider how $\psi^\dagger(x)$ might vary with these quantities, and we can thus produce bounds on the CRR for the best and the worst case under violations of Assumptions A1 and A2. This is the subject of the next subsection.

2.2.2 Partial identification of the CRR

We begin with the following lemma, which introduces an interesting connection to this work and inverse probability weighting (IPW) in causal inference.

Lemma 2.5. *Suppose the context X is discrete. Then*

$$\mathbb{E}[S(r) \mid D = d_1] = \mathbb{E}_{X|D=d_1} \left[\frac{1}{\Pr(R = r \mid D = d_1, X)} \mathbb{E}[\mathbf{1}\{R = r\} S \mid D = d_1, X] \right] \quad (8)$$

$$= \frac{1}{\Pr(D = d_1)} \sum_x \frac{\Pr(R = r, S = s_1, D = d_1, X = x)}{\Pr(R = r \mid D = d_1, X = x)} \quad (9)$$

Proof. Algebra. □

Equation (8) is not novel, and has been understood for decades in the causal literature (e.g.,

Robins et al., 2000). We emphasize this formulation however to bring about a different perspective on the problem. Fix an $x \in \mathcal{S}(X)$, say, a one mile stretch of Michigan Avenue in Chicago at 6PM on a Thursday in May 2025. Suppose, hypothetically, that we observe 100 police stops of Black drivers. Further suppose that Black drivers make up 25% of all dangerous drivers in this context. Equation (8) tells us that were everyone on the road Black, we should expect the police to have made 400 stops. Assume for the moment that officers don't lie. Then, we can now think of the benchmarking problem in the following sense: we need some measure of the dangerous driving population so as to deduce what proportion of that group are minorities. This we apply as a weight to the stop data to get the counterfactual count of stops were all drivers on the road Black.

Framing the problem this way is useful as it sharply highlights the two features of the data reality make the idealized description above fail. The first is that we do not know the weight to apply. Indeed, if cameras don't lie, then $\Pr(R = r \mid D = d_1, X = x) = \Pr(R = r \mid C = c_1, X = x)$ by arguments we discussed in Section 2.2.1. However, we do not wish to make this assumption. This implies that our observed measure $\Pr(R = r \mid C = c_1, X = x)$ is contaminated by some unknown share $1 - \theta_x$ of safe drivers, where $\theta_x := \Pr(D = d_1 \mid C = c_1, X = x)$ is the camera precision. Secondly, the counts to which we wish to apply the weight are also imperfect summaries of dangerous drivers, since officers might lie. In particular, some fraction $1 - \eta_x$ contaminates the officer stops with safe drivers where $\eta_x := \Pr(D = d_1 \mid S = s_1, X = x)$ is the precision of officer stops.

Starting from Lemma 2.5, we can derive bounds on $\psi^\dagger$ which are parameterized by η_x , and θ_x , which respectively quantify violations to Assumptions A1 and A2. We state this result in a Theorem.

Theorem 2.6 (Partial identification of CRR). *Under Assumption A4 only, we have that*

$$\begin{aligned} & \frac{\sum_x \Pr(S = s_1, X = x) \left\{ \Pr(R = r \mid S = s_1, X = x) + \eta_x - 1 \right\} \frac{\theta_x}{\Pr(R=r|C=c_1, X=x)}}{\sum_x \Pr(S = s_1, X = x) \Pr(R = r_0 \mid S = s_1, X = x) \frac{\theta_x}{\Pr(R=r_0|C=c_1, X=x) + \theta_x - 1}}} \\ & \leq \psi_r^\dagger \leq \\ & \frac{\sum_x \Pr(S = s_1, X = x) \Pr(R = r \mid S = s_1, X = x) \frac{\theta_x}{\Pr(R=r|C=c_1, X=x) + \theta_x - 1}}{\sum_x \Pr(S = s_1, X = x) \left\{ \Pr(R = r_0 \mid S = s_1, X = x) + \eta_x - 1 \right\} \frac{\theta_x}{\Pr(R=r_0|C=c_1, X=x)}} \end{aligned}$$

for $1 - \theta_x < \min_r \Pr(R = r \mid C = c_1, X = x)$ and $1 - \eta_x < \min_r \Pr(R = r \mid S = s_1, X = x)$.

Proof. Begin with Equation (9) of Lemma 2.5. Note that a nice mathematical property of using risk ratios is that base rates such as $\Pr(D = d_1)$ often cancel out, and so do not need to be computed (see e.g. Zhao et al. (2022) for a related example of this property.) This is the case here and so we need not pay attention to this term.

Consider the expectation in the numerator of the summand. Indeed

$$\Pr(R = r, S = s_1, D = d_1, X = x) \tag{10}$$

$$= \Pr(S = s_1, R = r, X = x) - \Pr(S = s_1, R = r, D = d_0, X = x) \tag{11}$$

$$= \Pr(S = s_1, R = r, X = x) \tag{12}$$

$$- \Pr(R = r \mid D = d_0, S = s_1, X = x) \Pr(D = d_0 \mid S = s_1, X = x) \Pr(S = s_1 \mid X = x) \tag{13}$$

$$= \Pr(S = s_1, X = x) \left\{ \Pr(R = r \mid S = s_1, X = x) \right. \tag{14}$$

$$\left. - \Pr(R = r \mid D = d_0, S = s_1, X = x) \Pr(D = d_0 \mid S = s_1, X = x) \right\} \tag{15}$$

$$= \Pr(S = s_1, X = x) \left\{ \Pr(R = r \mid S = s_1, X = x) - \pi_x^r (1 - \eta_x) \right\}, \tag{16}$$

where we define $\pi_x^r := \Pr(R = r \mid D = d_0, S = s_1, X = x)$ to be the prevalence of safe drivers of race r in the stop data.

Next, consider the denominator of the summand. Indeed,

$$\Pr(R = r \mid D = d_1, X = x) = \Pr(R = r \mid D = d_1, C = c_1, X = x) \quad (17)$$

$$= \frac{\Pr(R = r, D = d_1 \mid C = c_1, X = x)}{\Pr(D = d_1 \mid C = c_1, X = x)} \quad (18)$$

$$= \frac{\Pr(R = r \mid C = c_1, X = x) - \rho_x^r(1 - \theta_x)}{\theta_x}, \quad (19)$$

where we define $\rho_x^r := \Pr(R = r \mid D = d_0, X = x)$ to be equal to the proportion of safe drivers on the road with race r . This algebra shows that $\mathbb{E}[S(r) \mid D = d_1]$ is proportional to

$$\Pr(S = s_1, X = x) \left\{ \Pr(R = r \mid S = s_1, X = x) - \pi_x^r(1 - \eta_x) \right\} \times \left\{ \frac{\theta_x}{\Pr(R = r \mid C = c_1, X = x) - \rho_x^r(1 - \theta_x)} \right\}. \quad (20)$$

Now, for fixed θ_x and η_x , this term is decreasing in π_x^r and increasing in ρ_x^r . To maximize $\psi_r^\dagger$ for a fixed focal group r relative to the reference group r_0 , we maximize the numerator contribution of $\mathbb{E}[S(r) \mid D = d_1]$ by sending $\pi_x^r \rightarrow 0$ and $\rho_x^r \rightarrow 1$, while respecting the simplex constraints by allocating the remaining mass to the nonfocal racial categories. Similarly, we minimize the denominator contribution of $\mathbb{E}[S(r_0) \mid D = d_1]$ by sending $\pi_x^{r_0} \rightarrow 1$ and $\rho_x^{r_0} \rightarrow 0$, again allocating the remaining mass across the remaining racial categories. These choices yield the stated upper bound.

Symmetrically, to minimize $\psi_r^\dagger$, we minimize the numerator contribution of $\mathbb{E}[S(r) \mid D = d_1]$ by sending $\pi_x^r \rightarrow 1$ and $\rho_x^r \rightarrow 0$, while maximizing the denominator contribution of $\mathbb{E}[S(r_0) \mid D = d_1]$ by sending $\pi_x^{r_0} \rightarrow 0$ and $\rho_x^{r_0} \rightarrow 1$, with the remaining probability mass distributed over the other categories. These choices yield the stated lower bound.

The constraints on θ_x and η_x arise because $\Pr(R = r \mid D = d_1, X = x)$ and $\Pr(R = r, S = s_1, D = d_1, X = x)$ must remain positive for all admissible values of π_x^r and ρ_x^r , respectively. $\square$

Let us unpack this result. For simplicity of exposition, temporarily let R be binary and

suppose $R = r_1$ corresponds to Black drivers. We focus on the lower bound; the upper bound is similar. The structure of the lower bound becomes clearer once we interpret its constituent factors. The term

$$\Pr(S = s_1, X = x) \left\{ \Pr(R = r_1 \mid S = s_1, X = x) + \eta_x - 1 \right\} \quad (21)$$

can be viewed as an effective count of dangerous Black drivers stopped at location x . If officers were perfectly precise ($\eta_x = 1$), this reduces to $\Pr(R = r_1, S = s_1, X = x)$, which is proportional to the count of stopped Black drivers in x . When $\eta_x < 1$, the observed stop population is contaminated by safe drivers. To obtain a lower bound on racial bias *against dangerous drivers*, the miscaptured drivers are allocated adversarially: every safe driver wrongly stopped at x is Black. This explains the absence of η_x in the denominator of the lower bound because no White driver is unjustifiably stopped. As a result, the inferred prevalence of dangerous Black drivers in the stop data is minimized subject to a positivity restriction: we cannot subtract so many safe drivers that no remaining Black drivers were driving dangerously.

Now consider the camera data. The fraction $1 - \theta_x$ of camera captures corresponds to safe drivers. For the lower bound, the least favorable allocation assigns all such false positive detections to White drivers. This allocation implies

$$\Pr(R = r_1 \mid D = d_1, X = x) = \frac{\Pr(R = r_1 \mid C = c_1, X = x)}{\theta_x} \quad (22)$$

and

$$\Pr(R = r_0 \mid D = d_1, X = x) = \frac{\Pr(R = r_0 \mid C = c_1, X = x) - (1 - \theta_x)}{\theta_x}. \quad (23)$$

Thus, at the lower bound, the prevalence of Black drivers among dangerous drivers is at its highest and for White drivers its lowest. Inverting these terms produces the smallest IPW weight for Black drivers, and the largest for White drivers. Recalling Lemma 2.5, applying these weights on the numerator and denominator of the risk ratio has the impact

of decreasing the risk ratio, leading to less evidence of discrimination against dangerous drivers.

Combining the two allocations minimizes the inferred number of counterfactual stops for dangerous Black drivers and maximizes those of White dangerous drivers, thereby producing the lower bound on $\psi_{r_1}^{\dagger}$. The upper bound follows by reversing the allocations: all false positive officer stops are assigned to White drivers, while all false positive camera captures are assigned to Black drivers. Since the estimand conditions on the dangerous driving behavior of the motorist, the positivity conditions in Theorem 2.6 ensure that dangerous drivers in fact exist in the data generating processes for the best and the worst case, so that the bound is well defined. The resulting interval in Theorem 2.6 is thus the identified set for the causal risk ratio under Assumption A4, conditional on each η_x and θ_x .

2.2.3 Overlap considerations and substitution effects

In the above analysis, for X to sufficiently control for confounding by context we require overlap in the camera and officer stops. In the most favorable of data environments, the analyst possess the geographic coordinates of each incident, along with its recorded time measured to the minute.⁷ There, by imposing models for $\Pr(R = r \mid S = s_1, X = x)$ and $\Pr(R = r \mid C = c_1, X = x)$ the analyst can smooth over the context to consider incidents nearby in space and time as ‘similarly situated’ (cf. Section 2.4). However, the typical data environment for analysis in policing studies is not so favorable, and coarsening strategies are often used to define a suitable discrete X (or fixed effect) for confounding control (Iacus et al., 2012; Gelman et al., 2007). For example, in our empirical analysis of the Chicago Police Department, we were only able to obtain the location of police stops at the level of the police beat, while the camera tickets had precise geographic coordinates, forcing

⁷Additionally, features such as the type of road (arterial, highway, residential, etc.) are desirable, however, this is often attainable from other sources once the geographic information of the traffic incident is obtained.

the analyst to spatially coarsen the context to at least the level of the police beat. In this article we call such coarsening strategies, ‘blocking strategies’, so that we may refer to the resulting state space of X by ‘blocks’.

When X is discrete, its granularity, and thus the standard for ‘sufficient’ overlap, is chosen by the analyst. The goal is to ensure that camera citations and discretionary stops occur in comparable contexts, so that civilians may be considered similarly situated. Finer blocking is generally preferable because it better aligns the drivers and driving behaviors observed by officers and cameras, thereby reducing bias from the nonrandom placement of cameras, which have been found to be concentrated in minority neighborhoods ([Hopkins and Sanchez, 2022](#)). In Chicago, for example, a blocking strategy might compare officer stops and camera citations within the half mile road segments surrounding a camera, during the same hour, day of the week, month, and year. This locality reduces bias but also statistical power, particularly if departments deliberately station officers away from cameras. When such a finely grained adjustment is infeasible, X should be coarsened cautiously to preserve comparability between officer and camera encounters.

In urbanized jurisdictions, we recommend maintaining a geographically compact definition of X , large enough to include both cameras and nearby discretionary stops, but not so large as to span substantially different driving environments. This is because the demography and geography of the region’s neighborhoods can change rapidly. To increase power, temporal coarsening is advisable. Although hour by hour comparisons are ideal in principle, data constraints may justify comparing stops made on a given weekday to discretionary stops made on the same day. For less urbanized jurisdictions, using a broader spatial context is reasonable, however, careful attention should be paid to the types of roadway contained within that context, as driving behavior can change substantially across them. Overall, it is important to note that any effects computed via an analyst’s chosen blocking strategy

are to be interpreted within the context of what we call *feasible* blocks, those blocks in which there is sufficient overlap in officer and camera tickets. In Section 4 (estimation) and Section 5 (empirical applications) we will demonstrate different approaches to estimating effects depending on the granularity of X afforded to us by the data.

Substitution effects A related issue in the data generating process is the phenomenon that officer presence on the road may influence driving behavior such that a camera ticket is not issued. Consider a motorist driving down Michigan Avenue in Chicago at 6PM on a Thursday in May 2025. Within this x we assume are both an officer and a camera. Suppose that the driver encounters the police officer first, and is stopped due to dangerous driving. Hopefully, this encounter chastises the civilian into correcting their behavior. However, now, this driver will no longer be captured by the camera. If the officer’s decision to stop is racially biased, then cameras are no longer a valid benchmark, since this behavioral phenomenon violates the exclusion condition B2. Interestingly, this phenomenon is not symmetric. Encountering a camera first is likely not to affect driving behavior as strongly as an encounter with an officer, or perhaps at all. A street on which a camera is not clearly identified or does not alert the driver to their capture could cause such behavior. Alternatively, one may be willing to assume that drivers are less sensitive to the presence of cameras than that of officers. Under this assumption, cameras remain a valid benchmark. To what extent does this phenomenon affect identification? We find empirically that a unit is almost never recorded as stopped by both an officer and a camera, which suggests that this phenomenon is not uncommon. To intuitively understand the effects of this substitution, consider again the IPW formulation of the estimand from the previous subsection. If officers are racially biased such that they remove minority drivers from the road at higher rates than White drivers, then the observed IPW weight in camera stops $\Pr(R = r_1 \mid C = c_1, X = x)$ underestimates the desired weight $\Pr(R = r_1 \mid D = d_1, X = x)$. Naïvely making that

substitution applies a larger (smaller) weight to the Black (White) stops than we should, inflating estimates of discrimination. Were officers biased against White drivers, a symmetric argument illustrates that we would underestimate the magnitude of racial bias. In this sense, when racially biased officer encounters precede camera exposure, the use of cameras as benchmarks is anti-conservative for the risk ratio but, notably, will never imply the wrong type of discrimination (i.e. $\text{sign}(\psi^\dagger(x) - 1)$ does not change). See Appendix C for a complete formalization, proof and discussion of this observation.

A simple robustness strategy for mitigating this phenomenon is to, for each x , remove one camera detection for every observed officer stop. This adjustment ensures that the camera benchmark is not disproportionately deprived of units who would have been recorded by the camera had they not first encountered an officer. We acknowledge, however, that this procedure has important limitations. In particular, (i) it does not account for the possibility that the corrective effect of an officer stop may differ by race, (ii) nor does it offer an avenue to test changes in baseline dangerousness D induced by the mere presence of officers. Concerning (i), as we seldom see units arrive in officer and camera data in the same x , it is difficult to generate empirical exploratory evidence for this phenomenon, and thus we are hesitant to formulate a prior on this behavior should it vary by race. Although, an alternative appreciation of this observation is that corrective behavior differential by race does not exist, which is plausible given the lasting effect being stopped by an officer can have for any individual. We can mitigate the issue posed by (ii) by considering a fine context x within which to compare officers and cameras, so that behavior is unlikely to change within this context. However, often encountered data limitations may mean this is not always possible. Moreover, it raises modeling concerns regarding how to separate deterrence from enforcement exposure. For these reasons, we adopt the above deletion rule as a minimal sensitivity adjustment rather than a full behavioral model, which is left to

future work.

This concludes the discussion of identification on the use of cameras as a benchmark. We discuss issues of estimation in Section 4. Camera data is not available in all jurisdictions, and in many regions in the U.S. the use of traffic cameras is banned ([Insurance Institute for Highway Safety, 2026](#)). Next, we turn to a similar benchmark in police checkpoints.

2.3 Checkpoints

Checkpoints are fixed locations at which officers stop all drivers or a prespecified subset of drivers according to a standardized protocol. Examples include sobriety (DUI) checkpoints, vehicle safety and equipment inspections, license and registration checks, and, in some jurisdictions, insurance or emissions checks. Some checkpoints stop every vehicle, while others employ systematic or randomized sampling (e.g., stopping every k th car or a fixed fraction of vehicles) in order to manage congestion. The operation of police checkpoints implies that the stopping rule is specified in advance and does not permit individualized discretion. Hence, the issuance of a ticket at a checkpoint is predictive of dangerous driving, and uncorrelated with race except through the propensity of the civilian to drive dangerously. This implies that checkpoints may serve as a valid benchmark for discretionary officer stops. Furthermore, checkpoints are typically designed to catch drivers exhibiting a binary property of dangerous driving e.g. blood alcohol level exceedance of some threshold or an equipment violation. We thus maintain that dangerous driving, D , is binary.

On matters of identification, checkpoints are largely no different from automated cameras and thus all results from Section 2.2 apply. Furthermore, the benchmarks are similar in that they both suffer from a substitution problem. The nature of the problem for checkpoints is of two kinds: first, police departments are unlikely to assign discretionary police stops to locations that a checkpoint is present, as this is a waste of resources. Second, if checkpoint

enforcement is announced in advance of its implementation, drivers anticipating a checkpoint lying along their usual route may adjust their driving pattern on the day of the checkpoint to avoid an encounter with the police. As factors such as awareness of the announcement and aversion to police encounters plausibly varies in degree by race, in a manner separate from differential rates of dangerous driving, a found disparity may be in part explained by the induced change in driving patterns.

From the perspective of identification this poses a problem primarily with the choice of context X . We wish to ensure that checkpoint and discretionary stops are comparable in context, so that civilians may be considered ‘similarly situated’. However, since we do not anticipate officers assigned to regular enforcement will be very close to checkpoints spatially, this use of this benchmark mechanically forces the analysts to coarsen X to make this comparison feasible, in the manner described in Section 2.2.3. Moreover, checkpoints are deployed infrequently and for select periods within the day, leading to a low volume of checkpoint records.

Overall, while a valid benchmark, common data constraints make checkpoint analyses difficult to employ in practice owing to the credibility of confounding control by X . Due to space constraints, we relegate the empirical demonstration of this analysis to Appendix, with a detailed discussion of the tradeoffs regarding the level of coarsening in the blocks across various jurisdictions.

2.4 Crashes

It is uncontroversial to claim that drivers who are *at-fault* in a crash form a valid benchmark population from which to compare officers. A driver who is at-fault in a crash is driving dangerously, which meets the relevance condition B1. Furthermore, there are no aspects of one’s race which affect the likelihood of causing a crash conditional on driving behavior and

contextual conditions, meeting the exclusion condition B2.

However, unlike going through a redlight, not all crashes are equal. Some collisions involve only one vehicle, while others may lead to the hospitalization or even death of the drivers involved. The act of getting into a vehicle collision does not imply a binary type of driving behavior which led to the collision. Many driving behaviors can induce a crash, and this crash may itself vary significantly in severity. Therefore, it is no longer reasonable to assume that D is a binary variable, and before returning to identification strategies we must first analyze how our framework is altered by this concern.

Consider instead that there exists some unmeasurable, high dimensional quantity D^* which contains all possible dangerous behaviors a driver could exhibit on the road, e.g. unsafe lane change, tailgaiting, poor vehicle condition etc. To be precise, let $I \in \{i_0, i_1\}$ indicate involvement in a crash. Let $F \in \{f_0, f_1\}$, where $F = f_1$ indicates fault for a crash and $F = f_0$ denotes the negation, such that $F = f_F(I, D^*, X, \varepsilon_F)$ where ε_F is some idiosyncratic error. Then, define $C := \mathbf{1}\{I = i_1, F = f_1\}$ such that we interpret $C = c_1$ as the event that a motorist is involved in a crash and at-fault for it, and $C = c_0$ if either they are not involved in a crash or, if they were, they did not cause it. The addition of variables I and F are chiefly for conceptual precision as they completely mediate the path from D^* to C in the core DAG of Figure 2b with binary D replaced by D^* , so there is no loss of generality (see Figure 3). Now, this causal model implies that there exists an unspecified function f_C such that

$$C = f_C(D^*, X, \varepsilon_C) \tag{24}$$

where ε_C is some idiosyncratic error. Relation (24) implies that there exists a random variable D such that $D := \mathbb{E}[\mathbf{1}\{C = c_1\} \mid D^*, X] = \Pr(C = c_1 \mid D^*, X)$ which lies in the unit interval $[0, 1]$. We can interpret D as a summary of the ‘dangerousness’ of a person’s driving behavior required to cause a crash. We will now take this as a definition.

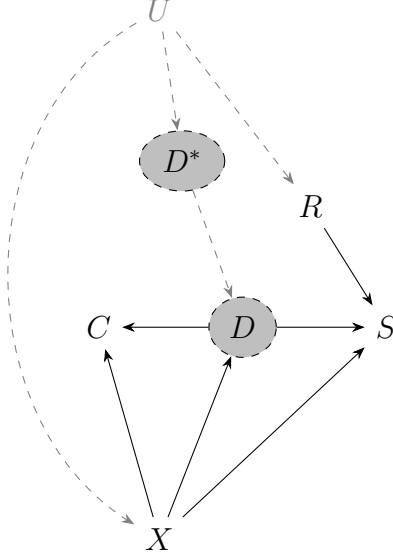


Figure 3: Redefining dangerousness. High-dimensional dangerous behavior D^* feeds into crash risk $D := \Pr(C = c_1 \mid D^*, X)$. Assumption A5 is reflected in that D^* causes S only through crash risk D .

Definition 2.7 (Crash risk). For a valid crash benchmark C binary, we define a driver’s *crash risk* by the scalar variable D , where $D := \Pr(C = c_1 \mid D^*, X)$.

In other words, we define driving as ‘dangerous’ to the extent that it likely causes a crash. We discuss the usefulness of this definition shortly. In Figure 3 we illustrate the position of crash risk D in the data generating process.

We now show how we may adapt the strategy of the camera analysis to obtain causal estimates of discrimination when we use crashes as a benchmark. The causal estimand of interest adapts as follows, we seek for each $x \in \mathcal{S}(X)$

$$\psi_{r,D}^\dagger(x) = \frac{\Pr(S(r) = s_1 \mid D, X = x)}{\Pr(S(r_0) = s_1 \mid D, X = x)} \quad (25)$$

which we may aggregate to obtain a global estimand $\psi_{r,D}^\dagger = \Pr(S(r) = s_1 \mid D) / \Pr(S(r_0) = s_1 \mid D)$. Note that $\psi_{r,D}^\dagger$ in (25) is a random variable, as it depends on the crash risk, D . However, this only poses a challenge for estimation, as by Definition 2.1 we must have

that officers do not discriminate if and only if $\psi_{r,D}^\dagger = 1$ almost surely. That is, race should no further influence an officer’s decision once driving behavior and context are known, no matter the level of crash risk. In what follows, we will develop a test of the global null that $\psi_{r,D}^\dagger = 1$.

To begin, we will start with some nomenclature which will permeate the remainder of this section.

Assumption A5 (Stop rule). Assume that, conditional on D , R and X , officer stops occur with probability $h_R(D, X) := \Pr(S = s_1 \mid D, R, X)$, where we call $h_R(D, X)$ an officer stop rule.

Let us motivate Assumption A5. In our empirical context where crash data is available, we study traffic enforcement on Texan highways, which is administered by the Texas Highway Patrol (THP). The THP is a division of the Department of Public Safety within the state of Texas, whose mission statement includes “detering, detecting and interdicting public safety threats on roadways” ([Texas Department of Public Safety, 2026](#)). Thus, it is reasonable—and favorable to the officer—to model an officer’s decision to stop a vehicle as a function of the context and its threat to public safety, which we encapsulate in crash risk D , and, under the null, is independent of race. The implications of different specifications of $h_R(D, X)$ we will discuss throughout this section. Furthermore, Assumption A5 reveals that the null hypothesis of no discrimination is equivalent to the independence of the stop rule, h_R , on race, R , conditional on D and X .

Next, akin to Section 2.2.1, we show under what conditions the odds ratio is informative of discrimination. Consider that the stop rule is linear in crash risk, so that $h_R(D, X) = \gamma_R(X)D$. That is, within some X , officers pull over drivers at a rate proportional to their crash risk, where the constant of proportionality may depend on race, R . Under this linear probability model (LPM), we have the following proposition.

Proposition 2.8. *Suppose, for some function $\gamma_R(X)$, that the officer stop rule is linear in crash risk, $h_R(D, X) = \gamma_R(X)D$. Then, the null hypothesis of no discrimination by race implies that $\gamma_r(X) = \gamma_{r_0}(X)$ and*

$$\text{odds}_r[R \mid S = s_1, X] = \text{odds}_r[R \mid C = c_1, X]$$

almost surely, where $C = c_1$ indicates the group of at-fault drivers in a crash.

Proof. By definition, $\psi_{r,D}^\dagger(X) = 1$ implies that $\gamma_r(X) = \gamma_{r_0}(X)$. We can go further, however. Indeed

$$\Pr(S = s_1 \mid R = r, X) = \mathbb{E}[\Pr(S = s_1 \mid R = r, D, X) \mid R = r, X] = \gamma_r(X)\mathbb{E}[D \mid R = r, X],$$

similarly

$$\Pr(C = c_1 \mid R = r, X) = \mathbb{E}[\Pr(C = c_1 \mid R = r, D, X) \mid R = r, X] = \mathbb{E}[D \mid R = r, X],$$

where we use that C is a valid benchmark so by the exclusion condition [B2](#), $C \perp\!\!\!\perp R \mid D, X$.

Therefore

$$\begin{aligned} \frac{\gamma_r(X)}{\gamma_{r_0}(X)} &= \frac{\Pr(S = s_1 \mid R = r, X) \Pr(C = c_1 \mid R = r_0, X)}{\Pr(S = s_1 \mid R = r_0, X) \Pr(C = c_1 \mid R = r, X)} \\ &= \frac{\Pr(R = r \mid S = s_1, X) \Pr(R = r_0 \mid C = c_1, X)}{\Pr(R = r_0 \mid S = s_1, X) \Pr(R = r \mid C = c_1, X)}. \end{aligned}$$

The above display shows that no discrimination implies that $\text{odds}_r[R \mid S = s_1, X] = \text{odds}_r[R \mid C = c_1, X]$. $\square$

We make two remarks on this result. First, we highlight the intuition. Even if officers have a different sensitivity to crash risk than its absolute measure, D , an unbiased stopping rule cannot extract relatively more Black drivers than White drivers compared to crashes, since the stopping rule is *proportional to risk*. Second, we see that once again the benchmarking framework we present in this article provides a causal interpretation of the odds ratio. Using

a linear stop rule, we find that the conditional causal risk ratio equals $\gamma_r(X)/\gamma_{r_0}(X)$ which the proof above shows is equal to the odds ratio. The linear stop rule is the continuous analog of Assumption A1 (officers don't lie), in that no unit is pulled over whose crash risk is equal to zero, thus stops reflect dangerous behavior on the road. Should a scholar be willing to make this assumption, Proposition 2.8 permits the use of the odds ratio as a causal measure of the parameter of interest $\psi_{r,D}^\dagger(X)$.

But should a scholar be willing to make this assumption? Suppose we find that the odds ratio does not equal unity. This could be evidence of two distinct phenomena. If the LPM is correct, then this finding implies that $\gamma_r(X) \neq \gamma_{r_0}(X)$ which indicates that civilian drivers are subject to *disparate treatment*—officers perceive and thus treat the same behavior differently depending on race. Another explanation for this finding could be that the LPM is not in fact accurate. For example, officers could exhibit a nonlinear stop rule $h_R(D, X) = \lambda(X)\mathbf{1}\{D > \bar{d}\}, \lambda(X) \leq 1$. Such a rule is consistent with officers only paying dutiful attention to offenses about some personally egregious threshold, e.g. speeding 20mph over the limit. Were such a rule applied, we should expect a higher proportion of minorities in the data as this is simply a manifestation of the disparity in baseline dangerousness. Notably, this rule does not depend on race, and is thus not indicative of disparate treatment. However, as the choice of $\bar{d}$ may disproportionately pick up more civilians of a given group, depending on latent crash risk, failing the test of no discrimination may be evidence of the *disparate impact* of a race neutral policy.

Both patterns of police behavior are illegal (U.S. Supreme Court, 1886, 1971). Failing this test is therefore strongly indicative of officer malpractice, however, we should aim to be more precise about which. Currently, Proposition 2.8 is limited in that we have imposed a restrictive functional form on the stop rule. In part, this choice is enforced by the fact that we have a binary proxy for a continuous confounder which means that the

proxy does not satisfy usual completeness conditions for identification (Wang et al., 2018; Tchetgen Tchetgen et al., 2024). To elicit more flexibility in the model, we need a proxy which explains greater variation in D . Moreover, by relaxing the LPM formulation, we may be more confident in the type of discrimination we detect. In Texas, the Texas Department of Transportation records the severity of the crashes identified includes, e.g., indicators for if the crash caused property damage, injury or a fatality. In what follows, we describe a novel falsification test for racial discrimination under a more flexibly modeled stop rule, using this more granular benchmark.

2.4.1 Falsification test for racial bias using crash severities

To abandon the linear probability model assumption, we will introduce a class of stop rules which permit a wide variety of functional forms, but are crucially independent of race. We will develop a falsification test that the officer stop rule belongs to this class.

To motivate our choice of this class, as described, we leverage an interesting property of the crash data in our empirical setting, its *severity*. Let $C \in \{c_0, k_1, \dots, k_K\}$, where c_0 denotes that the motorist was not involved in an at-fault crash and $k_1 < \dots < k_K$ index increasingly severe at-fault crashes. For concreteness, in the Texas data, $K = 5$, and $k_1, \dots, k_5$ correspond respectively to “not injured,” “possible injury,” “suspected minor injury,” “suspected serious injury,” and “fatal injury”. Similarly to the binary case, define D as the likelihood of causing an at-fault crash, $D := \Pr(C \geq k_1 \mid D^*, X)$, and maintain Assumption A5.

Owing once again to the characterization of police officers as responsive to crash risk, we motivate the ensuing construction by first considering a loss function based on crash severity of the form

$$\ell(C, X) = \sum_{\kappa=1}^K \lambda_{\kappa}(X) \mathbf{1}\{C \geq k_{\kappa}\}, \quad (26)$$

where $\lambda_\kappa \geq 0$ captures how much the officer wishes to avoid crashes of severity at least k_κ in context X . For instance, one might imagine that λ_κ increases with κ . For a driver with dangerousness level d in context x , the officer does not observe the severity of any prospective crash. Instead, we might consider a stopping rule consistent with the expected loss associated with the range of possible crash severities, namely

$$\mathbb{E}[S \mid D = d, X = x] := \mathbb{E}[\ell(C, x) \mid D = d, X = x] \quad (27)$$

$$= \sum_{\kappa=1}^K \lambda_\kappa(x) \Pr(C \geq k_\kappa \mid D = d, X = x). \quad (28)$$

Write $b_\kappa(d, x) := \Pr(C \geq k_\kappa \mid D = d, X = x)$. Think of these functions as generators for any reasonable officer stop rule which is based on a response to crash risk. By allowing the form of the b_κ 's to be flexible, we can use this formulation to define a richer class of stop rules than the linear probability model.

Specifically, for each severity index $\kappa \in \{1, \dots, K\}$, define the class of stop rules

$$\mathcal{H} := \left\{ h : h(d, x) = \sum_{\kappa=1}^K \lambda_\kappa(x) b_\kappa(d, x), \lambda_\kappa \in \mathbb{R}^+, \sum_{\kappa=1}^K \lambda_\kappa \leq 1, b_\kappa \in [0, 1] \text{ for each } \kappa \right\}, \quad (29)$$

where the coefficients are unspecified functions of x except that they must be positive and lie on or inside the simplex for each x . We now wish to derive observable implications of belonging to the class $\mathcal{H}$. Should the data not agree with these implications, we falsify the null hypothesis of no discrimination, as no function in $\mathcal{H}$ depends on race. We build the test in this fashion in order to provide a more robust strategy for detecting disparate treatment, as opposed to disparate impact, since the basis functions b_κ are constrained only by axioms of probability. The key implication is in the following proposition.

Proposition 2.9. *Suppose an officer stopping rule $h(D, X)$ belongs to class $\mathcal{H}$. Then*

$$\min_{1 \leq \kappa \leq K} \Pr(R = r \mid C \geq k_\kappa, X) \leq \Pr(R = r \mid S = s_1, X) \leq \max_{1 \leq \kappa \leq K} \Pr(R = r \mid C \geq k_\kappa, X) \quad (30)$$

almost surely.

Proof. Suppress conditioning on X for brevity. By definition $\Pr(S = s_1 \mid R = r) = \mathbb{E}[h(D) \mid R = r] = \mathbb{E}\left[\sum_{\kappa=1}^K \lambda_{\kappa} b_{\kappa}(D) \mid R = r\right] = \sum_{\kappa=1}^K \lambda_{\kappa} \Pr(C \geq k_{\kappa} \mid R = r)$, which implies that $\Pr(R = r \mid S = s_1) \propto \Pr(S = s_1 \mid R = r) \Pr(R = r) = \sum_{\kappa=1}^K \lambda_{\kappa} \Pr(C \geq k_{\kappa} \mid R = r) \Pr(R = r) = \sum_{\kappa=1}^K \lambda_{\kappa} \Pr(C \geq k_{\kappa}, R = r)$, which we normalize to attain $\Pr(R = r \mid S = s_1) = \sum_{\kappa=1}^K \lambda_{\kappa} \Pr(C \geq k_{\kappa}, R = r) / \sum_{r \in \mathcal{R}} \sum_{\kappa'=1}^K \lambda_{\kappa'} \Pr(C \geq k_{\kappa'}, R = r) = \sum_{\kappa=1}^K w_{\kappa} \Pr(R = r \mid C \geq k_{\kappa})$ where $w_{\kappa} = \lambda_{\kappa} \Pr(C \geq k_{\kappa}) / \sum_{\kappa'=1}^K \lambda_{\kappa'} \Pr(C \geq k_{\kappa'})$. As this is a convex combination ($\sum_{\kappa=1}^K w_{\kappa} = 1$), the result follows. $\square$

The intuition behind this result is subtle but quite straightforward. A necessary component of the proof above is the following fact that under the null, the observed proportion of a given racial group among officer stops is a weighted combination of the observed proportions within successive tails of the crash risk distribution. The weights w_{κ} describe how much an officer is sensitive to crashes of severity at least k_{κ} , normalized by how often those crashes occur. To grasp how these elements produce the conclusion, consult Figure 4 for a concrete example.

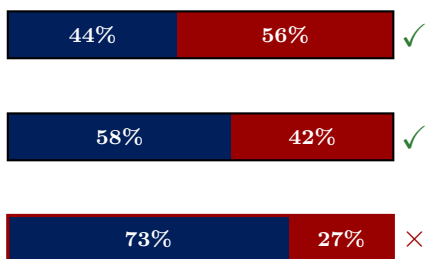
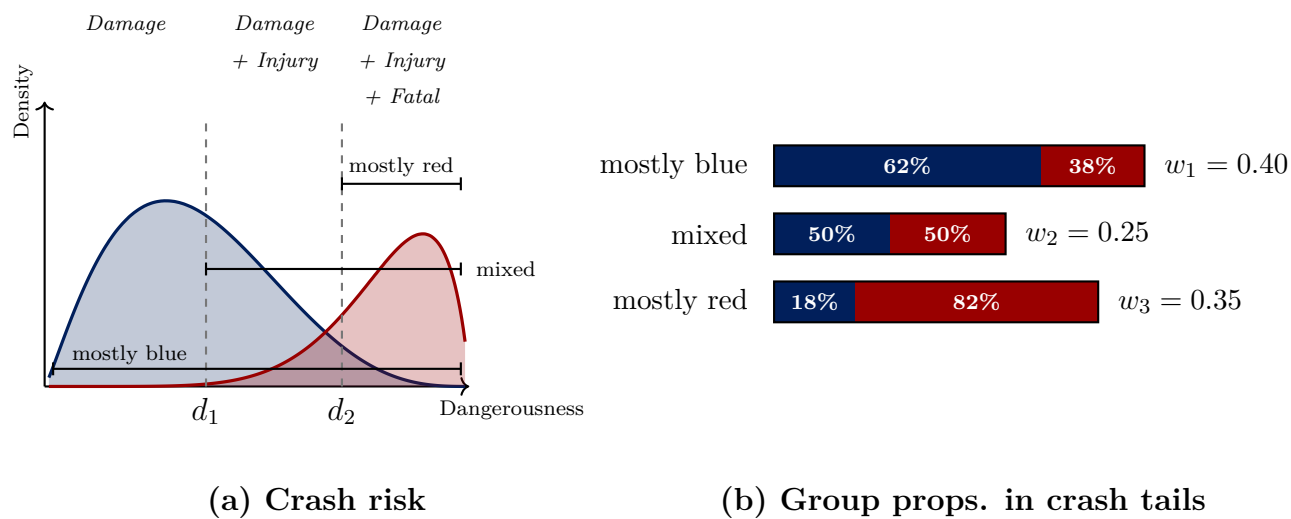
Example 2.10. *We work in a world of blue and red drivers: blue drivers mostly drive with low risk, while red drivers are disproportionately dangerous, as shown in panel (a). Suppose for the sake of exposition that a crash causes property ‘damage’ when crash risk does not exceed d_1 , causes damage or ‘injury’ between risks d_1 and d_2 and causes damage, injury, or a ‘fatal’ity when risk exceeds d_2 . Panel (a) shows that mostly blue drivers occupy any kind of crash, there is mixed representation among damaging and injurious crashes and mostly red drivers cause fatal crashes. Panel (b) writes this visual fact numerically, along with the attention weights officers pay to these tails; the closeness of w_1 and w_3 here indicate that while rarer than crashes that at least cause damage, the behavior that could lead to a fatal crash is a significant draw of police attention. The essence of the result in Proposition 2.9 is that the proportions of blue and red drivers we see in the data should be*

the combination of the three blue bars in panel (b). If we knew the weights, w_κ , we should find $\Pr(R = \text{blue} \mid S = s_1) = 0.4(0.62) + 0.25(0.50) + 0.35(0.18) = 0.436$, which is the first bar of panel (c). As red drivers dominate the most dangerous part of the crash risk distribution, it makes sense to find that they appear the most in officer stops. The problem is we do not know these weights. Due to the convexity, however, we know that racial prevalence among officer stops must lie between the extremes of the panels in (b), which means the second bar of panel (c) is consistent with no racial bias but the third bar is not.

What went wrong in the third bar of panel (c)? The added granularity of the crash severities permits us to examine how minorities are more strongly or weakly represented among the distribution of crash risk. The core intuition this example reveals is that if officers' primary mission is the protection of public safety, then they cannot mine more minorities from the dangerous driving population than they maximally appear across this distribution. Otherwise, officers possess a greater sensitivity to minorities than their driving behavior would imply, hence falsifying the null of no discrimination. This could be motivated by inaccurate stereotypes of poor driving by minority drivers, for instance. We find evidence in support of this cognitive bias in the empirical application, see Section 5.2.1.

We conveyed the intuition of Example 2.10 under the assumption that officers prioritize safety. That is, even if officers are racially biased, the functional form of their stop rule does not substantially deviate from $\mathcal{H}$. However, it could be that the officer stop rule admits a pathological shape which differentially but picks up minorities relative to Whites without racial prejudice. While $\mathcal{H}$ is sufficiently complex to render this unlikely, we leave to future work the development of an approach to determine to what degree these pathological stops are present in the data.

What remains is inference on the test implied by (30). Generally, this task is not trivial due to the nonregularity of the max function; we discuss our chosen approach in Section 4.



(c) Group props. in officer stops

Figure 4: Schematic illustrating the falsification test for racial bias using crash severities.

Consult the main text for a walkthrough of each diagram.

3 Race is missing, mismeasured, or unavailable

Throughout this analysis, we assumed that civilian race was (i) available in both datasets and (ii) where available, accurately recorded. In practice, policing scholars often find that at least one of these conditions does not hold. For example, automated traffic camera systems may only issue penalties to owners of the vehicle based on information tied to license plates, and so their race may not be a feature in the database. Regarding (ii), there have been many reported incidents of officers manipulating their recording of race in the data in response to increased oversight or political pressure. For instance, in the wake of George Floyd’s murder in 2020, Minneapolis PD was found to have recorded race in civilian encounters at a rate almost half that prior to the killing ([United States Department of Justice, Civil Rights Division and United States Attorney’s Office, District of Minnesota, Civil Division, 2023](#), pg 41). Relevant to our empirical examples in Texas, reports have found the officers frequently label Hispanic civilians in encounters as White (see [Luh \(2023\)](#) and references therein). A failure to account for mismeasurement of officer-reported race to this extent could engender severe underestimation of racial disparities. In the next section, we discuss how it is possible to correct for this measurement problem using proxies of race, such as civilian surname.

3.1 The problem with BISG

Unavailability of race data has long been an issue in the social sciences, and attempts at providing a solution using surnames have been evolving for nearly two decades. Starting with the work of [Fiscella and Fremont \(2006\)](#), the idea is simple. For a given civilian, let r be their race (which is unknown), and l their last name. A straightforward application of Bayes’ rule implies that

$$p(r \mid l) \propto p(l \mid r)p(r). \quad (31)$$

Every term on the right hand side of (31) is calculable from census data. Hence, given an individual’s surname, one can derive a posterior distribution over their race using data from the census, by computing (31) for each candidate race r and renormalizing. The accuracy of these predictions can be improved by also conditioning on a unit’s home address, as this more narrowly confines the relevant cells of the census data, which gives rise to Bayesian Improved Surname Geocoding (BISG) (Fiscella and Fremont, 2006; Elliott et al., 2009; Imai and Khanna, 2016). Suppose g represents a unit’s zip code, then

$$p(r \mid l, g) \propto p(l, g \mid r)p(r) \stackrel{!}{=} p(l \mid r)p(g \mid r)p(r) \quad (32)$$

where the equality assumes that surname and home address are independent within levels of race. Similarly to the relation in (32), all terms after the equality are estimable from census data. Zip code is only a specific example of additional proxy of race one could leverage to improve upon (32). For example, related work has also included first name into the estimation yielding Bayesian Improved Firstname Surname Geocoding (Voicu, 2018), and others choices may be determined by the domain (Decter-Frain, 2022).

BISG works well, assuming three conditions hold. The first is that the census data are correct. These numbers are typically inserted without error, and are consequently assumed to reflect the true prevalence of race, name and surname given name. The second is that surname is independent of all other proxies of race, within levels of race, so that the marginals on the right hand side of (32) may be computed. The third is that the dataset of interest is a random sample of the census population. A violation of this assumption implies the existence of some selection variable d such that the analyst is actually interested in $p(r \mid l, d)$. Then

$$p(r \mid l, d) \propto p(l \mid r, d)p(r \mid d) \quad (33)$$

and it is generally no longer true that the quantities on the right hand side of (33) are estimable from the census. Naïvely applying BSA or BISG assumes that $p(r \mid l, d) = p(r \mid l)$.

The first two assumptions have been studied in the literature. Comprehensive strategies for dealing with measurement error in census counts have been proposed (Imai et al., 2022). McCartan et al. (2025) develop sensitivity analyses for violations of the second assumption in the downstream estimation of disparities. Surprisingly, we find little to no discussion of the third assumption, even though violations of this condition can have substantial consequences. To give an example within our empirical context, suppose that in the population, people with the last name “Booker” are twice as likely to be Black than White. However, in this population, White people (say) are twice as likely to speed. Then,

$$\frac{\Pr(\text{Black} \mid \text{“Booker”}, \text{speeder})}{\Pr(\text{White} \mid \text{“Booker”}, \text{speeder})} = \underbrace{\frac{\Pr(\text{Black} \mid \text{“Booker”})}{\Pr(\text{White} \mid \text{“Booker”})}}_2 \times \underbrace{\frac{\Pr(\text{speeder} \mid \text{Black})}{\Pr(\text{speeder} \mid \text{White})}}_{\frac{1}{2}} = 1. \quad (34)$$

That is, if police officers select units on their propensity to drive dangerously, then within stop data, the analyst should expect the prevalence of minorities among Bookers to be much closer to that of Whites than within the population, given the underlying differences in driving behaviors. Imputing posteriors without accounting for selection would grossly underestimate the presence of White drivers in the data, which could lead to inflated estimates of discrimination. Of course, were the correction factor $p(d \mid r)$ known, we solve the problem. However, in general, this factor is not known.

This selection problem is conceptually related to issues identified in recent work on estimating disparities using BISG, namely the BIRDiE methodology of McCartan et al. (2025). Their proposed BIRDiE methodology, however, is not directly applicable in our setting. BIRDiE is designed for situations in which race enters as a covariate in a regression for a downstream outcome, and the goal is to estimate the effect of race on that outcome when race is observed only through proxy measures such as surnames and zip code. Conceptually similar to the aforementioned problem, the authors highlight that if BISG prediction errors are correlated

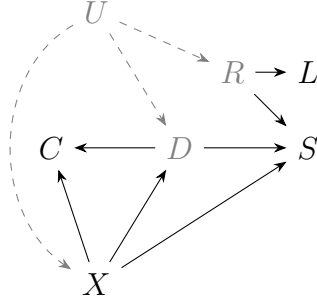


Figure 5: Last name is a proxy for race. Much like the benchmark C is a proxy for dangerous driving, last name, L , is a proxy for race, R , when race is unavailable.

with the outcome of interest, estimation of disparities is biased. For example, if BISG is more likely to misclassify a Black civilian as White when stopped by police, one would underestimate the size of discrimination against Black civilians. Their identification strategy treats surname as an instrumental variable and thus corrects for the bias that arises when the race probabilities produced by BISG are correlated with the outcome.

Our use case differs from theirs in two primary ways: (i) we are primarily interested in the BISG probabilities themselves, as they appear as elements of identified functionals earlier in this article and (ii) our outcomes suffer from sample selection. The central difficulty in our setting thus lies in how selection into the stop data weakens the predictive power of name for race, rather than in how race enters a downstream outcome model. Were data on vehicles police officers did not stop available, we could incorporate a similar analysis to these authors. Nevertheless, we can still make progress, and in fact, our solution shares some similarities with theirs.

3.2 Correcting BISG under selection

Much like how benchmarks are a proxy for dangerous driving, names are a proxy for race, see Figure 5. In particular, last name, L , satisfies the exclusion condition that name is independent of all other variables conditional on race. BISG methodology concentrates on

the quantity $\Pr(R = r \mid L = l)$, which is not invariant to domain shift: $\Pr(R = r \mid L = l) \neq \Pr(R = r \mid L = l, D = d_1)$. Consider instead the inverse quantity, $\Pr(L = l \mid R = r)$. Immediately, by the exclusion condition, we observe that this quantity is indeed invariant to domain shift, as $L \perp\!\!\!\perp D \mid R$, and this condition remains true for proxies of D . We have the following proposition.

Proposition 3.1. *Define L as civilian surname. Let $A \in \{\text{stop}, \text{camera}\}$ be the source of the civilian record. Assume the surname distribution given race, $\Pr(L = \ell \mid R = r)$, is known from the census. When race R is unmeasured, the quantity $\Pr(R = r \mid A = a, X = x)$ is identified for every r , a and x as the solution to the following linear system*

$$\Pr(L = \ell \mid A = a, X = x) = \sum_{r \in \mathcal{R}} q_{\ell r} \Pr(R = r \mid A = a, X = x), \quad (35)$$

for each context $x \in \mathcal{S}(X)$.

Proof. Apply the law of total probability to $\Pr(L = \ell \mid A = a, X = x)$, followed by the fact that $L \perp\!\!\!\perp (A, X) \mid R$. Notice that as $\Pr(L = \ell \mid A = a, X = x)$ is observed data and $\Pr(L = \ell \mid R = r)$ is assumed known from the census, $\Pr(R = r \mid A = a, X = x)$ is identified. $\square$

Intuitively, the surnames we see in our records are a mixture of surnames by race within our records, weighted by the prevalence of that race within the data. But, once race is known, selection into the sample is no further predictive of a civilian’s surname because the factors driving selection are unrelated to how surnames are assigned. The desired proportions $\Pr(R = r \mid A = a, X = x)$ may be obtained by solving linear system (35). We discuss estimation strategies in Section 4.

3.3 Officer misreporting of race

Let $\tilde{R} \in \mathcal{S}(R)$ be officer-reported race, the label an officer records for the civilian's race. Throughout, R denotes self-reported race, by which we mean the race we estimate the civilian would self report. The procedure outlined in the previous section identifies the prevalence of self-reported race in the stop data, $\Pr(R = r \mid S = s_1)$, and the data already contains $\Pr(\tilde{R} = \tilde{r} \mid S = s_1)$: comparing these quantities would crudely reveal to what extent officers are misreporting race. To be more precise about the degree of distortion in the data, a desirable target quantity is the sensitivity, or recall, of officer reports, $\Pr(\tilde{R} = \tilde{r} \mid R, S = s_1)$. That is, for stopped civilians whose self-reported race is R , how often do officers record their race correctly? Of particular interest is the quantity $\Pr(\tilde{R} = \text{White} \mid R = \text{minority}, S = s_1)$, as large values of this quantity may lead to severe underestimation of disparities.

Identifying this quantity follows directly from Proposition 3.1, which we apply within each officer-reported race $\tilde{r}$. Suppress X for brevity. Indeed, we have the following two stage procedure

$$\Pr(L = \ell \mid \tilde{R} = \tilde{r}, S = s_1) = \sum_{r \in \mathcal{R}} \Pr(L = \ell \mid R = r) \Pr(R = r \mid \tilde{R} = \tilde{r}, S = s_1) \quad (36)$$

$$\Pr(\tilde{R} = \tilde{r} \mid R = r, S = s_1) \propto \Pr(R = r \mid \tilde{R} = \tilde{r}, S = s_1) \Pr(\tilde{R} = \tilde{r} \mid S = s_1) \quad (37)$$

assuming in (36) that $L \perp\!\!\!\perp \tilde{R} \mid R, S$. The independence condition required for this calculation is quite strong. It is violated when, for instance, if an officer stops a Black civilian, they are no more likely to record them as White than Hispanic if their surname is Hernández. Given recent scrutiny into officer reporting of race, it is plausible that this assumption may not hold in practice. We are currently working on developing a sensitivity analysis to violations of this assumption based on reasonable monotonicity constraints based on the perceived cultural association between a civilian's name and their race. Although, in practice the nature of these relaxations will depend on institutional context, i.e. exactly how race is

recorded and reported by jurisdiction.

Another useful implication of this procedure is that it gives another avenue to correct estimates of officer-reported race without the use of surname data. Here and throughout, we call $\Pr(\tilde{R} = \tilde{r} \mid R = r, S = s_1, X = x)$ the forward misreporting mechanism, whose diagonal entries give the sensitivity, or recall, for each racial category, and we call $\Pr(R = r \mid \tilde{R} = \tilde{r}, S = s_1, X = x)$ the reverse misreporting mechanism, whose diagonal entries give the precision, or positive predictive value, for each racial category. Since

$$\Pr(\tilde{R} = \tilde{r} \mid S = s_1) = \sum_{r \in \mathcal{R}} \Pr(\tilde{R} = \tilde{r} \mid R = r, S = s_1) \Pr(R = r \mid S = s_1), \quad (38)$$

if the forward misreporting mechanism $\Pr(\tilde{R} = \tilde{r} \mid R = r, S = s_1)$ is known or assumed, analysts may compute $\Pr(R = r \mid S = s_1)$ by solving system (38), analogously to (35). Alternatively, if the reverse misreporting mechanism $\Pr(R = r \mid \tilde{R} = \tilde{r}, S = s_1)$ is known, then we can directly obtain

$$\Pr(R = r \mid S = s_1) = \sum_{\tilde{r} \in \mathcal{S}(\tilde{R})} \Pr(R = r \mid \tilde{R} = \tilde{r}, S = s_1) \Pr(\tilde{R} = \tilde{r} \mid S = s_1). \quad (39)$$

In practice, it is unlikely that the sensitivity or the precision of officer-reported race will be provided with finely grained X , if known, in which case relations (38) and (39) assume that the $\Pr(\tilde{R} = \tilde{r} \mid R = r, S = s_1)$ and $\Pr(R = r \mid \tilde{R} = \tilde{r}, S = s_1)$ may respectively only vary at the coarsest level of X available (e.g. by year). In such cases, it is advisable to conduct a sensitivity analysis by reanalyzing the data over a range of possible error mechanisms, examples of which we provide in Section 5.

Next, we turn to general estimation approaches for each of the empirical benchmarking strategies we have discussed in this article, while being careful to account for the measurement error described above.

4 Estimation

4.1 Cameras and checkpoints

All identified functionals of the data in Theorem 2.6 are relatively simple, consisting of elementary combinations of proportions within X , so what remains is to specify how these categorical proportions depend on X .

Recall that X contains spatial and temporal information for each police or benchmark record. At the most granular level these include the precise time and date the record was made, its geographic coordinates and other environmental features, such as the type of road. Should the data be sufficiently granular, more advanced estimation strategies such as double machine learning could be leveraged to reduce reliance on parametric structure (see, e.g. Section 4.2). In practice, this granularity is seldom enjoyed. For example, public Chicago PD data records the location of stops only at the beat level, while the corresponding traffic camera data contains the geographic coordinates of the camera. This mismatch forces the analyst to choose a modeling strategy which ensures the comparison between officer and benchmark data is as tight as possible given the constraints. A defensible choice which addresses this issue is coarsened exact matching (Iacus et al., 2012).

Using the temporal information the data contains, we construct variables for hour of day, day of week, month, and year. Once the coarsened variables have been specified, we define a set of discrete blocks $\mathcal{S}(X)$ given by the Cartesian product of the spatial and temporal grouping variables. This process also defines a set of what we call *feasible* blocks, which are the $X \in \mathcal{S}(X)$ that are shared across stop and benchmark data; all other data is dropped. Data sacrifice is a known drawback of matching approaches (Stuart, 2010), but as discussed in Section 2.2.3, a compact definition of X is required to ensure that compared units are similarly situated.

Turning to the upper and lower bounds of $\psi^\dagger$ in Theorem 2.6, if race is accurately reported, estimation may proceed by straightforwardly estimating $\widehat{\Pr}(R \mid A = a, X)$ for each $A \in \{\text{stop}, \text{camera}\}$ via the empirical mean. Whenever civilian surname is available in source a , we recommend computing this proportion using an estimation strategy pursuant to the identification result of Proposition 3.1. Several choices are available. As (35) is a linear system, we could run least squares regression. However, since the number of surnames in the data is typically large (with $|\mathcal{S}(L)| \gg |\mathcal{S}(R)|$), we avoid this option due to computational expense. Instead, we proceed by maximum likelihood estimation via a multinomial likelihood, which we carry out via a much faster EM algorithm, and standard normal theory for maximum likelihood estimation applies. Details are given in Appendix E. Should the analyst not have access to surname data, but know the precision $\Pr(\tilde{R} = \tilde{r} \mid R = r, S = s_1)$ or accuracy $\Pr(R = r \mid \tilde{R} = \tilde{r}, s_1)$ from other sources or previous literature, she could manually correct the empirical estimates $\widehat{\Pr}(\tilde{R} = \tilde{r} \mid S = s_1, X)$, we additionally document how to adjust the estimation procedure accordingly in Appendix E.

Once obtaining the racial proportions by source, replace $\Pr(X = x \mid S = s_1)$ with its sample analogue. Notice that because the bound is a ratio, the unknown rate of stops $\Pr(S = s_1)$ cancels from the numerator and denominator. Uncertainty is obtained by bootstrap. Care must be taken to ensure that the chosen sensitivity parameters (θ_x, η_x) satisfy the feasibility restrictions $1 - \theta_x < \min_r \Pr(R = r \mid C = c_1, X = x)$ and $1 - \eta_x < \min_r \Pr(R = r \mid S = s_1, X = x)$ for all x . In practice, these constraints may be enforced by truncation and trimming, where the term $\Pr(R = r \mid S = s_1, X = x) + \eta_x - 1$ is replaced by zero if it is negative, and the adjusted IPW weights replaced by the bounds of some prespecified interval should they fall outside of it, in this work we choose this interval to be $[1/0.9, 1/0.1]$. IPW trimming is a commonly used strategy in the literature to tackle (i) positivity violations, (ii) variance inflation due to small propensity scores (Lee et al., 2011). In this work, we

encounter the first challenge, but it is mechanical, as it is a natural artifact of the sensitivity analysis formulation. We acknowledge that one major drawback of trimming is that the procedure generally changes the estimand, as it may rebalance the target population in an unpredictable manner. However, our context is conceptually nonstandard in that regard. As the vector (η_x, θ_x) increases in each component, the bounds describe a world in which officers and cameras increasingly lie, and we control or eliminate x 's where the amount of lying is inconsistent with what the data imply is possible. Our bounds then average over the contexts where the data permit these levels of officer or camera miscapture, which describe the best and the worst possible cases for discrimination in the jurisdiction.

4.2 Crashes

A major advantage of the coarsening strategy described in the Section 4.1 is its simplicity and ease of implementation. Another is that it enforces tight comparisons by design, even if subject to researcher discretion. However, it possesses two primary limitations. First, the method is highly subject to data granularity. The blocks $\mathcal{S}(X)$ may only be as fine as the data available. Should the raw data be too coarse, comparing units within groups without nuance may extract shallower comparisons than are possible with more flexible modeling strategies. In Chicago, we only have access to police beat which is around a square mile in area. In Texas, we possess geographic coordinates for crashes and stops, meaning that a coarsening of the data would amount to dividing Texan highways into segments of m miles long, m small. A stop on a highway that is $m + 0.1$ miles away from a crash does not enter a consequentially different environment, but the lack of smoothness in matching leads to this undesirable property.

Given the more complex nature of X , it is natural to consider methods that allow the relationship between X and the quantities of interest to vary more smoothly than is possible

under coarse blocking. At the same time, the dependence of (R, S, C) on X is not intrinsically of scientific interest; rather, it is a nuisance feature required only to adjust for contextual differences between officer and benchmark. This motivates the use of modern semiparametric estimation strategies developed in the causal inference literature, which separate the target parameter from the nuisance structure elements of the problem, permitting these to be flexibly modeled, including via machine learning (Robins and Rotnitzky, 1995; Chernozhukov et al., 2018). Double/debiased machine learning (DML) as it is termed, guarantees regular, Wald type asymptotic properties for the estimator of the discrimination parameter of interest (Kennedy, 2024), while reducing risk of model misspecification.

In this work, we implement a DML approach to estimate the conditional odds ratio under the assumptions of Proposition 2.8 using estimating equations derived in Tchetgen Tchetgen et al. (2010), which we adapt to accommodate mismeasurement in the recording of race.⁸

We summarize the strategy here; additional details are provided in Appendix F. Let $R \in \{r_0, r_1, \dots, r_{J-1}\}$, where $J := |\mathcal{R}|$ is the number of racial categories, denote self-reported race, with r_0 a designated baseline category, which in our application is White drivers. Let $A \in \{c, s\}$ indicate the source of the record, where c is the benchmark source and s is the officer-stop source, and let X denote the observed context.

The method begins with a parameterization of the conditional association between A and

⁸Liu et al. (2021) study DML for the logistic partially linear model when the outcome is binary. Race, however, takes several levels, leading us to adopt the approach of Tchetgen Tchetgen et al. (2010), who derive estimating equations for discrete outcomes of arbitrary cardinality. Liu et al. (2021) partially motivate their approach by noting that it avoids estimating the conditional density $f(A \mid R = \text{White}, X)$. In our application, A is binary and R has relatively few levels, so estimating this nuisance function is not prohibitively demanding.

R given X . For each $r \neq r_0$, define the odds ratio

$$\gamma(r, a, x) = \frac{\Pr(R = r \mid A = a, X = x) / \Pr(R = r_0 \mid A = a, X = x)}{\Pr(R = r \mid A = c, X = x) / \Pr(R = r_0 \mid A = c, X = x)}. \quad (40)$$

We impose the model $\log \gamma(r, a, x; \psi) = \mathbf{1}\{a = s\} \psi_r$, $r \neq r_0$, so that $\gamma(r, c, x; \psi) = 1$ and $\gamma(r, s, x; \psi) = \exp(\psi_r)$. The parameter vector $\psi = (\psi_r : r \neq r_0)$ is the object of interest. In particular, $\exp(\psi_r)$ compares the odds of race r relative to White drivers among officer-stop records with the corresponding odds among benchmark records, conditional on X . To complete the model, define the nuisance functions $f_0(r \mid x) = \Pr(R = r \mid A = c, X = x)$ and $g_0(a \mid x) = \Pr(A = a \mid R = r_0, X = x)$. Let $h(r, a \mid x; \psi, f_0, g_0) := \Pr(R = r, A = a \mid X = x)$ denote the full data conditional distribution of self-reported race and record source. The odds ratio parameterization implies

$$h(r, a \mid x; \psi, f_0, g_0) = \frac{\gamma(r, a, x; \psi) f_0(r \mid x) g_0(a \mid x)}{\sum_{\bar{r}} \sum_{\bar{a}} \gamma(\bar{r}, \bar{a}, x; \psi) f_0(\bar{r} \mid x) g_0(\bar{a} \mid x)}. \quad (41)$$

where the pair (f_0, g_0) is variation independent. This representation follows from the symmetry of the conditional odds ratio and imposes no parametric restriction on either the baseline race distribution f_0 or the baseline source mechanism g_0 .

When race is measured accurately, the estimating equation of [Tchetgen Tchetgen et al. \(2010\)](#) is evaluated under the full data law h using indicators of the observed value of R . In some of our analyses, however, self-reported race is not observed directly. Instead, we observe an imperfect proxy of it, W , which may be a surname or officer-reported race. For records from source a , we characterize the measurement process by $t_a(w \mid r, x) = \Pr(W = w \mid R = r, A = a, X = x)$. The observed data conditional law is then obtained by marginalizing the full data law h over self-reported race

$$\Pr(W = w, A = a \mid X = x) = \sum_r t_a(w \mid r, x) h(r, a \mid x; \psi, f_0, g_0), \quad (42)$$

which imposes no additional assumptions on the data generating process. Consequently, it

does not alter the definition of the target parameter or impose any restriction between f_0 and g_0 .

To adapt the estimating equation, let $T_a(x)$ denote the matrix with entries $t_a(w \mid r, x)$. When $T_a(x)$ has full column rank, let $B_a(x)$ be a left inverse satisfying $B_a(x)T_a(x) = I_J$. We replace the unobserved vector of indicators for self-reported race with $Z^\dagger = B_A(X)e_A(W)$, where $e_A(W)$ is the one-hot vector associated with the observed race signal in the relevant source. By construction, $\mathbb{E}[Z^\dagger \mid R, A, X] = e(R)$. Consequently, replacing the indicator for self-reported race with $Z^\dagger$ preserves the population expectation of the original estimating equation under h . The corrected equation thus targets the same parameter ψ and retains the same variation independent nuisance functions (f_0, g_0) , while accounting for measurement error in race that differs by source.

When using surnames to learn unmeasured race, $T_a = Q$, the matrix of surname likelihoods given race taken from the census as described in Section 3.2. If instead officer-reported race is used, the entries of T_s are the forward misreporting probabilities. Because these are generally not point identified, we repeat the analysis over a prespecified collection of plausible matrices and interpret the resulting estimates as a sensitivity analysis to the degree of misreporting.

When R and A have finite support, [Tchetgen Tchetgen et al. \(2010\)](#) derive an estimating equation from the efficient influence function under the full data law h . The resulting moment condition is Neyman orthogonal with respect to f_0 and g_0 and is doubly robust in the sense that its expectation is zero at the true value of ψ if either nuisance function is correctly specified. Because the corrected race indicator has the same conditional expectation as the indicator for self-reported race, the adjusted moment has the same population expectation as the full data moment when the measurement operator is correctly specified. It thus inherits the corresponding orthogonality and double robustness properties with respect to

f_0 and g_0 .

Both nuisance functions may depend flexibly on a complex set of covariates X , so we estimate them using machine learning methods and cross fitting. Under standard regularity conditions, the resulting DML estimator permits $\sqrt{n}$ -consistent and asymptotically normal inference for ψ_r and the odds ratio $\exp(\psi_r)$. We apply this approach in Section 5.2.1 and examine the extent to which officer misreporting of race can affect results.

4.2.1 Inference for the falsification test

Here, we summarize the inference procedure we will adopt for the falsification test of Proposition 2.9 which was derived to avoid the assumption that stop intensity is proportional to crash risk. As the max and min operators are nonregular, we employ a simple approximate strategy based on that proposed in Levis et al. (2025) which uses the softmax function to approximate the maximum and minimum. As this function is smooth, inference becomes easier. For ease of implementation, we will employ a parametric method to estimate the racial proportions (including when race is mismeasured) in both sources and use the delta method for uncertainty quantification and testing. Full details are provided in Appendix G.

5 Applications

We now turn to the empirical applications. To illustrate the malleability of the methodology, we will study racial disparities in enforcement for the camera and crashes benchmarks we have proposed above. The use of checkpoints is discussed in Appendix D.

5.1 Cameras: Chicago

We demonstrate the camera benchmark in Chicago, a city that pairs traditional traffic enforcement by officers with one of the largest automated camera networks in the United

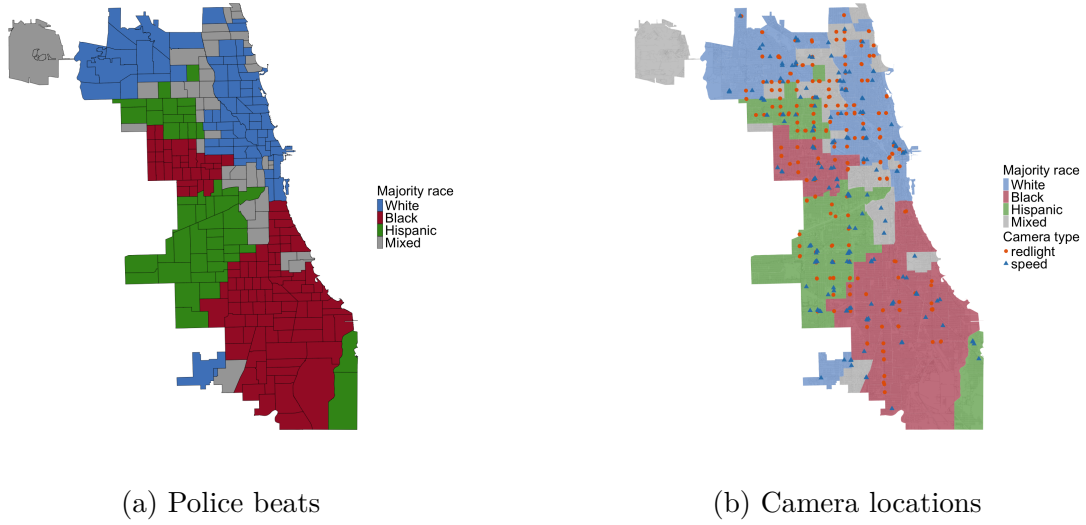


Figure 6: Chicago beat borders and demographics. (a) Police beats (b) Locations of speed and redlight cameras overlaid on the road network. Each beat is colored by the majority race residing in that beat according to the census, uncolored beats have no such majority. Redlight cameras are placed at intersections, while speed cameras are placed near schools.

States. Beginning in the early 2000s and expanding steadily thereafter, the City of Chicago deployed high visibility speed and redlight cameras across major arterial corridors, school zones, and other high risk segments of the road network ([City of Chicago Department of Transportation, 2023, 2025](#)). Public data on Chicago traffic stops covers the period 2017–2023 and thus this will be our period of study, within this time frame our data contains 184 speed cameras and 316 redlight across the City of Chicago, see Figure 6. Camera citations are triggered mechanically by detecting excess speed against a local speed limit or passing through an intersection on a red light. Crucially, the process is fully automated: no officer observes or interacts with the driver, and no human discretion enters the citation decision.

Each camera is tied to a fixed geographic location, whose coordinates are recorded in the data, along with information about the vehicle, its registered owner, and moving violation

Year	Redlight		Speed	
	Officer	Camera	Officer	Camera
2017	28,610	472,317	2,166	394,332
2018	46,033	500,124	1,646	370,437
2019	52,226	484,589	1,148	363,257
2020	28,078	484,804	440	240,246
2021	27,536	601,249	679	1,164,074
2022	39,599	620,278	494	807,270
2023	39,887	657,357	446	532,591

Table 2: Counts of officer stops and camera stops in the data by year, after filtering blocks by the coarsened exact matching procedure.

at the moment the threshold is crossed. Citations based on camera captures are processed centrally and mailed to the owner, and the underlying data include time measured to the second, date, location, plate state, and owner zip code. Across the period of study, we recorded around 4.49 million redlight camera citations and 12.7 million speeding citations. Importantly, this dataset does not record the self-reported race of the driver, however, it does contain their name. We instead estimate race proportions within X via the EM procedure detailed in Appendix E under the considerations of Section 3. A careful reader may be concerned that, because citations are issued to the registered owner of the vehicle, the race of the cited individual may differ from the race of the person driving at the time of the violation. We view this concern as limited in magnitude. In many cases, the driver is likely to be related to or otherwise closely associated with the registered owner and therefore of similar racial background. Moreover, our corrected BISG procedure does not deterministically assign race to any individual encounter. Instead, it returns the overall proportions of each

racial group in the data within a block x . By incorporating this uncertainty directly into the estimation procedure, the method attenuates the impact of any mismatch between the registered owner and the actual driver, thereby mitigating potential bias from this source.

Officer-initiated traffic stop data for Chicago are drawn from publicly available administrative records released under the Illinois Traffic and Pedestrian Stop Statistical Study Act (625 ILCS 5/11-212) ([Illinois General Assembly, 2003](#)). Since 2004, law enforcement agencies in Illinois have been required to report detailed information on each traffic stop to the state, including the police beat in which the stop occurred, date and time, model and make of the vehicle, stated reason for the stop, the name and race of the driver as recorded by the officer, and other features auxiliary to our application such as an initiated search. For Chicago, these records have been compiled and made publicly available by WBEZ/Chicago Public Media and the Investigative Project on Race and Equity ([WBEZ Chicago and Investigative Project on Race and Equity, 2023](#)). A limitation of the public data is that it only contains officer-reported race of the driver, which is subject to measurement error, as discussed in Section 3.3. We sought driver surnames through Freedom of Information Act requests to both the Chicago Police Department and the state of Illinois in order to apply the correction in Section 3, but both requests were denied.

To mitigate this issue, we appeal to a recently published report documenting statistical evidence supporting a lawsuit filed by the American Civil Liberties Union against the CPD for accusations of widespread racial bias in policing [FORTHCOMING]. There, it was discovered that officers systematically misreport race over the study period: as many as 1 in 4 drivers whom officers say are White are in fact minorities, see Table 3. We will use this data to correct empirical proportions of officer reported race. There are two drawbacks of this data released by the report, the first is that it is not reported at the level of granularity we have for the public data, and secondly the uncertainty in the estimates is unknown. To

accommodate this restriction, we take the estimates without error but perform a sensitivity analysis, by recomputing the results assuming increasingly accurate reporting by officers.⁹

In particular, let M_{rev}^t be the reverse misreporting matrix whose entry $[M_{\text{rev}}^t]_{r,\tilde{r}}$ is the probability that a motorist whom an officer reports as race $\tilde{r}$ would self-report race r in year t , that is $\Pr(R = r \mid \tilde{R} = \tilde{r}, S = s_1)$. Define the adjusted matrix

$$\tilde{M}_{\text{rev}}^t(\alpha) = (1 - \alpha)M_{\text{rev}}^t + \alpha I_J. \quad (43)$$

For $0 \leq \alpha \leq 1$, the adjusted matrix interpolates between the reverse misreporting matrix estimated by the ACLU and perfect reporting. At $\alpha = 0$, $\tilde{M}_{\text{rev}}^t(\alpha)$ treats the misreporting rates as those reported by the ACLU. At $\alpha = 1$, $\tilde{M}_{\text{rev}}^t(\alpha)$ pretends officer reports are perfectly precise, so that a motorist's self-reported race always coincides with the officer's label. We can interpret $\tilde{M}_{\text{rev}}^t(\alpha)$ as having artificially corrected $100\alpha\%$ of officer misreporting errors.¹⁰ For $\alpha < 0$, $\tilde{M}_{\text{rev}}^t(\alpha)$ extrapolates away from the identity matrix and represents worse officer misreporting than estimated by the ACLU; we retain such values only when $\tilde{M}_{\text{rev}}^t(\alpha)$ remains a valid stochastic matrix. Denote by $\psi_r^\dagger(\alpha)$ the target causal risk ratio comparing stops of violators of race r to White drivers where the misreporting rate is corrected at rate α .

We implement this correction as described in Section 3.3 and give additional details as

⁹Absent this correction, were an analyst forced to proceed with officer-reported race as given, it is important to note that as (i) it is easier to mistakenly misreport a minority of lighter complexion as White than the reverse, and (ii) there is little institutional incentive to overrepresent the presence of minority drivers in the data, we expect measurement error to inflate the count of White drivers in the data. Such behavior attenuates the size of discrimination against minorities, meaning their findings would be conservative in that they likely understate the true magnitudes of racial disparities.

¹⁰This interpretation is valid because, column by column, equation (43) raises the diagonal entry to $[M_{\text{rev}}^t]_{\tilde{r},\tilde{r}} + \alpha(1 - [M_{\text{rev}}^t]_{\tilde{r},\tilde{r}})$ and scales every off-diagonal entry by $(1 - \alpha)$. The weight the column assigns to races other than $\tilde{r}$ —the rate at which the label $\tilde{r}$ is wrong—shrinks by exactly a factor of $(1 - \alpha)$. Thus α is the fraction of stops mislabeled relative to self-reported race that we assume away, holding fixed which races those stops could otherwise have been.

Year	$p(b \mid \tilde{b}, s_1)$	$p(h \mid \tilde{h}, s_1)$	$p(w \mid \tilde{w}, s_1)$	Year	$p(\tilde{b} \mid b, s_1)$	$p(\tilde{h} \mid h, s_1)$	$p(\tilde{w} \mid w, s_1)$
2017	0.991	1.000	0.753	2017	0.998	0.899	0.995
2018	0.989	1.000	0.757	2018	0.999	0.901	1.000
2019	0.988	1.000	0.756	2019	0.999	0.906	0.993
2020	0.990	1.000	0.744	2020	0.997	0.927	0.992
2021	0.992	1.000	0.760	2021	0.999	0.925	1.000
2022	0.991	1.000	0.764	2022	0.999	0.923	1.000
2023	0.990	1.000	0.760	2023	0.999	0.926	0.995

Table 3: Mismatch between officer-reported ($\tilde{r}$) and self-reported (r) race by year among the stopped. Symbols b, h and w are shorthand for Black, Hispanic and White respectively. Left, taken directly from the ACLU report, gives the precision of the officer reports. Right, computed from values in the left table using Bayes’ rule after estimating $p(r \mid s_1)$ by (38), gives the sensitivity of officer reports.

they pertain to this particular example in Appendix E. Across the period of study, we count 638,889 stops for moving violations other than speeding, of which approximately 55% belong to (as reported by the officer) Black civilians, 26% Hispanic and 14% White. Additionally, there were 73,115 stops for speeding violations of which approximately 55% belong to Black civilians, 17% Hispanic and 25% White. We classify the race of the civilians in the remaining encounters as “other”.

As the camera data do not contain any reported race of the driver, and the officer stop records lack precise incident location data, we are limited in the granularity of the covariates we can use for confounding control. We accordingly proceed via coarsened exact matching (CEM), as motivated and described in Sections 2.2.3 and 4.1. Specifically, we defined discrete spatiotemporal blocks constructed from the intersection of calendar quarter-by-year (e.g., January–March 2020), day type (weekday versus weekend), police work shift (eight hour intervals), and police beat (approximately one square mile in area). We further filter blocks to those within which there were at least one hundred camera stops for increased

power. Each police stop was matched exactly to camera citations occurring within the same block. This procedure alleviates concerns about differential rates of exposure to the driving population that are observed due to the nonrandom placement of cameras throughout the city. By design, CEM restricts attention to regions of common support and reduces reliance on assumptions about functional form that would otherwise be required to model highly nonlinear patterns in enforcement intensity across space and time. Consequently, aggregate results should be interpreted as measuring discrimination over the geographic distribution of camera locations, however, this is expansive, see Figure 6. After filtering the datasets by the matching procedure the counts of stops and citations remaining are in Table 2.

5.1.1 Results: Chicago

We first pool the results across years and subsequently discuss yearly trends. To account for potential officer misreporting by race, we apply the algorithm in Appendix E using the left table in Table 3. We begin with moving offenses other than speeding. If neither officers nor cameras ever miscapture an offense ($\theta_x = \eta_x = 1$ for all x), Black drivers are more likely than White drivers to be stopped relative to the camera benchmark ($\psi^\dagger = 1.220 (1.176, 1.261)$), corresponding to an estimated disparity of approximately 22%. Hispanic drivers are also substantially more likely than White drivers to be stopped ($\psi^\dagger = 1.384 (1.350, 1.422)$), corresponding to an estimated disparity of approximately 39%. In Table 4 we demonstrate why the correction for officer misreporting is necessary. We find that were we to use officer reported race ($\alpha = 1$) directly, we would have erroneously concluded the presence of anti-White bias. Moreover, we show that even if the analyst believed that the results in Table 3 implausible, correcting 50% of officer errors in misreported race does not invalidate a conclusion of racial bias against Black and Hispanic drivers.

These results, owing to Theorem 2.6, are sensitive to the rates of officer and camera miscapture. The sensitivity results in Tables 13 and 14 show that increasing rates of

Year	Race (r)	$\psi_r^\dagger(\alpha = 0)$	$\psi_r^\dagger(\alpha = 1)$	$\psi_r^\dagger(\alpha = 0.5)$	$\psi_r^\dagger(\alpha = -0.5)$
All years	Black	1.220 (1.176, 1.261)	0.929 (0.895, 0.960)	1.055 (1.016, 1.090)	1.449 (1.396, 1.497)
		1.384 (1.350, 1.422)	0.905 (0.882, 0.932)	1.112 (1.084, 1.144)	1.759 (1.718, 1.807)
	Hispanic				

Table 4: Redlight camera benchmarking analysis in Chicago. The causal risk ratio is denoted by $\psi_r^\dagger$ for each race r (baseline group is White). Estimates assume officers and cameras do not lie, i.e. $\theta_x = \eta_x = 1$. Values $\alpha = 1, 0, -0.5$ correspond respectively to treating officer-reported race as accurate, applying the ACLU correction, and extrapolating beyond the correction to worse misreporting behavior.

miscapture widen the identified set for $\psi_r^\dagger$, in some cases sufficiently to include the null value of one, but equally could indicate greater prevalence of discrimination. For instance, we find that if there are no camera miscaptures, the results for Hispanic motorists are robust to 4% of officer stopped civilians being falsely charged. Interpreting these sensitivity results is inherently difficult because the true miscapture rates are unknown and nontrivial to quantify, and are themselves emblematic of another kind of discrimination (i.e. disparities in false charges). We do not impose any strict structure on these quantities because we lack sufficiently strong prior information to make an empirically grounded choice in that regard. Here, it is easiest to view this sensitivity analysis through the lens of data quality: does our inference of racial bias *among violators of the law* persist when the data set is contaminated by those who are not. We find that when miscapture rates are small, i.e. of the order 2%, this is the case. However, this analysis will benefit from future work which examines what magnitude of officer and camera miscapture we should expect; we discuss this further in the Section 6.

Figure 7 presents yearly estimates of $\psi^\dagger$ at $\theta = \eta = 1$. For Hispanic drivers, the estimated disparity is present throughout the study period, reaches its highest level in 2020, and

subsequently decreases, although it remains above one through 2023. The Black–White comparison exhibits a similar pattern, where estimates peak in 2019 but attenuate towards the null in later years. We note here that the timing of the decline overlaps with the COVID-19 pandemic and the aftermath of the killing of George Floyd in May 2020, interactions between which may explain the decrease in the risk ratio following 2020.

Regarding stops for speeding offenses, the pooled citywide estimates are below unity for both Black ($\psi^\dagger = 0.471$ (0.415, 0.564)) and Hispanic ($\psi^\dagger = 0.710$ (0.643, 0.813)) drivers, indicating that White drivers are more likely than minorities to be pulled over for speeding, across the study period. However, these results average over notably heterogeneous years, where for Black drivers, there is no detectable bias from 2020 through 2022, before a disparity re-emerges in 2023. In contrast, we find Hispanic drivers are over 50% more likely to be stopped in 2020, 2022 and 2023. These results should be interpreted in light of the relatively small sample sizes reported in Table 2. Speeding is policed far less frequently than other low level offenses in Chicago, so year to year variation in the concentration of speeding enforcement may induce substantial changes in which drivers are exposed to police scrutiny, driving the observed heterogeneity.

Mechanisms The results above reveal heterogeneity in Chicago that varies across racial groups and across offense types, rather than a uniform citywide pattern of discrimination against both Black and Hispanic drivers. In the analysis of moving offenses other than speeding, the pooled citywide estimates indicate disparities for both Black and Hispanic drivers, with the disparity larger for Hispanic drivers. Chicago is highly segregated by both race and socioeconomic status, and a single citywide estimand may thus obscure significant spatial variation in how disparities arise, potentially masking heterogeneity that is informative about underlying mechanisms (Ba et al., 2021).

Indeed, a substantial body of work has found that racial disparities in policing are heavily

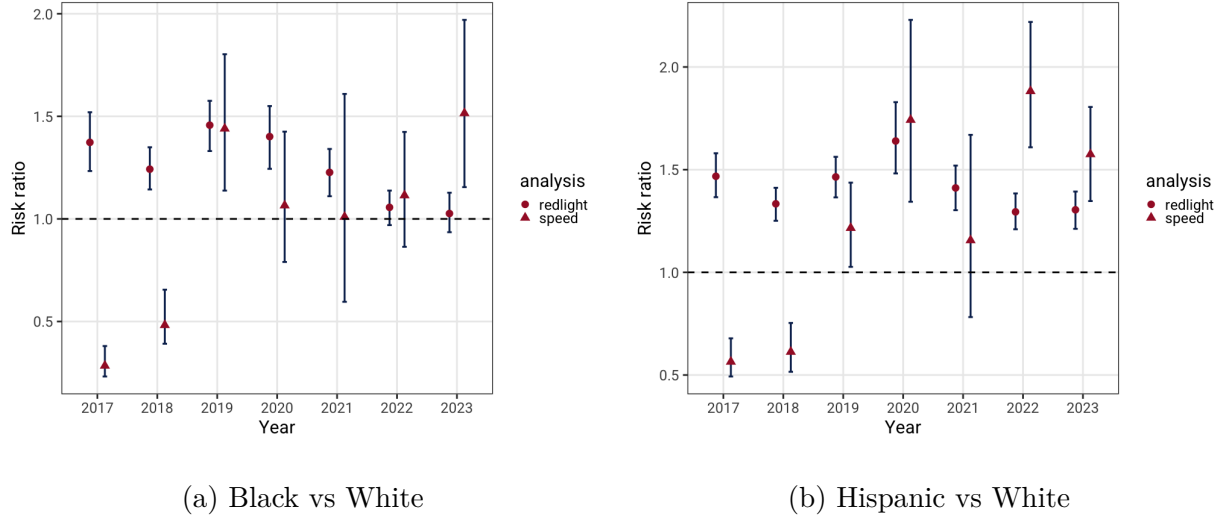


Figure 7: Yearly estimates of the causal risk ratio $\psi^\dagger$ setting $\theta = \eta = 1$, shown separately for moving offenses other than speeding (circles) and speeding offenses (triangles), error bars are 95% confidence intervals computed by the algorithm given in Appendix E.

influenced by the geography of the jurisdiction. One mechanism, often described as “out-of-place” policing, holds that officers use race as a cue for incongruence with the perceived social character of a neighborhood, generating heightened scrutiny when an individual appears not to suit local expectations (Meehan and Ponder, 2002; Supreme Court of Arizona, 1975). This behavior is a symptom of the legacy of slavery and segregation in the United States. “In-place” policing, on the other hand, has been proposed as an alternative mechanism for producing racial disparities in stops, whereby officers conduct more stops within minority neighborhoods because such areas are treated as higher crime environments and are therefore targeted for enforcement aimed at detecting contraband, where often low level offenses such as speeding are used as a ‘pretext’ to search a civilian (Gelman et al., 2007; Petrocelli et al., 2003; Epp et al., 2014; Pierson et al., 2020). Empirical work directly contrasting these predictions shows that disparities can arise in both contexts, indicating that neither mechanism alone is sufficient to explain discrimination (Hannon et al., 2021). This finding is reinforced by more recent studies which have examined the borders of racially and

socioeconomically distinct neighborhoods, and documented elevated police activity in such boundary zones even after adjusting for crime rates, particularly for low level discretionary enforcement (Neil and Legewie, 2024; Schenker et al., 2023; Verrilli et al., 2025).

Legally speaking, the framework governing investigative stops grants officers quite some leeway. The US supreme court verdict on the *Terry v. Ohio* case authorizes brief detentions based on “reasonable suspicion,” a standard that depends on officers’ interpretation of contextual cues (U.S. Supreme Court, 1968). Decisions based on these cues may use factors such as location or presence in a high crime area to enter the “totality of the circumstances” to justify the stop (U.S. Supreme Court, 2000), and minor traffic violations provide an objective legal basis for a stop regardless of subjective motive (U.S. Supreme Court, 1996; Rushin and Edwards, 2021). These doctrines formalize a setting in which discretionary judgments about space and behavior are legally actionable. However, a synthesis of the scholarly literature demonstrates that the manner in which these judgments are made is highly variable both within and across American jurisdictions.

From a causal perspective, these accounts generate testable predictions about effect heterogeneity by neighborhood within Chicago. In particular, we estimate the magnitude of $\psi^\dagger$ after restricting the context X to neighborhoods in which a single racial group comprises more than half of residents according to the Census. If officers behave in a manner consistent with out-of-place theory, disparities against minority drivers should be more pronounced in predominantly White neighborhoods. For moving offenses other than speeding Table 5 shows that the results instead provide evidence more consistent with in-place policing theory for both racial groups. For Black drivers, the largest disparity occurs in majority-Black beats, where the risk ratio is 2.822 (2.606, 3.048), while the pooled estimate in majority-White beats is close to one. For Hispanic drivers, the largest disparities likewise occur in majority-minority neighborhoods, with estimates of 2.081 (1.864, 2.315) in majority-Black

beats and 2.037 (1.882, 2.210) in majority-Hispanic beats. These patterns persist by year and remain robust to even 5% rates of officer and camera miscapture, see Table 15.

Beat majority	Race (r)	$\psi_r^\dagger(\alpha = 0)$	$\psi_r^\dagger(\alpha = 1)$	$\psi_r^\dagger(\alpha = 0.5)$	$\psi_r^\dagger(\alpha = -0.5)$
Black	Black	2.811	2.163	2.443	3.314
		(2.610, 3.027)	(2.006, 2.330)	(2.267, 2.632)	(3.078, 3.568)
Black	Hispanic	2.067	1.409	1.694	2.579
		(1.844, 2.314)	(1.247, 1.590)	(1.505, 1.903)	(2.309, 2.877)
Hispanic	Black	1.204	0.916	1.040	1.429
		(1.096, 1.317)	(0.834, 1.003)	(0.947, 1.139)	(1.303, 1.564)
Hispanic	Hispanic	2.043	1.493	1.730	2.474
		(1.891, 2.235)	(1.380, 1.636)	(1.601, 1.894)	(2.292, 2.704)
White	Black	0.968	0.733	0.834	1.152
		(0.917, 1.016)	(0.694, 0.770)	(0.790, 0.876)	(1.091, 1.209)
White	Hispanic	1.174	0.723	0.917	1.527
		(1.130, 1.214)	(0.694, 0.750)	(0.882, 0.950)	(1.472, 1.578)

Table 5: Redlight camera analysis in Chicago by beat demographics. Estimates assume officers and cameras do not lie, i.e. $\theta_x = \eta_x = 1$.

Next consider speeding offenses. Across the study period, Black drivers in majority-White beats are estimated to be 2.578 (2.174, 3.033) times as likely as White drivers to be stopped for speeding relative to the camera benchmark, an increase of 158%. Hispanic drivers in majority-White beats have a similarly large estimate of 2.063 (1.879, 2.246), or 106% higher. For Black and Hispanic drivers, the corresponding estimates in majority-minority beats indicate that they are less likely to be stopped for speeding than White drivers. All estimates in majority-White beats persist by year, and are robust to increased levels of officer and camera miscapture, see Table 16. Relative to the redlight results, this provides considerably clearer evidence consistent with an out-of-place policing mechanism in speeding enforcement, particularly for Black drivers.

Beat majority	Race (r)	$\psi_r^\dagger(\alpha = 0)$	$\psi_r^\dagger(\alpha = 1)$	$\psi_r^\dagger(\alpha = 0.5)$	$\psi_r^\dagger(\alpha = -0.5)$
Black	Black	0.724 (0.595, 0.862)	0.552 (0.453, 0.658)	0.626 (0.514, 0.746)	0.859 (0.707, 1.023)
Black	Hispanic	1.538 (1.257, 1.896)	1.033 (0.826, 1.301)	1.251 (1.012, 1.557)	1.934 (1.595, 2.364)
Hispanic	Black	0.293 (0.259, 0.353)	0.222 (0.196, 0.267)	0.253 (0.223, 0.304)	0.350 (0.310, 0.421)
Hispanic	Hispanic	0.475 (0.425, 0.542)	0.275 (0.242, 0.319)	0.361 (0.321, 0.415)	0.634 (0.570, 0.718)
White	Black	2.578 (2.174, 3.033)	1.955 (1.647, 2.302)	2.223 (1.874, 2.616)	3.069 (2.590, 3.609)
White	Hispanic	2.063 (1.879, 2.246)	1.300 (1.164, 1.433)	1.629 (1.472, 1.784)	2.665 (2.440, 2.888)

Table 6: Speed camera analysis in Chicago by beat demographics. Estimates assume officers and cameras do not lie, i.e. $\theta_x = \eta_x = 1$.

5.2 Crashes: Texas

Texas banned the use of road traffic cameras in 2019, making an up to date benchmark analysis using cameras impossible. Another benchmark is needed, here, as we will show, crashes make for a sound choice. Here, we will conduct a statewide analysis of the Texas Highway Patrol whose stop data is publicly available online.

Crash data for Texas are made publicly available by the Texas Department of Transportation, which maintains comprehensive administrative records on all reported motor vehicle accidents statewide. These data include detailed information on the circumstances of each crash, such as the geographic coordinates, date and time, number of vehicles involved, and driver level contributing factors, as well as officer reported race for drivers involved. We use these contributing factors to construct a crash benchmark intended to capture drivers whose behavior plausibly reflects dangerous driving that could cause a crash, which satisfies the relevance condition [B1](#). A standardized list of contributing factors is recorded by the Texas

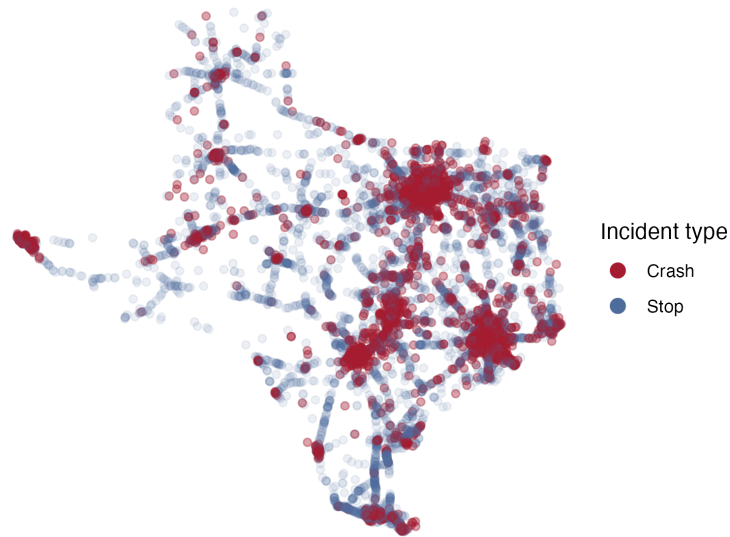


Figure 8: Sample locations of stops and crashes on Texan highways. Crashes tend to concentrate in urban areas.

Department of Transportation,¹¹ and their literal descriptions are contained in the data. We remove crashes where it is plausible neither driver could have been at fault, for instance, where an animal is on the road; pedestrian collisions are also excluded.

A potential threat to validity is that crash investigation coding may be biased—for example if investigating officers more likely record contributing factors that indicate dangerous driving for minority drivers conditional on the same underlying driving behavior, violating the benchmark exclusion condition B2. West (2018) finds evidence consistent with differential leniency in determinations made after a crash, by officer–driver race pairings. Directionally, if such bias inflates the share of minority drivers coded with contributing factors that indicate dangerous driving, then the benchmark denominator is shifted toward minorities and any estimated anti-minority enforcement disparities would be conservative.

¹¹See here, <https://www.txdot.gov/content/dam/docs/division/trf/crash-records/2024/21.pdf>

For officer stops, the primary data source used is records at the incident level from the Texas Highway Patrol. These administrative data provide detailed information on individual traffic stops and citations, including the date and time of the encounter, geographic coordinates, the reported race of the civilian, the literal violation recorded by the officer, and the type of ticket issued. We summarize raw stops and crashes data with corresponding racial proportions in Table 7.

Year	Stops					Crashes				
	<i>N</i>	Black	Hisp.	Other	White	<i>N</i>	Black	Hisp.	Other	White
2016	1,164,425	0.100	0.349	0.028	0.523	476,079	0.167	0.348	0.035	0.450
2017	1,344,896	0.101	0.365	0.026	0.508	457,614	0.170	0.340	0.037	0.454
2018	1,528,998	0.103	0.368	0.025	0.504	460,271	0.173	0.343	0.038	0.446
2019	1,472,161	0.101	0.385	0.024	0.490	478,881	0.175	0.362	0.038	0.425
2020	938,465	0.109	0.401	0.024	0.466	402,530	0.190	0.372	0.032	0.406
2021	831,182	0.099	0.445	0.023	0.432	475,009	0.188	0.375	0.035	0.402
2022	868,395	0.105	0.415	0.027	0.453	488,876	0.179	0.389	0.040	0.391
2023	373,587	0.109	0.381	0.033	0.477	495,130	0.173	0.392	0.044	0.391
2024	752,277	0.101	0.450	0.035	0.414	491,153	0.169	0.405	0.048	0.378

Table 7: Yearly stop and crash counts and race proportions in Texas data.

5.2.1 Results: Texas

We remind the reader that our results act as a test of the global null that officers do not discriminate against Texan drivers within X , assuming that under the null officers carry out their duties by sensitivity to the propensity of drivers to jeopardize road safety. We present the analysis faceted by year, the entirety of the state, and the four largest metro areas of

Texas (Houston, Dallas–Fort Worth, Austin, San Antonio) because the highest density of crashes occur in these more urban regions.

The results are visualized in Figure 9. At first glance, it would appear that across Texas as a whole, we find strong evidence of anti-White bias over time. First note that there is substantial spatial heterogeneity in this finding. For instance, in San Antonio, there is strong evidence for discriminatory traffic enforcement against both minority groups, which peaked in recent years. Results from Houston suggest that anti-minority bias in stops has been decreasing since 2016. In Austin, Dallas, and Texas as a whole, indications of anti-White bias appear to exist both before and after the murder of George Floyd, implying that our evidence is not an overcorrection by police departments in response to the political pressure in the wake of the murder.

These results are atypical. Some studies have claimed to have detected anti-White bias for some policing outcomes such as traffic stops and searches and police use of force ([Antonovics and Knight, 2009](#); [Sadler et al., 2012](#); [Novak and Chamlin, 2012](#); [James et al., 2013](#); [Rosenfeld et al., 2012](#); [Pierson et al., 2020](#)). However, the strength of our evidence in some years and metro areas outweighs that of past work, calling for further investigation. Indeed, as discussed in Section 3, we hypothesize these conclusions may stem from widespread misreporting of driver race in Texas. The data we obtained from the Texas Highway Patrol does not contain civilian surname. However, in prior work from some of the authors of this article ([Ba et al., 2021](#)), data was obtained by FOIA from the Houston PD which does contain civilian name, and we can utilize this data to get a sense of how severe the misreporting problem could be.¹² In Table 8, we document estimates of the officer

¹²An artifact of this data is that, for reasons unknown to us, no row in the data is recorded as belonging to a Hispanic driver, while all other standard (according to Houston PD) racial categories are labeled. The vast majority of the rows in the data labeled NA have Hispanic surnames, however. To address this problem, we compared our counts of drivers with a given race to the official reports released by Houston

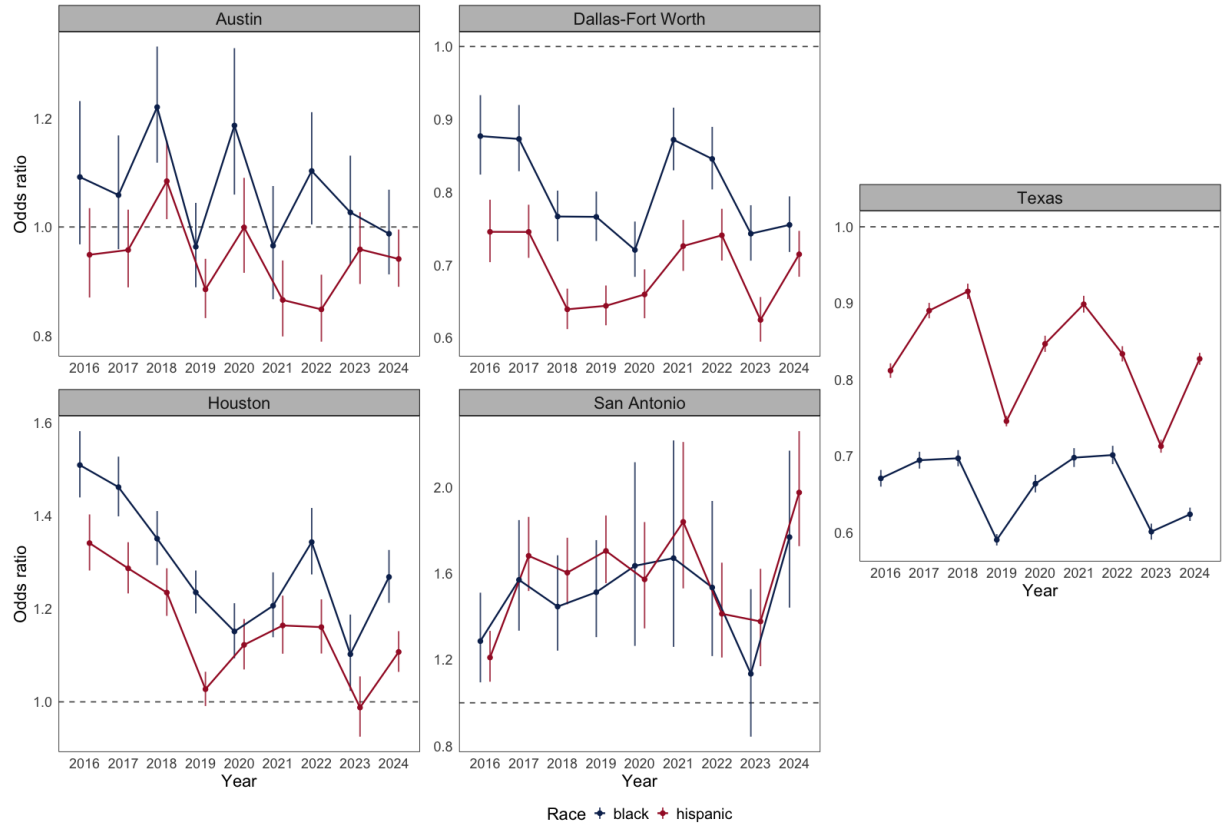


Figure 9: Odds ratios computed via DML in the four largest metro areas in Texas, Austin, Houston, Dallas–Fort Worth and San Antonio, along with their 95% confidence intervals.

misreporting rate computed via the method outlined in Section 3.3.

For Hispanic drivers particularly, these estimates are alarming. For instance, in 2017, it is estimated that over 7 in 10 Hispanic drivers and 1 in 10 Black drivers stopped are labeled as White. The rate of misreporting has decreased over time, suggesting that a series of journalistic efforts and related policy changes may have been successful. However, the prevalence of this practice remains common enough for estimates of discrimination to be significantly deflated. Based on the figures in this table, we apply the corrective procedure described in Section 4 to Houston data.

PD and found ours to be similar to theirs for all non-Hispanic racial groups. Hence, we calibrated our missing records to match the publicly available proportions of Hispanic drivers, and adjusted the remaining proportions accordingly.

Year	$p(\tilde{w} \mid s_1)$	$p(w \mid s_1)$	$p(\tilde{r} \mid s_1)$	$p(r \mid s_1)$	$p(\tilde{w} \mid r, s_1)$
$\tilde{r} = \text{Black}$					
2017	0.544	0.137	0.271	0.291	0.101
2018	0.562	0.129	0.264	0.273	0.082
2019	0.381	0.105	0.282	0.266	0.054
2020	0.310	0.146	0.327	0.278	0.048
$\tilde{r} = \text{Hispanic}$					
2017	0.544	0.137	0.108	0.506	0.720
2018	0.562	0.129	0.100	0.541	0.740
2019	0.381	0.105	0.253	0.564	0.450
2020	0.310	0.146	0.286	0.516	0.282

Table 8: Extent of officer misreporting in Houston PD by year. The rightmost column records how often Black and Hispanic motorists were labeled as White in stop data.

In the years where officer misreporting was so extreme (2017 and 2018), the corrected estimates were implausibly large—sometimes infinite—and unstable since the misreporting rates implied a very small fraction of stops belonged to White drivers. In part this is due to the coarse nature of the misreporting rate which we assume is constant by x . To mitigate this problem, we conduct a sensitivity analysis, almost identically to the cameras analysis, where we artificially increase the sensitivity (as opposed to the precision) of officer reports to varying degrees.¹³ In particular, let M_{for}^t be the forward misreporting matrix whose entry

¹³The DML estimator derived for Texas relies on the sensitivity $\Pr(\tilde{R} = \tilde{r} \mid R = r, S = s_1)$ of the officer labels, and we could compute a representative value of it from the Houston PD. In Chicago, it was more straightforward to work with $\Pr(R = r \mid \tilde{R} = \tilde{r}, S = s_1)$, since it was provided directly in the ACLU report, which explains the slight differences in how the sensitivity analysis is conducted, and why we change notation from α to β , since the parameter is conceptually similar but mathematically distinct.

$[M_{\text{for}}^t]_{\tilde{r},r}$ is the probability that an officer reports race $\tilde{r}$ for a motorist who would self-report race r in year t . Define the adjusted matrix

$$\widetilde{M}_{\text{for}}^t(\beta) = (1 - \beta)M_{\text{for}}^t + \beta I_J \quad (44)$$

for $\beta \in [0, 1]$. The adjusted matrix interpolates between the misreporting rates we estimate and perfect sensitivity. At $\beta = 0$, $\widetilde{M}_{\text{for}}^t(\beta)$ treats the misreporting rates as those estimated in Table 8. At $\beta = 1$, $\widetilde{M}_{\text{for}}^t(\beta)$ pretends officers are perfectly sensitive. We can, with a slight abuse of the English term ‘error’, equally interpret $\widetilde{M}_{\text{for}}^t(\beta)$ as having artificially corrected $100\beta\%$ of officer misreporting errors as we did in Chicago. In the main text, we present results for $\beta = 0.4$, but consider other values of β in the Appendix; qualitative conclusions are unchanged.

In Table 9, we show that failing to correct for officer misreporting underestimates racial bias by an order of magnitude. For instance, under the linear probability model assumption for the officer stopping rule, the results from 2018 imply that Hispanic drivers were nearly 2400% more likely than White drivers to be stopped by officers for the same road traffic violations, compared to a 24% increase, neglecting to account for errors in measurement.

We augment these results with those from the falsification test described in Section 2.4.1 which do not rely on the linear probability model assumption to derive a test of racial bias. The results are presented in Table 10, having corrected for officer misreporting. The test is carried out by comparing the proportion in the first numerical column $\Pr(R = r \mid S = s_1)$, to the maximum and minimum of the remaining numerical columns. For example, in 2017, 57% of officer stops were of Hispanic motorists, but Hispanics are most represented among fatal crashes at a rate of 52%. Within the sample, this is evidence of racial bias against Hispanic motorists, however, it is not statistically significant, as noted by the ‘Reject’ column.

Some interesting patterns emerge. White drivers are systematically underrepresented among

Year	Race	Odds ratio	
		$\beta = 1$	$\beta = 0.4$
2017	Black	1.461 (1.399, 1.527)	8.617 (7.169, 10.359)
	Hispanic	1.287 (1.233, 1.343)	13.296 (10.895, 16.226)
2018	Black	1.351 (1.294, 1.410)	14.490 (11.805, 17.785)
	Hispanic	1.235 (1.185, 1.287)	24.884 (20.068, 30.857)
2019	Black	1.235 (1.190, 1.282)	2.116 (2.013, 2.224)
	Hispanic	1.027 (0.991, 1.065)	2.647 (2.506, 2.796)
2020	Black	1.151 (1.094, 1.212)	1.454 (1.366, 1.548)
	Hispanic	1.122 (1.070, 1.178)	2.141 (2.012, 2.279)

Table 9: Estimates of the odds ratio in Houston, before and after correcting for misreporting of race, using the procedure described in Appendix F.

stops, relative to Black and Hispanic drivers. Their share of stops falls below their share of at-fault drivers at every level of crash severity we consider, and in 2017 the falsification test rejects on this basis. Black and Hispanic motorists, by contrast, only outweigh White drivers among the most serious crashes. Setting those aside, the point estimates convey systematic evidence of bias against minority drivers, especially Hispanics, although these comparisons do not attain statistical significance. Therefore, within our sample, White drivers are not stopped nearly as often as officer attention to road safety should imply. Put another way, the composition of stops in Houston looks much as it would were every minority motorist as dangerous as the drivers involved in the worst crashes, and every White motorist safer than those involved in the least severe.

One interpretation of these results is that officers penalize Black and Hispanic motorists

by treating them like the worst drivers among the group, in contrast to their treatment of White drivers, where officers devalue the safety risk posed by White drivers for more common dangerous driving behaviors. Two accounts of officer behavior can produce such a pattern, and our estimates lend support to each in a different part of the picture. Officers who hold biased priors about the propensity of certain drivers to jeopardize road safety may assess minority motorists as though they were drawn from the most dangerous tail of their group. We observe descriptive evidence of this behavior, as our estimates find the share of minority stops close to where minorities are most represented across the distribution of crash severity. Miscalibration such as this has been referred to, particularly in economics, as inaccurate statistical discrimination ([Bohren et al., 2025](#)), adapting prominent theories of how decision makers may discriminate in the presence of imperfect information ([Phelps, 1972](#); [Arrow, 1973](#)). We do not claim that this reasoning rationalizes officer behavior, as the miscalibration may itself be driven by prejudice or support of negative stereotypes ([Bordalo et al., 2016](#)).

A second potential mechanism is in-group leniency. Where officers discount the danger posed by drivers who share their own group, common dangerous behaviors by White motorists attract proportionately less intervention, depressing the White share of stops below what any level of crash severity warrants. [West \(2018\)](#) documents evidence consistent with such leniency in determinations made after a crash, by officer–driver race pairing, and the mechanism is plausible here, since a majority of commissioned officers at the Texas Department of Public Safety are White (55%, [Texas Department of Public Safety, 2022](#)). These accounts complement one another in that the first concerns how minority motorists are assessed, the second how White motorists are excused, and the asymmetry we document bears the signature of both.

We only apply this correction to Houston data because there is an empirical basis upon

which to select the misreporting rates. Future work will reapply the procedure producing the misreporting rates in Table 8 for other jurisdictions, and reapply the method we develop here.

Year	Race (r)	$\Pr(R = r \mid S = s_1)$	$\Pr(R = r \mid C \geq k_1)$	$\Pr(R = r \mid C \geq k_2)$	$\Pr(R = r \mid C \geq k_3)$	$\Pr(R = r \mid C \geq k_4)$	$\Pr(R = r \mid C \geq k_5)$	Reject
2017	Black	0.265 (0.248, 0.282)	0.209 (0.193, 0.225)	0.210 (0.180, 0.240)	0.192 (0.146, 0.238)	0.198 (0.147, 0.249)	0.464 (0.431, 0.497)	No
	Hispanic	0.573 (0.547, 0.599)	0.280 (0.262, 0.297)	0.281 (0.247, 0.314)	0.301 (0.246, 0.357)	0.258 (0.186, 0.329)	0.519 (0.484, 0.553)	No
	White	0.111 (0.091, 0.132)	0.462 (0.443, 0.482)	0.482 (0.445, 0.519)	0.486 (0.425, 0.547)	0.627 (0.557, 0.697)	0.443 (0.404, 0.482)	Yes
2018	Black	0.252 (0.237, 0.267)	0.217 (0.202, 0.233)	0.223 (0.191, 0.255)	0.235 (0.183, 0.287)	0.275 (0.176, 0.373)	0.367 (0.345, 0.390)	No
	Hispanic	0.573 (0.547, 0.598)	0.273 (0.256, 0.290)	0.291 (0.255, 0.327)	0.290 (0.235, 0.344)	0.399 (0.338, 0.461)	0.442 (0.411, 0.474)	No
	White	0.134 (0.113, 0.155)	0.473 (0.454, 0.492)	0.452 (0.412, 0.492)	0.437 (0.375, 0.499)	0.375 (0.255, 0.494)	0.477 (0.446, 0.507)	No
2019	Black	0.232 (0.218, 0.245)	0.206 (0.192, 0.220)	0.208 (0.181, 0.235)	0.210 (0.166, 0.253)	0.203 (0.108, 0.298)	0.427 (0.415, 0.440)	No
	Hispanic	0.448 (0.427, 0.468)	0.306 (0.289, 0.322)	0.312 (0.278, 0.346)	0.320 (0.264, 0.376)	0.508 (0.357, 0.658)	0.573 (0.560, 0.585)	No
	White	0.265 (0.247, 0.283)	0.446 (0.428, 0.463)	0.440 (0.403, 0.477)	0.449 (0.391, 0.507)	0.366 (0.203, 0.530)	0.414 (0.402, 0.426)	No
2020	Black	0.224 (0.205, 0.242)	0.232 (0.212, 0.251)	0.249 (0.209, 0.290)	0.223 (0.159, 0.288)	0.193 (0.113, 0.273)	0.329 (0.286, 0.371)	No
	Hispanic	0.450 (0.425, 0.475)	0.316 (0.294, 0.338)	0.308 (0.265, 0.350)	0.323 (0.249, 0.397)	0.339 (0.238, 0.440)	0.462 (0.439, 0.485)	No
	White	0.286 (0.264, 0.307)	0.402 (0.379, 0.426)	0.395 (0.350, 0.440)	0.399 (0.321, 0.476)	0.527 (0.397, 0.658)	0.426 (0.387, 0.465)	No

Notes: Each cell reports the point estimate on the first line and its 95% confidence interval in parentheses on the second line. “Reject” indicates rejection of the upper-bound component of the falsification test, except for White drivers, where it means a rejection of the lower-bound component.

Table 10: Severity falsification test for Houston, correcting for misreported race by the procedure described in Section G.

6 Discussion

We begin by discussing the results of our empirical analysis. Across two large US jurisdictions, our proposed framework uncovered substantial evidence of racial bias in traffic enforcement against Black and Hispanic civilians, using two simple benchmarks for officer behavior.

Using automated traffic cameras in Chicago, we find that minority drivers are systematically more likely to be stopped by police than their White counterparts, and that these disparities vary sharply with neighborhood demography. This provides new evidence of the role that policing practices tied to place can play in racial discrimination within US jurisdictions. For moving offenses other than speeding, Black drivers are approximately 182% more likely than White drivers to be stopped in majority-Black neighborhoods, while Hispanic drivers are approximately 104% more likely than White drivers to be stopped in majority-Hispanic neighborhoods. For speeding offenses, by contrast, the largest estimates of racial disparity are concentrated in majority-White neighborhoods.

We highlight that one drawback of our analysis is that the data do not contain the speeds of the drivers who were cited. Speed cameras may be triggered for smaller excess speeds than might incite an officer response. If White drivers disproportionately occupy the lower tail of the excess speed distribution—similar to our findings in Texas—then our estimates understate the magnitude of discrimination for speeding offenses, although, without additional data, this cannot be verified and the reverse pattern is equally plausible. Generally speaking, an analysis with complete data would examine to what degree excess speed predicts an officer response, and conduct the same analysis within reasonably defined tails of the speeding distribution.

These contrasting patterns by type of enforcement are revealing. In majority-minority neighborhoods, low level moving offenses other than speeding disproportionately expose mi-

minority drivers to police intervention, whereas in majority-White beats, speeding enforcement disproportionately falls on minority drivers. The former pattern is consistent with officers using common low level violations as sustained opportunities for intervention in minority neighborhoods, including potentially pretextual stops, and accords with anecdotal accounts of such practices, including those described in the ACLU lawsuit ([American Civil Liberties Union of Illinois, 2023](#)). Speeding, by contrast, is rarely enforced and does not appear to be a major focus of departmental activity. Yet when officers do enforce speeding laws, the resulting disparities are concentrated in White neighborhoods, suggesting that speeding violations may provide a legal basis for targeting minorities in these parts of the city.

In Texas, we use drivers at-fault in a crash to similarly find evidence of racial bias in stops carried out by the Texas Highway Patrol. At first, our results did not appear to show evidence of racial bias due to the impacts of widespread misreporting of civilian race. Our results thus underscore a key takeaway from this work: conclusions about racial bias are highly sensitive to how race is measured. Our investigation of police stops in Houston revealed that the use of officer reported labels for race severely underestimated the size of discrimination, in some years by an order of magnitude. Relaxing some assumptions underlying the use of the odds ratio for this analysis, the results for the falsification test based on crash severities point in the same direction: minority drivers, especially Hispanic drivers, are often more prevalent among police stops than would be implied by their representation across levels of dangerous driving behavior, whereas White drivers are systematically underrepresented among stops in the same sense.

Our theory underlying these results is not without limitations. First consider crashes. The test we derive in Proposition [2.8](#) relies on a stop rule that assumes that stop rates for minority and White drivers differ by a constant of proportionality. We abandon this limited linear probability model assumption for a richer class of functions of crash risk,

generating a simple falsification test for racial bias based on the severity of the crash. It is not always the case that an analyst possesses the severity levels of the crash, which may limit the applicability of this test. Beyond this point, these tests assume that (arbitrary transformations of) crash risk, D , sufficiently describes what captures officer attention on the road. While broad in the nature of driving behavior it captures, it may not be inclusive of all aspects of traffic enforcement relevant to officers, which may be subject to internal department imperatives. Any analyst making use of crashes data should interrogate this assumption as it pertains to their department of interest.

Next, cameras. Our main result in Theorem 2.6 derives an identification interval parameterized by the rates at which officers and cameras miscapture drivers. Because D is unobserved in principle, these miscapture rates are not directly estimable: the motorist’s behavior prior to police contact is never fully observed. Consequently, sensitivity analysis for bounds on $\psi^\dagger$ can be challenging to interpret on their face, though large estimated disparities, such as those observed in our Chicago heterogeneity analysis, eliminate this concern. Our analysis does not explore racial discrimination in miscapture rates, although it is likely to exist. We refrain from doing so because we lack sufficient empirical evidence to impose this structure credibly. As with crashes, measuring dangerousness directly before capture could better elucidate the bounds by better characterizing the population of safe drivers (cf. the proof of Theorem 2.6). Note though that, even if D were measured in the benchmark dataset, it would remain unavailable in police data, so our identification strategy would still be required.

Encouragingly, new data sources have emerged in the policing literature that may mitigate some of these challenges. Recent scholarship has employed telematic data—vehicle movement data recorded by GPS at fine granularity—to construct benchmarks against which to evaluate police stops. These data show promise because they allow analysts to construct

more accurate representations of the driving population than earlier methods (Knox and Mummolo, 2025). For example, Aggarwal et al. (2025) use high frequency telemetry data to characterize the population of (Lyft rideshare) drivers exceeding the speed limit, generating a benchmark with a credible proxy for D . One might conclude that such datasets eliminate many concerns raised in this work. While telematics significantly improve measurement of the at-risk driving population, important limitations remain. These can be grouped into issues of data quality and methodological application.

First, consider the data. Race and surname are typically unavailable. Race continues to be inferred using home Census block group demographics, which neglects the selection problem discussed in Section 3. Moreover, telematic samples are rarely representative of the driving population, and even less so of the violating population. For example, Aggarwal et al. (2025) rely on Lyft drivers, who may overrepresent minorities in certain cities, while Cai et al. (2022) use vehicle installed devices, which are thus likely correlated with a population of higher income containing a larger share of White drivers. Finally, such datasets are difficult to access and rarely available to police departments or independent scholars. There remains a need for robust methodology that does not rely on proprietary, high quality data.

Second, consider the application. Although data quality has improved, methodological foundations often have not. Despite similar benchmark data, recent studies vary dramatically in empirical design. Xu et al. (2024) and Gong et al. (2025) use camera and crash benchmarks to compute expected racial shares and compare these to stop counts. Aggarwal et al. (2025) infer stops among Lyft drivers and estimate the ATE of race on citation probability conditional on speed. Cai et al. (2022) compare speeding rates at the neighborhood level across racially distinct regions. We summarize these approaches in Table 11.

Gong et al. (2025) defend their design by appeal to legal statute. In our view, consistency with statutory language is neither a scientific justification nor a settled matter of law.

Although [Gong et al.](#) acknowledge that a benchmark should reflect the population at risk, their implementation includes drivers who may not be at risk of the relevant enforcement action. In principle, their benchmark differs from that of [Alpert et al. \(2004\)](#) only in degree: it may better describe the population of drivers, but it does not isolate the population of violators. Authors often characterize such findings as evidence of disparate impact rather than profiling. While this distinction is conceptually valid, two concerns remain. First, the authors state in their abstract that their goal is to “[establish] a reliable counterfactual for evaluating racial disparities in traffic stops using telemetric data to estimate the racial composition of motorists”. This statement is misleading insofar as it suggests that officers ought to stop minority motorists in proportion to the crashes benchmark they adopt which combines violators and nonviolators, even though the authors themselves later acknowledge that this is not the appropriate normative standard. Second, the design does not fully exploit the available behavioral information in the data to isolate the relevant counterfactual comparison among similarly situated motorists. Third, their reliance on statutory language is misplaced. For example, a DOJ manual states, “To prove... systemic discrimination... a plaintiff... can present statistical evidence of similarly situated individuals not in the protected class who were treated better...” ([U.S. Department of Justice, Civil Rights Division, 2015](#), pg 22). Instead, it uses a proxy that does not ensure comparability among drivers at the level relevant to the enforcement decision. Invoking statutory language does not remedy this conceptual loss. Legal standards articulate evidentiary principles; they do not determine whether a particular empirical construction satisfies those principles. Although [Xu et al. \(2024\)](#) do not rely on this appeal, the empirical strategy they adopt additionally suffers from this conceptual shortcoming.

[Cai et al. \(2022\)](#) test whether officers enforce traffic laws more aggressively in minority beats where speeding prevalence is no greater than in White beats. This choice is motivated

by the absence of race in the benchmark data. Analytically, they make strong parametric assumptions. Specifically, they estimate a negative binomial regression of stop counts on neighborhood demographic composition, controlling for the proportion of driving time exceeding 15 kilometers per hour over the speed limit, which is a limited summary of the available data. Moreover, their headline quantity, the coefficient on demographic composition, does not correspond to an estimand relevant to policy. It captures how aggregate enforcement intensity varies with neighborhood racial composition, conflating deployment patterns, exposure, and officer choices. Consequently, it cannot be directly interpreted as a parameter of discriminatory treatment and as a measure of disparate impact it has limited inferential content.

Ultimately, we argue that inconsistencies in this emerging literature stem from the absence of a clear causal foundation of the benchmarking test. A rigorous framework capable of leveraging these new data sources in a manner consistent with discrimination case law remains underdeveloped. This article may be used as a starting point, grounded in data accessible to most analysts and policymakers. As telemetric datasets become more widely available, future work will extend this causal framework to accommodate them.

Benchmark	Telemetry	Variables	Unit of Analysis	Model / Design	Implied Estimand	Limitations
Lyft drivers in Florida (Aggarwal et al., 2025)	Mobile phone GPS data at the driver level, high frequency	R, S, X, V (speed)	Driver $\times$ time inter- val	FE / DML regression: $S = \alpha + \tau R + f(V, X) + \varepsilon$	$\mathbb{E}[S \mid R = 1, V, X] - \mathbb{E}[S \mid R = 0, V, X]$	Population restricted to rideshare drivers; does not isolate enforce- ment risk among violators explic- itly
Road users and cam- era citations in Chicago metro (Xu et al., 2024)	Synthetic mobile phone GPS mobility data	R, S, C, X	Road segment	Share gap between stops and roadway exposure or camera citations	$\frac{\Pr(R = r \mid S = s_1, X)}{\Pr(R = r \mid \text{on road}, X)},$ $\frac{\Pr(R = r \mid S = s_1, X)}{\Pr(R = r \mid C = c_1, X)}$	Aggregates violators and nonvio- lators; conflates exposure, and en- forcement decisions
Drivers in commercial telematics across 10 U.S. cities (Cai et al., 2022)	Vehicle installed GPS data at the driver level, high frequency	N_L (stop count), μ_L (driving time), d_L (speeding share), r_L (demographic share)	Beat $\times$ year	Negative binomial re- gression: $\log E[N_L] =$ $\log \mu_L + \beta_r r_L + \beta_d d_L + \text{FE}$	β_r	Individual heterogeneity reduced to artificially scalar d_L ; does not represent counterfactual con- trast among similarly situated mo- torists; estimand not relevant to policy
Crashes in Massachusetts (Gong et al., 2025)	Mobile phone telemetric GPS data calibrated to crashes	R, S, X	Highway $\times$ town $\times$ hour cell	Stops vs roadway expo- sure	$\Pr(R = r \mid S = s_1, X) -$ $\Pr(R = r \mid \text{on road}, X)$	Combines crashes with vehicles both not-at-fault and at-fault; tar- gets disparity from roadway com- position; conflates exposure, de- ployment and enforcement deci- sions; race imputation based on Census block groups

Table 11: Recent studies of discrimination using telemetry data. The table describes the benchmark used, the telemetry data worked with, the unit of analysis and analytical strategy, implicit estimand where available and a summary of their limitations à la Section 1.

7 Conclusion

In this work, we develop a formal causal framework for quantifying the magnitude of disparate treatment of minority motorists by police officers conducting traffic enforcement in the United States. Traditional efforts in this vein aim to construct a “benchmark” for policing activity, which is assumed to form an appropriate comparison set for the stops made by officers. Studies typically claim that if the distribution of officer stops differs significantly from this benchmark, this constitutes evidence of racial profiling. However, for decades scholars have wrestled with the question of which benchmark to choose. As recent research has shown, when scholars examine the same substantive question using the same stop data, two seemingly innocuous benchmark choices can yield conclusions in opposite directions.

This article argues that this instability stems most fundamentally from a failure to define the causal estimand of interest. Imprecision in the scientific question has produced a body of research with a wide range of empirical approaches aimed at the same substantive goal, yielding contradictory conclusions and, at best, unclear—and at worst, incorrect—guidance for evaluating traffic enforcement.

We address this problem by first elaborating the manner in which data are generated in stop and benchmark data. This analysis revealed that benchmarks in the conventional sense are simply examples of negative control outcomes, which we take as a definition. In so doing, we provide explicit conditions under which one can evaluate the validity of a benchmark, and find that camera and checkpoint captures, and drivers who cause road traffic accidents serve as common examples. Next, we specify a causal estimand frequently sought by researchers, policymakers, and stakeholders: the causal risk ratio conditional on the group of violators, which quantifies the magnitude of racial discrimination in enforcement. We show that standard approaches used to approximate this quantity—most notably, the computation of

an odds ratio—coincide with $\psi^\dagger$ only in special cases. Specifically, equivalence requires that both officers and benchmarks perfectly measure violating behavior and that the quality of these measures is homogeneous across space and time. Of course, such conditions do not hold in practice. To accommodate violations of these assumptions, we show that $\psi^\dagger$ can be partially identified under weaker conditions, and we present resulting bounds that reveal troubling evidence of discrimination in Chicago.

In addition to cameras, crashes have long been thought to possess properties required of a valid benchmark. We show that, under careful conditions, crashes can be used to compute $\psi^\dagger$. First, the benchmark must be restricted to at-fault crash participants so that it reflects dangerous driving. Second, under the assumption that officers stop drivers according to their threat to road safety, $\psi^\dagger$ is identified via an odds ratio. Using flexible machine learning techniques and modern semiparametric theory, we further show how high dimensional covariates can be incorporated into an unbiased estimation framework for $\psi^\dagger$.

Using our framework, we uncover evidence of racial bias in multiple large U.S. jurisdictions, including Chicago and Texas. We also provide evidence in support of various mechanisms which could explain the found disparities, including policing tied to place and racial stereotyping, which will help to drive future reform efforts.

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A Connection to Kuroki and Pearl (2014)

In this first appendix, we show an alternative view of Proposition 2.3 using ideas from the seminal work by Kuroki and Pearl (2014), which gives algebraic identification conditions for causal effects when an analyst has access to an imperfect measure of a known confounding variable. For brevity, we will sometimes use the notational shorthand v_w to denote the event that variable $V = w$ where $w \in \mathcal{S}(V)$. Let V and V' be discrete with supports $\mathcal{S}(V) = \{v_1, \dots, v_m\}$ and $\mathcal{S}(V') = \{v'_1, \dots, v'_{m'}\}$. By convention, we define for any instantiations w, w' the column vectors $\mathbf{P}(w, \mathbf{V} \mid w') := (p(w, v_1 \mid w'), \dots, p(w, v_m \mid w'))^\top$ and $\mathbf{P}(w \mid \mathbf{V}, w') := (p(w \mid v_1, w'), \dots, p(w \mid v_m, w'))^\top$, both contained in the space $[0, 1]^m$. Further, define $\mathbf{P}(\mathbf{V} \mid \mathbf{V}', w) \in [0, 1]^{m \times m'}$ to be the matrix whose elements are given by $\mathbf{P}(\mathbf{V} \mid \mathbf{V}', w)_{i,j} = p(v_i \mid v'_j, w)$; note that this matrix is column stochastic. We have the following proposition.

Proposition A.1 (Proxy inversion, (Kuroki and Pearl, 2014)). *In the DAG of Figure 2b, the factual variables are linked by the following set of linear systems*

$$\mathbf{P}(s, \mathbf{C} \mid r, x) = \mathbf{P}(\mathbf{C} \mid \mathbf{D}, x) \mathbf{P}(s, \mathbf{D} \mid r, x)$$

$$\mathbf{P}(\mathbf{C} \mid r, x) = \mathbf{P}(\mathbf{C} \mid \mathbf{D}, x) \mathbf{P}(\mathbf{D} \mid r, x)$$

Proof. Apply the rules of d-separation (Pearl, 2009) to the DAG in Figure 2b. $\square$

As the system shows, the observed data on the left hand side, is a linear map of the missing data, on the right hand side, where the map is governed by the quality of the proxy, $\mathbf{P}(\mathbf{C} \mid \mathbf{D}, x)$. The crux of the identification argument stems from the ability to invert each of these systems. Assumption A2, cameras don't lie, constrains the first column of this matrix. In particular

$$\mathbf{P}(\mathbf{C} \mid \mathbf{D}, x) = \begin{pmatrix} p(c_0 \mid d_0, x) & p(c_0 \mid d_1, x) \\ p(c_1 \mid d_0, x) & p(c_1 \mid d_1, x) \end{pmatrix} = \begin{pmatrix} 1 & p(c_0 \mid d_1, x) \\ 0 & p(c_1 \mid d_1, x) \end{pmatrix}, \quad (45)$$

which is invertible whenever $p(c_1 \mid d_1, x) > 0$. The causal risk ratio compares the two quantities $\Pr(S(r) = s_1 \mid D = d_1, X = x)$ for each r . Since (D, X) is sufficient for confounding control, $\Pr(S(r) = s_1 \mid D = d_1, X = x) = \Pr(S = s_1 \mid R = r, D = d_1, X = x)$, and consequently

$$\psi_r^\dagger(x) = \frac{\Pr(S(r) = s_1 \mid D = d_1, X = x)}{\Pr(S(r_0) = s_1 \mid D = d_1, X = x)} \quad (46)$$

$$= \frac{\Pr(S = s_1 \mid R = r, D = d_1, X = x)}{\Pr(S = s_1 \mid R = r_0, D = d_1, X = x)} \quad (47)$$

$$= \frac{\Pr(S = s_1, D = d_1 \mid R = r, X = x)}{\Pr(S = s_1, D = d_1 \mid R = r_0, X = x)} \frac{\Pr(D = d_1 \mid R = r_0, X = x)}{\Pr(D = d_1 \mid R = r, X = x)}. \quad (48)$$

Inverting the second system above using (45), we find that

$$\Pr(D = d_1 \mid R = r, X = x) = \frac{\Pr(C = c_1 \mid R = r, X = x)}{\Pr(C = c_1 \mid D = d_1, X = x)}. \quad (49)$$

Applying this results to the previous display, we arrive at

$$\psi_r^\dagger(x) = \frac{\Pr(S = s_1, D = d_1 \mid R = r, X = x)}{\Pr(S = s_1, D = d_1 \mid R = r_0, X = x)} \frac{\Pr(D = d_1 \mid R = r_0, X = x)}{\Pr(D = d_1 \mid R = r, X = x)} \quad (50)$$

$$= \frac{\Pr(S = s_1, D = d_1 \mid R = r, X = x)}{\Pr(S = s_1, D = d_1 \mid R = r_0, X = x)} \frac{\Pr(C = c_1 \mid R = r_0, X = x)}{\Pr(C = c_1 \mid R = r, X = x)}. \quad (51)$$

Under Assumption A1, $S = s_1$ implies $D = d_1$. Hence, this expression simplifies to

$$\psi_r^\dagger(x) = \frac{\Pr(S = s_1 \mid R = r, X = x)}{\Pr(S = s_1 \mid R = r_0, X = x)} \frac{\Pr(C = c_1 \mid R = r_0, X = x)}{\Pr(C = c_1 \mid R = r, X = x)}. \quad (52)$$

Applying Bayes' rule to each factor yields

$$\frac{\Pr(S = s_1 \mid R = r, X = x)}{\Pr(S = s_1 \mid R = r_0, X = x)} = \frac{\Pr(R = r \mid S = s_1, X = x)}{\Pr(R = r_0 \mid S = s_1, X = x)} \frac{\Pr(R = r_0 \mid X = x)}{\Pr(R = r \mid X = x)}, \quad (53)$$

$$\frac{\Pr(C = c_1 \mid R = r_0, X = x)}{\Pr(C = c_1 \mid R = r, X = x)} = \frac{\Pr(R = r_0 \mid C = c_1, X = x)}{\Pr(R = r \mid C = c_1, X = x)} \frac{\Pr(R = r \mid X = x)}{\Pr(R = r_0 \mid X = x)}. \quad (54)$$

The marginal racial proportions cancel, giving

$$\psi_r^\dagger(x) = \frac{\Pr(R = r \mid S = s_1, X = x) / \Pr(R = r_0 \mid S = s_1, X = x)}{\Pr(R = r \mid C = c_1, X = x) / \Pr(R = r_0 \mid C = c_1, X = x)}. \quad (55)$$

which is the odds ratio found in the main text.

B Nonidentifiability of causal risk ratio from odds ratio

Recall the causal DAG of Figure 2b. U is an unmeasured confounder and all other variables are binary and the bidirected edge represents unobserved confounding. For the sake of simplicity we suppress conditioning on X .

We begin with a definition of identifiability of a causal parameter from a statistical functional.

Definition B.1 (Identification (Pearl, 2009, Def. 3.1.1)). Let $\mathcal{M}$ denote a class of causal models compatible with a given graph. Let P_M denote the full data law of any $M \in \mathcal{M}$. A causal functional $Q(P_M)$ is identifiable from a statistical functional $T(P_M)$ relative to $\mathcal{M}$ if there exists a mapping f such that

$$Q(P_M) = f(T(P_M)) \quad \text{for all } M \in \mathcal{M}.$$

Equivalently, Q is not identifiable from T if there exist $M_1, M_2 \in \mathcal{M}$ such that $T(P_{M_1}) = T(P_{M_2})$ but $Q(P_{M_1}) \neq Q(P_{M_2})$.

The causal estimand of interest is the causal risk ratio among units with $D = d_1$,

$$\psi^\dagger = \frac{\Pr(S(r_1) = s_1 \mid D = d_1)}{\Pr(S(r_0) = s_1 \mid D = d_1)}.$$

Because conditioning on D blocks the backdoor path $R \leftarrow U \rightarrow D \rightarrow S$, we have $\Pr(S(r) = s_1 \mid D = d_1) = \Pr(S = s_1 \mid R = r, D = d_1)$, and hence

$$\psi^\dagger = \frac{\Pr(S = s_1 \mid R = r_1, D = d_1)}{\Pr(S = s_1 \mid R = r_0, D = d_1)}.$$

Suppose that the analyst has available data the two conditional distributions $\Pr(R \mid C = c_1)$ and $\Pr(R \mid S = s_1)$ due to selection. We will show that $T(P) = \text{odds}_r[R \mid S = s_1] / \text{odds}_r[R \mid C = c_1]$ cannot identify $\psi^\dagger$ unless $\Pr(D = d_1 \mid C = c_1) = \Pr(D = d_1 \mid S = s_1) = 1$.

Proposition 2.3 proves that whenever these two conditions hold, $\psi^\dagger$ is identified from the odds ratio. In this Appendix, we wish to show the converse, that whenever the odds ratio identifies $\psi^\dagger$, it must be that $\Pr(D = d_1 \mid C = c_1) = \Pr(D = d_1 \mid S = s_1) = 1$. It is easier to do this by contrapositive. We will show that if at least one of $\Pr(D = d_1 \mid S = s_1)$ or $\Pr(D = d_1 \mid C = c_1)$ is not unity, then the odds ratio does not identify $\psi^\dagger$. To establish this, we construct two compatible models that agree on T but disagree on $\psi^\dagger$.

Let $U \sim \text{Bernoulli}(1/2)$ and define $R := U$ and $D := \mathbf{1}\{U \neq E\}$, where $E \sim \text{Bernoulli}(e)$ is independent noise. This induces correlation between R and D . Let $a_d^C := \Pr(C = c_1 \mid D = d)$ and $s_{rd} := \Pr(S = s_1 \mid R = r, D = d)$. Under this parameterization, $\psi^\dagger = s_{11}/s_{01}$. We are free to work with such a parametrization because the full data law factorizes as

$$\Pr(S, D, C, R, U) = \Pr(S \mid D, R) \Pr(C \mid D) \Pr(D \mid U) \Pr(R \mid U).$$

The distribution $\Pr(R \mid C = c_1)$ depends only on (e, a_0^C, a_1^C) and is thus invariant to the choice of s_{rd} . Meanwhile, $\Pr(R \mid S = s_1)$ equals

$$\frac{\Pr(R = r_1, D = d_0)s_{10} + \Pr(R = r_1, D = d_1)s_{11}}{\sum_{r,d} \Pr(R = r, D = d)s_{rd}},$$

which depends on $(e, s_{00}, s_{10}, s_{01}, s_{11})$. For fixed $(e, s_{10}, s_{01}, s_{11})$ and any target value $p^* \in (0, 1)$, the equation $\Pr(R = r_1 \mid S = s_1) = p^*$ is linear in s_{00} and hence admits a unique solution provided the resulting s_{00} lies in $[0, 1]$.

Now fix (e, a_0^C, a_1^C) and a target p^* . Choose two pairs $(s_{01}^{(1)}, s_{11}^{(1)})$ and $(s_{01}^{(2)}, s_{11}^{(2)})$ such that $s_{11}^{(1)}/s_{01}^{(1)} \neq s_{11}^{(2)}/s_{01}^{(2)}$. For each model, select s_{10} arbitrarily and solve for s_{00} so that $\Pr(R = r_1 \mid S = s_1) = p^*$. By construction, the two models agree on $\Pr(R \mid C = c_1)$ and on $\Pr(R \mid S = s_1)$, and hence on $T(P)$. However, they yield distinct values of $\psi^\dagger$.

For concreteness, consider the following numerical example. Let $e = 0.2$, $a_0^C = 0.2$, and $a_1^C = 0.8$, which fixes $\Pr(R \mid C = c_1)$. Choose a target value $\Pr(R = r_1 \mid S = s_1) = 0.55$. Define Model M_1 by $(s_{11}, s_{01}, s_{10}) = (0.8, 0.2, 0.2)$ and solve for s_{00} so that $\Pr(R = r_1 \mid S =$

$s_1) = 0.55$, yielding $s_{00} = 0.645$. Define Model M_2 by $(s_{11}, s_{01}, s_{10}) = (0.6, 0.4, 0.7)$ and again solve for s_{00} to obtain $s_{00} = 0.370$.

By construction, both models yield the same values of $\Pr(R \mid C = c_1)$ and $\Pr(R \mid S = s_1)$, and therefore the same value of $T(P)$. However, the causal risk ratios differ:

$$\psi_{(1)}^\dagger = \frac{0.8}{0.2} = 4, \quad \psi_{(2)}^\dagger = \frac{0.6}{0.4} = 1.5.$$

Hence, by Definition B.1, the causal risk ratio $\psi^\dagger = \Pr(S(r_1) = s_1 \mid D = d_1) / \Pr(S(r_0) = s_1 \mid D = d_1)$ is not generally identifiable from the odds ratio $T(P) = \text{odds}_r[R \mid S = s_1] / \text{odds}_r[R \mid C = c_1]$. The above discussion shows that whenever the conditions of Proposition 2.3 do not hold, $\psi^\dagger$ is not identified by the odds ratio. We are done.

C Additional discussion of officer–camera substitution

In this appendix, we formalize the phenomenon that officer presence on the road may substitute camera tickets in the data generating process, leading to the risk ratio being anti-conservative for racial bias as discussed in Section 2.2. To begin, let D denote dangerous driving in the absence of any prior enforcement (baseline conduct), and let D' denote dangerous driving at the time a camera would record the unit. We can think of D' as the analogue of D after enforcement, allowing for the possibility that an officer stop changes behavior. We make the following three assumptions.

Assumption A6 (Chastening). $D'(S = s_1, D) = d'_0$.

Assumption A7 (Monotonicity). $D'(S = s_0, D) \leq D$.

Assumption A8 (Minority vigilance). $\Pr(D' = d'_1 \mid R = r_1, D = d_1, S = s_0, X = x) \leq \Pr(D' = d'_1 \mid R = r_0, D = d_1, S = s_0, X = x)$.

Assumption A6 makes formal that a civilian will not continue to drive dangerously should

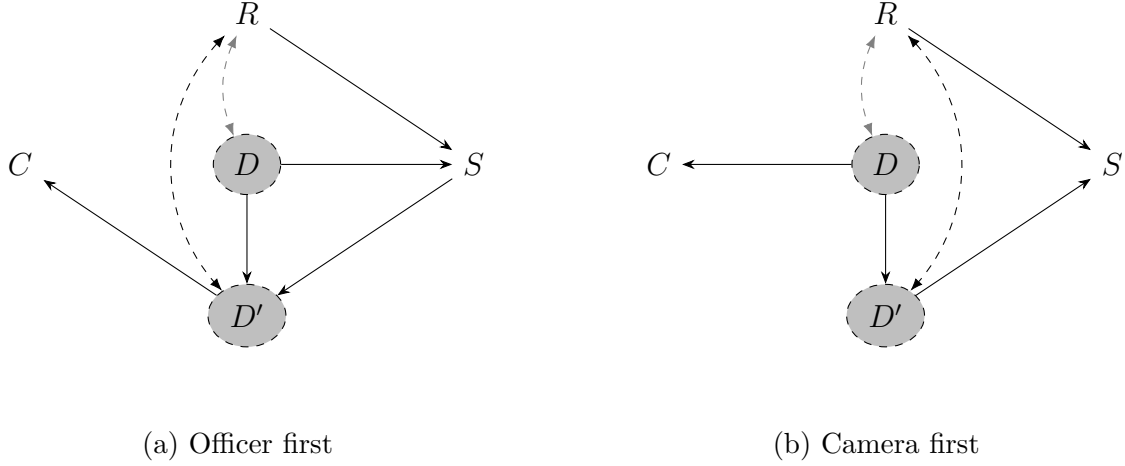


Figure 10: Officer camera substitution (context X suppressed for clarity). (a) Officer first: officers respond to dangerous D which induces new dangerousness D' which only cameras see. (b) Camera first: a mirror of (a), under the assumption that cameras do not affect subsequent dangerousness. We observe an unknown mixture of these phenomena (Bonvini and Kennedy, 2022).

an officer stop them. Assumption A7 implies that if an officer ignores a civilian motorist, their subsequent driving behavior cannot be worse than their original driving behavior. This accounts for the fact that even if an officer ignores a driver, the likely act of *seeing* the officer should induce a positive response in driving behavior. Last, Assumption A8 assumes that minorities are no more likely than White drivers to continue driving dangerously should an officer ignore them. This condition embodies the notion that minority drivers are likely to be more sensitive to the presence of police officers on the road lest a stop result in a potentially violent altercation (Geller et al., 2014; Bor et al., 2018).

At present, our target causal risk ratio at location x is defined in terms of D ,

$$\psi^\dagger(x) = \frac{\Pr(S = s_1 \mid R = r_1, D = d_1, X = x)}{\Pr(S = s_1 \mid R = r_0, D = d_1, X = x)}.$$

Now suppose that, within x , units may encounter an officer before reaching the camera, and that the officer’s decision rule is anti-Black such that $\psi^\dagger(x) > 1$. Among drivers dangerous

at baseline ($D = d_1$), Black drivers are thus selectively removed from the driving population prior to potential camera capture more often than White drivers. Combining this selection with the other assumptions implies that the composition of dangerous drivers observed by the camera is tilted toward White drivers: $\Pr(R = r_1 \mid D' = d'_1, X = x) < \Pr(R = r_1 \mid D = d_1, X = x)$. Suppose cameras don't lie, so that $\Pr(R = r_1 \mid D' = d'_1, X = x) = \Pr(R = r_1 \mid C = c_1, X = x)$. Looking at the IPW representation of $\psi^\dagger(x)$ in (8), we see that if we naïvely substituted $\Pr(R = r_1 \mid D = d_1, X = x)$ with the attenuated camera share $\Pr(R = r_1 \mid C = c_1, X = x)$ we would apply a larger (smaller) weight to the Black (White) stops than we should, inflating estimates of discrimination. Were the discrimination anti-White, a symmetric argument illustrates that we would underestimate $\psi^\dagger(x)$ without adjustment. In this sense, when racially biased officer encounters precede camera exposure, the use of cameras as benchmarks is anti-conservative for the risk ratio but, notably, will never imply the wrong type of discrimination (i.e. $\text{sign}(\psi^\dagger(x) - 1)$ does not change).

To illustrate the anti-conservative result formally, we leverage the following proposition. All results hold conditional on context X and so we omit it from the notation for brevity. For simplicity of exposition assume R is binary.

Proposition C.1. *Assume the data generating process in which the officer is encountered first, given by Figure 10a. Under Assumptions A6-A8, whenever officers exhibit bias against racial group r_1 relative to r_0 , i.e.*

$$\Pr(S = s_1 \mid D = d_1, R = r_1) > \Pr(S = s_1 \mid D = d_1, R = r_0), \quad (56)$$

it follows that

$$\Pr(R = r_1 \mid D' = d'_1) < \Pr(R = r_1 \mid D = d_1). \quad (57)$$

Proof. Assumption A6 implies that dangerous driving at the camera, $D' = d'_1$, can only

occur if the driver is not stopped:

$$\begin{aligned}\Pr(D' = d'_1 \mid D = d_1, R = r_1) &= \Pr(D' = d'_1, S = s_0 \mid D = d_1, R = r_1) \\ &= \Pr(S = s_0 \mid D = d_1, R = r_1) \Pr(D' = d'_1 \mid D = d_1, S = s_0, R = r_1),\end{aligned}\quad (58)$$

and analogously for group r_0 . Note that officer bias (56) may be equivalently expressed $\Pr(S = s_0 \mid D = d_1, R = r_1) < \Pr(S = s_0 \mid D = d_1, R = r_0)$. Assumption A8 states that, among drivers dangerous at baseline who are not stopped, group r_1 is no more likely to remain dangerous at the camera than group r_0 :

$$\Pr(D' = d'_1 \mid D = d_1, S = s_0, R = r_1) \leq \Pr(D' = d'_1 \mid D = d_1, S = s_0, R = r_0). \quad (59)$$

Combining these displays, we obtain

$$\Pr(D' = d'_1 \mid D = d_1, R = r_1) = \Pr(S = s_0 \mid D = d_1, R = r_1) \Pr(D' = d'_1 \mid D = d_1, S = s_0, R = r_1) \quad (60)$$

$$< \Pr(S = s_0 \mid D = d_1, R = r_0) \Pr(D' = d'_1 \mid D = d_1, S = s_0, R = r_0) \quad (61)$$

$$= \Pr(D' = d'_1 \mid D = d_1, R = r_0). \quad (62)$$

Now let $p_{r_1} = \Pr(R = r_1 \mid D = d_1)$ and $p_{r_0} = \Pr(R = r_0 \mid D = d_1)$. By Bayes' rule,

$$\frac{\Pr(R = r_1 \mid D' = d'_1, D = d_1)}{\Pr(R = r_0 \mid D' = d'_1, D = d_1)} = \frac{\Pr(D' = d'_1 \mid D = d_1, R = r_1)}{\Pr(D' = d'_1 \mid D = d_1, R = r_0)} \cdot \frac{p_{r_1}}{p_{r_0}}. \quad (63)$$

Since $\Pr(D' = d'_1 \mid D = d_1, R = r_1) < \Pr(D' = d'_1 \mid D = d_1, R = r_0)$, the factor $\Pr(D' = d'_1 \mid D = d_1, R = r_1) / \Pr(D' = d'_1 \mid D = d_1, R = r_0)$ is strictly less than one, and hence

$$\frac{\Pr(R = r_1 \mid D' = d'_1, D = d_1)}{\Pr(R = r_0 \mid D' = d'_1, D = d_1)} < \frac{\Pr(R = r_1 \mid D = d_1)}{\Pr(R = r_0 \mid D = d_1)}. \quad (64)$$

As R is binary, this inequality of odds is equivalent to

$$\Pr(R = r_1 \mid D' = d'_1, D = d_1) < \Pr(R = r_1 \mid D = d_1). \quad (65)$$

By monotonicity A7, $D' = d'_1 \implies D = d_1$, meaning that $\Pr(R = r_1 \mid D' = d'_1, D = d_1) = \Pr(R = r_1 \mid D' = d'_1)$, and thus $\Pr(R = r_1 \mid D' = d'_1) < \Pr(R = r_1 \mid D = d_1)$, which establishes the claim. $\square$

The immediate consequence of Proposition C.1 is that whenever officers are seen first, as $\Pr(R = r_1 \mid C = c_1)$ is used to approximate $\Pr(R = r_1 \mid D' = d'_1)$, we will tend to exaggerate discrimination against minority drivers and understate discrimination against White drivers. The above proof illustrates that this is because in the case of anti-minority bias, the weight $\Pr(R = r_1 \mid C = c_1)$ is smaller than it should be while the reverse is true for anti-White bias. Indeed, the impact of this behavior on Theorem 2.6 is readily seen, for the same reasoning shows that the bounds will be pushed upwards (downwards) for anti-minority (-White) bias. To deal with this issue, we propose a minimal sensitivity adjustment in the main text, and devote analysis of this general phenomenon to future work.

D Checkpoints Application

In this appendix, we demonstrate an application of a third valid benchmark, sobriety checkpoints. We consider several US jurisdictions served by: Chicago Police Department (years covered: 2014–2018), San Diego County Sheriff Department (2010–2026), Tucson Police Department (2007–2010), Pima County Sheriff Department (2008–2009), and all local law enforcement agencies within the entire Commonwealth of Virginia (2020–2024).

The analysis uses DUI arrest data from several jurisdictions in which checkpoint involvement can be identified. For jurisdictions in Chicago, San Diego, and Arizona, checkpoint arrests were identified from public checkpoint announcements and press releases, then matched to arrest records using the date, time, and location of the arrest.¹⁴ In Virginia, the

¹⁴An example in San Diego: <https://www.sdsheriff.gov/Home/Components/News/News/917/>

source data come from the statewide traffic stop database, which records stops by law enforcement agencies across the Commonwealth, including whether a stop was associated with a checkpoint and whether it resulted in an arrest. These data were restricted to stops that resulted in DUI arrests. Throughout stopped vehicles are filtered to those that led to a DUI arrest because such an action makes it plausible that suspected impaired driving was the reason motivating the officer to stop the driver.

We highlight some important features of the data. Checkpoint deployment and subsequently checkpoint arrests are rare. For example, in Chicago, of 47,837 recorded DUI related stops, only 153 (0.32%) were from checkpoints. In Virginia, only 498 from 138,261 originated from a checkpoint. Moreover, in Chicago, San Diego, and Arizona, some checkpoints may be missed if no press release or public notice was issued, and some matches of arrests to checkpoints remain subject to uncertainty when arrests occurred near, but not exactly at, the reported checkpoint location. Virginia does not suffer from this issue as the checkpoint indicator is recorded directly in the statewide stop data.

Estimation proceeds by coarsened exact matching on blocks, X , that define the local comparison between checkpoint and noncheckpoint DUI arrests. Similar to the camera analysis, blocks limit feasible comparisons with a coarse spatiotemporal context. A block is *feasible* only if it contains sufficient support for the comparison: at least one checkpoint observation, at least one stop, at least one driver of each racial group for each binary comparison. Denote by b the number of feasible blocks and n denotes the number of observations contained in those feasible blocks. Due to the many limitations of the acquired data and identification challenges discussed in Section 2.3, we will proceed for the purposes of illustration and omit the sensitivity analysis due to miscaptures by checkpoints and officers. We contain our results in Table 12 with the blocking strategy chosen for that region; next we discuss the general tensions underlying this choice.

In Section 2.3, we describe the limitations of using checkpoint data as a means of comparison for officer stops, as making for a ‘similarly situated’ group of units via observed contextual factors becomes more challenging. An additional common data challenge in the use of checkpoints is their rarity. Checkpoints are seldom deployed for enforcement (relative to discretionary alternatives), and consequently, few arrests are made. In tandem with the identification challenge induced by substitution, this poses a problem for statistical power in detecting discrimination: the more granular the context X used, the smaller the number of viable comparison units. This tension forces several considerations for the analyst. Foremost is the bias–variance tradeoff—a more granular X makes for the cleanest causal comparison, however, weakens confidence in the point estimate due to small samples. Moreover, a finer X reduces the number of blocks containing a feasible comparison, limiting the breadth of the conclusion over the jurisdiction studied, as data may only be usable within certain geographies. Coarsening X will increase power, however, also merits more careful justification that the units captured by the benchmark and the police are similarly situated. It is also important to remark that, by the exclusion criteria outlined above, changing the blocking strategy X affects the estimand: the resulting contrast is no longer taken over the same set of comparable checkpoint and noncheckpoint stops. Instead, each blocking strategy defines its own target population defined by the exclusion criteria. Most often, coarsened blocks will contain all units of finer blocks, but this is not guaranteed. Therefore, we highlight that, in large part due to the noncollapsibility of the odds ratio, it is not a-priori possible to qualitatively predict how point estimates may change as X is coarsened. In Figure 11, we depict how excluding fewer stops near a checkpoint by increasing the size of the spatial inclusion set can affect both the power and qualitative conclusion of the result.

Ideally, data would contain spatial coordinates and hourly recording of all incidents, us-

ing which the analyst may specify well defined, restrictive regions in which to compare discretionary and nondiscretionary stops informed by domain knowledge of the regional geography. However, choices of X in limited data environments should be guided by the enforcement context. Police beats are generally an attractive blocking choice because they reflect operational boundaries and are frequently drawn around neighborhood or demographic divisions. Hence, conditioning on beat does well to compare stops occurring in more similar enforcement environments, especially in dense urban regions. In jurisdictions with sparse physical geography, the choice of spatial blocking strategy may depend on the road structure. For example, expanding a radius around a checkpoint may not produce a meaningfully more credible comparison group if doing so incorporates mostly farmland. In such cases, should the region served be sufficiently small and homogeneous (as judged by the analyst), the agency operating in the jurisdiction may serve as a meaningful spatial blocking unit. Temporally, to account for driving patterns across the day that differ due to differences across race such as occupation, grouping units as finely as the data allow into morning, afternoon, evening and night periods is advised.

While sound in theory, due to the practical problems common datasets possess, sobriety checkpoints are the weakest identification strategy for detecting racial discrimination via the framework presented in this paper. Should this be the only data available for the analyst's study of interest, a careful balance of the aforementioned tradeoffs is crucial for obtaining credible inferences about discrimination.

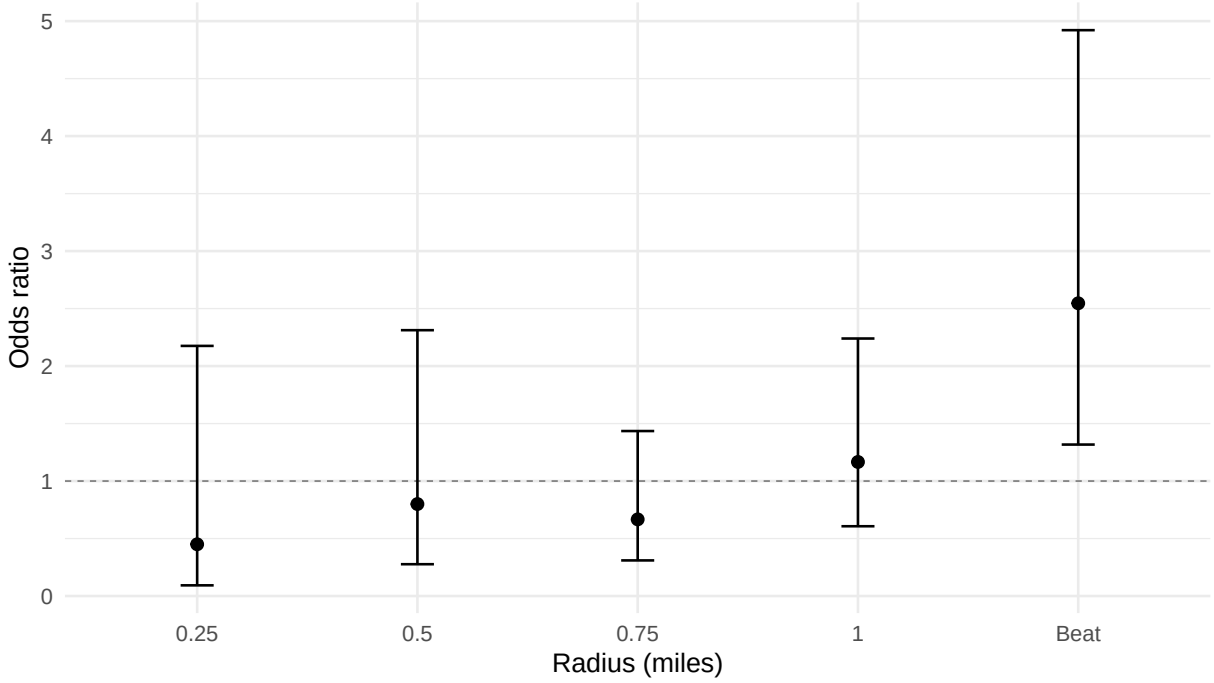


Figure 11: Odds ratios as they vary with blocking granularity. A dot marks the point estimates of the causal risk ratio $\psi^\dagger$ assuming neither officers nor checkpoints lie, accompanied by its 95% CI. The horizontal axis increases the circular radius around which we compare stops to each checkpoint in the blocking strategy (with TOD, DOW, Quarter, Year temporal blocks). The fourth estimate is not a circular area but beats defined by the Sheriff’s office and whose numerical value is contained in Table 12.

Region	Block	b	n	$\psi^\dagger$ (95% CI)
<i>Comparison: Black vs White</i>				
Chicago	Beat, TOD, DOW, Month, Year	5	28	0.38 (0.13, 1.13)
San Diego	Beat, TOD, DOW, Quarter, Year	5	33	0.36 (0.08, 1.55)
Tucson	Agency, TOD, DOW, Month, Year	0	0	–
Pima County	Agency, TOD, DOW, Month, Year	0	0	–
Commonwealth of Virginia	Agency, DOW, Month, Year	185	29140	1.73 (1.45, 2.06)
<i>Comparison: Hispanic vs White</i>				
Chicago	Beat, TOD, DOW, Month, Year	4	20	3.45 (0.31, 39.01)
San Diego	Beat, TOD, DOW, Quarter, Year	18	97	2.55 (1.32, 4.92)
Tucson	Agency, TOD, DOW, Month, Year	10	122	5.47 (2.67, 11.20)
Pima County	Agency, TOD, DOW, Month, Year	12	150	1.33 (0.84, 2.11)
Commonwealth of Virginia	Agency, DOW, Month, Year	161	28988	1.31 (1.08, 1.60)

Beat: Administrative region defined by police department for patrol operations

TOD: Time of Day, grouped into blocks: 4AM-12PM, 12PM-8PM, 8PM-4AM

DOW: Day of the Week, grouped into weekdays (Mon–Thu) and weekend (Fri–Sun)

Agency: Law enforcement body serving the region. Acts as a proxy for geography as certain regions may contain more than one nonoverlapping law enforcement agency

Table 12: Estimates of causal risk ratio $\psi^\dagger$ by region and blocking strategy using checkpoints as a benchmark under the assumption that neither checkpoints nor officers lie. The number of feasible blocks is given by b and n the total sample size across those blocks.

E Estimation Details for Camera Analysis

E.1 Maximum likelihood estimation for BISG correction in camera data

We give details on the maximum likelihood strategy used to estimate $\Pr(R = r \mid C = c_1, X = x)$ for discrete blocks X using surnames observed in the camera record. The procedure is repeated for each $x \in \mathcal{S}(X)$, so we suppress conditioning on x for exposition. Let $\mathcal{L}$ denote the set of recorded surnames.

For each camera observation $i \in \{1, \dots, n\}$ in block x , we observe a surname $L_i \in \mathcal{L}$ and an optional user supplied weight $w_i > 0$ (typically $w_i \equiv 1$). Let $\mathcal{R} = \{r_0, \dots, r_{J-1}\}$ denote the race categories, with $J := |\mathcal{R}|$, and let $R_i \in \mathcal{R}$ be unobserved. Define the camera block composition $\pi_r^{(c)} := \Pr(R = r \mid C = c_1, X = x)$ for $r \in \mathcal{R}$, collected in $\pi^{(c)} \in \Delta^{J-1}$.

Using census data, we take as known the conditional surname distribution given race,

$$q_{\ell r} := \Pr(L = \ell \mid R = r), \quad \ell \in \mathcal{L}, \quad r \in \mathcal{R},$$

which does not depend on the camera record. By the law of total probability,

$$\Pr(L = \ell \mid C = c_1, X = x) = \sum_{r \in \mathcal{R}} q_{\ell r} \pi_r^{(c)}. \quad (66)$$

We maximize the weighted log likelihood

$$\ell_w(\pi^{(c)}) = \sum_{i=1}^n w_i \log \left(\sum_{r \in \mathcal{R}} q_{L_i r} \pi_r^{(c)} \right). \quad (67)$$

For computation, we collapse to unique surnames. Let $\ell_1, \dots, \ell_N$ denote the distinct surnames in the camera data within block x , and define aggregated weights $w_j = \sum_{i: L_i = \ell_j} w_i$ for $j = 1, \dots, N$. Then

$$\ell_w(\pi^{(c)}) = \sum_{j=1}^N w_j \log \left(\sum_{r \in \mathcal{R}} q_{\ell_j r} \pi_r^{(c)} \right).$$

We maximize ℓ_w using the EM algorithm. Introduce latent race indicators $Z_{jr} \in \{0, 1\}$ for each unique surname ℓ_j , with $\sum_{r \in \mathcal{R}} Z_{jr} = 1$. At iteration t , the E-step computes

$$\tau_{jr}^{(t)} = \frac{\pi_r^{(c,t)} q_{\ell_j r}}{\sum_{s \in \mathcal{R}} \pi_s^{(c,t)} q_{\ell_j s}},$$

and then the M-step computes

$$\pi_r^{(c,t+1)} = \frac{\sum_{j=1}^N w_j \tau_{jr}^{(t)}}{\sum_{j=1}^N w_j}, \quad r \in \mathcal{R}.$$

Iterations continue until $\max_r |\pi_r^{(t+1)} - \pi_r^{(t)}| < \varepsilon$. Because ℓ_w is concave in π over the simplex, EM converges to the global maximizer.

To propagate uncertainty in the camera race composition, we use a parametric approximation based on the observed information. Let $\hat{\pi}^{(c)}$ denote the EM estimate and let $\hat{\Sigma}^{(c)}$ denote an estimate of its covariance obtained by inverting the negative Hessian of $-\ell_w(\pi^{(c)})$ at $\hat{\pi}^{(c)}$. We draw $\{\hat{\pi}^{(c,b)}\}_{b=1}^B$ by sampling a $(J-1)$ dimensional normal distribution for the nondominant components, reconstructing the remaining component so the entries sum to one, and rejecting any draws that leave the simplex.

E.2 Combining camera BISG correction with bounds

We now describe how the camera BISG correction is combined with the bounding functional, matching our implementation. For a given analysis type (redlight or speeding) and year specification, we first restrict to blocks x that appear in both the stop and camera records and meet a minimum camera sample size threshold. We obtain $\widehat{\Pr}(R = r \mid C = c_1, X = x)$ from the EM procedure above.

In the stop data, self-reported race R is not observed. We observe officer-reported race $\tilde{R}$ and use external data from an ACLU report to provide a correction for potential misreporting. Let the reverse misreporting matrix $M_{\text{rev},x}$, with entries $[M_{\text{rev},x}]_{r,\tilde{r}} = \Pr(R = r \mid \tilde{R} = \tilde{r}, S =$

$s_1, X = x$), contain the precision of officer labels. We have these rates at the level of the year only.

For each block x , let $\tilde{n}_{\tilde{r}}(x)$ be the observed count with officer-reported race $\tilde{r}$, and let $\tilde{p}_x(\tilde{r}) = \Pr(\tilde{R} = \tilde{r} \mid S = s_1, X = x)$ be the empirical proportions of officer-reported race. We correct this quantity deterministically using the ACLU report data, so that $p_x(r) = \Pr(R = r \mid S = s_1, X = x) = \sum_{\tilde{r} \in \mathcal{S}(\tilde{R})} [M_{\text{rev},x}]_{r,\tilde{r}} \tilde{p}_x(\tilde{r})$. To propagate sampling uncertainty from officer stops, we draw the proportions of officer-reported race from a Dirichlet distribution with pseudocount parameters $\tilde{n}_{\tilde{r}}(x) + \alpha_0$, and map every draw through the same $M_{\text{rev},x}$. These reverse misreporting probabilities are treated as known, so this step adds no additional uncertainty from estimating the misreporting mechanism.

Fix $\theta \in (0, 1]$ and a minority group $r \in \{\text{Black}, \text{Hispanic}\}$. We impose the camera feasibility constraint by dropping any block x with

$$1 - \theta \geq \min_r \widehat{\Pr}(R = r \mid C = c_1, X = x).$$

On the remaining blocks, we form the IPW weights by x that enter the bounds by combining θ with the camera proportions for race m and for White, and we further trim blocks so that all resulting IPW weights lie in the interval $[1/0.9, 1/0.1]$. Let $\mathcal{X}_\theta$ denote the set of blocks retained after these steps.

We weight blocks according to their stop sample size. Writing $n(x)$ for the number of stops in block x , we set

$$\hat{w}_x \propto n(x), \quad \text{so that} \quad \hat{w}_x = \frac{n(x)}{\sum_{x' \in \mathcal{X}_\theta} n(x')}, \quad x \in \mathcal{X}_\theta.$$

For each η in a grid, we then evaluate the lower and upper bound functionals. The η dependence enters through the terms $\Pr(R = r \mid S = s_1, X = x) + \eta - 1$ and $\Pr(R = r_0 \mid S = s_1, X = x) + \eta - 1$; in our implementation these are truncated at zero via $\max\{0, \cdot\}$ to meet the conditions of the bound in Theorem 2.6.

Finally, we propagate uncertainty by combining camera draws with the corrected stop Dirichlet draws. For each combination of a camera draw $\widehat{\text{Pr}}(R \mid C = c_1, X = x)$ and a stop draw, the latter is first generated using officer-reported race and then mapped through the fixed reverse misreporting matrix $M_{\text{rev},x}$ to obtain $\widehat{\text{Pr}}(R \mid S = s_1, X = x)$. We rerun the same trimming and weighting steps for each combination of draws and recompute the full bound curve over η (and over a grid of θ). This yields a sampling distribution of bound curves from which we summarize uncertainty. We also repeat the same procedure after restricting to subsets of blocks defined by the majority race in each beat, taken from census demographics.

F DML details for crashes analysis

F.1 General approach

The unit of analysis is a vehicle-mile that resulted in a traffic encounter. The officer and benchmark records are stacked to form the analysis sample, and $A \in \{s, c\}$ denotes the source of an observed record. We estimate a finite dimensional parameter ψ indexing a semi-parametric conditional odds ratio model relating categorical race R to source A , conditional on covariates X , following [Tchetgen Tchetgen et al. \(2010\)](#). Let $\mathcal{R} = \{r_0, r_1, \dots, r_{J-1}\}$ with $J := |\mathcal{R}|$, where r_0 is the baseline category, and let $R^{\text{ind}} = \{\mathbf{1}\{R = r_1\}, \dots, \mathbf{1}\{R = r_{J-1}\}\}^\top$ denote the $d = J - 1$ indicators for the categories other than the baseline.

The conditional odds ratio function $\gamma(r, a; \psi)$ satisfies $\gamma(r, c; \psi) = 1$ and $\gamma(r_0, a; \psi) = 1$. For $r_j \neq r_0$, we specify $\gamma(r_j, s; \psi) = \exp(\psi_j)$, so that

$$\exp(\psi_j) = \frac{\Pr(R = r_j \mid A = s, X = x) / \Pr(R = r_0 \mid A = s, X = x)}{\Pr(R = r_j \mid A = c, X = x) / \Pr(R = r_0 \mid A = c, X = x)}. \quad (68)$$

Thus, ψ_j compares race category r_j , relative to r_0 , between encountered vehicle-miles from the officer source and from the benchmark source with the same context.

The model leaves unrestricted the baseline race distribution $f_0(r \mid x) := \Pr(R = r \mid A = c, X = x)$ and the baseline source mechanism $g_0(a \mid x) := \Pr(A = a \mid R = r_0, X = x)$. Together with γ , these determine the full data conditional law

$$h(r, a \mid x; \psi, f_0, g_0) = \frac{\gamma(r, a; \psi) f_0(r \mid x) g_0(a \mid x)}{\sum_{\bar{r} \in \mathcal{R}} \sum_{\bar{a} \in \{s, c\}} \gamma(\bar{r}, \bar{a}; \psi) f_0(\bar{r} \mid x) g_0(\bar{a} \mid x)}, \quad (69)$$

where nuisance functions f_0 and g_0 are variation independent. We interpret $g_0(s \mid x)$ as the probability that an encountered vehicle-mile comes from the officer source among White encountered vehicle-miles with context $X = x$ in the stacked analysis sample, and $f_0(r \mid x)$ as the proportion of at-fault crashes caused by members of racial group r .

Let $\mu_a(x) = \mathbb{E}[R^{\text{ind}} \mid A = a, X = x]$, $V_a(x) = \text{Var}[R^{\text{ind}} \mid A = a, X = x]$, and $p_a(x) = \Pr(A = a \mid X = x)$, where each operator is defined with respect to the density h in (69) and dependence on (ψ, f_0, g_0) is suppressed. Define $\bar{V}(x) = \sum_a p_a(x) V_a(x)$. For a $d \times d$ matrix valued function $k(A, X)$, let

$$\Delta_k(x) = \left\{ \sum_{a \in \{s, c\}} p_a(x) k(a, x) V_a(x) \right\} \bar{V}(x)^{-1}. \quad (70)$$

As shown in [Tchetgen Tchetgen et al. \(2010\)](#), the full data estimating function is

$$U_i^{\text{full}}(\psi; f_0, g_0) = \{k(A_i, X_i) - \Delta_k(X_i)\} \{R_i^{\text{ind}} - \mu_{A_i}(X_i)\}. \quad (71)$$

For $\gamma(r_j, s; \psi) = \exp(\psi_j)$, we use $k(A_i, X_i) = \mathbf{1}\{A_i = s\} I_d$, giving $\Delta_k(x) = p_s(x) V_s(x) \bar{V}(x)^{-1}$. The target ψ_0 solves $\mathbb{E}[U_i^{\text{full}}(\psi; f_0, g_0)] = 0$. At $\psi = \psi_0$, this moment is doubly robust in that it is unbiased if either the working model for f_0 or the working model for g_0 is correct.

F.2 Accounting for measurement error

We now allow self-reported race to be unobserved in one or both sources. Let W denote the observed proxy for race. Depending on the source and scenario, W may equal self-reported race R , surname L , or officer-reported race $\tilde{R}$.

For source a , define the measurement operator $T_a(x) = [t_a(w \mid r, x)]$, where $t_a(w \mid r, x) = \Pr(W = w \mid R = r, A = a, X = x)$. These operators are treated as fixed inputs. When race is measured accurately, $T_a(x) = I_J$. When surname is observed, $T_a = Q$, where $[Q]_{\ell r} = q_{\ell r} = \Pr(L = \ell \mid R = r)$ is obtained from the Census; the exclusion restriction of Section 3.2 means this operator does not depend on x . When officer-reported race is observed, $T_s(x) = \Lambda_s(x; \chi)$, where $[\Lambda_s(x; \chi)]_{\tilde{r}, r} = \Pr_\chi(\tilde{R} = \tilde{r} \mid R = r, A = s, X = x)$. The index χ denotes one assumed forward misreporting matrix, and a sensitivity analysis is conducted over several values of χ .

The observed data law is obtained by applying the measurement operator to the full data law

$$p(w, a \mid x; \psi, f_0, g_0) = \sum_{r \in \mathcal{R}} t_a(w \mid r, x) h(r, a \mid x; \psi, f_0, g_0). \quad (72)$$

Because each column of $T_a(x)$ sums to one, this transformation maps every valid full data law h to a valid observed data law. Measurement error changes how the nuisance functions and estimating equation are implemented, but it does not impose compatibility restrictions between f_0 and g_0 .

Assume that each relevant $T_a(x)$ has full column rank, and let $B_a(x)$ be a fixed left inverse satisfying $B_a(x)T_a(x) = I_J$. For accurately measured race, $B_a(x) = I_J$. For officer-reported race, $B_s(x; \chi) = \Lambda_s(x; \chi)^{-1}$ when the matrix is nonsingular. For surname, one possible left inverse is $B_Q = (Q^\top Q)^{-1}Q^\top$ when Q has full column rank.

Let $e_a(W_i)$ be the one-hot vector for the observed signal in source a , and define the corrected race indicator

$$Z_i^\dagger = B_{A_i}(X_i)e_{A_i}(W_i). \quad (73)$$

Since $\mathbb{E}[e_a(W_i) \mid R_i = r, A_i = a, X_i = x] = T_a(x)e(r)$, the correction satisfies

$$\mathbb{E}[Z_i^\dagger \mid R_i, A_i, X_i] = e(R_i). \quad (74)$$

Let $Z_i^{\dagger, \text{ind}}$ denote the nonbaseline components of $Z_i^{\dagger}$. The corrected estimating function is

$$U_i^{\dagger}(\psi; f_0, g_0) = \{k(A_i, X_i) - \Delta_k(X_i)\} \{Z_i^{\dagger, \text{ind}} - \mu_{A_i}(X_i)\}. \quad (75)$$

For every candidate pair (f_0, g_0) , $\mathbb{E}[U_i^{\dagger}(\psi; f_0, g_0) \mid R_i, A_i, X_i] = U_i^{\text{full}}(\psi; f_0, g_0)$. Hence the corrected and full data moments have the same population expectation. When the measurement operator is correctly specified, the corrected equation solves for the same ψ_0 and inherits the original double robustness and Neyman orthogonality properties with respect to f_0 and g_0 .

We remark that the components of $Z_i^{\dagger}$ may be negative or exceed one. They should not be projected onto the probability simplex before entering the estimating equation, since doing so would generally destroy the conditional unbiasedness property above.

F.3 Estimation of the nuisance functions

The nuisances we are required to estimate are $f_0(r \mid x)$ and $g_0(a \mid x)$. We now discuss how to do so.

Because c is the baseline source, $f_0(r \mid x) = \Pr(R = r \mid A = c, X = x)$. In the benchmark source, the observed signal satisfies $\Pr(W = w \mid A = c, X = x) = \sum_{r \in \mathcal{R}} t_c(w \mid r, x) f_0(r \mid x)$.

We accordingly estimate f_0 by minimizing

$$\mathcal{L}_f(f_0) = - \sum_{i: A_i = c} \log \left\{ \sum_{r \in \mathcal{R}} t_c(W_i \mid r, X_i) f_0(r \mid X_i) \right\}, \quad (76)$$

subject to $f_0(r \mid x) \geq 0$ and $\sum_{r \in \mathcal{R}} f_0(r \mid x) = 1$.

With accurately measured benchmark race, this reduces to ordinary multinomial log loss. When race is unobserved and surnames are used as proxy, the loss becomes $-\sum_{i: A_i = c} \log \{ \sum_{r \in \mathcal{R}} q_{L_i r} f_0(r \mid X_i) \}$. The model may be fitted using multinomial regression, generalized additive models, trees fitted by likelihood, boosting, or empirical proportions

within sufficiently large blocks. Should the analyst be concerned with overlap in X as discussed in the main text, we note that any coarsening or discretization of X used for estimation should be applied consistently throughout the analysis, in particular to the estimation of g_0 below.

The second nuisance is $g_0(a | x) = \Pr(A = a | R = r_0, X = x)$. Let $Z_{i0}^\dagger = e_0^\top Z_i^\dagger$ be the corrected indicator for the baseline race; we do not reuse D here, which denotes crash risk. Thus, $Z_{i0}^\dagger = \mathbf{1}\{R_i = r_0\}$ assuming no mismeasurement of race, $Z_{i0}^\dagger = e_0^\top B_Q e(L_i)$ using surnames as a proxy, and $Z_{i0}^\dagger = e_0^\top \Lambda_s(X_i; \chi)^{-1} e(\tilde{R}_i)$ using officer-reported race as a proxy.

Because $\mathbb{E}[Z_{i0}^\dagger | R_i, A_i, X_i] = \mathbf{1}\{R_i = r_0\}$, we have $\mathbb{E}[Z_{i0}^\dagger \mathbf{1}\{A_i = a\} | X_i = x] = \Pr(A_i = a, R_i = r_0 | X_i = x)$ and $\mathbb{E}[Z_{i0}^\dagger | X_i = x] = \Pr(R_i = r_0 | X_i = x)$. Hence,

$$g_0(a | x) = \frac{\mathbb{E}[Z_{i0}^\dagger \mathbf{1}\{A_i = a\} | X_i = x]}{\mathbb{E}[Z_{i0}^\dagger | X_i = x]}. \quad (77)$$

Equivalently, $g_0(s | x)$ is identified by the conditional moment

$$\mathbb{E}[Z_{i0}^\dagger \{\mathbf{1}\{A_i = s\} - g_0(s | X_i)\} | X_i] = 0. \quad (78)$$

When race is accurately measured, g_0 may be estimated using any standard binary response learner applied to the encounters involving White drivers, including logistic regression, generalized additive models, random forests, or boosting. When measurement error is present, $Z_{i0}^\dagger$ may take values outside $[0, 1]$, so it should not be used as a frequency weight in an ordinary binary likelihood. One option is to estimate the conditional moment directly via generalized method of moments, generalized random forests ([Athey et al., 2019](#)), or kernel methods. Alternatively, one may separately estimate $m_s(x) = \mathbb{E}[Z_{i0}^\dagger \mathbf{1}\{A_i = s\} | X_i = x]$ and $m_0(x) = \mathbb{E}[Z_{i0}^\dagger | X_i = x]$ using regression methods and form $\hat{g}_0(s | x) = \hat{m}_s(x)/\hat{m}_0(x)$. In practice, estimation is restricted to regions where $\hat{m}_0(x)$ is bounded away from zero, and the final fitted probabilities may be truncated away from zero and one for numerical stability.

As is standard practice to remove bias from overfitting, we use cross fitting to generate valid inference. The sample is partitioned into V folds; we reserve K for the number of crash severity thresholds. For each held-out fold, f_0 and g_0 are fitted separately using the remaining observations. The fixed measurement operators are then used to construct $Z_i^\dagger$ in the held out fold. These cross fitted nuisance estimates determine $\hat{h}$, $\hat{\mu}_a$, $\hat{V}_a$, and $\hat{\Delta}_k$.

The estimator $\hat{\psi}$ solves

$$\frac{1}{n} \sum_{i=1}^n \left\{ k(A_i, X_i) - \hat{\Delta}_k^{(-k(i))}(X_i; \psi) \right\} \left\{ Z_i^{\dagger, \text{ind}} - \hat{\mu}_{A_i}^{(-k(i))}(X_i; \psi) \right\} = 0. \quad (79)$$

Let $\hat{U}_i^\dagger(\psi)$ denote the corresponding cross fitted score, let $\bar{U}^\dagger(\psi) = n^{-1} \sum_i \hat{U}_i^\dagger(\psi)$, and define $\hat{G} = \partial \bar{U}^\dagger(\psi) / \partial \psi^\top|_{\psi=\hat{\psi}}$ and $\hat{\Sigma}^\dagger = n^{-1} \sum_i \hat{U}_i^\dagger(\hat{\psi}) \hat{U}_i^\dagger(\hat{\psi})^\top$. The estimated covariance matrix is

$$\hat{V}^\dagger = \frac{1}{n} \hat{G}^{-1} \hat{\Sigma}^\dagger \hat{G}^{-\top}. \quad (80)$$

Under standard regularity conditions, and sufficiently well conditioned mismeasurement operators T_a , $\sqrt{n}(\hat{\psi} - \psi_0)$ is asymptotically normal and standard Wald inference applies.

F.4 Adjustments depending on the data environment

We next discuss five scenarios a scholar is likely to encounter in this analysis depending on the data available to them, and we will specify how the above procedure adapts to each. The scenarios differ only in the measurement operators and the resulting forms of $Z_i^\dagger$ and D_i . The target ψ and nuisance functions (f_0, g_0) remain unchanged.

Scenario 1: accurate race in both sources. Here, $T_s(x) = T_c(x) = I_J$, so $Z_i^\dagger = e(R_i)$ and $Z_{i0}^\dagger = \mathbf{1}\{R_i = r_0\}$. The benchmark nuisance f_0 is estimated using an ordinary multinomial model, and g_0 is estimated via some binary response model, e.g. logistic regression. The corrected score equals the original full data score.

Scenario 2: unobserved race in one source, surname used as proxy. Suppose surname is used in source $a^\star$ and race is accurate in the other source a' . Then $T_{a^\star} = Q$ and $T_{a'}(x) = I_J$, so

$$Z_i^\dagger = \begin{cases} B_Q e(L_i), & A_i = a^\star, \\ e(R_i), & A_i = a'. \end{cases} \quad (81)$$

The corresponding baseline component is $Z_{i0}^\dagger = e_0^\top B_Q e(L_i)$ in source $a^\star$ and $Z_{i0}^\dagger = \mathbf{1}\{R_i = r_0\}$ in source a' . If the benchmark source uses surname, f_0 is fitted with the surname-mixture loss; otherwise it is fitted with ordinary multinomial loss.

Scenario 3: unobserved race in both sources, surname used as proxy.

Here, $T_s = T_c = Q$, so $Z_i^\dagger = B_Q e(L_i)$ and $Z_{i0}^\dagger = e_0^\top B_Q e(L_i)$ for every observation. The benchmark nuisance f_0 is estimated using the surname mixture loss, while g_0 is estimated from the corrected conditional moment based on $Z_{i0}^\dagger$.

Scenario 4: known misreporting in police data, accurate benchmark race. For each sensitivity value χ , let $T_s(x; \chi) = \Lambda_s(x; \chi)$ and $T_c(x) = I_J$. Then

$$Z_i^\dagger(\chi) = \begin{cases} \Lambda_s(X_i; \chi)^{-1} e(\tilde{R}_i), & A_i = s, \\ e(R_i), & A_i = c. \end{cases} \quad (82)$$

The corresponding baseline component is $Z_{i0}^\dagger(\chi) = e_0^\top \Lambda_s(X_i; \chi)^{-1} e(\tilde{R}_i)$ in the officer source and $Z_{i0}^\dagger(\chi) = \mathbf{1}\{R_i = r_0\}$ in the benchmark source. The nuisance f_0 is fitted using ordinary multinomial loss, and the complete analysis is repeated over the prespecified set of plausible forward misreporting matrices.

Scenario 5: known misreporting in stops, unobserved race in benchmark. For each sensitivity value χ , let $T_s(x; \chi) = \Lambda_s(x; \chi)$ and $T_c = Q$. Then

$$Z_i^\dagger(\chi) = \begin{cases} \Lambda_s(X_i; \chi)^{-1} e(\tilde{R}_i), & A_i = s, \\ B_Q e(L_i), & A_i = c. \end{cases} \quad (83)$$

The corresponding baseline component is $Z_{i0}^\dagger(\chi) = e_0^\top \Lambda_s(X_i; \chi)^{-1} e(\tilde{R}_i)$ in the officer source and $Z_{i0}^\dagger(\chi) = e_0^\top B_Q e(L_i)$ in the benchmark source. The benchmark nuisance f_0 is estimated using the surname mixture loss, and the complete DML analysis is repeated over the prespecified forward misreporting matrices.

G Inference for crashes falsification test details

Here we discuss statistical inference on the inequalities derived in Proposition 2.9. To avoid the nonregularity introduced by the pointwise minimum and maximum over levels of crash severity, we replace these functions with smooth log-sum-exp approximations. This produces a conservative relaxation of the original inequalities for which plug-in estimation and standard inference by the delta method are available (Levis et al., 2025).

For each severity index $\kappa \in \{1, \dots, K\}$, define $q_{r\kappa}(x) := \Pr(R = r \mid C \geq k_\kappa, X = x)$ and let $q_r(x) = (q_{r1}(x), \dots, q_{rK}(x))$. After averaging over the covariate distribution among stopped drivers, define the lower and upper endpoints and the race share among stopped drivers as

$$\underline{\mu}_r := \mathbb{E} \left[\min_{1 \leq \kappa \leq K} q_{r\kappa}(X) \mid S = s_1 \right], \quad \bar{\mu}_r := \mathbb{E} \left[\max_{1 \leq \kappa \leq K} q_{r\kappa}(X) \mid S = s_1 \right], \quad (84)$$

and write $\pi_r := \Pr(R = r \mid S = s_1)$. Under the null of no discrimination, where an officer stopping rule belongs to class $\mathcal{H}$ (as described in main text), an immediate corollary of Proposition 2.9 is $\underline{\mu}_r \leq \pi_r \leq \bar{\mu}_r$.

For a fixed smoothing parameter $\nu > 0$, and for a positive vector z of K elements define

the soft minimum and soft maximum functions as

$$f_{\nu}^{-}(z) := -\frac{1}{\nu} \log \left(\sum_{\kappa=1}^K \exp\{-\nu z_{\kappa}\} \right), \quad f_{\nu}^{+}(z) := \frac{1}{\nu} \log \left(\sum_{\kappa=1}^K \exp\{\nu z_{\kappa}\} \right). \quad (85)$$

These functions satisfy

$$\min_{1 \leq \kappa \leq K} z_{\kappa} - \frac{\log K}{\nu} \leq f_{\nu}^{-}(z) \leq \min_{1 \leq \kappa \leq K} z_{\kappa}, \quad \max_{1 \leq \kappa \leq K} z_{\kappa} \leq f_{\nu}^{+}(z) \leq \max_{1 \leq \kappa \leq K} z_{\kappa} + \frac{\log K}{\nu}. \quad (86)$$

We accordingly define the smoothed endpoints

$$\underline{\mu}_{r,\nu} := \mathbb{E} \left[f_{\nu}^{-} \{q_r(X)\} \mid S = s_1 \right], \quad \bar{\mu}_{r,\nu} := \mathbb{E} \left[f_{\nu}^{+} \{q_r(X)\} \mid S = s_1 \right]. \quad (87)$$

Because $\underline{\mu}_{r,\nu} \leq \underline{\mu}_r$ and $\bar{\mu}_r \leq \bar{\mu}_{r,\nu}$, it follows that a rejection of the null hypothesis

$$H_{0,r}^{\nu} : \quad \underline{\mu}_{r,\nu} \leq \pi_r \leq \bar{\mu}_{r,\nu}. \quad (88)$$

also rejects the original inequality. Failure to reject should be interpreted relative to the chosen degree of smoothing, since the smoothed interval can be wider than the sharp interval by at most $(\log K)/\nu$ at each endpoint.

We select ν by specifying an interpretable tolerance $\varepsilon > 0$ and setting $\nu = (\log K)/\varepsilon$. Under this choice, the deterministic approximation error at either endpoint is no greater than ε on the probability scale. In the Texas application, where $K = 5$, this gives $\nu = \log(5)/\varepsilon$.

Let $\mathcal{D}_B = \{(R_j, C_j, X_j)\}_{j=1}^{n_B}$ denote the crash benchmark sample. For each severity threshold k_{κ} , we estimate $q_{r\kappa}(x)$ using a finite dimensional parametric model $q_{r\kappa}(x; \gamma_{r\kappa})$. For example, when $\mathbf{1}\{R = r\}$ is binary, we may use the logistic specification $q_{r\kappa}(x; \gamma_{r\kappa}) = \Lambda\{f_{\kappa}(x)^{\top} \gamma_{r\kappa}\}$, where $\Lambda(u) = \exp(u)/(1 + \exp(u))$, with $\nabla_{\gamma_{r\kappa}} q_{r\kappa}(x; \gamma_{r\kappa}) = q_{r\kappa}(x; \gamma_{r\kappa})\{1 - q_{r\kappa}(x; \gamma_{r\kappa})\} f_{\kappa}(x)$, estimated among benchmark observations satisfying $C \geq k_{\kappa}$. Here, $f_{\kappa}(x)$ is a prespecified vector of transformations of X .

Let $\gamma_r = (\gamma_{r1}^{\top}, \dots, \gamma_{rK}^{\top})^{\top}$ collect the parameters across severity thresholds, and let $\hat{\gamma}_r$ denote their joint estimator. Because the events $\{C \geq k_{\kappa} : 1 \leq \kappa \leq K\}$ are nested, the estimators

$\hat{\gamma}_{r1}, \dots, \hat{\gamma}_{rK}$ are generally correlated. Their covariance matrix must therefore be estimated jointly.

Define $Y_{rj} := \mathbf{1}\{R_j = r\}$ and $H_{\kappa j} := \mathbf{1}\{C_j \geq k_\kappa\}$; we do not reuse A , which denotes the data source. The score contribution for severity threshold k_κ is $U_{r\kappa,j}(\gamma_{r\kappa}) = H_{\kappa j} f_\kappa(X_j) \{Y_{rj} - q_{r\kappa}(X_j; \gamma_{r\kappa})\}$, where we reserve ψ for the target estimand. Stacking these scores over κ gives $U_{r,j}(\gamma_r)$. We estimate the covariance matrix of $\hat{\gamma}_r$ using $\widehat{\text{Var}}(\hat{\gamma}_r) = \frac{1}{n_B} \hat{A}_r^{-1} \hat{B}_r \hat{A}_r^{-\top}$, where

$$\hat{A}_r = -\frac{1}{n_B} \sum_{j=1}^{n_B} \nabla_{\gamma_r} U_{r,j}(\hat{\gamma}_r), \quad \hat{B}_r = \frac{1}{n_B} \sum_{j=1}^{n_B} U_{r,j}(\hat{\gamma}_r) U_{r,j}(\hat{\gamma}_r)^\top. \quad (89)$$

The off-diagonal blocks of $\hat{B}_r$ account for the fact that a benchmark observation may contribute to several nested severity regressions.

Let $\hat{q}_{r\kappa}(x) := q_{r\kappa}(x; \hat{\gamma}_{r\kappa})$ and $\hat{q}_r(x) = (\hat{q}_{r1}(x), \dots, \hat{q}_{rK}(x))$. The derivatives of the soft minimum and soft maximum with respect to the probabilities at each severity threshold are

$$\hat{w}_{r\kappa,\nu}^-(x) := \frac{\exp\{-\nu \hat{q}_{r\kappa}(x)\}}{\sum_{\kappa'=1}^K \exp\{-\nu \hat{q}_{r\kappa'}(x)\}}, \quad \hat{w}_{r\kappa,\nu}^+(x) := \frac{\exp\{\nu \hat{q}_{r\kappa}(x)\}}{\sum_{\kappa'=1}^K \exp\{\nu \hat{q}_{r\kappa'}(x)\}}. \quad (90)$$

The gradients of the smoothed endpoints with respect to γ_r are estimated over the empirical distribution of covariates in the stop sample:

$$\hat{G}_{r,\nu}^- = \frac{1}{n_S} \sum_{i=1}^{n_S} \sum_{\kappa=1}^K \hat{w}_{r\kappa,\nu}^-(X_i) \nabla_{\gamma_r} q_{r\kappa}(X_i; \hat{\gamma}_{r\kappa}), \quad \hat{G}_{r,\nu}^+ = \frac{1}{n_S} \sum_{i=1}^{n_S} \sum_{\kappa=1}^K \hat{w}_{r\kappa,\nu}^+(X_i) \nabla_{\gamma_r} q_{r\kappa}(X_i; \hat{\gamma}_{r\kappa}), \quad (91)$$

where $\nabla_{\gamma_r} q_{r\kappa}(X_i; \hat{\gamma}_{r\kappa})$ is zero in the components of γ_r corresponding to severity thresholds other than κ . Define the $2 \times \dim(\gamma_r)$ Jacobian

$$\hat{J}_{r,\nu}^B = \begin{pmatrix} \hat{G}_{r,\nu}^{-\top} \\ -\hat{G}_{r,\nu}^{+\top} \end{pmatrix}. \quad (92)$$

The negative sign in the second row arises because the upper discrepancy is defined as the racial proportion in stops minus the upper bound of the interval.

G.0.1 Inference When Race Is Correctly Measured

Let $\mathcal{D}_S = \{(R_i, X_i) : S_i = s_1\}_{i=1}^{n_S}$ denote the stop sample. We use its empirical covariate distribution to estimate expectations under $X \mid S = s_1$. The plug-in estimators of the smoothed endpoints and the race share among stopped drivers are

$$\hat{\mu}_{r,\nu}^- = \frac{1}{n_S} \sum_{i=1}^{n_S} f_\nu^- \{\hat{q}_r(X_i)\}, \quad \hat{\mu}_{r,\nu}^+ = \frac{1}{n_S} \sum_{i=1}^{n_S} f_\nu^+ \{\hat{q}_r(X_i)\}, \quad \hat{\pi}_r = \frac{1}{n_S} \sum_{i=1}^{n_S} \mathbf{1}\{R_i = r\}. \quad (93)$$

Define the estimated lower and upper ‘discrepancies’ as $\hat{\Delta}_{r,-} = \hat{\mu}_{r,\nu}^- - \hat{\pi}_r$ and $\hat{\Delta}_{r,+} = \hat{\pi}_r - \hat{\mu}_{r,\nu}^+$. Equivalently, let

$$\hat{d}_{r,i} = \begin{pmatrix} f_\nu^- \{\hat{q}_r(X_i)\} - \mathbf{1}\{R_i = r\} \\ \mathbf{1}\{R_i = r\} - f_\nu^+ \{\hat{q}_r(X_i)\} \end{pmatrix}, \quad \hat{\Delta}_r = \begin{pmatrix} \hat{\Delta}_{r,-} \\ \hat{\Delta}_{r,+} \end{pmatrix} = \frac{1}{n_S} \sum_{i=1}^{n_S} \hat{d}_{r,i}. \quad (94)$$

Conditional on the fitted crash regressions, the covariance of $\hat{\Delta}_r$ in the stop sample is estimated by the covariance matrix of the sample mean

$$\hat{\Omega}_r^S = \frac{1}{n_S(n_S - 1)} \sum_{i=1}^{n_S} (\hat{d}_{r,i} - \hat{\Delta}_r) (\hat{d}_{r,i} - \hat{\Delta}_r)^\top. \quad (95)$$

Uncertainty from estimating the crash regressions is propagated by the delta method. Assuming that the stop and benchmark samples are independent, the total covariance estimator is

$$\hat{\Omega}_r = \hat{\Omega}_r^S + \hat{J}_{r,\nu}^B \widehat{\text{Var}}(\hat{\gamma}_r) \hat{J}_{r,\nu}^{B\top}. \quad (96)$$

The diagonal entries of $\hat{\Omega}_r$ are the estimated variances of $\hat{\Delta}_{r,-}$ and $\hat{\Delta}_{r,+}$.

G.0.2 Inference When Race Is Mismeasured

Parametric MLE Let $\tilde{R}$ denote officer-reported race and let R denote self-reported race. We assume that the forward misreporting matrix $M_{\text{for}} = (m_{\tilde{r}r})_{\tilde{r}, r \in \mathcal{R}}$, where $m_{\tilde{r}r} := \Pr(\tilde{R} = \tilde{r} \mid R = r, S = s_1)$, is known and does not vary with X . Let $p_r(x; \xi) := \Pr(R = r \mid$

$S = s_1, X = x$) denote a parametric model for the conditional distribution of self-reported race. The observed conditional distribution of officer-reported race is

$$\Pr(\tilde{R} = \tilde{r} \mid S = s_1, X = x; \xi) = \sum_{r \in \mathcal{R}} m_{\tilde{r}r} p_r(x; \xi). \quad (97)$$

For observation i in the stop sample, the observed data log likelihood contribution is $\ell_i(\xi) = \log\{\sum_{r \in \mathcal{R}} m_{\tilde{R}_i r} p_r(X_i; \xi)\}$. Let $s_i(\xi) = \nabla_\xi \ell_i(\xi)$ denote the score, and let $\hat{\xi}$ be the maximum likelihood estimator. Define

$$\hat{A}_\xi = -\frac{1}{n_S} \sum_{i=1}^{n_S} \nabla_\xi s_i(\hat{\xi}), \quad \hat{B}_\xi = \frac{1}{n_S} \sum_{i=1}^{n_S} s_i(\hat{\xi}) s_i(\hat{\xi})^\top, \quad \widehat{\text{Var}}(\hat{\xi}) = \frac{1}{n_S} \hat{A}_\xi^{-1} \hat{B}_\xi \hat{A}_\xi^{-\top}. \quad (98)$$

The corrected share of race r among stopped drivers is estimated by

$$\hat{\pi}_r^{\text{cor}} = \frac{1}{n_S} \sum_{i=1}^{n_S} p_r(X_i; \hat{\xi}). \quad (99)$$

The corresponding estimated discrepancies are $\hat{\Delta}_{r,-}^{\text{cor}} = \hat{\mu}_{r,\nu} - \hat{\pi}_r^{\text{cor}}$ and $\hat{\Delta}_{r,+}^{\text{cor}} = \hat{\pi}_r^{\text{cor}} - \hat{\mu}_{r,\nu}$.

Because $\hat{\xi}$ and the empirical averages defining the two discrepancies are estimated from the same stop sample, their covariance must be included. This can be done directly by treating the discrepancies and ξ as a joint system of smooth estimating equations.

Let $\delta = (\delta_-, \delta_+)^\top$, let $\zeta = (\delta^\top, \xi^\top)^\top$, and define

$$g_{r,i}(\zeta) = \begin{pmatrix} f_\nu^- \{\hat{q}_r(X_i)\} - p_r(X_i; \xi) - \delta_- \\ p_r(X_i; \xi) - f_\nu^+ \{\hat{q}_r(X_i)\} - \delta_+ \\ s_i(\xi) \end{pmatrix}. \quad (100)$$

The estimator $\hat{\zeta}_r = ((\hat{\Delta}_{r,-}^{\text{cor}}, \hat{\Delta}_{r,+}^{\text{cor}})^\top, \hat{\xi}^\top)^\top$ satisfies $n_S^{-1} \sum_{i=1}^{n_S} g_{r,i}(\hat{\zeta}_r) = 0$.

Define the Jacobian and empirical covariance matrices

$$\hat{A}_{\zeta,r} = -\frac{1}{n_S} \sum_{i=1}^{n_S} \nabla_\zeta g_{r,i}(\hat{\zeta}_r), \quad \hat{B}_{\zeta,r} = \frac{1}{n_S} \sum_{i=1}^{n_S} g_{r,i}(\hat{\zeta}_r) g_{r,i}(\hat{\zeta}_r)^\top, \quad \widehat{\text{Var}}(\hat{\zeta}_r) = \frac{1}{n_S} \hat{A}_{\zeta,r}^{-1} \hat{B}_{\zeta,r} \hat{A}_{\zeta,r}^{-\top}. \quad (101)$$

Let $\widehat{\Omega}_r^{S,\text{cor}}$ denote the upper left 2×2 block of $\widehat{\text{Var}}(\widehat{\zeta}_r)$. This block is the stop-sample covariance matrix of the corrected lower and upper discrepancies. It incorporates sampling variation in the empirical distribution of X , uncertainty from estimating ξ , and the covariance between these two sources.

Because the forward misreporting matrix is known, no variance term is required for estimation of M_{for} . Assuming that the stop and crash benchmark samples are independent, the total covariance matrix is

$$\widehat{\Omega}_r^{\text{cor}} = \widehat{\Omega}_r^{S,\text{cor}} + \widehat{J}_{r,\nu}^B \widehat{\text{Var}}(\widehat{\gamma}_r) \widehat{J}_{r,\nu}^{B\top}. \quad (102)$$

Compared to the case where race is measured accurately, only the variance estimation from the stops data changes, where the covariance is instead obtained from the joint estimating system in equation (100).

Estimation via coarsening. As an alternative to parametrically modeling $\Pr(R = r \mid S = s_1, X = x)$, we may replace X by a discrete blocking strategy and estimate the distribution of self-reported race separately within each block. Let $\tilde{p}(x) = \{\Pr(\tilde{R} = \tilde{r} \mid S = s_1, X = x)\}_{\tilde{r} \in \mathcal{S}(\tilde{R})}$ and $p(x) = \{\Pr(R = r \mid S = s_1, X = x)\}_{r \in \mathcal{R}}$. Because the known forward misreporting matrix M_{for} is constant in X , these probabilities satisfy

$$\tilde{p}(x) = M_{\text{for}} p(x), \quad p(x) = M_{\text{for}}^{-1} \tilde{p}(x). \quad (103)$$

Let $n_x = \sum_{i=1}^{n_S} \mathbf{1}\{X_i = x\}$ and estimate the observed race distribution within block x by $\widehat{\tilde{p}}(x) = n_x^{-1} \sum_{i: X_i = x} e_{\tilde{R}_i}$, where $e_{\tilde{r}}$ is the unit vector for category $\tilde{r}$. The corrected race distribution within each block is then $\widehat{p}(x) = M_{\text{for}}^{-1} \widehat{\tilde{p}}(x)$. Averaging these estimates over the empirical distribution of X gives

$$\widehat{\pi}^{\text{block}} = \sum_x \frac{n_x}{n_S} \widehat{p}(x) = M_{\text{for}}^{-1} \left\{ \frac{1}{n_S} \sum_{i=1}^{n_S} e_{\tilde{R}_i} \right\}. \quad (104)$$

The second equality follows because the same forward misreporting matrix applies in every block. For race category r , define $\omega_{ri} = e_r^\top M_{\text{for}}^{-1} e_{\tilde{R}_i}$, so that $\widehat{\pi}_r^{\text{block}} = n_S^{-1} \sum_{i=1}^{n_S} \omega_{ri}$.

The corresponding contributions at the observation level to the two estimated inequality discrepancies are

$$\hat{d}_{r,i}^{\text{block}} = \begin{pmatrix} f_{\nu}^{-}\{\hat{q}_r(X_i)\} - \omega_{ri} \\ \omega_{ri} - f_{\nu}^{+}\{\hat{q}_r(X_i)\} \end{pmatrix}, \quad \hat{\Delta}_r^{\text{block}} = \frac{1}{n_S} \sum_{i=1}^{n_S} \hat{d}_{r,i}^{\text{block}}. \quad (105)$$

The covariance matrix is estimated with

$$\hat{\Omega}_r^{S,\text{block}} = \frac{1}{n_S(n_S - 1)} \sum_{i=1}^{n_S} \left(\hat{d}_{r,i}^{\text{block}} - \hat{\Delta}_r^{\text{block}} \right) \left(\hat{d}_{r,i}^{\text{block}} - \hat{\Delta}_r^{\text{block}} \right)^{\top}, \quad (106)$$

from which we edit (102) to give the following $\hat{\Omega}_r^{S,\text{block}}$:

$$\hat{\Omega}_r^{\text{block}} = \hat{\Omega}_r^{S,\text{block}} + \hat{J}_{r,\nu}^B \widehat{\text{Var}}(\hat{\gamma}_r) \hat{J}_{r,\nu}^{B\top}. \quad (107)$$

G.0.3 Testing Procedure

For correctly measured race, let $\hat{V}_{r,-}$ and $\hat{V}_{r,+}$ denote the first and second diagonal entries of $\hat{\Omega}_r$. For mismeasured race, let them denote the corresponding diagonal entries of $\hat{\Omega}_r^{\text{cor}}$, and replace the estimated discrepancies by their corrected counterparts. In either case, form the one sided Wald statistics

$$T_{r,-} = \frac{\hat{\Delta}_{r,-}}{\sqrt{\hat{V}_{r,-}}}, \quad T_{r,+} = \frac{\hat{\Delta}_{r,+}}{\sqrt{\hat{V}_{r,+}}}, \quad \text{reject } H_{0,r}^{\nu} \text{ if } T_{r,-} > z_{1-\alpha_{\text{test}}/2} \text{ or } T_{r,+} > z_{1-\alpha_{\text{test}}/2}. \quad (108)$$

For the analysis with mismeasured race, $\hat{\Delta}_{r,-}$ and $\hat{\Delta}_{r,+}$ in equation (108) are understood to be $\hat{\Delta}_{r,-}^{\text{cor}}$ and $\hat{\Delta}_{r,+}^{\text{cor}}$. Here, z_p denotes the p th quantile of the standard normal distribution. We write α_{test} for the test level to distinguish it from the Chicago reporting correction α . At level $\alpha_{\text{test}} = 0.05$, the critical value adjusted using the Bonferroni correction is $z_{0.975} \simeq 1.96$.

A positive value of $T_{r,-}$ indicates that the race share among stopped drivers lies below the smoothed lower endpoint, whereas a positive value of $T_{r,+}$ indicates that it lies above the smoothed upper endpoint. If the null is tested simultaneously for J_{test} race categories, a direct correction for the family wise error rate replaces the critical value by $z_{1-\alpha_{\text{test}}/(2J_{\text{test}})}$.

H Sensitivity Tables

Here, we state the robustness of the found disparities to variation in data quality. In Chicago, we report how the bounds on the causal risk ratio widen as officer ($\theta \in \{0.90, 0.92, \dots, 1.00\}$) and camera ($\eta \in \{0.90, 0.92, \dots, 1.00\}$) miscaptures increase, correcting officer labels for civilian race at the estimated rate ($\alpha = 0$, cf equation (43)) from the ACLU report. Tables of these quantities at different levels of α are available on request. In Texas, we show sensitivity of the odds ratio to officer misreporting of race at different levels of β , see equation (44).

H.1 Cameras: Chicago

Table 13: Camera analysis (redlight): bounds on the causal risk ratio $\psi_r^\dagger$ by race, year, and the sensitivity parameters θ and η .

Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	All years	0.90	0.90	(0.369, 2.306) (0.334, 2.527)
Black	All years	0.90	0.92	(0.411, 2.097) (0.373, 2.291)
Black	All years	0.90	0.94	(0.455, 1.919) (0.415, 2.091)
Black	All years	0.90	0.96	(0.500, 1.766) (0.458, 1.921)
Black	All years	0.90	0.98	(0.547, 1.634) (0.502, 1.774)
Black	All years	0.90	1.00	(0.594, 1.519) (0.546, 1.645)
Black	All years	0.92	0.90	(0.417, 2.178) (0.381, 2.359)
Black	All years	0.92	0.92	(0.464, 1.976) (0.427, 2.137)
Black	All years	0.92	0.94	(0.515, 1.804) (0.475, 1.947)
Black	All years	0.92	0.96	(0.567, 1.657) (0.526, 1.784)
Black	All years	0.92	0.98	(0.620, 1.529) (0.577, 1.644)

Continued on next page

Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	All years	0.92	1.00	(0.675, 1.418) (0.629, 1.523)
Black	All years	0.94	0.90	(0.471, 2.084) (0.439, 2.222)
Black	All years	0.94	0.92	(0.525, 1.885) (0.491, 2.005)
Black	All years	0.94	0.94	(0.582, 1.716) (0.546, 1.821)
Black	All years	0.94	0.96	(0.641, 1.570) (0.603, 1.663)
Black	All years	0.94	0.98	(0.702, 1.443) (0.661, 1.526)
Black	All years	0.94	1.00	(0.765, 1.333) (0.721, 1.407)
Black	All years	0.96	0.90	(0.539, 2.032) (0.504, 2.156)
Black	All years	0.96	0.92	(0.600, 1.830) (0.562, 1.938)
Black	All years	0.96	0.94	(0.665, 1.658) (0.624, 1.753)
Black	All years	0.96	0.96	(0.732, 1.508) (0.689, 1.593)
Black	All years	0.96	0.98	(0.802, 1.378) (0.757, 1.453)
Black	All years	0.96	1.00	(0.874, 1.264) (0.826, 1.330)
Black	All years	0.98	0.90	(0.641, 2.081) (0.604, 2.185)
Black	All years	0.98	0.92	(0.709, 1.861) (0.670, 1.949)
Black	All years	0.98	0.94	(0.782, 1.671) (0.741, 1.747)
Black	All years	0.98	0.96	(0.860, 1.505) (0.816, 1.571)
Black	All years	0.98	0.98	(0.940, 1.359) (0.894, 1.417)
Black	All years	0.98	1.00	(1.023, 1.230) (0.975, 1.280)
Black	All years	1.00	0.90	(0.785, 2.245) (0.751, 2.343)
Black	All years	1.00	0.92	(0.862, 1.986) (0.826, 2.069)
Black	All years	1.00	0.94	(0.945, 1.760) (0.906, 1.830)
Black	All years	1.00	0.96	(1.032, 1.559) (0.992, 1.618)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	All years	1.00	0.98	(1.125, 1.380) (1.082, 1.430)
Black	All years	1.00	1.00	1.220 (1.176, 1.261)
Black	2017	0.90	0.90	(0.357, 2.174) (0.253, 2.822)
Black	2017	0.90	0.92	(0.399, 1.993) (0.288, 2.573)
Black	2017	0.90	0.94	(0.444, 1.838) (0.325, 2.357)
Black	2017	0.90	0.96	(0.490, 1.703) (0.364, 2.176)
Black	2017	0.90	0.98	(0.537, 1.585) (0.405, 2.014)
Black	2017	0.90	1.00	(0.586, 1.481) (0.445, 1.874)
Black	2017	0.92	0.90	(0.437, 2.204) (0.334, 2.776)
Black	2017	0.92	0.92	(0.488, 2.017) (0.378, 2.520)
Black	2017	0.92	0.94	(0.541, 1.856) (0.425, 2.302)
Black	2017	0.92	0.96	(0.597, 1.716) (0.474, 2.115)
Black	2017	0.92	0.98	(0.654, 1.593) (0.525, 1.952)
Black	2017	0.92	1.00	(0.713, 1.484) (0.578, 1.811)
Black	2017	0.94	0.90	(0.519, 2.198) (0.422, 2.626)
Black	2017	0.94	0.92	(0.578, 2.006) (0.475, 2.387)
Black	2017	0.94	0.94	(0.640, 1.840) (0.530, 2.177)
Black	2017	0.94	0.96	(0.705, 1.695) (0.589, 1.998)
Black	2017	0.94	0.98	(0.772, 1.568) (0.648, 1.841)
Black	2017	0.94	1.00	(0.842, 1.455) (0.710, 1.701)
Black	2017	0.96	0.90	(0.623, 2.236) (0.520, 2.670)
Black	2017	0.96	0.92	(0.691, 2.030) (0.581, 2.416)
Black	2017	0.96	0.94	(0.762, 1.851) (0.646, 2.196)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2017	0.96	0.96	(0.837, 1.694) (0.713, 2.001)
Black	2017	0.96	0.98	(0.915, 1.555) (0.784, 1.829)
Black	2017	0.96	1.00	(0.996, 1.431) (0.857, 1.672)
Black	2017	0.98	0.90	(0.741, 2.310) (0.637, 2.702)
Black	2017	0.98	0.92	(0.816, 2.080) (0.706, 2.425)
Black	2017	0.98	0.94	(0.896, 1.878) (0.780, 2.181)
Black	2017	0.98	0.96	(0.981, 1.699) (0.859, 1.962)
Black	2017	0.98	0.98	(1.070, 1.538) (0.941, 1.768)
Black	2017	0.98	1.00	(1.163, 1.394) (1.027, 1.591)
Black	2017	1.00	0.90	(0.906, 2.532) (0.799, 2.871)
Black	2017	1.00	0.92	(0.988, 2.246) (0.875, 2.536)
Black	2017	1.00	0.94	(1.076, 1.992) (0.956, 2.243)
Black	2017	1.00	0.96	(1.170, 1.765) (1.044, 1.975)
Black	2017	1.00	0.98	(1.269, 1.559) (1.136, 1.736)
Black	2017	1.00	1.00	1.373 (1.233, 1.520)
Black	2018	0.90	0.90	(0.334, 2.085) (0.265, 2.546)
Black	2018	0.90	0.92	(0.374, 1.906) (0.301, 2.314)
Black	2018	0.90	0.94	(0.415, 1.753) (0.337, 2.118)
Black	2018	0.90	0.96	(0.458, 1.622) (0.376, 1.952)
Black	2018	0.90	0.98	(0.502, 1.507) (0.415, 1.807)
Black	2018	0.90	1.00	(0.546, 1.407) (0.455, 1.682)
Black	2018	0.92	0.90	(0.377, 1.970) (0.310, 2.366)
Black	2018	0.92	0.92	(0.422, 1.798) (0.352, 2.147)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2018	0.92	0.94	(0.470, 1.651) (0.395, 1.962)
Black	2018	0.92	0.96	(0.519, 1.524) (0.439, 1.802)
Black	2018	0.92	0.98	(0.570, 1.413) (0.484, 1.663)
Black	2018	0.92	1.00	(0.622, 1.316) (0.531, 1.544)
Black	2018	0.94	0.90	(0.442, 1.920) (0.378, 2.252)
Black	2018	0.94	0.92	(0.494, 1.748) (0.426, 2.044)
Black	2018	0.94	0.94	(0.549, 1.601) (0.477, 1.866)
Black	2018	0.94	0.96	(0.606, 1.473) (0.529, 1.713)
Black	2018	0.94	0.98	(0.664, 1.361) (0.583, 1.578)
Black	2018	0.94	1.00	(0.724, 1.264) (0.638, 1.460)
Black	2018	0.96	0.90	(0.526, 1.906) (0.459, 2.151)
Black	2018	0.96	0.92	(0.584, 1.729) (0.515, 1.942)
Black	2018	0.96	0.94	(0.646, 1.576) (0.573, 1.764)
Black	2018	0.96	0.96	(0.712, 1.443) (0.634, 1.609)
Black	2018	0.96	0.98	(0.779, 1.326) (0.698, 1.474)
Black	2018	0.96	1.00	(0.849, 1.223) (0.764, 1.355)
Black	2018	0.98	0.90	(0.641, 1.976) (0.566, 2.213)
Black	2018	0.98	0.92	(0.707, 1.781) (0.628, 1.990)
Black	2018	0.98	0.94	(0.778, 1.612) (0.694, 1.796)
Black	2018	0.98	0.96	(0.852, 1.462) (0.764, 1.628)
Black	2018	0.98	0.98	(0.930, 1.329) (0.837, 1.474)
Black	2018	0.98	1.00	(1.010, 1.209) (0.913, 1.336)
Black	2018	1.00	0.90	(0.823, 2.227) (0.745, 2.460)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2018	1.00	0.92	(0.897, 1.985) (0.815, 2.186)
Black	2018	1.00	0.94	(0.976, 1.772) (0.890, 1.946)
Black	2018	1.00	0.96	(1.061, 1.580) (0.970, 1.730)
Black	2018	1.00	0.98	(1.150, 1.403) (1.055, 1.532)
Black	2018	1.00	1.00	1.242 (1.144, 1.349)
Black	2019	0.90	0.90	(0.444, 2.641) (0.335, 3.439)
Black	2019	0.90	0.92	(0.488, 2.396) (0.375, 3.098)
Black	2019	0.90	0.94	(0.534, 2.188) (0.416, 2.822)
Black	2019	0.90	0.96	(0.582, 2.009) (0.458, 2.578)
Black	2019	0.90	0.98	(0.630, 1.855) (0.501, 2.366)
Black	2019	0.90	1.00	(0.678, 1.720) (0.543, 2.183)
Black	2019	0.92	0.90	(0.517, 2.580) (0.410, 3.143)
Black	2019	0.92	0.92	(0.569, 2.331) (0.457, 2.821)
Black	2019	0.92	0.94	(0.622, 2.121) (0.505, 2.553)
Black	2019	0.92	0.96	(0.678, 1.940) (0.555, 2.323)
Black	2019	0.92	0.98	(0.734, 1.784) (0.607, 2.126)
Black	2019	0.92	1.00	(0.791, 1.647) (0.659, 1.957)
Black	2019	0.94	0.90	(0.598, 2.542) (0.499, 3.039)
Black	2019	0.94	0.92	(0.657, 2.287) (0.553, 2.714)
Black	2019	0.94	0.94	(0.718, 2.070) (0.610, 2.444)
Black	2019	0.94	0.96	(0.782, 1.884) (0.669, 2.211)
Black	2019	0.94	0.98	(0.847, 1.721) (0.729, 2.005)
Black	2019	0.94	1.00	(0.914, 1.579) (0.790, 1.825)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2019	0.96	0.90	(0.692, 2.515) (0.595, 2.943)
Black	2019	0.96	0.92	(0.758, 2.252) (0.657, 2.624)
Black	2019	0.96	0.94	(0.828, 2.028) (0.723, 2.351)
Black	2019	0.96	0.96	(0.900, 1.833) (0.791, 2.113)
Black	2019	0.96	0.98	(0.975, 1.662) (0.862, 1.901)
Black	2019	0.96	1.00	(1.052, 1.512) (0.933, 1.720)
Black	2019	0.98	0.90	(0.813, 2.583) (0.717, 2.940)
Black	2019	0.98	0.92	(0.887, 2.295) (0.788, 2.604)
Black	2019	0.98	0.94	(0.966, 2.048) (0.862, 2.316)
Black	2019	0.98	0.96	(1.049, 1.830) (0.940, 2.059)
Black	2019	0.98	0.98	(1.135, 1.636) (1.022, 1.832)
Black	2019	0.98	1.00	(1.223, 1.465) (1.106, 1.631)
Black	2019	1.00	0.90	(0.993, 2.836) (0.888, 3.143)
Black	2019	1.00	0.92	(1.076, 2.494) (0.967, 2.751)
Black	2019	1.00	0.94	(1.164, 2.193) (1.051, 2.410)
Black	2019	1.00	0.96	(1.257, 1.922) (1.141, 2.101)
Black	2019	1.00	0.98	(1.356, 1.676) (1.234, 1.823)
Black	2019	1.00	1.00	1.457 (1.331, 1.576)
Black	2020	0.90	0.90	(0.466, 2.888) (0.303, 4.084)
Black	2020	0.90	0.92	(0.511, 2.589) (0.338, 3.609)
Black	2020	0.90	0.94	(0.558, 2.338) (0.375, 3.221)
Black	2020	0.90	0.96	(0.607, 2.127) (0.416, 2.900)
Black	2020	0.90	0.98	(0.658, 1.946) (0.457, 2.630)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2020	0.90	1.00	(0.709, 1.790) (0.501, 2.399)
Black	2020	0.92	0.90	(0.520, 2.690) (0.373, 3.615)
Black	2020	0.92	0.92	(0.571, 2.408) (0.415, 3.194)
Black	2020	0.92	0.94	(0.624, 2.172) (0.459, 2.851)
Black	2020	0.92	0.96	(0.679, 1.971) (0.506, 2.564)
Black	2020	0.92	0.98	(0.736, 1.799) (0.555, 2.321)
Black	2020	0.92	1.00	(0.795, 1.651) (0.605, 2.112)
Black	2020	0.94	0.90	(0.585, 2.584) (0.447, 3.352)
Black	2020	0.94	0.92	(0.642, 2.304) (0.496, 2.960)
Black	2020	0.94	0.94	(0.702, 2.068) (0.549, 2.639)
Black	2020	0.94	0.96	(0.765, 1.868) (0.604, 2.368)
Black	2020	0.94	0.98	(0.830, 1.696) (0.661, 2.134)
Black	2020	0.94	1.00	(0.897, 1.548) (0.721, 1.930)
Black	2020	0.96	0.90	(0.665, 2.574) (0.538, 3.156)
Black	2020	0.96	0.92	(0.730, 2.279) (0.596, 2.768)
Black	2020	0.96	0.94	(0.798, 2.030) (0.658, 2.446)
Black	2020	0.96	0.96	(0.870, 1.817) (0.724, 2.177)
Black	2020	0.96	0.98	(0.945, 1.634) (0.793, 1.936)
Black	2020	0.96	1.00	(1.022, 1.475) (0.863, 1.733)
Black	2020	0.98	0.90	(0.783, 2.672) (0.662, 3.148)
Black	2020	0.98	0.92	(0.856, 2.344) (0.730, 2.744)
Black	2020	0.98	0.94	(0.934, 2.065) (0.803, 2.404)
Black	2020	0.98	0.96	(1.016, 1.824) (0.879, 2.113)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2020	0.98	0.98	(1.102, 1.615) (0.959, 1.856)
Black	2020	0.98	1.00	(1.191, 1.433) (1.041, 1.634)
Black	2020	1.00	0.90	(0.938, 2.867) (0.815, 3.312)
Black	2020	1.00	0.92	(1.020, 2.485) (0.890, 2.844)
Black	2020	1.00	0.94	(1.108, 2.158) (0.973, 2.453)
Black	2020	1.00	0.96	(1.201, 1.874) (1.058, 2.112)
Black	2020	1.00	0.98	(1.299, 1.623) (1.149, 1.812)
Black	2020	1.00	1.00	1.401 (1.244, 1.550)
Black	2021	0.90	0.90	(0.458, 2.962) (0.343, 3.849)
Black	2021	0.90	0.92	(0.503, 2.642) (0.382, 3.388)
Black	2021	0.90	0.94	(0.551, 2.374) (0.423, 3.018)
Black	2021	0.90	0.96	(0.600, 2.150) (0.466, 2.707)
Black	2021	0.90	0.98	(0.651, 1.961) (0.510, 2.445)
Black	2021	0.90	1.00	(0.703, 1.799) (0.555, 2.226)
Black	2021	0.92	0.90	(0.486, 2.626) (0.382, 3.304)
Black	2021	0.92	0.92	(0.536, 2.343) (0.425, 2.924)
Black	2021	0.92	0.94	(0.587, 2.107) (0.471, 2.604)
Black	2021	0.92	0.96	(0.641, 1.908) (0.519, 2.342)
Black	2021	0.92	0.98	(0.697, 1.739) (0.569, 2.123)
Black	2021	0.92	1.00	(0.754, 1.594) (0.620, 1.935)
Black	2021	0.94	0.90	(0.533, 2.434) (0.436, 2.887)
Black	2021	0.94	0.92	(0.587, 2.169) (0.484, 2.556)
Black	2021	0.94	0.94	(0.645, 1.946) (0.536, 2.285)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2021	0.94	0.96	(0.705, 1.758) (0.590, 2.053)
Black	2021	0.94	0.98	(0.768, 1.598) (0.648, 1.858)
Black	2021	0.94	1.00	(0.832, 1.461) (0.707, 1.691)
Black	2021	0.96	0.90	(0.590, 2.311) (0.503, 2.701)
Black	2021	0.96	0.92	(0.650, 2.050) (0.558, 2.382)
Black	2021	0.96	0.94	(0.714, 1.831) (0.616, 2.116)
Black	2021	0.96	0.96	(0.782, 1.646) (0.679, 1.892)
Black	2021	0.96	0.98	(0.853, 1.488) (0.745, 1.701)
Black	2021	0.96	1.00	(0.926, 1.351) (0.813, 1.539)
Black	2021	0.98	0.90	(0.686, 2.298) (0.593, 2.650)
Black	2021	0.98	0.92	(0.753, 2.026) (0.656, 2.320)
Black	2021	0.98	0.94	(0.825, 1.797) (0.723, 2.047)
Black	2021	0.98	0.96	(0.901, 1.601) (0.795, 1.813)
Black	2021	0.98	0.98	(0.982, 1.433) (0.870, 1.614)
Black	2021	0.98	1.00	(1.066, 1.287) (0.950, 1.441)
Black	2021	1.00	0.90	(0.807, 2.352) (0.715, 2.650)
Black	2021	1.00	0.92	(0.880, 2.054) (0.783, 2.299)
Black	2021	1.00	0.94	(0.959, 1.800) (0.857, 2.005)
Black	2021	1.00	0.96	(1.043, 1.582) (0.936, 1.752)
Black	2021	1.00	0.98	(1.133, 1.392) (1.021, 1.532)
Black	2021	1.00	1.00	1.227 (1.111, 1.341)
Black	2022	0.90	0.90	(0.371, 2.275) (0.282, 2.905)
Black	2022	0.90	0.92	(0.413, 2.069) (0.317, 2.623)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2022	0.90	0.94	(0.458, 1.894) (0.355, 2.383)
Black	2022	0.90	0.96	(0.503, 1.744) (0.395, 2.177)
Black	2022	0.90	0.98	(0.550, 1.614) (0.435, 2.006)
Black	2022	0.90	1.00	(0.597, 1.502) (0.475, 1.857)
Black	2022	0.92	0.90	(0.390, 2.020) (0.317, 2.424)
Black	2022	0.92	0.92	(0.436, 1.836) (0.358, 2.183)
Black	2022	0.92	0.94	(0.485, 1.679) (0.401, 1.985)
Black	2022	0.92	0.96	(0.536, 1.544) (0.445, 1.816)
Black	2022	0.92	0.98	(0.588, 1.428) (0.491, 1.672)
Black	2022	0.92	1.00	(0.641, 1.327) (0.537, 1.548)
Black	2022	0.94	0.90	(0.411, 1.837) (0.345, 2.177)
Black	2022	0.94	0.92	(0.461, 1.664) (0.391, 1.962)
Black	2022	0.94	0.94	(0.515, 1.517) (0.440, 1.781)
Black	2022	0.94	0.96	(0.571, 1.392) (0.491, 1.626)
Black	2022	0.94	0.98	(0.628, 1.283) (0.543, 1.493)
Black	2022	0.94	1.00	(0.686, 1.189) (0.597, 1.378)
Black	2022	0.96	0.90	(0.458, 1.746) (0.400, 1.999)
Black	2022	0.96	0.92	(0.513, 1.576) (0.452, 1.797)
Black	2022	0.96	0.94	(0.573, 1.431) (0.507, 1.628)
Black	2022	0.96	0.96	(0.635, 1.306) (0.565, 1.483)
Black	2022	0.96	0.98	(0.699, 1.199) (0.625, 1.358)
Black	2022	0.96	1.00	(0.765, 1.105) (0.686, 1.249)
Black	2022	0.98	0.90	(0.538, 1.766) (0.480, 1.967)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2022	0.98	0.92	(0.602, 1.584) (0.540, 1.759)
Black	2022	0.98	0.94	(0.670, 1.428) (0.604, 1.584)
Black	2022	0.98	0.96	(0.741, 1.294) (0.671, 1.433)
Black	2022	0.98	0.98	(0.815, 1.176) (0.741, 1.300)
Black	2022	0.98	1.00	(0.892, 1.073) (0.813, 1.183)
Black	2022	1.00	0.90	(0.650, 1.858) (0.585, 2.037)
Black	2022	1.00	0.92	(0.723, 1.651) (0.653, 1.803)
Black	2022	1.00	0.94	(0.800, 1.473) (0.726, 1.604)
Black	2022	1.00	0.96	(0.882, 1.318) (0.804, 1.430)
Black	2022	1.00	0.98	(0.968, 1.180) (0.886, 1.276)
Black	2022	1.00	1.00	1.056 (0.970, 1.138)
Black	2023	0.90	0.90	(0.288, 2.012) (0.206, 2.623)
Black	2023	0.90	0.92	(0.328, 1.828) (0.240, 2.365)
Black	2023	0.90	0.94	(0.371, 1.673) (0.277, 2.148)
Black	2023	0.90	0.96	(0.415, 1.540) (0.315, 1.964)
Black	2023	0.90	0.98	(0.461, 1.425) (0.353, 1.805)
Black	2023	0.90	1.00	(0.506, 1.326) (0.392, 1.672)
Black	2023	0.92	0.90	(0.312, 1.824) (0.240, 2.233)
Black	2023	0.92	0.92	(0.358, 1.658) (0.280, 2.017)
Black	2023	0.92	0.94	(0.406, 1.516) (0.322, 1.835)
Black	2023	0.92	0.96	(0.456, 1.395) (0.364, 1.681)
Black	2023	0.92	0.98	(0.507, 1.291) (0.408, 1.549)
Black	2023	0.92	1.00	(0.559, 1.200) (0.452, 1.435)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2023	0.94	0.90	(0.349, 1.723) (0.280, 2.046)
Black	2023	0.94	0.92	(0.401, 1.562) (0.327, 1.846)
Black	2023	0.94	0.94	(0.456, 1.425) (0.378, 1.676)
Black	2023	0.94	0.96	(0.514, 1.308) (0.431, 1.533)
Black	2023	0.94	0.98	(0.573, 1.207) (0.485, 1.408)
Black	2023	0.94	1.00	(0.632, 1.118) (0.540, 1.301)
Black	2023	0.96	0.90	(0.399, 1.679) (0.332, 1.958)
Black	2023	0.96	0.92	(0.458, 1.516) (0.386, 1.763)
Black	2023	0.96	0.94	(0.521, 1.376) (0.444, 1.592)
Black	2023	0.96	0.96	(0.587, 1.256) (0.505, 1.447)
Black	2023	0.96	0.98	(0.655, 1.151) (0.568, 1.322)
Black	2023	0.96	1.00	(0.725, 1.061) (0.631, 1.214)
Black	2023	0.98	0.90	(0.477, 1.727) (0.413, 1.993)
Black	2023	0.98	0.92	(0.545, 1.548) (0.476, 1.780)
Black	2023	0.98	0.94	(0.618, 1.393) (0.545, 1.595)
Black	2023	0.98	0.96	(0.696, 1.259) (0.617, 1.434)
Black	2023	0.98	0.98	(0.776, 1.141) (0.691, 1.294)
Black	2023	0.98	1.00	(0.858, 1.039) (0.767, 1.173)
Black	2023	1.00	0.90	(0.584, 1.838) (0.521, 2.073)
Black	2023	1.00	0.92	(0.662, 1.631) (0.594, 1.831)
Black	2023	1.00	0.94	(0.747, 1.449) (0.673, 1.617)
Black	2023	1.00	0.96	(0.837, 1.289) (0.758, 1.431)
Black	2023	1.00	0.98	(0.930, 1.149) (0.846, 1.268)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2023	1.00	1.00	1.026 (0.935, 1.128)
Hispanic	All years	0.90	0.90	(0.495, 2.905) (0.457, 3.135)
Hispanic	All years	0.90	0.92	(0.544, 2.642) (0.504, 2.844)
Hispanic	All years	0.90	0.94	(0.594, 2.418) (0.551, 2.595)
Hispanic	All years	0.90	0.96	(0.644, 2.225) (0.599, 2.384)
Hispanic	All years	0.90	0.98	(0.695, 2.059) (0.647, 2.202)
Hispanic	All years	0.90	1.00	(0.745, 1.915) (0.695, 2.045)
Hispanic	All years	0.92	0.90	(0.572, 2.722) (0.538, 2.893)
Hispanic	All years	0.92	0.92	(0.627, 2.469) (0.592, 2.620)
Hispanic	All years	0.92	0.94	(0.684, 2.255) (0.645, 2.387)
Hispanic	All years	0.92	0.96	(0.740, 2.071) (0.700, 2.189)
Hispanic	All years	0.92	0.98	(0.797, 1.912) (0.754, 2.018)
Hispanic	All years	0.92	1.00	(0.854, 1.774) (0.809, 1.870)
Hispanic	All years	0.94	0.90	(0.647, 2.549) (0.618, 2.677)
Hispanic	All years	0.94	0.92	(0.708, 2.306) (0.677, 2.417)
Hispanic	All years	0.94	0.94	(0.770, 2.099) (0.737, 2.197)
Hispanic	All years	0.94	0.96	(0.832, 1.923) (0.797, 2.009)
Hispanic	All years	0.94	0.98	(0.895, 1.770) (0.858, 1.847)
Hispanic	All years	0.94	1.00	(0.958, 1.637) (0.918, 1.707)
Hispanic	All years	0.96	0.90	(0.732, 2.422) (0.702, 2.527)
Hispanic	All years	0.96	0.92	(0.799, 2.184) (0.767, 2.275)
Hispanic	All years	0.96	0.94	(0.867, 1.981) (0.833, 2.061)
Hispanic	All years	0.96	0.96	(0.936, 1.806) (0.900, 1.877)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	All years	0.96	0.98	(1.005, 1.656) (0.967, 1.718)
Hispanic	All years	0.96	1.00	(1.074, 1.525) (1.034, 1.580)
Hispanic	All years	0.98	0.90	(0.837, 2.379) (0.808, 2.464)
Hispanic	All years	0.98	0.92	(0.911, 2.134) (0.881, 2.207)
Hispanic	All years	0.98	0.94	(0.987, 1.924) (0.955, 1.987)
Hispanic	All years	0.98	0.96	(1.063, 1.743) (1.030, 1.798)
Hispanic	All years	0.98	0.98	(1.140, 1.585) (1.105, 1.633)
Hispanic	All years	0.98	1.00	(1.218, 1.448) (1.180, 1.490)
Hispanic	All years	1.00	0.90	(0.954, 2.397) (0.928, 2.477)
Hispanic	All years	1.00	0.92	(1.037, 2.134) (1.009, 2.204)
Hispanic	All years	1.00	0.94	(1.122, 1.907) (1.093, 1.967)
Hispanic	All years	1.00	0.96	(1.208, 1.709) (1.178, 1.761)
Hispanic	All years	1.00	0.98	(1.296, 1.536) (1.264, 1.580)
Hispanic	All years	1.00	1.00	1.384 (1.350, 1.422)
Hispanic	2017	0.90	0.90	(0.498, 2.798) (0.387, 3.498)
Hispanic	2017	0.90	0.92	(0.549, 2.562) (0.431, 3.180)
Hispanic	2017	0.90	0.94	(0.601, 2.358) (0.475, 2.909)
Hispanic	2017	0.90	0.96	(0.654, 2.182) (0.519, 2.673)
Hispanic	2017	0.90	0.98	(0.707, 2.029) (0.564, 2.468)
Hispanic	2017	0.90	1.00	(0.760, 1.894) (0.607, 2.289)
Hispanic	2017	0.92	0.90	(0.588, 2.712) (0.488, 3.223)
Hispanic	2017	0.92	0.92	(0.647, 2.477) (0.541, 2.925)
Hispanic	2017	0.92	0.94	(0.707, 2.274) (0.594, 2.671)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2017	0.92	0.96	(0.768, 2.099) (0.647, 2.454)
Hispanic	2017	0.92	0.98	(0.829, 1.946) (0.701, 2.266)
Hispanic	2017	0.92	1.00	(0.890, 1.812) (0.755, 2.101)
Hispanic	2017	0.94	0.90	(0.671, 2.591) (0.580, 2.986)
Hispanic	2017	0.94	0.92	(0.738, 2.363) (0.641, 2.711)
Hispanic	2017	0.94	0.94	(0.806, 2.166) (0.703, 2.476)
Hispanic	2017	0.94	0.96	(0.875, 1.995) (0.765, 2.271)
Hispanic	2017	0.94	0.98	(0.944, 1.846) (0.827, 2.096)
Hispanic	2017	0.94	1.00	(1.014, 1.715) (0.890, 1.941)
Hispanic	2017	0.96	0.90	(0.760, 2.490) (0.684, 2.796)
Hispanic	2017	0.96	0.92	(0.835, 2.264) (0.755, 2.533)
Hispanic	2017	0.96	0.94	(0.911, 2.069) (0.826, 2.303)
Hispanic	2017	0.96	0.96	(0.989, 1.898) (0.899, 2.105)
Hispanic	2017	0.96	0.98	(1.066, 1.748) (0.971, 1.932)
Hispanic	2017	0.96	1.00	(1.144, 1.617) (1.044, 1.780)
Hispanic	2017	0.98	0.90	(0.864, 2.444) (0.789, 2.684)
Hispanic	2017	0.98	0.92	(0.947, 2.211) (0.866, 2.421)
Hispanic	2017	0.98	0.94	(1.031, 2.007) (0.945, 2.191)
Hispanic	2017	0.98	0.96	(1.117, 1.829) (1.026, 1.990)
Hispanic	2017	0.98	0.98	(1.203, 1.671) (1.107, 1.813)
Hispanic	2017	0.98	1.00	(1.290, 1.533) (1.188, 1.659)
Hispanic	2017	1.00	0.90	(0.994, 2.471) (0.916, 2.707)
Hispanic	2017	1.00	0.92	(1.085, 2.217) (1.002, 2.420)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2017	1.00	0.94	(1.178, 1.994) (1.092, 2.169)
Hispanic	2017	1.00	0.96	(1.274, 1.797) (1.183, 1.948)
Hispanic	2017	1.00	0.98	(1.371, 1.622) (1.274, 1.752)
Hispanic	2017	1.00	1.00	1.468 (1.366, 1.580)
Hispanic	2018	0.90	0.90	(0.441, 2.571) (0.362, 3.036)
Hispanic	2018	0.90	0.92	(0.489, 2.355) (0.405, 2.767)
Hispanic	2018	0.90	0.94	(0.538, 2.171) (0.448, 2.542)
Hispanic	2018	0.90	0.96	(0.587, 2.011) (0.492, 2.346)
Hispanic	2018	0.90	0.98	(0.635, 1.872) (0.535, 2.175)
Hispanic	2018	0.90	1.00	(0.684, 1.750) (0.579, 2.027)
Hispanic	2018	0.92	0.90	(0.519, 2.450) (0.449, 2.817)
Hispanic	2018	0.92	0.92	(0.573, 2.241) (0.500, 2.565)
Hispanic	2018	0.92	0.94	(0.629, 2.062) (0.551, 2.350)
Hispanic	2018	0.92	0.96	(0.684, 1.907) (0.604, 2.169)
Hispanic	2018	0.92	0.98	(0.740, 1.772) (0.655, 2.010)
Hispanic	2018	0.92	1.00	(0.796, 1.654) (0.707, 1.871)
Hispanic	2018	0.94	0.90	(0.596, 2.324) (0.532, 2.588)
Hispanic	2018	0.94	0.92	(0.657, 2.122) (0.589, 2.358)
Hispanic	2018	0.94	0.94	(0.719, 1.949) (0.647, 2.159)
Hispanic	2018	0.94	0.96	(0.781, 1.799) (0.706, 1.986)
Hispanic	2018	0.94	0.98	(0.844, 1.668) (0.765, 1.837)
Hispanic	2018	0.94	1.00	(0.906, 1.552) (0.823, 1.706)
Hispanic	2018	0.96	0.90	(0.685, 2.238) (0.623, 2.448)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2018	0.96	0.92	(0.753, 2.037) (0.687, 2.222)
Hispanic	2018	0.96	0.94	(0.822, 1.864) (0.753, 2.028)
Hispanic	2018	0.96	0.96	(0.892, 1.714) (0.819, 1.860)
Hispanic	2018	0.96	0.98	(0.962, 1.583) (0.886, 1.714)
Hispanic	2018	0.96	1.00	(1.032, 1.467) (0.952, 1.586)
Hispanic	2018	0.98	0.90	(0.783, 2.190) (0.719, 2.363)
Hispanic	2018	0.98	0.92	(0.859, 1.984) (0.791, 2.138)
Hispanic	2018	0.98	0.94	(0.936, 1.806) (0.863, 1.942)
Hispanic	2018	0.98	0.96	(1.014, 1.650) (0.936, 1.771)
Hispanic	2018	0.98	0.98	(1.092, 1.513) (1.010, 1.621)
Hispanic	2018	0.98	1.00	(1.170, 1.393) (1.084, 1.490)
Hispanic	2018	1.00	0.90	(0.893, 2.213) (0.831, 2.370)
Hispanic	2018	1.00	0.92	(0.977, 1.990) (0.913, 2.126)
Hispanic	2018	1.00	0.94	(1.065, 1.795) (0.996, 1.913)
Hispanic	2018	1.00	0.96	(1.153, 1.622) (1.080, 1.725)
Hispanic	2018	1.00	0.98	(1.243, 1.469) (1.166, 1.558)
Hispanic	2018	1.00	1.00	1.334 (1.251, 1.412)
Hispanic	2019	0.90	0.90	(0.490, 2.925) (0.406, 3.611)
Hispanic	2019	0.90	0.92	(0.541, 2.657) (0.452, 3.254)
Hispanic	2019	0.90	0.94	(0.593, 2.430) (0.499, 2.959)
Hispanic	2019	0.90	0.96	(0.645, 2.236) (0.546, 2.710)
Hispanic	2019	0.90	0.98	(0.698, 2.068) (0.593, 2.497)
Hispanic	2019	0.90	1.00	(0.751, 1.923) (0.640, 2.314)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2019	0.92	0.90	(0.588, 2.843) (0.509, 3.300)
Hispanic	2019	0.92	0.92	(0.647, 2.575) (0.564, 2.968)
Hispanic	2019	0.92	0.94	(0.706, 2.348) (0.620, 2.686)
Hispanic	2019	0.92	0.96	(0.767, 2.154) (0.675, 2.448)
Hispanic	2019	0.92	0.98	(0.827, 1.985) (0.730, 2.245)
Hispanic	2019	0.92	1.00	(0.888, 1.839) (0.785, 2.068)
Hispanic	2019	0.94	0.90	(0.672, 2.695) (0.604, 3.056)
Hispanic	2019	0.94	0.92	(0.738, 2.434) (0.665, 2.747)
Hispanic	2019	0.94	0.94	(0.804, 2.213) (0.727, 2.486)
Hispanic	2019	0.94	0.96	(0.871, 2.022) (0.789, 2.263)
Hispanic	2019	0.94	0.98	(0.939, 1.858) (0.852, 2.067)
Hispanic	2019	0.94	1.00	(1.006, 1.714) (0.914, 1.900)
Hispanic	2019	0.96	0.90	(0.759, 2.564) (0.685, 2.846)
Hispanic	2019	0.96	0.92	(0.831, 2.308) (0.753, 2.553)
Hispanic	2019	0.96	0.94	(0.904, 2.090) (0.822, 2.302)
Hispanic	2019	0.96	0.96	(0.978, 1.902) (0.892, 2.088)
Hispanic	2019	0.96	0.98	(1.052, 1.738) (0.962, 1.900)
Hispanic	2019	0.96	1.00	(1.127, 1.595) (1.033, 1.740)
Hispanic	2019	0.98	0.90	(0.866, 2.527) (0.796, 2.803)
Hispanic	2019	0.98	0.92	(0.946, 2.263) (0.872, 2.501)
Hispanic	2019	0.98	0.94	(1.028, 2.037) (0.949, 2.243)
Hispanic	2019	0.98	0.96	(1.111, 1.841) (1.027, 2.019)
Hispanic	2019	0.98	0.98	(1.195, 1.669) (1.106, 1.823)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2019	0.98	1.00	(1.279, 1.518) (1.185, 1.651)
Hispanic	2019	1.00	0.90	(0.989, 2.586) (0.915, 2.812)
Hispanic	2019	1.00	0.92	(1.080, 2.301) (1.002, 2.493)
Hispanic	2019	1.00	0.94	(1.174, 2.053) (1.090, 2.218)
Hispanic	2019	1.00	0.96	(1.269, 1.833) (1.181, 1.971)
Hispanic	2019	1.00	0.98	(1.367, 1.637) (1.272, 1.751)
Hispanic	2019	1.00	1.00	1.465 (1.365, 1.562)
Hispanic	2020	0.90	0.90	(0.588, 3.524) (0.440, 4.461)
Hispanic	2020	0.90	0.92	(0.642, 3.170) (0.484, 3.971)
Hispanic	2020	0.90	0.94	(0.697, 2.873) (0.530, 3.555)
Hispanic	2020	0.90	0.96	(0.752, 2.621) (0.575, 3.215)
Hispanic	2020	0.90	0.98	(0.808, 2.406) (0.620, 2.930)
Hispanic	2020	0.90	1.00	(0.864, 2.221) (0.665, 2.690)
Hispanic	2020	0.92	0.90	(0.678, 3.308) (0.548, 4.014)
Hispanic	2020	0.92	0.92	(0.738, 2.967) (0.600, 3.569)
Hispanic	2020	0.92	0.94	(0.800, 2.682) (0.653, 3.194)
Hispanic	2020	0.92	0.96	(0.862, 2.440) (0.706, 2.883)
Hispanic	2020	0.92	0.98	(0.924, 2.234) (0.759, 2.620)
Hispanic	2020	0.92	1.00	(0.986, 2.056) (0.813, 2.402)
Hispanic	2020	0.94	0.90	(0.779, 3.164) (0.659, 3.742)
Hispanic	2020	0.94	0.92	(0.846, 2.826) (0.718, 3.316)
Hispanic	2020	0.94	0.94	(0.914, 2.542) (0.778, 2.964)
Hispanic	2020	0.94	0.96	(0.983, 2.303) (0.839, 2.668)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2020	0.94	0.98	(1.053, 2.099) (0.900, 2.415)
Hispanic	2020	0.94	1.00	(1.122, 1.923) (0.960, 2.202)
Hispanic	2020	0.96	0.90	(0.893, 3.087) (0.788, 3.601)
Hispanic	2020	0.96	0.92	(0.968, 2.741) (0.857, 3.172)
Hispanic	2020	0.96	0.94	(1.044, 2.451) (0.927, 2.818)
Hispanic	2020	0.96	0.96	(1.120, 2.205) (0.998, 2.521)
Hispanic	2020	0.96	0.98	(1.197, 1.995) (1.068, 2.264)
Hispanic	2020	0.96	1.00	(1.274, 1.815) (1.139, 2.042)
Hispanic	2020	0.98	0.90	(1.018, 3.066) (0.910, 3.558)
Hispanic	2020	0.98	0.92	(1.101, 2.703) (0.987, 3.121)
Hispanic	2020	0.98	0.94	(1.186, 2.398) (1.065, 2.755)
Hispanic	2020	0.98	0.96	(1.272, 2.138) (1.144, 2.439)
Hispanic	2020	0.98	0.98	(1.358, 1.915) (1.224, 2.170)
Hispanic	2020	0.98	1.00	(1.445, 1.724) (1.303, 1.937)
Hispanic	2020	1.00	0.90	(1.152, 3.112) (1.031, 3.543)
Hispanic	2020	1.00	0.92	(1.245, 2.720) (1.117, 3.087)
Hispanic	2020	1.00	0.94	(1.341, 2.387) (1.205, 2.700)
Hispanic	2020	1.00	0.96	(1.440, 2.102) (1.296, 2.369)
Hispanic	2020	1.00	0.98	(1.539, 1.854) (1.388, 2.078)
Hispanic	2020	1.00	1.00	1.640 (1.482, 1.829)
Hispanic	2021	0.90	0.90	(0.609, 3.775) (0.485, 4.720)
Hispanic	2021	0.90	0.92	(0.661, 3.362) (0.532, 4.159)
Hispanic	2021	0.90	0.94	(0.715, 3.020) (0.581, 3.703)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2021	0.90	0.96	(0.770, 2.734) (0.629, 3.328)
Hispanic	2021	0.90	0.98	(0.825, 2.493) (0.678, 3.018)
Hispanic	2021	0.90	1.00	(0.880, 2.288) (0.727, 2.754)
Hispanic	2021	0.92	0.90	(0.664, 3.310) (0.565, 3.888)
Hispanic	2021	0.92	0.92	(0.720, 2.950) (0.616, 3.437)
Hispanic	2021	0.92	0.94	(0.777, 2.651) (0.669, 3.069)
Hispanic	2021	0.92	0.96	(0.836, 2.401) (0.723, 2.763)
Hispanic	2021	0.92	0.98	(0.895, 2.189) (0.777, 2.510)
Hispanic	2021	0.92	1.00	(0.953, 2.009) (0.830, 2.294)
Hispanic	2021	0.94	0.90	(0.719, 2.950) (0.624, 3.381)
Hispanic	2021	0.94	0.92	(0.780, 2.627) (0.680, 2.991)
Hispanic	2021	0.94	0.94	(0.842, 2.359) (0.738, 2.671)
Hispanic	2021	0.94	0.96	(0.904, 2.134) (0.797, 2.406)
Hispanic	2021	0.94	0.98	(0.968, 1.943) (0.855, 2.182)
Hispanic	2021	0.94	1.00	(1.031, 1.780) (0.913, 1.990)
Hispanic	2021	0.96	0.90	(0.799, 2.753) (0.707, 3.076)
Hispanic	2021	0.96	0.92	(0.865, 2.443) (0.768, 2.712)
Hispanic	2021	0.96	0.94	(0.933, 2.185) (0.831, 2.414)
Hispanic	2021	0.96	0.96	(1.002, 1.969) (0.895, 2.165)
Hispanic	2021	0.96	0.98	(1.071, 1.786) (0.960, 1.956)
Hispanic	2021	0.96	1.00	(1.141, 1.629) (1.025, 1.777)
Hispanic	2021	0.98	0.90	(0.891, 2.637) (0.807, 2.914)
Hispanic	2021	0.98	0.92	(0.963, 2.328) (0.876, 2.559)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2021	0.98	0.94	(1.039, 2.072) (0.948, 2.267)
Hispanic	2021	0.98	0.96	(1.115, 1.856) (1.020, 2.022)
Hispanic	2021	0.98	0.98	(1.193, 1.672) (1.094, 1.815)
Hispanic	2021	0.98	1.00	(1.271, 1.515) (1.167, 1.639)
Hispanic	2021	1.00	0.90	(0.984, 2.568) (0.901, 2.830)
Hispanic	2021	1.00	0.92	(1.065, 2.253) (0.977, 2.470)
Hispanic	2021	1.00	0.94	(1.149, 1.989) (1.057, 2.172)
Hispanic	2021	1.00	0.96	(1.235, 1.767) (1.138, 1.920)
Hispanic	2021	1.00	0.98	(1.323, 1.576) (1.220, 1.704)
Hispanic	2021	1.00	1.00	1.411 (1.303, 1.520)
Hispanic	2022	0.90	0.90	(0.509, 3.105) (0.416, 3.743)
Hispanic	2022	0.90	0.92	(0.560, 2.809) (0.460, 3.355)
Hispanic	2022	0.90	0.94	(0.611, 2.559) (0.505, 3.032)
Hispanic	2022	0.90	0.96	(0.663, 2.346) (0.550, 2.763)
Hispanic	2022	0.90	0.98	(0.715, 2.164) (0.595, 2.533)
Hispanic	2022	0.90	1.00	(0.767, 2.006) (0.640, 2.339)
Hispanic	2022	0.92	0.90	(0.569, 2.780) (0.490, 3.169)
Hispanic	2022	0.92	0.92	(0.625, 2.515) (0.540, 2.851)
Hispanic	2022	0.92	0.94	(0.681, 2.292) (0.592, 2.586)
Hispanic	2022	0.92	0.96	(0.737, 2.101) (0.642, 2.362)
Hispanic	2022	0.92	0.98	(0.794, 1.938) (0.694, 2.171)
Hispanic	2022	0.92	1.00	(0.851, 1.796) (0.745, 2.007)
Hispanic	2022	0.94	0.90	(0.623, 2.503) (0.556, 2.794)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2022	0.94	0.92	(0.683, 2.260) (0.611, 2.515)
Hispanic	2022	0.94	0.94	(0.743, 2.055) (0.666, 2.280)
Hispanic	2022	0.94	0.96	(0.803, 1.881) (0.722, 2.081)
Hispanic	2022	0.94	0.98	(0.864, 1.731) (0.778, 1.911)
Hispanic	2022	0.94	1.00	(0.925, 1.602) (0.833, 1.764)
Hispanic	2022	0.96	0.90	(0.691, 2.314) (0.622, 2.540)
Hispanic	2022	0.96	0.92	(0.756, 2.083) (0.682, 2.281)
Hispanic	2022	0.96	0.94	(0.821, 1.888) (0.743, 2.063)
Hispanic	2022	0.96	0.96	(0.886, 1.723) (0.804, 1.878)
Hispanic	2022	0.96	0.98	(0.952, 1.580) (0.865, 1.718)
Hispanic	2022	0.96	1.00	(1.018, 1.457) (0.926, 1.580)
Hispanic	2022	0.98	0.90	(0.783, 2.221) (0.716, 2.415)
Hispanic	2022	0.98	0.92	(0.853, 1.991) (0.783, 2.157)
Hispanic	2022	0.98	0.94	(0.925, 1.796) (0.850, 1.940)
Hispanic	2022	0.98	0.96	(0.996, 1.629) (0.919, 1.755)
Hispanic	2022	0.98	0.98	(1.068, 1.485) (0.987, 1.595)
Hispanic	2022	0.98	1.00	(1.141, 1.359) (1.055, 1.456)
Hispanic	2022	1.00	0.90	(0.899, 2.214) (0.830, 2.393)
Hispanic	2022	1.00	0.92	(0.976, 1.971) (0.905, 2.125)
Hispanic	2022	1.00	0.94	(1.055, 1.765) (0.980, 1.899)
Hispanic	2022	1.00	0.96	(1.134, 1.587) (1.057, 1.704)
Hispanic	2022	1.00	0.98	(1.214, 1.432) (1.133, 1.535)
Hispanic	2022	1.00	1.00	1.295 (1.210, 1.384)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2023	0.90	0.90	(0.485, 2.769) (0.405, 3.437)
Hispanic	2023	0.90	0.92	(0.531, 2.519) (0.445, 3.096)
Hispanic	2023	0.90	0.94	(0.578, 2.307) (0.485, 2.805)
Hispanic	2023	0.90	0.96	(0.625, 2.126) (0.526, 2.563)
Hispanic	2023	0.90	0.98	(0.672, 1.970) (0.566, 2.356)
Hispanic	2023	0.90	1.00	(0.718, 1.834) (0.607, 2.177)
Hispanic	2023	0.92	0.90	(0.551, 2.540) (0.475, 2.976)
Hispanic	2023	0.92	0.92	(0.603, 2.306) (0.520, 2.684)
Hispanic	2023	0.92	0.94	(0.654, 2.108) (0.566, 2.436)
Hispanic	2023	0.92	0.96	(0.706, 1.939) (0.611, 2.228)
Hispanic	2023	0.92	0.98	(0.757, 1.793) (0.657, 2.049)
Hispanic	2023	0.92	1.00	(0.809, 1.666) (0.702, 1.897)
Hispanic	2023	0.94	0.90	(0.620, 2.352) (0.553, 2.640)
Hispanic	2023	0.94	0.92	(0.676, 2.130) (0.603, 2.381)
Hispanic	2023	0.94	0.94	(0.732, 1.942) (0.654, 2.163)
Hispanic	2023	0.94	0.96	(0.788, 1.780) (0.704, 1.976)
Hispanic	2023	0.94	0.98	(0.844, 1.642) (0.754, 1.817)
Hispanic	2023	0.94	1.00	(0.900, 1.522) (0.805, 1.680)
Hispanic	2023	0.96	0.90	(0.703, 2.246) (0.639, 2.506)
Hispanic	2023	0.96	0.92	(0.764, 2.025) (0.695, 2.257)
Hispanic	2023	0.96	0.94	(0.824, 1.837) (0.752, 2.040)
Hispanic	2023	0.96	0.96	(0.885, 1.676) (0.808, 1.854)
Hispanic	2023	0.96	0.98	(0.946, 1.537) (0.865, 1.694)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2023	0.96	1.00	(1.007, 1.417) (0.921, 1.557)
Hispanic	2023	0.98	0.90	(0.814, 2.235) (0.738, 2.458)
Hispanic	2023	0.98	0.92	(0.881, 2.003) (0.799, 2.196)
Hispanic	2023	0.98	0.94	(0.947, 1.804) (0.861, 1.971)
Hispanic	2023	0.98	0.96	(1.014, 1.633) (0.923, 1.778)
Hispanic	2023	0.98	0.98	(1.081, 1.486) (0.984, 1.610)
Hispanic	2023	0.98	1.00	(1.148, 1.358) (1.046, 1.467)
Hispanic	2023	1.00	0.90	(0.934, 2.268) (0.863, 2.467)
Hispanic	2023	1.00	0.92	(1.007, 2.016) (0.932, 2.184)
Hispanic	2023	1.00	0.94	(1.081, 1.798) (1.001, 1.941)
Hispanic	2023	1.00	0.96	(1.155, 1.609) (1.071, 1.731)
Hispanic	2023	1.00	0.98	(1.230, 1.446) (1.142, 1.549)
Hispanic	2023	1.00	1.00	1.305 (1.212, 1.393)

Table 14: Camera analysis (speed): bounds on the causal risk ratio $\psi_r^\dagger$ by race, year, and the sensitivity parameters θ and η .

Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	All years	0.90	0.90	(0.748, 4.754) (0.497, 7.115)
Black	All years	0.90	0.92	(0.809, 4.219) (0.542, 6.233)
Black	All years	0.90	0.94	(0.872, 3.764) (0.590, 5.453)
Black	All years	0.90	0.96	(0.938, 3.375) (0.640, 4.819)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	All years	0.90	0.98	(1.006, 3.041) (0.691, 4.290)
Black	All years	0.90	1.00	(1.077, 2.753) (0.745, 3.836)
Black	All years	0.92	0.90	(0.837, 4.333) (0.590, 6.138)
Black	All years	0.92	0.92	(0.906, 3.863) (0.643, 5.423)
Black	All years	0.92	0.94	(0.977, 3.462) (0.697, 4.792)
Black	All years	0.92	0.96	(1.050, 3.117) (0.754, 4.261)
Black	All years	0.92	0.98	(1.127, 2.817) (0.812, 3.821)
Black	All years	0.92	1.00	(1.206, 2.557) (0.875, 3.429)
Black	All years	0.94	0.90	(0.858, 3.740) (0.615, 5.043)
Black	All years	0.94	0.92	(0.929, 3.339) (0.669, 4.464)
Black	All years	0.94	0.94	(1.003, 2.997) (0.727, 3.987)
Black	All years	0.94	0.96	(1.080, 2.701) (0.788, 3.575)
Black	All years	0.94	0.98	(1.160, 2.445) (0.851, 3.221)
Black	All years	0.94	1.00	(1.243, 2.222) (0.917, 2.915)
Black	All years	0.96	0.90	(0.488, 1.745) (0.250, 3.271)
Black	All years	0.96	0.92	(0.528, 1.569) (0.273, 2.865)
Black	All years	0.96	0.94	(0.570, 1.420) (0.297, 2.531)
Black	All years	0.96	0.96	(0.614, 1.292) (0.322, 2.259)
Black	All years	0.96	0.98	(0.659, 1.181) (0.347, 2.027)
Black	All years	0.96	1.00	(0.706, 1.084) (0.373, 1.830)
Black	All years	0.98	0.90	(0.311, 0.815) (0.246, 1.043)
Black	All years	0.98	0.92	(0.336, 0.746) (0.267, 0.947)
Black	All years	0.98	0.94	(0.362, 0.686) (0.288, 0.863)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	All years	0.98	0.96	(0.388, 0.633) (0.310, 0.792)
Black	All years	0.98	0.98	(0.416, 0.587) (0.333, 0.730)
Black	All years	0.98	1.00	(0.443, 0.546) (0.355, 0.676)
Black	All years	1.00	0.90	(0.329, 0.702) (0.286, 0.864)
Black	All years	1.00	0.92	(0.356, 0.643) (0.310, 0.786)
Black	All years	1.00	0.94	(0.384, 0.592) (0.335, 0.719)
Black	All years	1.00	0.96	(0.412, 0.547) (0.361, 0.661)
Black	All years	1.00	0.98	(0.441, 0.507) (0.387, 0.609)
Black	All years	1.00	1.00	0.471 (0.415, 0.564)
Black	2017	0.90	0.90	(0.665, 4.462) (0.260, 9.846)
Black	2017	0.90	0.92	(0.723, 3.872) (0.294, 8.150)
Black	2017	0.90	0.94	(0.784, 3.401) (0.329, 6.917)
Black	2017	0.90	0.96	(0.846, 3.024) (0.364, 6.002)
Black	2017	0.90	0.98	(0.910, 2.716) (0.400, 5.290)
Black	2017	0.90	1.00	(0.975, 2.458) (0.439, 4.707)
Black	2017	0.92	0.90	(0.666, 3.481) (0.284, 6.949)
Black	2017	0.92	0.92	(0.726, 3.093) (0.316, 5.983)
Black	2017	0.92	0.94	(0.788, 2.770) (0.348, 5.235)
Black	2017	0.92	0.96	(0.853, 2.501) (0.380, 4.631)
Black	2017	0.92	0.98	(0.919, 2.274) (0.416, 4.149)
Black	2017	0.92	1.00	(0.987, 2.081) (0.453, 3.755)
Black	2017	0.94	0.90	(0.674, 2.880) (0.323, 5.098)
Black	2017	0.94	0.92	(0.737, 2.586) (0.358, 4.528)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2017	0.94	0.94	(0.802, 2.338) (0.396, 4.040)
Black	2017	0.94	0.96	(0.869, 2.128) (0.437, 3.642)
Black	2017	0.94	0.98	(0.938, 1.948) (0.479, 3.309)
Black	2017	0.94	1.00	(1.010, 1.792) (0.522, 3.018)
Black	2017	0.96	0.90	(0.334, 1.219) (0.133, 2.973)
Black	2017	0.96	0.92	(0.368, 1.104) (0.149, 2.641)
Black	2017	0.96	0.94	(0.403, 1.008) (0.166, 2.360)
Black	2017	0.96	0.96	(0.440, 0.925) (0.182, 2.140)
Black	2017	0.96	0.98	(0.478, 0.854) (0.199, 1.946)
Black	2017	0.96	1.00	(0.516, 0.792) (0.216, 1.779)
Black	2017	0.98	0.90	(0.170, 0.458) (0.123, 0.602)
Black	2017	0.98	0.92	(0.189, 0.425) (0.138, 0.556)
Black	2017	0.98	0.94	(0.208, 0.396) (0.153, 0.517)
Black	2017	0.98	0.96	(0.227, 0.371) (0.168, 0.482)
Black	2017	0.98	0.98	(0.247, 0.349) (0.184, 0.451)
Black	2017	0.98	1.00	(0.267, 0.329) (0.200, 0.424)
Black	2017	1.00	0.90	(0.181, 0.396) (0.143, 0.543)
Black	2017	1.00	0.92	(0.201, 0.368) (0.160, 0.500)
Black	2017	1.00	0.94	(0.221, 0.343) (0.178, 0.464)
Black	2017	1.00	0.96	(0.242, 0.321) (0.195, 0.433)
Black	2017	1.00	0.98	(0.263, 0.302) (0.213, 0.405)
Black	2017	1.00	1.00	0.284 (0.232, 0.380)
Black	2018	0.90	0.90	(0.530, 3.513) (0.138, 10.057)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2018	0.90	0.92	(0.577, 3.122) (0.161, 8.654)
Black	2018	0.90	0.94	(0.626, 2.795) (0.187, 7.328)
Black	2018	0.90	0.96	(0.679, 2.519) (0.217, 6.409)
Black	2018	0.90	0.98	(0.734, 2.282) (0.249, 5.645)
Black	2018	0.90	1.00	(0.793, 2.078) (0.286, 4.988)
Black	2018	0.92	0.90	(0.623, 3.324) (0.251, 7.474)
Black	2018	0.92	0.92	(0.675, 2.971) (0.281, 6.470)
Black	2018	0.92	0.94	(0.731, 2.671) (0.316, 5.620)
Black	2018	0.92	0.96	(0.789, 2.412) (0.351, 4.938)
Black	2018	0.92	0.98	(0.851, 2.189) (0.389, 4.360)
Black	2018	0.92	1.00	(0.915, 1.994) (0.427, 3.874)
Black	2018	0.94	0.90	(0.717, 3.201) (0.371, 5.766)
Black	2018	0.94	0.92	(0.777, 2.861) (0.413, 5.038)
Black	2018	0.94	0.94	(0.841, 2.567) (0.457, 4.417)
Black	2018	0.94	0.96	(0.908, 2.313) (0.506, 3.913)
Black	2018	0.94	0.98	(0.978, 2.092) (0.558, 3.466)
Black	2018	0.94	1.00	(1.052, 1.900) (0.613, 3.093)
Black	2018	0.96	0.90	(0.500, 1.832) (0.201, 4.189)
Black	2018	0.96	0.92	(0.539, 1.624) (0.219, 3.610)
Black	2018	0.96	0.94	(0.580, 1.451) (0.236, 3.174)
Black	2018	0.96	0.96	(0.622, 1.305) (0.255, 2.785)
Black	2018	0.96	0.98	(0.666, 1.182) (0.273, 2.486)
Black	2018	0.96	1.00	(0.712, 1.077) (0.292, 2.233)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2018	0.98	0.90	(0.387, 1.027) (0.274, 1.566)
Black	2018	0.98	0.92	(0.410, 0.913) (0.292, 1.357)
Black	2018	0.98	0.94	(0.435, 0.820) (0.311, 1.192)
Black	2018	0.98	0.96	(0.460, 0.743) (0.330, 1.060)
Black	2018	0.98	0.98	(0.485, 0.678) (0.350, 0.953)
Black	2018	0.98	1.00	(0.511, 0.624) (0.370, 0.865)
Black	2018	1.00	0.90	(0.357, 0.759) (0.285, 1.098)
Black	2018	1.00	0.92	(0.381, 0.684) (0.305, 0.973)
Black	2018	1.00	0.94	(0.406, 0.621) (0.326, 0.871)
Black	2018	1.00	0.96	(0.431, 0.567) (0.348, 0.788)
Black	2018	1.00	0.98	(0.456, 0.522) (0.369, 0.716)
Black	2018	1.00	1.00	0.482 (0.392, 0.655)
Black	2019	0.90	0.90	(1.327, 9.806) (0.530, 24.874)
Black	2019	0.90	0.92	(1.418, 8.311) (0.585, 20.041)
Black	2019	0.90	0.94	(1.512, 7.104) (0.641, 16.329)
Black	2019	0.90	0.96	(1.609, 6.116) (0.705, 13.506)
Black	2019	0.90	0.98	(1.710, 5.296) (0.768, 11.228)
Black	2019	0.90	1.00	(1.815, 4.615) (0.840, 9.435)
Black	2019	0.92	0.90	(1.586, 9.227) (0.823, 19.304)
Black	2019	0.92	0.92	(1.689, 7.904) (0.894, 15.925)
Black	2019	0.92	0.94	(1.796, 6.809) (0.965, 13.169)
Black	2019	0.92	0.96	(1.906, 5.894) (1.045, 11.019)
Black	2019	0.92	0.98	(2.020, 5.122) (1.124, 9.264)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2019	0.92	1.00	(2.138, 4.471) (1.205, 7.865)
Black	2019	0.94	0.90	(1.526, 7.147) (0.845, 13.268)
Black	2019	0.94	0.92	(1.624, 6.201) (0.909, 11.171)
Black	2019	0.94	0.94	(1.725, 5.402) (0.978, 9.495)
Black	2019	0.94	0.96	(1.830, 4.722) (1.049, 8.047)
Black	2019	0.94	0.98	(1.939, 4.138) (1.125, 6.904)
Black	2019	0.94	1.00	(2.051, 3.638) (1.207, 5.899)
Black	2019	0.96	0.90	(1.067, 4.152) (0.574, 7.559)
Black	2019	0.96	0.92	(1.134, 3.605) (0.615, 6.331)
Black	2019	0.96	0.94	(1.204, 3.151) (0.659, 5.364)
Black	2019	0.96	0.96	(1.276, 2.768) (0.705, 4.591)
Black	2019	0.96	0.98	(1.350, 2.442) (0.751, 3.973)
Black	2019	0.96	1.00	(1.426, 2.163) (0.798, 3.444)
Black	2019	0.98	0.90	(1.050, 3.385) (0.696, 5.295)
Black	2019	0.98	0.92	(1.106, 2.898) (0.739, 4.384)
Black	2019	0.98	0.94	(1.164, 2.498) (0.785, 3.670)
Black	2019	0.98	0.96	(1.223, 2.168) (0.832, 3.113)
Black	2019	0.98	0.98	(1.284, 1.891) (0.879, 2.654)
Black	2019	0.98	1.00	(1.347, 1.654) (0.927, 2.273)
Black	2019	1.00	0.90	(1.122, 2.939) (0.869, 3.951)
Black	2019	1.00	0.92	(1.182, 2.526) (0.920, 3.335)
Black	2019	1.00	0.94	(1.244, 2.183) (0.972, 2.837)
Black	2019	1.00	0.96	(1.307, 1.896) (1.026, 2.429)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2019	1.00	0.98	(1.373, 1.652) (1.082, 2.091)
Black	2019	1.00	1.00	1.440 (1.138, 1.803)
Black	2020	0.90	0.90	(0.898, 6.582) (0.320, 17.106)
Black	2020	0.90	0.92	(0.975, 5.548) (0.369, 13.793)
Black	2020	0.90	0.94	(1.055, 4.798) (0.421, 11.307)
Black	2020	0.90	0.96	(1.138, 4.194) (0.473, 9.416)
Black	2020	0.90	0.98	(1.225, 3.696) (0.527, 7.904)
Black	2020	0.90	1.00	(1.314, 3.279) (0.582, 6.664)
Black	2020	0.92	0.90	(1.097, 6.432) (0.450, 14.773)
Black	2020	0.92	0.92	(1.190, 5.551) (0.511, 12.080)
Black	2020	0.92	0.94	(1.287, 4.833) (0.575, 10.101)
Black	2020	0.92	0.96	(1.388, 4.241) (0.641, 8.491)
Black	2020	0.92	0.98	(1.494, 3.741) (0.713, 7.174)
Black	2020	0.92	1.00	(1.603, 3.317) (0.788, 6.096)
Black	2020	0.94	0.90	(1.214, 5.887) (0.574, 12.071)
Black	2020	0.94	0.92	(1.316, 5.095) (0.642, 9.969)
Black	2020	0.94	0.94	(1.422, 4.444) (0.714, 8.421)
Black	2020	0.94	0.96	(1.532, 3.901) (0.792, 7.147)
Black	2020	0.94	0.98	(1.647, 3.440) (0.873, 6.102)
Black	2020	0.94	1.00	(1.765, 3.046) (0.957, 5.225)
Black	2020	0.96	0.90	(1.002, 4.400) (0.545, 7.986)
Black	2020	0.96	0.92	(1.087, 3.771) (0.601, 6.636)
Black	2020	0.96	0.94	(1.175, 3.248) (0.661, 5.514)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2020	0.96	0.96	(1.267, 2.813) (0.726, 4.638)
Black	2020	0.96	0.98	(1.363, 2.451) (0.794, 3.929)
Black	2020	0.96	1.00	(1.462, 2.147) (0.862, 3.347)
Black	2020	0.98	0.90	(0.811, 2.949) (0.513, 4.751)
Black	2020	0.98	0.92	(0.878, 2.534) (0.564, 3.962)
Black	2020	0.98	0.94	(0.948, 2.186) (0.619, 3.334)
Black	2020	0.98	0.96	(1.021, 1.894) (0.676, 2.812)
Black	2020	0.98	0.98	(1.097, 1.650) (0.736, 2.394)
Black	2020	0.98	1.00	(1.176, 1.443) (0.796, 2.046)
Black	2020	1.00	0.90	(0.739, 2.199) (0.527, 3.254)
Black	2020	1.00	0.92	(0.799, 1.886) (0.575, 2.727)
Black	2020	1.00	0.94	(0.862, 1.625) (0.626, 2.296)
Black	2020	1.00	0.96	(0.927, 1.405) (0.678, 1.945)
Black	2020	1.00	0.98	(0.995, 1.221) (0.733, 1.663)
Black	2020	1.00	1.00	1.066 (0.790, 1.425)
Black	2021	0.90	0.90	(1.093, 7.616) (0.472, 16.693)
Black	2021	0.90	0.92	(1.176, 6.543) (0.525, 13.717)
Black	2021	0.90	0.94	(1.262, 5.654) (0.579, 11.383)
Black	2021	0.90	0.96	(1.352, 4.911) (0.634, 9.492)
Black	2021	0.90	0.98	(1.445, 4.289) (0.693, 7.970)
Black	2021	0.90	1.00	(1.541, 3.762) (0.753, 6.702)
Black	2021	0.92	0.90	(1.116, 6.128) (0.502, 11.836)
Black	2021	0.92	0.92	(1.200, 5.330) (0.559, 9.961)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2021	0.92	0.94	(1.287, 4.653) (0.619, 8.343)
Black	2021	0.92	0.96	(1.378, 4.081) (0.682, 7.111)
Black	2021	0.92	0.98	(1.473, 3.595) (0.745, 6.070)
Black	2021	0.92	1.00	(1.569, 3.181) (0.810, 5.222)
Black	2021	0.94	0.90	(1.007, 4.539) (0.170, 8.346)
Black	2021	0.94	0.92	(1.085, 3.987) (0.187, 7.135)
Black	2021	0.94	0.94	(1.166, 3.515) (0.207, 6.094)
Black	2021	0.94	0.96	(1.251, 3.112) (0.228, 5.253)
Black	2021	0.94	0.98	(1.339, 2.766) (0.251, 4.570)
Black	2021	0.94	1.00	(1.430, 2.468) (0.277, 3.979)
Black	2021	0.96	0.90	(0.828, 3.219) (0.259, 5.581)
Black	2021	0.96	0.92	(0.890, 2.813) (0.281, 4.746)
Black	2021	0.96	0.94	(0.956, 2.469) (0.307, 4.071)
Black	2021	0.96	0.96	(1.025, 2.178) (0.336, 3.522)
Black	2021	0.96	0.98	(1.096, 1.932) (0.367, 3.066)
Black	2021	0.96	1.00	(1.169, 1.721) (0.399, 2.676)
Black	2021	0.98	0.90	(0.747, 2.387) (0.317, 4.470)
Black	2021	0.98	0.92	(0.803, 2.094) (0.344, 3.811)
Black	2021	0.98	0.94	(0.860, 1.843) (0.373, 3.270)
Black	2021	0.98	0.96	(0.921, 1.629) (0.404, 2.807)
Black	2021	0.98	0.98	(0.984, 1.446) (0.437, 2.425)
Black	2021	0.98	1.00	(1.049, 1.289) (0.472, 2.098)
Black	2021	1.00	0.90	(0.725, 1.813) (0.407, 3.411)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2021	1.00	0.92	(0.776, 1.607) (0.440, 2.914)
Black	2021	1.00	0.94	(0.831, 1.427) (0.475, 2.497)
Black	2021	1.00	0.96	(0.888, 1.270) (0.513, 2.156)
Black	2021	1.00	0.98	(0.948, 1.133) (0.554, 1.862)
Black	2021	1.00	1.00	1.010 (0.596, 1.609)
Black	2022	0.90	0.90	(0.656, 4.189) (0.274, 8.535)
Black	2022	0.90	0.92	(0.712, 3.722) (0.312, 7.354)
Black	2022	0.90	0.94	(0.770, 3.326) (0.353, 6.343)
Black	2022	0.90	0.96	(0.831, 2.989) (0.398, 5.549)
Black	2022	0.90	0.98	(0.894, 2.701) (0.445, 4.894)
Black	2022	0.90	1.00	(0.960, 2.453) (0.495, 4.345)
Black	2022	0.92	0.90	(0.744, 3.927) (0.371, 7.007)
Black	2022	0.92	0.92	(0.809, 3.495) (0.420, 6.122)
Black	2022	0.92	0.94	(0.877, 3.127) (0.470, 5.361)
Black	2022	0.92	0.96	(0.949, 2.813) (0.525, 4.702)
Black	2022	0.92	0.98	(1.023, 2.543) (0.584, 4.167)
Black	2022	0.92	1.00	(1.102, 2.309) (0.646, 3.717)
Black	2022	0.94	0.90	(0.846, 3.799) (0.464, 6.337)
Black	2022	0.94	0.92	(0.922, 3.379) (0.519, 5.526)
Black	2022	0.94	0.94	(1.002, 3.021) (0.579, 4.840)
Black	2022	0.94	0.96	(1.087, 2.714) (0.644, 4.266)
Black	2022	0.94	0.98	(1.176, 2.450) (0.712, 3.780)
Black	2022	0.94	1.00	(1.270, 2.221) (0.784, 3.382)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2022	0.96	0.90	(0.912, 3.542) (0.550, 5.551)
Black	2022	0.96	0.92	(0.994, 3.135) (0.613, 4.813)
Black	2022	0.96	0.94	(1.081, 2.789) (0.682, 4.193)
Black	2022	0.96	0.96	(1.173, 2.494) (0.758, 3.682)
Black	2022	0.96	0.98	(1.271, 2.240) (0.831, 3.262)
Black	2022	0.96	1.00	(1.374, 2.021) (0.913, 2.901)
Black	2022	0.98	0.90	(0.834, 2.816) (0.556, 4.114)
Black	2022	0.98	0.92	(0.910, 2.477) (0.618, 3.559)
Black	2022	0.98	0.94	(0.991, 2.190) (0.682, 3.091)
Black	2022	0.98	0.96	(1.076, 1.945) (0.750, 2.709)
Black	2022	0.98	0.98	(1.167, 1.736) (0.824, 2.384)
Black	2022	0.98	1.00	(1.263, 1.555) (0.902, 2.115)
Black	2022	1.00	0.90	(0.740, 2.056) (0.539, 2.774)
Black	2022	1.00	0.92	(0.806, 1.803) (0.595, 2.406)
Black	2022	1.00	0.94	(0.876, 1.590) (0.656, 2.091)
Black	2022	1.00	0.96	(0.951, 1.407) (0.721, 1.831)
Black	2022	1.00	0.98	(1.030, 1.250) (0.790, 1.612)
Black	2022	1.00	1.00	1.115 (0.864, 1.424)
Black	2023	0.90	0.90	(0.850, 6.351) (0.279, 16.959)
Black	2023	0.90	0.92	(0.924, 5.579) (0.320, 13.844)
Black	2023	0.90	0.94	(1.001, 4.741) (0.365, 11.376)
Black	2023	0.90	0.96	(1.082, 4.127) (0.414, 9.484)
Black	2023	0.90	0.98	(1.168, 3.626) (0.464, 7.994)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2023	0.90	1.00	(1.258, 3.203) (0.523, 6.798)
Black	2023	0.92	0.90	(0.983, 5.987) (0.392, 13.732)
Black	2023	0.92	0.92	(1.067, 5.176) (0.445, 11.392)
Black	2023	0.92	0.94	(1.155, 4.505) (0.502, 9.494)
Black	2023	0.92	0.96	(1.248, 3.950) (0.565, 7.954)
Black	2023	0.92	0.98	(1.346, 3.485) (0.628, 6.785)
Black	2023	0.92	1.00	(1.448, 3.087) (0.700, 5.818)
Black	2023	0.94	0.90	(1.108, 5.542) (0.528, 11.067)
Black	2023	0.94	0.92	(1.200, 4.816) (0.594, 9.257)
Black	2023	0.94	0.94	(1.296, 4.210) (0.659, 7.801)
Black	2023	0.94	0.96	(1.397, 3.701) (0.732, 6.658)
Black	2023	0.94	0.98	(1.503, 3.272) (0.809, 5.678)
Black	2023	0.94	1.00	(1.613, 2.900) (0.891, 4.886)
Black	2023	0.96	0.90	(1.094, 4.571) (0.635, 7.700)
Black	2023	0.96	0.92	(1.183, 3.984) (0.701, 6.517)
Black	2023	0.96	0.94	(1.277, 3.490) (0.773, 5.562)
Black	2023	0.96	0.96	(1.376, 3.072) (0.846, 4.781)
Black	2023	0.96	0.98	(1.480, 2.717) (0.923, 4.128)
Black	2023	0.96	1.00	(1.588, 2.409) (1.010, 3.577)
Black	2023	0.98	0.90	(1.074, 3.725) (0.718, 5.636)
Black	2023	0.98	0.92	(1.164, 3.249) (0.790, 4.818)
Black	2023	0.98	0.94	(1.257, 2.848) (0.862, 4.147)
Black	2023	0.98	0.96	(1.357, 2.506) (0.943, 3.591)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2023	0.98	0.98	(1.461, 2.213) (1.030, 3.113)
Black	2023	0.98	1.00	(1.571, 1.960) (1.123, 2.709)
Black	2023	1.00	0.90	(1.032, 2.937) (0.751, 4.053)
Black	2023	1.00	0.92	(1.119, 2.555) (0.821, 3.488)
Black	2023	1.00	0.94	(1.210, 2.232) (0.895, 3.017)
Black	2023	1.00	0.96	(1.307, 1.957) (0.977, 2.606)
Black	2023	1.00	0.98	(1.408, 1.720) (1.064, 2.266)
Black	2023	1.00	1.00	1.515 (1.155, 1.970)
Hispanic	All years	0.90	0.90	(0.924, 5.804) (0.681, 7.660)
Hispanic	All years	0.90	0.92	(1.014, 5.201) (0.756, 6.794)
Hispanic	All years	0.90	0.94	(1.110, 4.680) (0.834, 6.042)
Hispanic	All years	0.90	0.96	(1.209, 4.226) (0.916, 5.398)
Hispanic	All years	0.90	0.98	(1.310, 3.830) (1.001, 4.842)
Hispanic	All years	0.90	1.00	(1.411, 3.483) (1.086, 4.355)
Hispanic	All years	0.92	0.90	(1.000, 5.198) (0.776, 6.518)
Hispanic	All years	0.92	0.92	(1.100, 4.666) (0.864, 5.797)
Hispanic	All years	0.92	0.94	(1.206, 4.206) (0.956, 5.167)
Hispanic	All years	0.92	0.96	(1.315, 3.805) (1.050, 4.631)
Hispanic	All years	0.92	0.98	(1.427, 3.455) (1.146, 4.170)
Hispanic	All years	0.92	1.00	(1.539, 3.147) (1.244, 3.768)
Hispanic	All years	0.94	0.90	(0.982, 4.275) (0.797, 5.101)
Hispanic	All years	0.94	0.92	(1.081, 3.836) (0.880, 4.542)
Hispanic	All years	0.94	0.94	(1.184, 3.457) (0.967, 4.064)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	All years	0.94	0.96	(1.292, 3.127) (1.056, 3.654)
Hispanic	All years	0.94	0.98	(1.401, 2.839) (1.147, 3.298)
Hispanic	All years	0.94	1.00	(1.510, 2.587) (1.238, 2.989)
Hispanic	All years	0.96	0.90	(0.623, 2.358) (0.405, 3.645)
Hispanic	All years	0.96	0.92	(0.690, 2.129) (0.454, 3.238)
Hispanic	All years	0.96	0.94	(0.760, 1.933) (0.504, 2.892)
Hispanic	All years	0.96	0.96	(0.832, 1.763) (0.555, 2.599)
Hispanic	All years	0.96	0.98	(0.905, 1.615) (0.606, 2.347)
Hispanic	All years	0.96	1.00	(0.978, 1.485) (0.657, 2.127)
Hispanic	All years	0.98	0.90	(0.404, 1.258) (0.339, 1.510)
Hispanic	All years	0.98	0.92	(0.455, 1.150) (0.383, 1.370)
Hispanic	All years	0.98	0.94	(0.507, 1.057) (0.429, 1.252)
Hispanic	All years	0.98	0.96	(0.561, 0.976) (0.475, 1.149)
Hispanic	All years	0.98	0.98	(0.615, 0.905) (0.522, 1.058)
Hispanic	All years	0.98	1.00	(0.670, 0.841) (0.569, 0.979)
Hispanic	All years	1.00	0.90	(0.432, 1.055) (0.386, 1.242)
Hispanic	All years	1.00	0.92	(0.485, 0.967) (0.435, 1.130)
Hispanic	All years	1.00	0.94	(0.539, 0.890) (0.485, 1.035)
Hispanic	All years	1.00	0.96	(0.595, 0.823) (0.537, 0.952)
Hispanic	All years	1.00	0.98	(0.652, 0.763) (0.590, 0.878)
Hispanic	All years	1.00	1.00	0.710 (0.643, 0.813)
Hispanic	2017	0.90	0.90	(0.829, 4.963) (0.390, 9.143)
Hispanic	2017	0.90	0.92	(0.915, 4.493) (0.448, 8.062)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2017	0.90	0.94	(1.005, 4.088) (0.507, 7.130)
Hispanic	2017	0.90	0.96	(1.096, 3.738) (0.569, 6.347)
Hispanic	2017	0.90	0.98	(1.188, 3.434) (0.631, 5.709)
Hispanic	2017	0.90	1.00	(1.280, 3.169) (0.696, 5.161)
Hispanic	2017	0.92	0.90	(0.856, 4.410) (0.467, 7.061)
Hispanic	2017	0.92	0.92	(0.951, 3.996) (0.535, 6.276)
Hispanic	2017	0.92	0.94	(1.050, 3.640) (0.602, 5.613)
Hispanic	2017	0.92	0.96	(1.150, 3.333) (0.669, 5.066)
Hispanic	2017	0.92	0.98	(1.252, 3.066) (0.736, 4.596)
Hispanic	2017	0.92	1.00	(1.353, 2.833) (0.804, 4.206)
Hispanic	2017	0.94	0.90	(0.811, 3.556) (0.554, 5.223)
Hispanic	2017	0.94	0.92	(0.904, 3.219) (0.627, 4.638)
Hispanic	2017	0.94	0.94	(1.000, 2.931) (0.701, 4.149)
Hispanic	2017	0.94	0.96	(1.097, 2.683) (0.777, 3.741)
Hispanic	2017	0.94	0.98	(1.196, 2.468) (0.852, 3.398)
Hispanic	2017	0.94	1.00	(1.295, 2.280) (0.926, 3.099)
Hispanic	2017	0.96	0.90	(0.502, 1.864) (0.264, 3.523)
Hispanic	2017	0.96	0.92	(0.561, 1.700) (0.301, 3.152)
Hispanic	2017	0.96	0.94	(0.621, 1.561) (0.338, 2.835)
Hispanic	2017	0.96	0.96	(0.682, 1.440) (0.375, 2.566)
Hispanic	2017	0.96	0.98	(0.743, 1.335) (0.412, 2.343)
Hispanic	2017	0.96	1.00	(0.804, 1.243) (0.449, 2.146)
Hispanic	2017	0.98	0.90	(0.314, 0.925) (0.250, 1.098)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2017	0.98	0.92	(0.356, 0.859) (0.286, 1.015)
Hispanic	2017	0.98	0.94	(0.398, 0.801) (0.321, 0.943)
Hispanic	2017	0.98	0.96	(0.441, 0.751) (0.357, 0.880)
Hispanic	2017	0.98	0.98	(0.483, 0.706) (0.392, 0.825)
Hispanic	2017	0.98	1.00	(0.526, 0.665) (0.428, 0.776)
Hispanic	2017	1.00	0.90	(0.340, 0.783) (0.290, 0.962)
Hispanic	2017	1.00	0.92	(0.384, 0.728) (0.330, 0.890)
Hispanic	2017	1.00	0.94	(0.429, 0.680) (0.370, 0.827)
Hispanic	2017	1.00	0.96	(0.474, 0.637) (0.411, 0.772)
Hispanic	2017	1.00	0.98	(0.519, 0.599) (0.452, 0.722)
Hispanic	2017	1.00	1.00	0.565 (0.493, 0.679)
Hispanic	2018	0.90	0.90	(0.893, 5.382) (0.470, 9.911)
Hispanic	2018	0.90	0.92	(0.982, 4.828) (0.536, 8.646)
Hispanic	2018	0.90	0.94	(1.075, 4.353) (0.604, 7.539)
Hispanic	2018	0.90	0.96	(1.172, 3.944) (0.674, 6.646)
Hispanic	2018	0.90	0.98	(1.271, 3.590) (0.744, 5.911)
Hispanic	2018	0.90	1.00	(1.370, 3.282) (0.815, 5.279)
Hispanic	2018	0.92	0.90	(0.971, 4.801) (0.631, 7.166)
Hispanic	2018	0.92	0.92	(1.069, 4.326) (0.709, 6.340)
Hispanic	2018	0.92	0.94	(1.173, 3.915) (0.790, 5.633)
Hispanic	2018	0.92	0.96	(1.280, 3.559) (0.876, 5.033)
Hispanic	2018	0.92	0.98	(1.390, 3.249) (0.964, 4.532)
Hispanic	2018	0.92	1.00	(1.499, 2.978) (1.053, 4.086)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2018	0.94	0.90	(1.019, 4.305) (0.724, 5.950)
Hispanic	2018	0.94	0.92	(1.119, 3.864) (0.805, 5.247)
Hispanic	2018	0.94	0.94	(1.225, 3.482) (0.889, 4.641)
Hispanic	2018	0.94	0.96	(1.334, 3.150) (0.978, 4.138)
Hispanic	2018	0.94	0.98	(1.446, 2.862) (1.064, 3.719)
Hispanic	2018	0.94	1.00	(1.557, 2.610) (1.154, 3.356)
Hispanic	2018	0.96	0.90	(0.626, 2.494) (0.247, 4.452)
Hispanic	2018	0.96	0.92	(0.695, 2.226) (0.289, 3.908)
Hispanic	2018	0.96	0.94	(0.768, 1.999) (0.332, 3.433)
Hispanic	2018	0.96	0.96	(0.844, 1.806) (0.375, 3.048)
Hispanic	2018	0.96	0.98	(0.920, 1.641) (0.419, 2.720)
Hispanic	2018	0.96	1.00	(0.997, 1.499) (0.462, 2.447)
Hispanic	2018	0.98	0.90	(0.323, 1.254) (0.228, 1.798)
Hispanic	2018	0.98	0.92	(0.373, 1.116) (0.272, 1.565)
Hispanic	2018	0.98	0.94	(0.429, 1.002) (0.318, 1.375)
Hispanic	2018	0.98	0.96	(0.486, 0.909) (0.365, 1.223)
Hispanic	2018	0.98	0.98	(0.544, 0.830) (0.413, 1.101)
Hispanic	2018	0.98	1.00	(0.602, 0.763) (0.461, 1.000)
Hispanic	2018	1.00	0.90	(0.336, 0.961) (0.267, 1.258)
Hispanic	2018	1.00	0.92	(0.387, 0.866) (0.313, 1.114)
Hispanic	2018	1.00	0.94	(0.442, 0.787) (0.363, 0.999)
Hispanic	2018	1.00	0.96	(0.499, 0.720) (0.414, 0.904)
Hispanic	2018	1.00	0.98	(0.556, 0.663) (0.465, 0.822)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2018	1.00	1.00	0.613 (0.516, 0.753)
Hispanic	2019	0.90	0.90	(1.026, 7.134) (0.507, 13.547)
Hispanic	2019	0.90	0.92	(1.125, 6.253) (0.583, 11.453)
Hispanic	2019	0.90	0.94	(1.232, 5.501) (0.663, 9.735)
Hispanic	2019	0.90	0.96	(1.345, 4.861) (0.751, 8.370)
Hispanic	2019	0.90	0.98	(1.464, 4.312) (0.841, 7.184)
Hispanic	2019	0.90	1.00	(1.585, 3.836) (0.932, 6.216)
Hispanic	2019	0.92	0.90	(1.100, 6.351) (0.636, 10.014)
Hispanic	2019	0.92	0.92	(1.210, 5.589) (0.728, 8.637)
Hispanic	2019	0.92	0.94	(1.330, 4.932) (0.823, 7.449)
Hispanic	2019	0.92	0.96	(1.457, 4.365) (0.928, 6.473)
Hispanic	2019	0.92	0.98	(1.590, 3.874) (1.036, 5.649)
Hispanic	2019	0.92	1.00	(1.727, 3.444) (1.145, 4.920)
Hispanic	2019	0.94	0.90	(1.060, 5.297) (0.696, 7.753)
Hispanic	2019	0.94	0.92	(1.169, 4.659) (0.783, 6.733)
Hispanic	2019	0.94	0.94	(1.286, 4.107) (0.878, 5.833)
Hispanic	2019	0.94	0.96	(1.411, 3.630) (0.979, 5.078)
Hispanic	2019	0.94	0.98	(1.543, 3.215) (1.084, 4.425)
Hispanic	2019	0.94	1.00	(1.677, 2.851) (1.189, 3.860)
Hispanic	2019	0.96	0.90	(0.844, 3.694) (0.511, 5.538)
Hispanic	2019	0.96	0.92	(0.932, 3.238) (0.576, 4.753)
Hispanic	2019	0.96	0.94	(1.027, 2.852) (0.647, 4.103)
Hispanic	2019	0.96	0.96	(1.129, 2.523) (0.723, 3.575)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2019	0.96	0.98	(1.237, 2.239) (0.808, 3.120)
Hispanic	2019	0.96	1.00	(1.347, 1.994) (0.894, 2.738)
Hispanic	2019	0.98	0.90	(0.690, 2.867) (0.521, 3.950)
Hispanic	2019	0.98	0.92	(0.766, 2.468) (0.585, 3.326)
Hispanic	2019	0.98	0.94	(0.849, 2.137) (0.655, 2.829)
Hispanic	2019	0.98	0.96	(0.942, 1.863) (0.738, 2.424)
Hispanic	2019	0.98	0.98	(1.047, 1.631) (0.828, 2.087)
Hispanic	2019	0.98	1.00	(1.157, 1.433) (0.921, 1.807)
Hispanic	2019	1.00	0.90	(0.727, 2.427) (0.589, 3.112)
Hispanic	2019	1.00	0.92	(0.806, 2.097) (0.659, 2.630)
Hispanic	2019	1.00	0.94	(0.893, 1.820) (0.736, 2.245)
Hispanic	2019	1.00	0.96	(0.991, 1.587) (0.825, 1.929)
Hispanic	2019	1.00	0.98	(1.101, 1.389) (0.924, 1.664)
Hispanic	2019	1.00	1.00	1.216 (1.027, 1.437)
Hispanic	2020	0.90	0.90	(0.958, 7.300) (0.368, 17.834)
Hispanic	2020	0.90	0.92	(1.057, 6.240) (0.430, 14.316)
Hispanic	2020	0.90	0.94	(1.164, 5.393) (0.502, 11.700)
Hispanic	2020	0.90	0.96	(1.276, 4.703) (0.578, 9.740)
Hispanic	2020	0.90	0.98	(1.390, 4.126) (0.653, 8.180)
Hispanic	2020	0.90	1.00	(1.505, 3.640) (0.735, 6.834)
Hispanic	2020	0.92	0.90	(1.128, 6.520) (0.543, 13.324)
Hispanic	2020	0.92	0.92	(1.240, 5.674) (0.624, 11.019)
Hispanic	2020	0.92	0.94	(1.360, 4.970) (0.710, 9.237)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2020	0.92	0.96	(1.484, 4.379) (0.805, 7.875)
Hispanic	2020	0.92	0.98	(1.610, 3.874) (0.900, 6.710)
Hispanic	2020	0.92	1.00	(1.738, 3.439) (0.998, 5.736)
Hispanic	2020	0.94	0.90	(1.234, 5.768) (0.710, 9.801)
Hispanic	2020	0.94	0.92	(1.351, 5.036) (0.801, 8.296)
Hispanic	2020	0.94	0.94	(1.476, 4.422) (0.897, 7.087)
Hispanic	2020	0.94	0.96	(1.605, 3.902) (0.996, 6.095)
Hispanic	2020	0.94	0.98	(1.737, 3.456) (1.101, 5.298)
Hispanic	2020	0.94	1.00	(1.870, 3.069) (1.203, 4.564)
Hispanic	2020	0.96	0.90	(1.214, 4.888) (0.796, 7.631)
Hispanic	2020	0.96	0.92	(1.321, 4.243) (0.882, 6.423)
Hispanic	2020	0.96	0.94	(1.435, 3.699) (0.971, 5.475)
Hispanic	2020	0.96	0.96	(1.552, 3.239) (1.067, 4.692)
Hispanic	2020	0.96	0.98	(1.672, 2.849) (1.164, 4.052)
Hispanic	2020	0.96	1.00	(1.793, 2.517) (1.260, 3.503)
Hispanic	2020	0.98	0.90	(1.160, 3.961) (0.824, 5.706)
Hispanic	2020	0.98	0.92	(1.258, 3.436) (0.903, 4.870)
Hispanic	2020	0.98	0.94	(1.360, 2.991) (0.985, 4.165)
Hispanic	2020	0.98	0.96	(1.466, 2.615) (1.074, 3.577)
Hispanic	2020	0.98	0.98	(1.573, 2.295) (1.163, 3.080)
Hispanic	2020	0.98	1.00	(1.682, 2.024) (1.254, 2.658)
Hispanic	2020	1.00	0.90	(1.213, 3.481) (0.896, 4.809)
Hispanic	2020	1.00	0.92	(1.312, 3.008) (0.980, 4.106)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2020	1.00	0.94	(1.416, 2.607) (1.067, 3.495)
Hispanic	2020	1.00	0.96	(1.523, 2.270) (1.157, 3.002)
Hispanic	2020	1.00	0.98	(1.632, 1.985) (1.251, 2.585)
Hispanic	2020	1.00	1.00	1.742 (1.344, 2.229)
Hispanic	2021	0.90	0.90	(0.896, 6.686) (0.396, 13.294)
Hispanic	2021	0.90	0.92	(0.984, 5.816) (0.456, 10.934)
Hispanic	2021	0.90	0.94	(1.079, 5.090) (0.521, 9.247)
Hispanic	2021	0.90	0.96	(1.178, 4.477) (0.595, 7.826)
Hispanic	2021	0.90	0.98	(1.281, 3.961) (0.669, 6.711)
Hispanic	2021	0.90	1.00	(1.386, 3.521) (0.741, 5.754)
Hispanic	2021	0.92	0.90	(1.021, 5.898) (0.504, 9.984)
Hispanic	2021	0.92	0.92	(1.119, 5.177) (0.581, 8.528)
Hispanic	2021	0.92	0.94	(1.223, 4.566) (0.652, 7.309)
Hispanic	2021	0.92	0.96	(1.332, 4.044) (0.733, 6.304)
Hispanic	2021	0.92	0.98	(1.444, 3.598) (0.815, 5.473)
Hispanic	2021	0.92	1.00	(1.558, 3.215) (0.896, 4.778)
Hispanic	2021	0.94	0.90	(1.023, 4.806) (0.221, 7.762)
Hispanic	2021	0.94	0.92	(1.121, 4.240) (0.247, 6.714)
Hispanic	2021	0.94	0.94	(1.224, 3.757) (0.276, 5.852)
Hispanic	2021	0.94	0.96	(1.333, 3.342) (0.304, 5.106)
Hispanic	2021	0.94	0.98	(1.444, 2.986) (0.332, 4.467)
Hispanic	2021	0.94	1.00	(1.557, 2.679) (0.363, 3.947)
Hispanic	2021	0.96	0.90	(0.932, 3.594) (0.382, 5.493)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2021	0.96	0.92	(1.019, 3.164) (0.423, 4.748)
Hispanic	2021	0.96	0.94	(1.110, 2.797) (0.466, 4.109)
Hispanic	2021	0.96	0.96	(1.206, 2.485) (0.512, 3.580)
Hispanic	2021	0.96	0.98	(1.303, 2.219) (0.558, 3.144)
Hispanic	2021	0.96	1.00	(1.402, 1.991) (0.604, 2.769)
Hispanic	2021	0.98	0.90	(0.807, 2.559) (0.423, 4.116)
Hispanic	2021	0.98	0.92	(0.882, 2.266) (0.467, 3.557)
Hispanic	2021	0.98	0.94	(0.960, 2.014) (0.514, 3.093)
Hispanic	2021	0.98	0.96	(1.041, 1.797) (0.563, 2.701)
Hispanic	2021	0.98	0.98	(1.125, 1.610) (0.614, 2.373)
Hispanic	2021	0.98	1.00	(1.209, 1.448) (0.664, 2.086)
Hispanic	2021	1.00	0.90	(0.772, 2.015) (0.509, 3.345)
Hispanic	2021	1.00	0.92	(0.843, 1.796) (0.561, 2.891)
Hispanic	2021	1.00	0.94	(0.918, 1.604) (0.614, 2.507)
Hispanic	2021	1.00	0.96	(0.996, 1.436) (0.669, 2.178)
Hispanic	2021	1.00	0.98	(1.076, 1.288) (0.725, 1.910)
Hispanic	2021	1.00	1.00	1.156 (0.782, 1.670)
Hispanic	2022	0.90	0.90	(1.230, 8.409) (0.651, 15.875)
Hispanic	2022	0.90	0.92	(1.336, 7.338) (0.725, 13.244)
Hispanic	2022	0.90	0.94	(1.445, 6.451) (0.805, 11.205)
Hispanic	2022	0.90	0.96	(1.557, 5.706) (0.887, 9.575)
Hispanic	2022	0.90	0.98	(1.671, 5.074) (0.967, 8.248)
Hispanic	2022	0.90	1.00	(1.785, 4.538) (1.047, 7.166)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2022	0.92	0.90	(1.351, 7.257) (0.849, 11.731)
Hispanic	2022	0.92	0.92	(1.466, 6.401) (0.942, 10.069)
Hispanic	2022	0.92	0.94	(1.584, 5.678) (1.034, 8.691)
Hispanic	2022	0.92	0.96	(1.705, 5.061) (1.130, 7.517)
Hispanic	2022	0.92	0.98	(1.828, 4.531) (1.228, 6.589)
Hispanic	2022	0.92	1.00	(1.951, 4.076) (1.326, 5.797)
Hispanic	2022	0.94	0.90	(1.439, 6.058) (1.014, 8.919)
Hispanic	2022	0.94	0.92	(1.557, 5.374) (1.115, 7.723)
Hispanic	2022	0.94	0.94	(1.680, 4.788) (1.221, 6.739)
Hispanic	2022	0.94	0.96	(1.805, 4.286) (1.325, 5.901)
Hispanic	2022	0.94	0.98	(1.932, 3.851) (1.429, 5.213)
Hispanic	2022	0.94	1.00	(2.059, 3.474) (1.535, 4.629)
Hispanic	2022	0.96	0.90	(1.450, 4.965) (1.111, 6.838)
Hispanic	2022	0.96	0.92	(1.568, 4.417) (1.216, 5.975)
Hispanic	2022	0.96	0.94	(1.690, 3.946) (1.324, 5.232)
Hispanic	2022	0.96	0.96	(1.815, 3.540) (1.435, 4.617)
Hispanic	2022	0.96	0.98	(1.941, 3.187) (1.547, 4.097)
Hispanic	2022	0.96	1.00	(2.068, 2.880) (1.656, 3.648)
Hispanic	2022	0.98	0.90	(1.369, 4.032) (1.096, 5.269)
Hispanic	2022	0.98	0.92	(1.481, 3.582) (1.194, 4.609)
Hispanic	2022	0.98	0.94	(1.596, 3.196) (1.296, 4.042)
Hispanic	2022	0.98	0.96	(1.714, 2.861) (1.402, 3.575)
Hispanic	2022	0.98	0.98	(1.834, 2.571) (1.509, 3.175)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2022	0.98	1.00	(1.954, 2.318) (1.615, 2.831)
Hispanic	2022	1.00	0.90	(1.310, 3.377) (1.087, 4.180)
Hispanic	2022	1.00	0.92	(1.418, 2.980) (1.186, 3.648)
Hispanic	2022	1.00	0.94	(1.531, 2.643) (1.289, 3.202)
Hispanic	2022	1.00	0.96	(1.647, 2.351) (1.395, 2.824)
Hispanic	2022	1.00	0.98	(1.764, 2.100) (1.501, 2.503)
Hispanic	2022	1.00	1.00	1.882 (1.609, 2.219)
Hispanic	2023	0.90	0.90	(1.070, 7.134) (0.534, 14.108)
Hispanic	2023	0.90	0.92	(1.163, 6.253) (0.601, 11.871)
Hispanic	2023	0.90	0.94	(1.259, 5.509) (0.670, 10.060)
Hispanic	2023	0.90	0.96	(1.358, 4.879) (0.740, 8.595)
Hispanic	2023	0.90	0.98	(1.458, 4.340) (0.813, 7.391)
Hispanic	2023	0.90	1.00	(1.558, 3.870) (0.884, 6.360)
Hispanic	2023	0.92	0.90	(1.092, 5.559) (0.662, 9.221)
Hispanic	2023	0.92	0.92	(1.191, 4.947) (0.737, 8.001)
Hispanic	2023	0.92	0.94	(1.294, 4.420) (0.816, 6.934)
Hispanic	2023	0.92	0.96	(1.399, 3.964) (0.897, 6.076)
Hispanic	2023	0.92	0.98	(1.505, 3.568) (0.976, 5.360)
Hispanic	2023	0.92	1.00	(1.611, 3.218) (1.058, 4.725)
Hispanic	2023	0.94	0.90	(1.057, 4.319) (0.735, 6.245)
Hispanic	2023	0.94	0.92	(1.150, 3.858) (0.812, 5.463)
Hispanic	2023	0.94	0.94	(1.247, 3.460) (0.894, 4.816)
Hispanic	2023	0.94	0.96	(1.347, 3.114) (0.975, 4.273)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2023	0.94	0.98	(1.447, 2.812) (1.055, 3.797)
Hispanic	2023	0.94	1.00	(1.548, 2.546) (1.138, 3.397)
Hispanic	2023	0.96	0.90	(1.082, 3.749) (0.824, 4.916)
Hispanic	2023	0.96	0.92	(1.174, 3.333) (0.902, 4.317)
Hispanic	2023	0.96	0.94	(1.270, 2.975) (0.983, 3.806)
Hispanic	2023	0.96	0.96	(1.368, 2.665) (1.065, 3.375)
Hispanic	2023	0.96	0.98	(1.467, 2.397) (1.150, 3.001)
Hispanic	2023	0.96	1.00	(1.566, 2.161) (1.234, 2.682)
Hispanic	2023	0.98	0.90	(1.089, 3.288) (0.869, 4.124)
Hispanic	2023	0.98	0.92	(1.180, 2.909) (0.948, 3.609)
Hispanic	2023	0.98	0.94	(1.275, 2.584) (1.029, 3.178)
Hispanic	2023	0.98	0.96	(1.372, 2.303) (1.115, 2.805)
Hispanic	2023	0.98	0.98	(1.470, 2.060) (1.200, 2.486)
Hispanic	2023	0.98	1.00	(1.568, 1.848) (1.285, 2.211)
Hispanic	2023	1.00	0.90	(1.090, 2.850) (0.908, 3.433)
Hispanic	2023	1.00	0.92	(1.182, 2.514) (0.992, 2.993)
Hispanic	2023	1.00	0.94	(1.278, 2.227) (1.079, 2.618)
Hispanic	2023	1.00	0.96	(1.376, 1.978) (1.169, 2.305)
Hispanic	2023	1.00	0.98	(1.476, 1.763) (1.258, 2.038)
Hispanic	2023	1.00	1.00	1.575 (1.347, 1.805)

Table 15: Camera analysis (redlight, beat-majority): bounds on the causal risk ratio $\psi_r^\dagger$ by beat majority, race, year, and the sensitivity parameters θ and η .

Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Beat majority: Black				
Black	All years	0.90	0.90	(2.966, 87.156) (1.282, 316.484)
Black	All years	0.90	0.95	(3.251, 23.244) (1.412, 63.911)
Black	All years	0.90	1.00	(3.527, 8.905) (1.542, 17.814)
Black	All years	0.95	0.90	(2.643, 22.318) (2.094, 36.553)
Black	All years	0.95	0.95	(2.877, 10.175) (2.288, 14.851)
Black	All years	0.95	1.00	(3.112, 4.811) (2.481, 6.085)
Black	All years	1.00	0.90	(2.417, 15.259) (2.241, 18.107)
Black	All years	1.00	0.95	(2.614, 6.654) (2.425, 7.516)
Black	All years	1.00	1.00	2.811 (2.610, 3.027)
Black	2017	0.90	0.90	(2.596, 142.548) (0.019, 817.616)
Black	2017	0.90	0.95	(3.443, 86.219) (0.081, 448.322)
Black	2017	0.90	1.00	(8.447, 21.000) (0.156, 125.070)
Black	2017	0.95	0.90	(2.634, 20.751) (1.401, 55.759)
Black	2017	0.95	0.95	(2.882, 9.649) (1.559, 20.468)
Black	2017	0.95	1.00	(3.131, 4.795) (1.711, 8.274)
Black	2017	1.00	0.90	(2.324, 13.091) (1.913, 19.664)
Black	2017	1.00	0.95	(2.517, 5.988) (2.079, 7.983)
Black	2017	1.00	1.00	2.711 (2.244, 3.245)
Black	2018	0.90	0.90	(3.985, 285.035) (0.385, 1239.113)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2018	0.90	0.95	(4.996, 104.234) (0.550, 551.294)
Black	2018	0.90	1.00	(10.030, 26.150) (0.693, 116.370)
Black	2018	0.95	0.90	(3.003, 28.447) (1.762, 77.244)
Black	2018	0.95	0.95	(3.268, 11.966) (1.938, 25.187)
Black	2018	0.95	1.00	(3.533, 5.466) (2.105, 8.827)
Black	2018	1.00	0.90	(2.715, 19.650) (2.286, 30.457)
Black	2018	1.00	0.95	(2.927, 8.047) (2.467, 10.791)
Black	2018	1.00	1.00	3.138 (2.647, 3.707)
Black	2019	0.90	0.90	(2.671, 157.759) (0.557, 740.733)
Black	2019	0.90	0.95	(3.231, 88.910) (0.693, 224.476)
Black	2019	0.90	1.00	(5.016, 11.976) (0.799, 38.601)
Black	2019	0.95	0.90	(3.345, 44.999) (2.064, 127.066)
Black	2019	0.95	0.95	(3.618, 15.952) (2.244, 32.381)
Black	2019	0.95	1.00	(3.891, 6.021) (2.424, 9.183)
Black	2019	1.00	0.90	(2.818, 23.648) (2.399, 36.236)
Black	2019	1.00	0.95	(3.037, 8.937) (2.588, 11.767)
Black	2019	1.00	1.00	3.255 (2.778, 3.847)
Black	2020	0.90	0.90	(2.098, 98.979) (0.166, 703.999)
Black	2020	0.90	0.95	(3.115, 170.879) (0.331, 619.972)
Black	2020	0.90	1.00	(8.144, 17.945) (0.533, 85.765)
Black	2020	0.95	0.90	(2.668, 107.302) (1.050, 439.875)
Black	2020	0.95	0.95	(2.941, 17.382) (1.168, 64.655)
Black	2020	0.95	1.00	(3.184, 4.909) (1.271, 10.921)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2020	1.00	0.90	(2.352, 16.024) (1.830, 26.746)
Black	2020	1.00	0.95	(2.544, 6.668) (1.985, 9.625)
Black	2020	1.00	1.00	2.735 (2.141, 3.433)
Black	2021	0.90	0.90	(1.713, 70.295) (0.162, 359.868)
Black	2021	0.90	0.95	(2.596, 51.608) (0.261, 280.160)
Black	2021	0.90	1.00	(7.221, 17.527) (0.364, 83.672)
Black	2021	0.95	0.90	(3.115, 83.335) (1.326, 264.721)
Black	2021	0.95	0.95	(3.393, 14.893) (1.450, 47.669)
Black	2021	0.95	1.00	(3.645, 5.619) (1.572, 12.278)
Black	2021	1.00	0.90	(2.207, 13.383) (1.752, 23.858)
Black	2021	1.00	0.95	(2.385, 5.802) (1.898, 8.625)
Black	2021	1.00	1.00	2.563 (2.043, 3.252)
Black	2022	0.90	0.90	(2.193, 137.960) (0.189, 948.450)
Black	2022	0.90	0.95	(3.030, 104.409) (0.308, 434.407)
Black	2022	0.90	1.00	(14.106, 34.845) (0.408, 124.321)
Black	2022	0.95	0.90	(2.047, 61.534) (0.943, 159.461)
Black	2022	0.95	0.95	(2.258, 8.644) (1.051, 24.823)
Black	2022	0.95	1.00	(2.461, 3.759) (1.169, 7.599)
Black	2022	1.00	0.90	(2.260, 14.008) (1.790, 22.537)
Black	2022	1.00	0.95	(2.454, 5.977) (1.950, 8.324)
Black	2022	1.00	1.00	2.648 (2.111, 3.282)
Black	2023	0.90	0.90	(1.244, 275.824) (0.000, 356.335)
Black	2023	0.90	0.95	(1.798, 133.984) (0.026, 216.641)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2023	0.90	1.00	(5.193, 12.923) (0.094, 76.401)
Black	2023	0.95	0.90	(1.866, 25.308) (0.784, 97.558)
Black	2023	0.95	0.95	(2.075, 7.332) (0.884, 21.040)
Black	2023	0.95	1.00	(2.279, 3.458) (0.986, 7.105)
Black	2023	1.00	0.90	(1.890, 10.577) (1.457, 17.443)
Black	2023	1.00	0.95	(2.068, 4.931) (1.600, 7.218)
Black	2023	1.00	1.00	2.246 (1.742, 2.891)
Hispanic	All years	0.90	0.90	(0.641, 37.458) (0.148, 164.640)
Hispanic	All years	0.90	0.95	(0.941, 10.954) (0.294, 35.599)
Hispanic	All years	0.90	1.00	(1.319, 3.718) (0.474, 8.679)
Hispanic	All years	0.95	0.90	(0.860, 12.415) (0.599, 22.433)
Hispanic	All years	0.95	0.95	(1.210, 5.625) (0.888, 8.915)
Hispanic	All years	0.95	1.00	(1.603, 2.738) (1.216, 3.727)
Hispanic	All years	1.00	0.90	(1.028, 11.049) (0.878, 13.954)
Hispanic	All years	1.00	0.95	(1.502, 4.727) (1.318, 5.581)
Hispanic	All years	1.00	1.00	2.067 (1.844, 2.314)
Hispanic	2017	0.90	0.90	(0.566, 51.601) (0.000, 266.408)
Hispanic	2017	0.90	0.95	(1.052, 35.563) (0.000, 146.667)
Hispanic	2017	0.90	1.00	(3.851, 10.211) (0.168, 63.291)
Hispanic	2017	0.95	0.90	(0.948, 14.932) (0.387, 45.338)
Hispanic	2017	0.95	0.95	(1.325, 5.944) (0.603, 13.882)
Hispanic	2017	0.95	1.00	(1.740, 2.838) (0.865, 4.970)
Hispanic	2017	1.00	0.90	(0.900, 8.617) (0.616, 15.216)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2017	1.00	0.95	(1.332, 3.931) (0.966, 5.887)
Hispanic	2017	1.00	1.00	1.863 (1.398, 2.454)
Hispanic	2018	0.90	0.90	(0.740, 68.103) (0.000, 405.351)
Hispanic	2018	0.90	0.95	(1.238, 30.685) (0.102, 213.759)
Hispanic	2018	0.90	1.00	(4.340, 11.332) (0.307, 38.902)
Hispanic	2018	0.95	0.90	(0.730, 12.104) (0.361, 36.398)
Hispanic	2018	0.95	0.95	(1.061, 5.064) (0.590, 11.503)
Hispanic	2018	0.95	1.00	(1.447, 2.373) (0.860, 3.963)
Hispanic	2018	1.00	0.90	(0.773, 10.264) (0.558, 16.870)
Hispanic	2018	1.00	0.95	(1.243, 4.225) (0.949, 5.968)
Hispanic	2018	1.00	1.00	1.823 (1.432, 2.276)
Hispanic	2019	0.90	0.90	(0.454, 43.743) (0.000, 172.052)
Hispanic	2019	0.90	0.95	(0.802, 12.828) (0.045, 61.037)
Hispanic	2019	0.90	1.00	(1.824, 4.786) (0.209, 17.109)
Hispanic	2019	0.95	0.90	(0.718, 18.605) (0.283, 58.450)
Hispanic	2019	0.95	0.95	(1.115, 6.976) (0.550, 15.979)
Hispanic	2019	0.95	1.00	(1.581, 2.714) (0.874, 4.575)
Hispanic	2019	1.00	0.90	(0.998, 17.893) (0.709, 31.845)
Hispanic	2019	1.00	0.95	(1.555, 6.320) (1.190, 9.169)
Hispanic	2019	1.00	1.00	2.229 (1.776, 2.756)
Hispanic	2020	0.90	0.90	(0.253, 17.277) (0.000, 116.236)
Hispanic	2020	0.90	0.95	(0.615, 12.010) (0.000, 66.729)
Hispanic	2020	0.90	1.00	(1.645, 4.966) (0.113, 23.608)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2020	0.95	0.90	(0.968, 113.451) (0.000, 412.428)
Hispanic	2020	0.95	0.95	(1.536, 32.765) (0.127, 179.536)
Hispanic	2020	0.95	1.00	(2.281, 4.174) (0.382, 12.187)
Hispanic	2020	1.00	0.90	(1.224, 15.635) (0.701, 32.694)
Hispanic	2020	1.00	0.95	(1.774, 5.986) (1.118, 10.120)
Hispanic	2020	1.00	1.00	2.426 (1.640, 3.417)
Hispanic	2021	0.90	0.90	(0.229, 27.326) (0.000, 103.403)
Hispanic	2021	0.90	0.95	(0.543, 23.485) (0.000, 102.542)
Hispanic	2021	0.90	1.00	(3.799, 11.918) (0.109, 34.880)
Hispanic	2021	0.95	0.90	(0.672, 71.410) (0.082, 203.531)
Hispanic	2021	0.95	0.95	(1.104, 12.791) (0.298, 33.337)
Hispanic	2021	0.95	1.00	(1.624, 2.805) (0.593, 6.564)
Hispanic	2021	1.00	0.90	(0.869, 11.539) (0.563, 25.204)
Hispanic	2021	1.00	0.95	(1.342, 4.533) (0.938, 7.658)
Hispanic	2021	1.00	1.00	1.927 (1.401, 2.630)
Hispanic	2022	0.90	0.90	(0.796, 79.058) (0.000, 488.848)
Hispanic	2022	0.90	0.95	(1.412, 49.700) (0.051, 288.657)
Hispanic	2022	0.90	1.00	(6.764, 18.808) (0.196, 83.251)
Hispanic	2022	0.95	0.90	(1.210, 89.794) (0.362, 248.111)
Hispanic	2022	0.95	0.95	(1.607, 9.278) (0.580, 25.551)
Hispanic	2022	0.95	1.00	(1.990, 3.499) (0.796, 7.411)
Hispanic	2022	1.00	0.90	(1.121, 10.852) (0.783, 19.244)
Hispanic	2022	1.00	0.95	(1.566, 4.445) (1.151, 6.507)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2022	1.00	1.00	2.066 (1.570, 2.675)
Hispanic	2023	0.90	0.90	(0.839, 61.273) (0.000, 421.412)
Hispanic	2023	0.90	0.95	(1.611, 53.406) (0.060, 218.760)
Hispanic	2023	0.90	1.00	(8.610, 25.323) (0.205, 174.880)
Hispanic	2023	0.95	0.90	(1.351, 33.707) (0.427, 154.873)
Hispanic	2023	0.95	0.95	(1.716, 8.960) (0.611, 33.464)
Hispanic	2023	0.95	1.00	(2.063, 3.494) (0.793, 8.872)
Hispanic	2023	1.00	0.90	(1.479, 10.491) (0.937, 18.315)
Hispanic	2023	1.00	0.95	(1.890, 4.870) (1.275, 7.407)
Hispanic	2023	1.00	1.00	2.341 (1.650, 3.072)

Beat majority: Hispanic

Black	All years	0.90	0.90	(0.464, 18.483) (0.077, 67.359)
Black	All years	0.90	0.95	(0.690, 6.188) (0.198, 19.064)
Black	All years	0.90	1.00	(0.943, 2.966) (0.354, 6.979)
Black	All years	0.95	0.90	(0.505, 4.628) (0.374, 6.421)
Black	All years	0.95	0.95	(0.723, 2.733) (0.562, 3.564)
Black	All years	0.95	1.00	(0.985, 1.759) (0.797, 2.186)
Black	All years	1.00	0.90	(0.594, 4.005) (0.522, 4.672)
Black	All years	1.00	0.95	(0.864, 2.118) (0.776, 2.381)
Black	All years	1.00	1.00	1.204 (1.096, 1.317)
Black	2017	0.90	0.90	(0.547, 78.848) (0.000, 245.416)
Black	2017	0.90	0.95	(0.907, 65.697) (0.000, 148.725)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2017	0.90	1.00	(1.991, 5.982) (0.026, 36.594)
Black	2017	0.95	0.90	(0.702, 6.262) (0.308, 14.177)
Black	2017	0.95	0.95	(0.984, 3.601) (0.495, 6.671)
Black	2017	0.95	1.00	(1.318, 2.313) (0.744, 3.735)
Black	2017	1.00	0.90	(0.701, 4.224) (0.489, 6.171)
Black	2017	1.00	0.95	(0.973, 2.271) (0.722, 3.055)
Black	2017	1.00	1.00	1.327 (1.034, 1.679)
Black	2018	0.90	0.90	(0.332, 17.277) (0.000, 66.132)
Black	2018	0.90	0.95	(0.583, 11.394) (0.044, 29.255)
Black	2018	0.90	1.00	(2.484, 7.073) (0.190, 20.776)
Black	2018	0.95	0.90	(0.494, 5.196) (0.209, 11.207)
Black	2018	0.95	0.95	(0.728, 2.853) (0.378, 5.243)
Black	2018	0.95	1.00	(1.020, 1.787) (0.608, 2.969)
Black	2018	1.00	0.90	(0.531, 3.833) (0.359, 5.675)
Black	2018	1.00	0.95	(0.777, 1.953) (0.568, 2.615)
Black	2018	1.00	1.00	1.117 (0.867, 1.409)
Black	2019	0.90	0.90	(0.385, 56.003) (0.000, 133.895)
Black	2019	0.90	0.95	(0.716, 30.589) (0.000, 88.747)
Black	2019	0.90	1.00	(1.372, 4.674) (0.084, 18.457)
Black	2019	0.95	0.90	(0.546, 5.479) (0.308, 10.601)
Black	2019	0.95	0.95	(0.785, 3.043) (0.489, 5.074)
Black	2019	0.95	1.00	(1.067, 1.926) (0.714, 2.815)
Black	2019	1.00	0.90	(0.576, 4.130) (0.410, 6.046)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2019	1.00	0.95	(0.861, 2.149) (0.651, 2.894)
Black	2019	1.00	1.00	1.230 (0.974, 1.554)
Black	2020	0.90	0.90	(0.466, 635.635) (0.000, 231.082)
Black	2020	0.90	0.95	(0.868, 24.353) (0.000, 105.582)
Black	2020	0.90	1.00	(3.312, 9.228) (0.118, 58.652)
Black	2020	0.95	0.90	(0.648, 9.526) (0.189, 35.524)
Black	2020	0.95	0.95	(0.887, 4.014) (0.332, 10.888)
Black	2020	0.95	1.00	(1.162, 2.080) (0.516, 4.747)
Black	2020	1.00	0.90	(0.654, 4.776) (0.428, 7.920)
Black	2020	1.00	0.95	(0.924, 2.383) (0.648, 3.548)
Black	2020	1.00	1.00	1.264 (0.933, 1.701)
Black	2021	0.90	0.90	(0.357, 93.813) (0.000, 314.284)
Black	2021	0.90	0.95	(0.904, 31.052) (0.000, 214.507)
Black	2021	0.90	1.00	(6.680, 19.316) (0.045, 75.097)
Black	2021	0.95	0.90	(0.500, 5.222) (0.093, 18.576)
Black	2021	0.95	0.95	(0.695, 2.764) (0.192, 7.939)
Black	2021	0.95	1.00	(0.936, 1.622) (0.353, 3.582)
Black	2021	1.00	0.90	(0.613, 3.223) (0.412, 5.136)
Black	2021	1.00	0.95	(0.854, 1.909) (0.608, 2.791)
Black	2021	1.00	1.00	1.163 (0.866, 1.557)
Black	2022	0.90	0.90	(0.343, 16.353) (0.000, 81.261)
Black	2022	0.90	0.95	(0.608, 9.833) (0.000, 58.535)
Black	2022	0.90	1.00	(1.855, 5.904) (0.094, 28.760)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2022	0.95	0.90	(0.337, 3.148) (0.133, 6.534)
Black	2022	0.95	0.95	(0.514, 1.930) (0.258, 3.397)
Black	2022	0.95	1.00	(0.720, 1.294) (0.415, 2.086)
Black	2022	1.00	0.90	(0.510, 3.377) (0.358, 4.930)
Black	2022	1.00	0.95	(0.781, 1.913) (0.586, 2.539)
Black	2022	1.00	1.00	1.099 (0.858, 1.370)
Black	2023	0.90	0.90	(0.693, 45.927) (0.000, 255.929)
Black	2023	0.90	0.95	(1.205, 33.799) (0.000, 206.683)
Black	2023	0.90	1.00	(4.172, 13.732) (0.143, 81.755)
Black	2023	0.95	0.90	(0.615, 7.109) (0.197, 19.797)
Black	2023	0.95	0.95	(0.838, 3.527) (0.338, 7.665)
Black	2023	0.95	1.00	(1.098, 1.998) (0.517, 3.623)
Black	2023	1.00	0.90	(0.680, 5.931) (0.478, 10.160)
Black	2023	1.00	0.95	(0.985, 2.624) (0.720, 3.918)
Black	2023	1.00	1.00	1.337 (1.008, 1.796)
Hispanic	All years	0.90	0.90	(1.193, 7.653) (0.801, 13.518)
Hispanic	All years	0.90	0.95	(1.313, 4.731) (0.891, 7.464)
Hispanic	All years	0.90	1.00	(1.433, 3.254) (0.980, 4.725)
Hispanic	All years	0.95	0.90	(1.267, 5.331) (1.098, 6.530)
Hispanic	All years	0.95	0.95	(1.396, 3.332) (1.212, 3.952)
Hispanic	All years	0.95	1.00	(1.525, 2.265) (1.325, 2.618)
Hispanic	All years	1.00	0.90	(1.714, 6.143) (1.584, 6.962)
Hispanic	All years	1.00	0.95	(1.878, 3.417) (1.738, 3.799)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	All years	1.00	1.00	2.043 (1.891, 2.235)
Hispanic	2017	0.90	0.90	(1.326, 17.177) (0.602, 51.249)
Hispanic	2017	0.90	0.95	(1.476, 5.340) (0.679, 12.577)
Hispanic	2017	0.90	1.00	(1.617, 3.429) (0.756, 6.428)
Hispanic	2017	0.95	0.90	(1.317, 5.258) (0.986, 8.308)
Hispanic	2017	0.95	0.95	(1.461, 3.345) (1.099, 4.745)
Hispanic	2017	0.95	1.00	(1.604, 2.316) (1.213, 3.063)
Hispanic	2017	1.00	0.90	(1.626, 5.672) (1.333, 7.587)
Hispanic	2017	1.00	0.95	(1.787, 3.198) (1.471, 4.018)
Hispanic	2017	1.00	1.00	1.949 (1.610, 2.332)
Hispanic	2018	0.90	0.90	(1.173, 8.338) (0.584, 21.588)
Hispanic	2018	0.90	0.95	(1.289, 4.695) (0.658, 10.127)
Hispanic	2018	0.90	1.00	(1.405, 3.170) (0.730, 6.209)
Hispanic	2018	0.95	0.90	(1.402, 6.006) (0.996, 9.740)
Hispanic	2018	0.95	0.95	(1.541, 3.709) (1.099, 5.545)
Hispanic	2018	0.95	1.00	(1.681, 2.496) (1.205, 3.499)
Hispanic	2018	1.00	0.90	(1.703, 6.297) (1.416, 8.496)
Hispanic	2018	1.00	0.95	(1.863, 3.395) (1.552, 4.252)
Hispanic	2018	1.00	1.00	2.024 (1.690, 2.392)
Hispanic	2019	0.90	0.90	(1.276, 22.473) (0.437, 44.345)
Hispanic	2019	0.90	0.95	(1.407, 5.775) (0.499, 14.930)
Hispanic	2019	0.90	1.00	(1.534, 3.500) (0.561, 7.832)
Hispanic	2019	0.95	0.90	(1.381, 6.465) (1.029, 10.207)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2019	0.95	0.95	(1.515, 3.737) (1.134, 5.507)
Hispanic	2019	0.95	1.00	(1.649, 2.449) (1.239, 3.478)
Hispanic	2019	1.00	0.90	(1.810, 6.932) (1.495, 9.359)
Hispanic	2019	1.00	0.95	(1.977, 3.651) (1.636, 4.636)
Hispanic	2019	1.00	1.00	2.144 (1.779, 2.593)
Hispanic	2020	0.90	0.90	(1.165, 710.931) (0.109, 194.279)
Hispanic	2020	0.90	0.95	(1.506, 18.548) (0.159, 62.913)
Hispanic	2020	0.90	1.00	(3.448, 8.405) (0.213, 34.737)
Hispanic	2020	0.95	0.90	(1.404, 7.604) (0.836, 18.224)
Hispanic	2020	0.95	0.95	(1.549, 4.180) (0.927, 8.147)
Hispanic	2020	0.95	1.00	(1.695, 2.603) (1.019, 4.371)
Hispanic	2020	1.00	0.90	(1.906, 7.624) (1.472, 11.993)
Hispanic	2020	1.00	0.95	(2.092, 4.027) (1.621, 6.023)
Hispanic	2020	1.00	1.00	2.278 (1.769, 3.187)
Hispanic	2021	0.90	0.90	(1.003, 45.982) (0.116, 82.574)
Hispanic	2021	0.90	0.95	(1.259, 11.598) (0.156, 53.715)
Hispanic	2021	0.90	1.00	(4.121, 10.129) (0.192, 41.993)
Hispanic	2021	0.95	0.90	(1.184, 4.470) (0.685, 7.812)
Hispanic	2021	0.95	0.95	(1.306, 2.960) (0.765, 4.699)
Hispanic	2021	0.95	1.00	(1.429, 2.105) (0.841, 3.103)
Hispanic	2021	1.00	0.90	(1.590, 4.706) (1.282, 6.315)
Hispanic	2021	1.00	0.95	(1.753, 2.932) (1.416, 3.773)
Hispanic	2021	1.00	1.00	1.917 (1.552, 2.376)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2022	0.90	0.90	(1.182, 15.036) (0.278, 67.536)
Hispanic	2022	0.90	0.95	(1.388, 7.315) (0.341, 26.524)
Hispanic	2022	0.90	1.00	(1.980, 4.790) (0.404, 13.179)
Hispanic	2022	0.95	0.90	(1.050, 4.275) (0.676, 7.703)
Hispanic	2022	0.95	0.95	(1.161, 2.742) (0.754, 4.493)
Hispanic	2022	0.95	1.00	(1.271, 1.915) (0.832, 2.860)
Hispanic	2022	1.00	0.90	(1.679, 5.624) (1.334, 7.874)
Hispanic	2022	1.00	0.95	(1.841, 3.316) (1.466, 4.441)
Hispanic	2022	1.00	1.00	2.003 (1.600, 2.528)
Hispanic	2023	0.90	0.90	(1.344, 29.642) (0.254, 174.100)
Hispanic	2023	0.90	0.95	(1.676, 15.509) (0.306, 69.132)
Hispanic	2023	0.90	1.00	(3.924, 8.435) (0.369, 32.036)
Hispanic	2023	0.95	0.90	(1.258, 6.341) (0.781, 14.288)
Hispanic	2023	0.95	0.95	(1.386, 3.671) (0.869, 6.886)
Hispanic	2023	0.95	1.00	(1.514, 2.300) (0.956, 3.722)
Hispanic	2023	1.00	0.90	(1.778, 7.599) (1.444, 10.546)
Hispanic	2023	1.00	0.95	(1.946, 3.826) (1.584, 4.917)
Hispanic	2023	1.00	1.00	2.114 (1.723, 2.523)
Beat majority: White				
Black	All years	0.90	0.90	(0.334, 2.109) (0.294, 2.352)
Black	All years	0.90	0.95	(0.442, 1.708) (0.396, 1.890)
Black	All years	0.90	1.00	(0.558, 1.430) (0.505, 1.574)
<i>Continued on next page</i>				

Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	All years	0.95	0.90	(0.425, 1.713) (0.389, 1.846)
Black	All years	0.95	0.95	(0.574, 1.390) (0.530, 1.492)
Black	All years	0.95	1.00	(0.735, 1.164) (0.682, 1.248)
Black	All years	1.00	0.90	(0.546, 1.434) (0.511, 1.514)
Black	All years	1.00	0.95	(0.745, 1.159) (0.703, 1.220)
Black	All years	1.00	1.00	0.968 (0.917, 1.016)
Black	2017	0.90	0.90	(0.326, 2.006) (0.214, 2.693)
Black	2017	0.90	0.95	(0.437, 1.660) (0.304, 2.193)
Black	2017	0.90	1.00	(0.558, 1.413) (0.404, 1.846)
Black	2017	0.95	0.90	(0.465, 1.796) (0.367, 2.166)
Black	2017	0.95	0.95	(0.626, 1.487) (0.507, 1.778)
Black	2017	0.95	1.00	(0.805, 1.266) (0.660, 1.502)
Black	2017	1.00	0.90	(0.603, 1.530) (0.505, 1.773)
Black	2017	1.00	0.95	(0.822, 1.262) (0.703, 1.452)
Black	2017	1.00	1.00	1.071 (0.932, 1.227)
Black	2018	0.90	0.90	(0.312, 2.022) (0.219, 2.570)
Black	2018	0.90	0.95	(0.424, 1.662) (0.314, 2.082)
Black	2018	0.90	1.00	(0.545, 1.408) (0.414, 1.747)
Black	2018	0.95	0.90	(0.411, 1.677) (0.330, 2.009)
Black	2018	0.95	0.95	(0.568, 1.383) (0.469, 1.646)
Black	2018	0.95	1.00	(0.738, 1.175) (0.622, 1.392)
Black	2018	1.00	0.90	(0.538, 1.410) (0.452, 1.617)
Black	2018	1.00	0.95	(0.748, 1.160) (0.642, 1.321)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2018	1.00	1.00	0.983 (0.856, 1.115)
Black	2019	0.90	0.90	(0.384, 2.328) (0.281, 3.033)
Black	2019	0.90	0.95	(0.500, 1.889) (0.380, 2.424)
Black	2019	0.90	1.00	(0.622, 1.584) (0.486, 2.012)
Black	2019	0.95	0.90	(0.517, 1.977) (0.424, 2.316)
Black	2019	0.95	0.95	(0.680, 1.606) (0.571, 1.864)
Black	2019	0.95	1.00	(0.856, 1.348) (0.730, 1.554)
Black	2019	1.00	0.90	(0.656, 1.648) (0.561, 1.876)
Black	2019	1.00	0.95	(0.876, 1.338) (0.764, 1.511)
Black	2019	1.00	1.00	1.122 (0.993, 1.262)
Black	2020	0.90	0.90	(0.397, 2.441) (0.243, 3.606)
Black	2020	0.90	0.95	(0.511, 1.949) (0.333, 2.817)
Black	2020	0.90	1.00	(0.636, 1.610) (0.435, 2.289)
Black	2020	0.95	0.90	(0.528, 2.047) (0.391, 2.680)
Black	2020	0.95	0.95	(0.686, 1.636) (0.521, 2.103)
Black	2020	0.95	1.00	(0.859, 1.352) (0.666, 1.717)
Black	2020	1.00	0.90	(0.683, 1.735) (0.546, 2.112)
Black	2020	1.00	0.95	(0.896, 1.381) (0.728, 1.661)
Black	2020	1.00	1.00	1.136 (0.936, 1.350)
Black	2021	0.90	0.90	(0.457, 2.853) (0.324, 3.899)
Black	2021	0.90	0.95	(0.575, 2.198) (0.421, 2.908)
Black	2021	0.90	1.00	(0.704, 1.769) (0.530, 2.290)
Black	2021	0.95	0.90	(0.544, 2.153) (0.439, 2.585)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2021	0.95	0.95	(0.693, 1.672) (0.569, 1.975)
Black	2021	0.95	1.00	(0.858, 1.351) (0.717, 1.580)
Black	2021	1.00	0.90	(0.654, 1.711) (0.560, 1.969)
Black	2021	1.00	0.95	(0.845, 1.322) (0.732, 1.503)
Black	2021	1.00	1.00	1.065 (0.932, 1.200)
Black	2022	0.90	0.90	(0.347, 2.108) (0.259, 2.788)
Black	2022	0.90	0.95	(0.452, 1.694) (0.349, 2.203)
Black	2022	0.90	1.00	(0.563, 1.412) (0.443, 1.813)
Black	2022	0.95	0.90	(0.393, 1.548) (0.329, 1.825)
Black	2022	0.95	0.95	(0.526, 1.251) (0.448, 1.464)
Black	2022	0.95	1.00	(0.670, 1.047) (0.575, 1.217)
Black	2022	1.00	0.90	(0.505, 1.298) (0.443, 1.450)
Black	2022	1.00	0.95	(0.679, 1.045) (0.603, 1.160)
Black	2022	1.00	1.00	0.871 (0.780, 0.962)
Black	2023	0.90	0.90	(0.250, 1.799) (0.173, 2.411)
Black	2023	0.90	0.95	(0.352, 1.449) (0.258, 1.903)
Black	2023	0.90	1.00	(0.460, 1.210) (0.351, 1.572)
Black	2023	0.95	0.90	(0.312, 1.433) (0.251, 1.689)
Black	2023	0.95	0.95	(0.452, 1.158) (0.377, 1.357)
Black	2023	0.95	1.00	(0.604, 0.969) (0.510, 1.131)
Black	2023	1.00	0.90	(0.415, 1.230) (0.352, 1.415)
Black	2023	1.00	0.95	(0.609, 0.990) (0.531, 1.134)
Black	2023	1.00	1.00	0.825 (0.726, 0.942)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	All years	0.90	0.90	(0.462, 2.680) (0.421, 2.908)
Hispanic	All years	0.90	0.95	(0.586, 2.178) (0.537, 2.354)
Hispanic	All years	0.90	1.00	(0.711, 1.827) (0.655, 1.971)
Hispanic	All years	0.95	0.90	(0.614, 2.132) (0.583, 2.244)
Hispanic	All years	0.95	0.95	(0.775, 1.733) (0.739, 1.818)
Hispanic	All years	0.95	1.00	(0.939, 1.455) (0.896, 1.523)
Hispanic	All years	1.00	0.90	(0.774, 1.729) (0.742, 1.797)
Hispanic	All years	1.00	0.95	(0.972, 1.402) (0.935, 1.453)
Hispanic	All years	1.00	1.00	1.174 (1.130, 1.214)
Hispanic	2017	0.90	0.90	(0.436, 2.452) (0.328, 3.121)
Hispanic	2017	0.90	0.95	(0.564, 2.033) (0.435, 2.551)
Hispanic	2017	0.90	1.00	(0.694, 1.732) (0.543, 2.151)
Hispanic	2017	0.95	0.90	(0.614, 2.108) (0.517, 2.439)
Hispanic	2017	0.95	0.95	(0.791, 1.749) (0.676, 2.013)
Hispanic	2017	0.95	1.00	(0.972, 1.490) (0.838, 1.707)
Hispanic	2017	1.00	0.90	(0.796, 1.779) (0.714, 1.993)
Hispanic	2017	1.00	0.95	(1.021, 1.471) (0.924, 1.639)
Hispanic	2017	1.00	1.00	1.249 (1.137, 1.387)
Hispanic	2018	0.90	0.90	(0.417, 2.382) (0.328, 2.915)
Hispanic	2018	0.90	0.95	(0.540, 1.969) (0.433, 2.380)
Hispanic	2018	0.90	1.00	(0.664, 1.675) (0.540, 2.010)
Hispanic	2018	0.95	0.90	(0.590, 2.002) (0.507, 2.279)
Hispanic	2018	0.95	0.95	(0.754, 1.658) (0.656, 1.873)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2018	0.95	1.00	(0.920, 1.412) (0.805, 1.586)
Hispanic	2018	1.00	0.90	(0.763, 1.668) (0.687, 1.827)
Hispanic	2018	1.00	0.95	(0.966, 1.378) (0.878, 1.504)
Hispanic	2018	1.00	1.00	1.172 (1.068, 1.275)
Hispanic	2019	0.90	0.90	(0.463, 2.713) (0.369, 3.362)
Hispanic	2019	0.90	0.95	(0.594, 2.207) (0.483, 2.693)
Hispanic	2019	0.90	1.00	(0.728, 1.855) (0.598, 2.243)
Hispanic	2019	0.95	0.90	(0.634, 2.188) (0.560, 2.440)
Hispanic	2019	0.95	0.95	(0.809, 1.793) (0.722, 1.986)
Hispanic	2019	0.95	1.00	(0.986, 1.514) (0.886, 1.670)
Hispanic	2019	1.00	0.90	(0.806, 1.775) (0.744, 1.924)
Hispanic	2019	1.00	0.95	(1.017, 1.456) (0.943, 1.572)
Hispanic	2019	1.00	1.00	1.230 (1.145, 1.324)
Hispanic	2020	0.90	0.90	(0.530, 3.127) (0.371, 4.207)
Hispanic	2020	0.90	0.95	(0.666, 2.501) (0.478, 3.276)
Hispanic	2020	0.90	1.00	(0.804, 2.069) (0.586, 2.666)
Hispanic	2020	0.95	0.90	(0.704, 2.479) (0.580, 2.960)
Hispanic	2020	0.95	0.95	(0.878, 1.983) (0.733, 2.334)
Hispanic	2020	0.95	1.00	(1.054, 1.639) (0.888, 1.909)
Hispanic	2020	1.00	0.90	(0.867, 1.969) (0.754, 2.235)
Hispanic	2020	1.00	0.95	(1.077, 1.570) (0.945, 1.763)
Hispanic	2020	1.00	1.00	1.292 (1.138, 1.440)
Hispanic	2021	0.90	0.90	(0.601, 3.762) (0.457, 4.972)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2021	0.90	0.95	(0.744, 2.896) (0.580, 3.689)
Hispanic	2021	0.90	1.00	(0.891, 2.328) (0.703, 2.902)
Hispanic	2021	0.95	0.90	(0.689, 2.581) (0.591, 2.936)
Hispanic	2021	0.95	0.95	(0.860, 2.014) (0.748, 2.259)
Hispanic	2021	0.95	1.00	(1.036, 1.635) (0.909, 1.817)
Hispanic	2021	1.00	0.90	(0.827, 1.964) (0.741, 2.166)
Hispanic	2021	1.00	0.95	(1.030, 1.530) (0.931, 1.671)
Hispanic	2021	1.00	1.00	1.239 (1.127, 1.345)
Hispanic	2022	0.90	0.90	(0.464, 2.846) (0.359, 3.469)
Hispanic	2022	0.90	0.95	(0.590, 2.285) (0.467, 2.738)
Hispanic	2022	0.90	1.00	(0.718, 1.902) (0.575, 2.252)
Hispanic	2022	0.95	0.90	(0.583, 2.086) (0.509, 2.336)
Hispanic	2022	0.95	0.95	(0.739, 1.687) (0.651, 1.875)
Hispanic	2022	0.95	1.00	(0.896, 1.411) (0.794, 1.560)
Hispanic	2022	1.00	0.90	(0.712, 1.623) (0.650, 1.774)
Hispanic	2022	1.00	0.95	(0.900, 1.308) (0.829, 1.421)
Hispanic	2022	1.00	1.00	1.091 (1.010, 1.180)
Hispanic	2023	0.90	0.90	(0.471, 2.637) (0.379, 3.237)
Hispanic	2023	0.90	0.95	(0.585, 2.130) (0.475, 2.571)
Hispanic	2023	0.90	1.00	(0.699, 1.782) (0.568, 2.121)
Hispanic	2023	0.95	0.90	(0.609, 2.052) (0.542, 2.297)
Hispanic	2023	0.95	0.95	(0.753, 1.654) (0.672, 1.843)
Hispanic	2023	0.95	1.00	(0.898, 1.381) (0.803, 1.532)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2023	1.00	0.90	(0.753, 1.658) (0.685, 1.815)
Hispanic	2023	1.00	0.95	(0.930, 1.331) (0.847, 1.452)
Hispanic	2023	1.00	1.00	1.107 (1.012, 1.205)

Table 16: Camera analysis (speed, beat-majority): bounds on the causal risk ratio $\psi_r^\dagger$ by beat majority, race, year, and the sensitivity parameters θ and η .

Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Beat majority: Black				
Black	All years	0.90	0.90	(0.792, 177.852) (0.000, 164.227)
Black	All years	0.90	0.95	(1.376, 83.760) (0.000, 149.203)
Black	All years	0.90	1.00	(7.786, 18.408) (0.080, 127.327)
Black	All years	0.95	0.90	(0.739, 3.493) (0.408, 7.636)
Black	All years	0.95	0.95	(0.855, 2.236) (0.491, 4.166)
Black	All years	0.95	1.00	(0.976, 1.510) (0.579, 2.427)
Black	All years	1.00	0.90	(0.556, 1.549) (0.450, 2.034)
Black	All years	1.00	0.95	(0.638, 1.045) (0.521, 1.303)
Black	All years	1.00	1.00	0.724 (0.595, 0.862)
Black	2017	0.90	0.90	(0.440, 10.218) (0.000, 82.671)
Black	2017	0.90	0.95	(0.674, 5.593) (0.017, 26.447)
Black	2017	0.90	1.00	(3.521, 9.322) (0.100, 45.270)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2017	0.95	0.90	(1.155, 19.141) (0.278, 96.011)
Black	2017	0.95	0.95	(1.390, 5.527) (0.359, 24.961)
Black	2017	0.95	1.00	(1.572, 2.369) (0.435, 6.520)
Black	2017	1.00	0.90	(0.638, 1.921) (0.432, 3.395)
Black	2017	1.00	0.95	(0.727, 1.219) (0.499, 1.895)
Black	2017	1.00	1.00	0.820 (0.570, 1.152)
Black	2018	0.90	0.90	(0.165, 6.793) (0.000, 22.094)
Black	2018	0.90	0.95	(0.413, 5.474) (0.000, 41.312)
Black	2018	0.90	1.00	(2.510, 5.837) (0.020, 46.762)
Black	2018	0.95	0.90	(0.843, 8.709) (0.218, 22.182)
Black	2018	0.95	0.95	(0.997, 2.946) (0.288, 9.959)
Black	2018	0.95	1.00	(1.161, 1.778) (0.372, 4.450)
Black	2018	1.00	0.90	(0.482, 1.311) (0.313, 2.272)
Black	2018	1.00	0.95	(0.556, 0.901) (0.369, 1.407)
Black	2018	1.00	1.00	0.635 (0.428, 0.908)
Black	2019	0.90	0.90	(0.938, 22.970) (0.028, 184.847)
Black	2019	0.90	0.95	(1.811, 28.975) (0.095, 182.933)
Black	2019	0.90	1.00	(11.843, 27.419) (0.178, 211.737)
Black	2019	0.95	0.90	(1.030, 6.370) (0.214, 16.734)
Black	2019	0.95	0.95	(1.171, 3.358) (0.251, 8.309)
Black	2019	0.95	1.00	(1.311, 1.999) (0.293, 4.127)
Black	2019	1.00	0.90	(0.650, 1.892) (0.395, 3.520)
Black	2019	1.00	0.95	(0.737, 1.229) (0.455, 2.032)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2019	1.00	1.00	0.827 (0.517, 1.235)
Black	2020	0.95	0.90	(0.529, 18.859) (0.000, 61.701)
Black	2020	0.95	0.95	(0.896, 11.280) (0.035, 48.376)
Black	2020	0.95	1.00	(2.576, 3.926) (0.073, 24.902)
Black	2020	1.00	0.90	(0.504, 1.561) (0.258, 3.589)
Black	2020	1.00	0.95	(0.586, 1.001) (0.315, 1.967)
Black	2020	1.00	1.00	0.671 (0.375, 1.138)
Black	2021	0.95	0.90	(0.544, 7.523) (0.045, 38.071)
Black	2021	0.95	0.95	(0.840, 5.661) (0.100, 32.391)
Black	2021	0.95	1.00	(18.032, 29.430) (0.164, 26.457)
Black	2021	1.00	0.90	(0.694, 2.426) (0.302, 6.708)
Black	2021	1.00	0.95	(0.805, 1.423) (0.374, 3.054)
Black	2021	1.00	1.00	0.920 (0.450, 1.674)
Black	2022	0.95	0.90	(0.375, 4.939) (0.000, 34.590)
Black	2022	0.95	0.95	(0.641, 6.760) (0.000, 25.724)
Black	2022	0.95	1.00	(3.813, 6.063) (0.049, 39.002)
Black	2022	1.00	0.90	(0.445, 1.457) (0.208, 3.415)
Black	2022	1.00	0.95	(0.519, 0.912) (0.256, 1.823)
Black	2022	1.00	1.00	0.598 (0.311, 1.040)
Black	2023	0.90	0.90	(0.412, 7.914) (0.000, 50.176)
Black	2023	0.90	0.95	(0.737, 5.145) (0.009, 36.853)
Black	2023	0.90	1.00	(3.033, 6.833) (0.074, 55.671)
Black	2023	0.95	0.90	(0.410, 17.994) (0.000, 96.656)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2023	0.95	0.95	(0.610, 6.967) (0.000, 23.277)
Black	2023	0.95	1.00	(1.476, 2.357) (0.038, 14.400)
Black	2023	1.00	0.90	(0.428, 1.272) (0.216, 2.571)
Black	2023	1.00	0.95	(0.506, 0.850) (0.273, 1.490)
Black	2023	1.00	1.00	0.591 (0.338, 0.934)
Hispanic	All years	0.90	0.90	(0.526, 629.799) (0.000, 201.860)
Hispanic	All years	0.90	0.95	(0.958, 72.643) (0.000, 165.765)
Hispanic	All years	0.90	1.00	(5.549, 16.799) (0.161, 131.151)
Hispanic	All years	0.95	0.90	(0.668, 4.141) (0.356, 9.082)
Hispanic	All years	0.95	0.95	(0.838, 2.586) (0.481, 4.808)
Hispanic	All years	0.95	1.00	(1.030, 1.718) (0.624, 2.782)
Hispanic	All years	1.00	0.90	(0.995, 3.427) (0.774, 4.669)
Hispanic	All years	1.00	0.95	(1.251, 2.260) (0.999, 2.927)
Hispanic	All years	1.00	1.00	1.538 (1.257, 1.896)
Hispanic	2017	0.90	0.90	(0.761, 20.539) (0.058, 115.466)
Hispanic	2017	0.90	0.95	(1.437, 15.511) (0.160, 99.627)
Hispanic	2017	0.90	1.00	(4.628, 14.226) (0.278, 68.833)
Hispanic	2017	0.95	0.90	(1.046, 23.036) (0.160, 125.773)
Hispanic	2017	0.95	0.95	(1.415, 7.479) (0.279, 34.227)
Hispanic	2017	0.95	1.00	(1.760, 3.002) (0.452, 9.129)
Hispanic	2017	1.00	0.90	(1.071, 4.030) (0.618, 7.831)
Hispanic	2017	1.00	0.95	(1.356, 2.530) (0.842, 4.231)
Hispanic	2017	1.00	1.00	1.678 (1.095, 2.526)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2018	0.90	0.90	(1.220, 23.673) (0.113, 139.683)
Hispanic	2018	0.90	0.95	(1.888, 17.920) (0.188, 114.816)
Hispanic	2018	0.90	1.00	(12.420, 33.391) (0.282, 176.613)
Hispanic	2018	0.95	0.90	(0.942, 10.475) (0.287, 25.012)
Hispanic	2018	0.95	0.95	(1.139, 3.692) (0.395, 11.034)
Hispanic	2018	0.95	1.00	(1.345, 2.134) (0.518, 4.903)
Hispanic	2018	1.00	0.90	(1.006, 3.460) (0.562, 6.697)
Hispanic	2018	1.00	0.95	(1.277, 2.315) (0.760, 3.955)
Hispanic	2018	1.00	1.00	1.585 (0.991, 2.415)
Hispanic	2019	0.90	0.90	(0.274, 17.526) (0.000, 113.934)
Hispanic	2019	0.90	0.95	(0.752, 26.627) (0.000, 125.947)
Hispanic	2019	0.90	1.00	(3.550, 10.483) (0.126, 71.615)
Hispanic	2019	0.95	0.90	(0.566, 5.427) (0.165, 14.610)
Hispanic	2019	0.95	0.95	(0.763, 2.799) (0.276, 7.244)
Hispanic	2019	0.95	1.00	(0.998, 1.675) (0.415, 3.628)
Hispanic	2019	1.00	0.90	(1.010, 3.810) (0.575, 7.116)
Hispanic	2019	1.00	0.95	(1.284, 2.418) (0.791, 4.086)
Hispanic	2019	1.00	1.00	1.602 (1.045, 2.434)
Hispanic	2020	0.95	0.90	(0.541, 32.456) (0.000, 64.126)
Hispanic	2020	0.95	0.95	(0.987, 12.383) (0.025, 66.836)
Hispanic	2020	0.95	1.00	(1.874, 3.091) (0.131, 14.463)
Hispanic	2020	1.00	0.90	(1.449, 7.205) (0.536, 20.141)
Hispanic	2020	1.00	0.95	(1.744, 3.403) (0.714, 7.970)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2020	1.00	1.00	2.064 (0.913, 3.940)
Hispanic	2021	0.95	0.90	(0.492, 11.487) (0.000, 30.335)
Hispanic	2021	0.95	0.95	(0.700, 4.208) (0.000, 23.710)
Hispanic	2021	0.95	1.00	(6.308, 10.723) (0.109, 17.469)
Hispanic	2021	1.00	0.90	(0.868, 3.407) (0.368, 8.698)
Hispanic	2021	1.00	0.95	(1.066, 1.993) (0.495, 4.057)
Hispanic	2021	1.00	1.00	1.286 (0.632, 2.192)
Hispanic	2022	0.95	0.90	(0.401, 4.941) (0.000, 26.746)
Hispanic	2022	0.95	0.95	(0.723, 7.045) (0.020, 31.898)
Hispanic	2022	0.95	1.00	(4.496, 7.876) (0.086, 53.820)
Hispanic	2022	1.00	0.90	(1.085, 4.091) (0.512, 9.883)
Hispanic	2022	1.00	0.95	(1.332, 2.491) (0.676, 5.043)
Hispanic	2022	1.00	1.00	1.605 (0.869, 2.814)
Hispanic	2023	0.90	0.90	(0.380, 12.045) (0.000, 81.063)
Hispanic	2023	0.90	0.95	(0.863, 7.894) (0.000, 40.280)
Hispanic	2023	0.90	1.00	(3.168, 9.466) (0.142, 33.994)
Hispanic	2023	0.95	0.90	(0.573, 29.503) (0.000, 151.859)
Hispanic	2023	0.95	0.95	(0.889, 7.370) (0.077, 38.006)
Hispanic	2023	0.95	1.00	(2.098, 3.574) (0.154, 22.148)
Hispanic	2023	1.00	0.90	(0.771, 2.434) (0.427, 4.955)
Hispanic	2023	1.00	0.95	(0.945, 1.626) (0.558, 2.876)
Hispanic	2023	1.00	1.00	1.131 (0.697, 1.796)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Beat majority: Hispanic				
Black	All years	0.90	0.90	(0.488, 15.491) (0.008, 54.578)
Black	All years	0.90	0.95	(0.658, 6.225) (0.039, 25.991)
Black	All years	0.90	1.00	(1.187, 3.187) (0.096, 11.607)
Black	All years	0.95	0.90	(0.292, 1.440) (0.092, 3.717)
Black	All years	0.95	0.95	(0.365, 1.017) (0.127, 2.365)
Black	All years	0.95	1.00	(0.448, 0.765) (0.165, 1.631)
Black	All years	1.00	0.90	(0.205, 0.424) (0.179, 0.525)
Black	All years	1.00	0.95	(0.248, 0.349) (0.218, 0.425)
Black	All years	1.00	1.00	0.293 (0.259, 0.353)
Black	2017	0.90	0.90	(0.303, 19.763) (0.000, 43.296)
Black	2017	0.90	0.95	(0.452, 6.728) (0.000, 22.287)
Black	2017	0.90	1.00	(0.822, 2.167) (0.040, 9.409)
Black	2017	0.95	0.90	(0.326, 2.612) (0.045, 7.923)
Black	2017	0.95	0.95	(0.409, 1.205) (0.081, 4.050)
Black	2017	0.95	1.00	(0.495, 0.825) (0.115, 2.410)
Black	2017	1.00	0.90	(0.107, 0.242) (0.086, 0.319)
Black	2017	1.00	0.95	(0.141, 0.205) (0.117, 0.266)
Black	2017	1.00	1.00	0.177 (0.148, 0.228)
Black	2018	0.90	0.90	(0.651, 70.500) (0.000, 119.906)
Black	2018	0.90	0.95	(1.233, 49.214) (0.000, 197.674)
Black	2018	0.90	1.00	(7.283, 18.502) (0.058, 137.580)
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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2018	0.95	0.90	(0.765, 9.971) (0.066, 48.281)
Black	2018	0.95	0.95	(0.921, 3.450) (0.100, 14.924)
Black	2018	0.95	1.00	(1.090, 1.836) (0.139, 6.105)
Black	2018	1.00	0.90	(0.299, 0.609) (0.235, 0.883)
Black	2018	1.00	0.95	(0.344, 0.478) (0.274, 0.667)
Black	2018	1.00	1.00	0.391 (0.314, 0.530)
Black	2019	0.90	0.90	(0.909, 41.040) (0.000, 472.135)
Black	2019	0.90	0.95	(1.583, 30.628) (0.000, 169.752)
Black	2019	0.90	1.00	(8.726, 26.354) (0.105, 188.137)
Black	2019	0.95	0.90	(0.475, 6.019) (0.054, 26.791)
Black	2019	0.95	0.95	(0.616, 2.984) (0.094, 13.420)
Black	2019	0.95	1.00	(0.778, 1.303) (0.138, 5.131)
Black	2019	1.00	0.90	(0.894, 2.323) (0.652, 3.637)
Black	2019	1.00	0.95	(1.001, 1.588) (0.739, 2.271)
Black	2019	1.00	1.00	1.116 (0.830, 1.487)
Black	2020	0.90	0.90	(0.358, 4.151) (0.000, 21.176)
Black	2020	0.90	0.95	(0.569, 3.969) (0.050, 36.491)
Black	2020	0.90	1.00	(4.428, 10.958) (0.101, 32.153)
Black	2020	0.95	0.90	(0.405, 5.533) (0.000, 32.818)
Black	2020	0.95	0.95	(0.696, 7.439) (0.000, 33.231)
Black	2020	0.95	1.00	(2.979, 4.816) (0.044, 26.268)
Black	2020	1.00	0.90	(0.403, 1.299) (0.196, 2.612)
Black	2020	1.00	0.95	(0.496, 0.866) (0.258, 1.611)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2020	1.00	1.00	0.599 (0.328, 1.029)
Black	2021	0.90	0.90	(0.435, 10.108) (0.000, 57.837)
Black	2021	0.90	0.95	(0.755, 12.138) (0.000, 70.236)
Black	2021	0.90	1.00	(4.354, 12.067) (0.037, 50.561)
Black	2021	0.95	0.90	(0.853, 9.430) (0.000, 57.500)
Black	2021	0.95	0.95	(1.139, 5.735) (0.013, 37.476)
Black	2021	0.95	1.00	(1.549, 2.658) (0.044, 11.635)
Black	2021	1.00	0.90	(0.601, 1.604) (0.187, 4.677)
Black	2021	1.00	0.95	(0.701, 1.136) (0.235, 2.883)
Black	2021	1.00	1.00	0.815 (0.299, 1.830)
Black	2022	0.90	0.90	(0.411, 35.119) (0.000, 307.972)
Black	2022	0.90	0.95	(0.567, 5.640) (0.000, 29.473)
Black	2022	0.90	1.00	(1.449, 3.772) (0.033, 21.098)
Black	2022	0.95	0.90	(0.626, 11.115) (0.062, 30.377)
Black	2022	0.95	0.95	(0.776, 2.837) (0.115, 10.526)
Black	2022	0.95	1.00	(0.993, 1.579) (0.190, 4.981)
Black	2022	1.00	0.90	(0.488, 1.467) (0.279, 2.399)
Black	2022	1.00	0.95	(0.606, 1.031) (0.371, 1.567)
Black	2022	1.00	1.00	0.749 (0.485, 1.083)
Black	2023	0.95	0.90	(0.624, 20.202) (0.000, 66.883)
Black	2023	0.95	0.95	(1.193, 16.000) (0.000, 75.332)
Black	2023	0.95	1.00	(4.565, 8.805) (0.052, 57.955)
Black	2023	1.00	0.90	(0.778, 2.626) (0.363, 5.626)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2023	1.00	0.95	(0.941, 1.699) (0.466, 3.247)
Black	2023	1.00	1.00	1.127 (0.590, 1.906)
Hispanic	All years	0.90	0.90	(0.593, 14.453) (0.035, 47.041)
Hispanic	All years	0.90	0.95	(0.762, 6.998) (0.096, 22.261)
Hispanic	All years	0.90	1.00	(1.253, 3.413) (0.171, 10.208)
Hispanic	All years	0.95	0.90	(0.459, 1.979) (0.171, 4.249)
Hispanic	All years	0.95	0.95	(0.546, 1.413) (0.228, 2.592)
Hispanic	All years	0.95	1.00	(0.635, 1.071) (0.287, 1.772)
Hispanic	All years	1.00	0.90	(0.274, 0.688) (0.239, 0.806)
Hispanic	All years	1.00	0.95	(0.372, 0.565) (0.330, 0.652)
Hispanic	All years	1.00	1.00	0.475 (0.425, 0.542)
Hispanic	2017	0.90	0.90	(0.387, 19.460) (0.000, 64.748)
Hispanic	2017	0.90	0.95	(0.583, 6.531) (0.016, 28.001)
Hispanic	2017	0.90	1.00	(1.484, 4.356) (0.111, 19.977)
Hispanic	2017	0.95	0.90	(0.503, 2.961) (0.153, 8.716)
Hispanic	2017	0.95	0.95	(0.594, 1.635) (0.219, 4.146)
Hispanic	2017	0.95	1.00	(0.688, 1.162) (0.283, 2.422)
Hispanic	2017	1.00	0.90	(0.266, 0.613) (0.223, 0.735)
Hispanic	2017	1.00	0.95	(0.356, 0.517) (0.306, 0.614)
Hispanic	2017	1.00	1.00	0.447 (0.388, 0.525)
Hispanic	2018	0.90	0.90	(0.864, 40.528) (0.000, 84.119)
Hispanic	2018	0.90	0.95	(1.569, 95.986) (0.037, 150.069)
Hispanic	2018	0.90	1.00	(8.133, 19.112) (0.125, 116.560)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2018	0.95	0.90	(0.626, 4.685) (0.154, 19.440)
Hispanic	2018	0.95	0.95	(0.748, 2.221) (0.225, 7.436)
Hispanic	2018	0.95	1.00	(0.872, 1.414) (0.299, 3.722)
Hispanic	2018	1.00	0.90	(0.238, 0.735) (0.177, 0.937)
Hispanic	2018	1.00	0.95	(0.352, 0.577) (0.283, 0.711)
Hispanic	2018	1.00	1.00	0.472 (0.391, 0.569)
Hispanic	2019	0.90	0.90	(0.333, 7.313) (0.000, 66.274)
Hispanic	2019	0.90	0.95	(0.575, 5.363) (0.000, 33.774)
Hispanic	2019	0.90	1.00	(4.716, 10.051) (0.078, 72.372)
Hispanic	2019	0.95	0.90	(0.529, 3.738) (0.091, 15.919)
Hispanic	2019	0.95	0.95	(0.651, 2.092) (0.158, 8.559)
Hispanic	2019	0.95	1.00	(0.778, 1.284) (0.226, 4.292)
Hispanic	2019	1.00	0.90	(0.384, 1.486) (0.268, 2.297)
Hispanic	2019	1.00	0.95	(0.514, 1.018) (0.374, 1.445)
Hispanic	2019	1.00	1.00	0.717 (0.548, 0.947)
Hispanic	2020	0.90	0.90	(0.354, 10.055) (0.000, 84.709)
Hispanic	2020	0.90	0.95	(0.681, 6.642) (0.029, 29.390)
Hispanic	2020	0.90	1.00	(2.972, 7.234) (0.119, 46.188)
Hispanic	2020	0.95	0.90	(0.666, 10.029) (0.000, 67.584)
Hispanic	2020	0.95	0.95	(0.969, 7.068) (0.032, 42.332)
Hispanic	2020	0.95	1.00	(2.406, 3.796) (0.106, 20.725)
Hispanic	2020	1.00	0.90	(0.937, 2.729) (0.509, 5.137)
Hispanic	2020	1.00	0.95	(1.092, 1.817) (0.622, 3.090)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2020	1.00	1.00	1.252 (0.737, 1.938)
Hispanic	2021	0.90	0.90	(0.407, 7.917) (0.000, 44.137)
Hispanic	2021	0.90	0.95	(0.699, 7.203) (0.013, 39.476)
Hispanic	2021	0.90	1.00	(7.944, 20.865) (0.080, 63.074)
Hispanic	2021	0.95	0.90	(0.600, 6.648) (0.032, 36.123)
Hispanic	2021	0.95	0.95	(0.835, 3.874) (0.057, 20.153)
Hispanic	2021	0.95	1.00	(1.097, 1.728) (0.082, 6.862)
Hispanic	2021	1.00	0.90	(0.351, 1.015) (0.148, 2.692)
Hispanic	2021	1.00	0.95	(0.436, 0.725) (0.196, 1.665)
Hispanic	2021	1.00	1.00	0.527 (0.245, 1.065)
Hispanic	2022	0.90	0.90	(0.533, 58.920) (0.000, 454.478)
Hispanic	2022	0.90	0.95	(0.818, 9.392) (0.049, 45.519)
Hispanic	2022	0.90	1.00	(3.582, 10.868) (0.118, 65.806)
Hispanic	2022	0.95	0.90	(0.665, 7.122) (0.086, 23.076)
Hispanic	2022	0.95	0.95	(0.815, 2.693) (0.148, 8.901)
Hispanic	2022	0.95	1.00	(0.973, 1.516) (0.222, 4.143)
Hispanic	2022	1.00	0.90	(0.600, 1.616) (0.404, 2.582)
Hispanic	2022	1.00	0.95	(0.707, 1.133) (0.488, 1.710)
Hispanic	2022	1.00	1.00	0.819 (0.577, 1.164)
Hispanic	2023	0.90	0.90	(0.266, 2.519) (0.002, 13.558)
Hispanic	2023	0.90	0.95	(0.536, 11.523) (0.041, 57.824)
Hispanic	2023	0.90	1.00	(2.645, 6.395) (0.067, 41.956)
Hispanic	2023	0.95	0.90	(0.699, 13.216) (0.022, 56.378)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2023	0.95	0.95	(1.111, 9.281) (0.060, 57.920)
Hispanic	2023	0.95	1.00	(4.263, 5.957) (0.106, 29.140)
Hispanic	2023	1.00	0.90	(0.665, 2.086) (0.364, 3.911)
Hispanic	2023	1.00	0.95	(0.779, 1.351) (0.442, 2.316)
Hispanic	2023	1.00	1.00	0.896 (0.525, 1.375)
Beat majority: White				
Black	All years	0.90	0.90	(0.802, 4.910) (0.500, 7.845)
Black	All years	0.90	0.95	(0.968, 3.684) (0.623, 5.654)
Black	All years	0.90	1.00	(1.149, 2.858) (0.756, 4.229)
Black	All years	0.95	0.90	(1.130, 4.246) (0.826, 5.674)
Black	All years	0.95	0.95	(1.368, 3.268) (1.020, 4.259)
Black	All years	0.95	1.00	(1.628, 2.582) (1.237, 3.313)
Black	All years	1.00	0.90	(1.778, 4.210) (1.457, 5.115)
Black	All years	1.00	0.95	(2.158, 3.252) (1.796, 3.884)
Black	All years	1.00	1.00	2.578 (2.174, 3.033)
Black	2017	0.90	0.90	(0.737, 4.952) (0.264, 12.343)
Black	2017	0.90	0.95	(0.899, 3.444) (0.351, 7.544)
Black	2017	0.90	1.00	(1.070, 2.624) (0.441, 5.360)
Black	2017	0.95	0.90	(0.849, 3.081) (0.422, 5.204)
Black	2017	0.95	0.95	(1.045, 2.434) (0.539, 3.997)
Black	2017	0.95	1.00	(1.254, 1.996) (0.669, 3.216)
Black	2017	1.00	0.90	(1.476, 3.379) (1.040, 4.739)
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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2017	1.00	0.95	(1.814, 2.667) (1.310, 3.656)
Black	2017	1.00	1.00	2.182 (1.617, 2.929)
Black	2018	0.90	0.90	(0.601, 4.272) (0.099, 14.498)
Black	2018	0.90	0.95	(0.737, 3.017) (0.155, 9.304)
Black	2018	0.90	1.00	(0.893, 2.243) (0.243, 6.514)
Black	2018	0.95	0.90	(0.998, 3.814) (0.467, 7.540)
Black	2018	0.95	0.95	(1.216, 2.953) (0.614, 5.514)
Black	2018	0.95	1.00	(1.462, 2.346) (0.787, 4.179)
Black	2018	1.00	0.90	(1.617, 3.884) (0.960, 5.793)
Black	2018	1.00	0.95	(2.014, 3.056) (1.281, 4.413)
Black	2018	1.00	1.00	2.463 (1.654, 3.472)
Black	2019	0.90	0.90	(1.395, 11.076) (0.459, 33.740)
Black	2019	0.90	0.95	(1.646, 6.913) (0.600, 18.112)
Black	2019	0.90	1.00	(1.921, 4.702) (0.762, 11.029)
Black	2019	0.95	0.90	(2.189, 9.075) (1.232, 17.367)
Black	2019	0.95	0.95	(2.545, 6.385) (1.506, 11.337)
Black	2019	0.95	1.00	(2.934, 4.608) (1.810, 7.709)
Black	2019	1.00	0.90	(3.491, 8.566) (2.067, 13.491)
Black	2019	1.00	0.95	(4.029, 6.225) (2.476, 9.434)
Black	2019	1.00	1.00	4.613 (2.939, 6.760)
Black	2020	0.90	0.90	(0.908, 6.370) (0.244, 17.768)
Black	2020	0.90	0.95	(1.114, 4.397) (0.374, 10.670)
Black	2020	0.90	1.00	(1.337, 3.230) (0.517, 7.030)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2020	0.95	0.90	(1.399, 6.093) (0.687, 11.991)
Black	2020	0.95	0.95	(1.708, 4.303) (0.907, 7.689)
Black	2020	0.95	1.00	(2.051, 3.160) (1.160, 5.187)
Black	2020	1.00	0.90	(1.941, 5.412) (1.112, 9.402)
Black	2020	1.00	0.95	(2.370, 3.859) (1.450, 6.212)
Black	2020	1.00	1.00	2.849 (1.842, 4.325)
Black	2021	0.90	0.90	(1.241, 8.479) (0.522, 19.729)
Black	2021	0.90	0.95	(1.477, 5.796) (0.671, 11.863)
Black	2021	0.90	1.00	(1.735, 4.098) (0.840, 7.458)
Black	2021	0.95	0.90	(1.212, 4.943) (0.680, 9.247)
Black	2021	0.95	0.95	(1.456, 3.602) (0.853, 6.215)
Black	2021	0.95	1.00	(1.729, 2.683) (1.050, 4.297)
Black	2021	1.00	0.90	(1.474, 3.705) (0.910, 5.807)
Black	2021	1.00	0.95	(1.783, 2.781) (1.154, 4.150)
Black	2021	1.00	1.00	2.134 (1.422, 3.036)
Black	2022	0.90	0.90	(0.695, 4.326) (0.281, 8.967)
Black	2022	0.90	0.95	(0.848, 3.278) (0.392, 6.269)
Black	2022	0.90	1.00	(1.018, 2.570) (0.519, 4.645)
Black	2022	0.95	0.90	(1.066, 4.237) (0.582, 7.067)
Black	2022	0.95	0.95	(1.316, 3.210) (0.775, 5.113)
Black	2022	0.95	1.00	(1.599, 2.501) (0.994, 3.813)
Black	2022	1.00	0.90	(1.546, 4.011) (0.975, 6.240)
Black	2022	1.00	0.95	(1.908, 3.006) (1.256, 4.464)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Black	2022	1.00	1.00	2.319 (1.580, 3.309)
Black	2023	0.90	0.90	(0.928, 7.107) (0.282, 19.960)
Black	2023	0.90	0.95	(1.134, 4.883) (0.396, 12.246)
Black	2023	0.90	1.00	(1.372, 3.486) (0.542, 7.753)
Black	2023	0.95	0.90	(1.506, 7.075) (0.742, 14.452)
Black	2023	0.95	0.95	(1.818, 4.928) (0.955, 9.049)
Black	2023	0.95	1.00	(2.169, 3.542) (1.213, 5.954)
Black	2023	1.00	0.90	(1.910, 5.146) (1.166, 8.389)
Black	2023	1.00	0.95	(2.321, 3.748) (1.469, 5.674)
Black	2023	1.00	1.00	2.792 (1.833, 4.018)
Hispanic	All years	0.90	0.90	(1.044, 6.378) (0.730, 8.679)
Hispanic	All years	0.90	0.95	(1.306, 4.856) (0.945, 6.372)
Hispanic	All years	0.90	1.00	(1.586, 3.789) (1.174, 4.810)
Hispanic	All years	0.95	0.90	(1.107, 4.274) (0.936, 5.078)
Hispanic	All years	0.95	0.95	(1.408, 3.322) (1.211, 3.876)
Hispanic	All years	0.95	1.00	(1.728, 2.643) (1.503, 3.040)
Hispanic	All years	1.00	0.90	(1.326, 3.295) (1.177, 3.647)
Hispanic	All years	1.00	0.95	(1.684, 2.577) (1.518, 2.827)
Hispanic	All years	1.00	1.00	2.063 (1.879, 2.246)
Hispanic	2017	0.90	0.90	(0.920, 5.529) (0.323, 11.572)
Hispanic	2017	0.90	0.95	(1.169, 4.255) (0.470, 8.224)
Hispanic	2017	0.90	1.00	(1.430, 3.402) (0.610, 6.242)
Hispanic	2017	0.95	0.90	(0.871, 3.479) (0.568, 4.949)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2017	0.95	0.95	(1.156, 2.779) (0.786, 3.818)
Hispanic	2017	0.95	1.00	(1.451, 2.293) (1.005, 3.075)
Hispanic	2017	1.00	0.90	(1.144, 2.859) (0.874, 3.508)
Hispanic	2017	1.00	0.95	(1.513, 2.291) (1.204, 2.770)
Hispanic	2017	1.00	1.00	1.893 (1.541, 2.262)
Hispanic	2018	0.90	0.90	(0.985, 5.747) (0.470, 11.390)
Hispanic	2018	0.90	0.95	(1.242, 4.409) (0.660, 7.950)
Hispanic	2018	0.90	1.00	(1.519, 3.495) (0.870, 5.876)
Hispanic	2018	0.95	0.90	(1.183, 4.368) (0.841, 5.975)
Hispanic	2018	0.95	0.95	(1.499, 3.407) (1.116, 4.470)
Hispanic	2018	0.95	1.00	(1.838, 2.729) (1.411, 3.484)
Hispanic	2018	1.00	0.90	(1.332, 3.283) (1.057, 4.015)
Hispanic	2018	1.00	0.95	(1.688, 2.576) (1.374, 3.087)
Hispanic	2018	1.00	1.00	2.071 (1.718, 2.440)
Hispanic	2019	0.90	0.90	(1.174, 7.837) (0.510, 16.424)
Hispanic	2019	0.90	0.95	(1.460, 5.659) (0.723, 10.649)
Hispanic	2019	0.90	1.00	(1.785, 4.199) (0.960, 7.198)
Hispanic	2019	0.95	0.90	(1.298, 5.685) (0.879, 8.261)
Hispanic	2019	0.95	0.95	(1.655, 4.180) (1.171, 5.790)
Hispanic	2019	0.95	1.00	(2.065, 3.115) (1.511, 4.152)
Hispanic	2019	1.00	0.90	(1.380, 3.867) (1.091, 5.011)
Hispanic	2019	1.00	0.95	(1.775, 2.910) (1.441, 3.693)
Hispanic	2019	1.00	1.00	2.223 (1.836, 2.772)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2020	0.90	0.90	(1.047, 7.938) (0.347, 21.867)
Hispanic	2020	0.90	0.95	(1.331, 5.402) (0.531, 12.656)
Hispanic	2020	0.90	1.00	(1.638, 3.856) (0.735, 7.885)
Hispanic	2020	0.95	0.90	(1.393, 6.034) (0.780, 10.368)
Hispanic	2020	0.95	0.95	(1.742, 4.304) (1.061, 6.868)
Hispanic	2020	0.95	1.00	(2.119, 3.166) (1.367, 4.743)
Hispanic	2020	1.00	0.90	(1.543, 4.260) (1.017, 6.149)
Hispanic	2020	1.00	0.95	(1.919, 3.102) (1.333, 4.233)
Hispanic	2020	1.00	1.00	2.325 (1.688, 3.066)
Hispanic	2021	0.90	0.90	(1.048, 7.725) (0.422, 17.422)
Hispanic	2021	0.90	0.95	(1.309, 5.415) (0.610, 10.603)
Hispanic	2021	0.90	1.00	(1.598, 3.962) (0.809, 6.980)
Hispanic	2021	0.95	0.90	(1.280, 5.227) (0.817, 7.935)
Hispanic	2021	0.95	0.95	(1.593, 3.895) (1.073, 5.615)
Hispanic	2021	0.95	1.00	(1.933, 2.972) (1.354, 4.101)
Hispanic	2021	1.00	0.90	(1.384, 3.447) (1.010, 4.412)
Hispanic	2021	1.00	0.95	(1.713, 2.636) (1.297, 3.261)
Hispanic	2021	1.00	1.00	2.065 (1.609, 2.498)
Hispanic	2022	0.90	0.90	(1.398, 9.093) (0.707, 18.226)
Hispanic	2022	0.90	0.95	(1.697, 6.547) (0.929, 11.740)
Hispanic	2022	0.90	1.00	(2.007, 4.900) (1.154, 8.011)
Hispanic	2022	0.95	0.90	(1.556, 5.789) (1.080, 8.705)
Hispanic	2022	0.95	0.95	(1.899, 4.397) (1.378, 6.206)

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Race (r)	Year	θ	η	$\psi_r^\dagger$ (95% CI)
Hispanic	2022	0.95	1.00	(2.257, 3.424) (1.679, 4.599)
Hispanic	2022	1.00	0.90	(1.848, 4.473) (1.446, 5.878)
Hispanic	2022	1.00	0.95	(2.267, 3.437) (1.823, 4.369)
Hispanic	2022	1.00	1.00	2.707 (2.218, 3.351)
Hispanic	2023	0.90	0.90	(1.211, 8.297) (0.552, 18.494)
Hispanic	2023	0.90	0.95	(1.463, 5.867) (0.726, 11.699)
Hispanic	2023	0.90	1.00	(1.726, 4.275) (0.908, 7.552)
Hispanic	2023	0.95	0.90	(1.126, 4.166) (0.791, 5.971)
Hispanic	2023	0.95	0.95	(1.373, 3.156) (0.997, 4.309)
Hispanic	2023	0.95	1.00	(1.631, 2.441) (1.211, 3.207)
Hispanic	2023	1.00	0.90	(1.304, 3.176) (1.052, 3.911)
Hispanic	2023	1.00	0.95	(1.602, 2.440) (1.313, 2.914)
Hispanic	2023	1.00	1.00	1.913 (1.586, 2.234)

H.2 Crashes: Texas

Year	Race	Odds ratio					
		$\beta = 1$	$\beta = 0$	$\beta = 0.2$	$\beta = 0.4$	$\beta = 0.6$	$\beta = 0.8$
2017	Black	1.461 (1.399, 1.527)	–	–	8.617 (7.169, 10.359)	2.850 (2.678, 3.033)	1.85 (1.764, 1.93)
	Hispanic	1.287 (1.233, 1.343)	–	–	13.296 (10.895, 16.226)	3.479 (3.248, 3.726)	1.89 (1.800, 1.98)
2018	Black	1.351 (1.294, 1.410)	–	–	14.490 (11.805, 17.785)	2.611 (2.458, 2.772)	1.69 (1.621, 1.76)
	Hispanic	1.235 (1.185, 1.287)	–	–	24.884 (20.068, 30.857)	3.421 (3.201, 3.657)	1.82 (1.739, 1.91)
2019	Black	1.235 (1.190, 1.282)	21.432 (14.785, 31.067)	4.804 (4.413, 5.229)	2.116 (2.013, 2.224)	1.635 (1.566, 1.706)	1.38 (1.335, 1.43)
	Hispanic	1.027 (0.991, 1.065)	44.230 (30.113, 64.966)	7.199 (6.538, 7.926)	2.647 (2.506, 2.796)	1.740 (1.665, 1.818)	1.29 (1.244, 1.34)
2020	Black	1.151 (1.094, 1.212)	2.174 (1.979, 2.389)	1.682 (1.565, 1.808)	1.454 (1.366, 1.548)	1.317 (1.244, 1.395)	1.22 (1.158, 1.28)
	Hispanic	1.122 (1.070, 1.178)	5.021 (4.535, 5.559)	3.020 (2.804, 3.253)	2.141 (2.012, 2.279)	1.651 (1.562, 1.745)	1.33 (1.272, 1.39)

Table 17: Estimates of the causal risk ratio in Houston across every adjustment strength β ; see Equation (??). β shrinks the estimated misclassification matrix toward the identity: $\beta = 0$ applies the estimated matrix in full and $\beta = 1$ takes officer-reported race as true, so β is the share of officer misreports assumed away. Entries are point estimates with 95% intervals beneath. Entries marked – were not estimated: at low β the estimated matrix is too ill-conditioned in the earlier years to invert stably.